

TEST-1

Skin Cancer

PART A

TEXT A

Skin cancer (melanoma of the skin) is the third most commonly diagnosed cancer in males (after prostate and bowel cancer) and females (after breast and bowel cancer). In Australia. In 2016, an estimated 13,280 new cases of melanoma will be diagnosed in Australia, and 1,770 people will die.

In Australia, between 1982 and 2016, the number of cases of skin cancer rose from 27 cases per 100,000 to an estimated 49 cases per 100,000. However, how much of this increase is due to a real increase in the underlying disease, and how much is due to improved detection methods, is unknown. The incidence of melanoma of the skin rose at around 5.0% per year during the 1980s, moderating to 2.8% per year after that up until 2010.

It is predicted that the initial rapid increase is partly attributable to individual behaviour and the use of solariums, resulting in increased exposure to solar ultraviolet radiation. The moderated trend after the 1980s is consistent with increased awareness of skin cancer and improved sun protective behaviours as a result of extensive skin cancer prevention programs dating back to the 1980s.

Melanoma is a commonly used term for skin cancer. Melanoma of the eye and of the ano-rectal area can also occur.

TEXT B

Over exposure to ultra violet (UV) light causes 95% of melanoma Skin cancer be prevented through skin protection and early identification.

Skin protection includes:

- seeking shade especially during summer peak hours of 10am-4pm
- wearing clothing that covers back, shoulders, arms and legs
- wearing a broad-brimmed hat
- wearing wrap round sunglasses

Early identification involves checking moles and freckles for changes that fall into 5 categories of ABCDE :

- Asymmetry when one-half of a mole or birthmark does not match the other
- Border irregularity when the edges are irregular, ragged, notched, or blurred
- Colour variation when the colour is not the same all over, but may have differing shades of brown or black, sometimes with patches of red, white, or blue
- Diameter of the mole is larger than 6 mm (about the size of a pencil eraser) or is growing larger
- Evolving in size, shape, colour, elevation, or another trait (such as itching, bleeding or crusting). (This last point is likely the strongest of all of the warning signs)

Melanoma of the skin can develop anywhere on the body including:

- scalp – so check through hair
- ears and nostrils
- underarms
- hands and nail beds
- soles of the feet

Specialist mole check centres are increasingly available to carry out a thorough check for early changes and monitor at risk moles and spots. Seeking medical advice if any of the ABCDE signs appear is vital.

TEXT C

Sunscreen use

Three national surveys during summers between 2003-04 and 2010-11 show that sun protection compliance while outdoors on the weekend during peak UV radiation hours was relatively low.

Table 1 Trends in adolescents' weekend sun protection behaviours and sunburn

Table 1 Trends in adolescents' weekend sun protection behaviours and sunburn

Adolescents (12 – 17), n= 2,718	2003-04	2006-07	2010-11	Significant changes in 2010-11 %
Respondents outdoors > 15 minutes	80%	82%	77%	↓ since 2006-07
Time spent outdoors (minutes)	110	111	112	no change
Hat used	38%	29%	23%	↓ since 2003-04
Sunscreen used (at least SPF15+)	37%	37%	37%	no change
¾ or long sleeved top worn	11%	9%	11%	no change
¾ or long leg cover worn	37%	30%	28%	↓ since 2003-04
Sunglasses	23%	24%	24%	no change
Stayed mostly in the shade	19%	20%	21%	no change
Two or more sun protective behaviours	29%	22%	24%	↓ since 2003-04
Weekend sunburn	25%	24%	21%	↓ since 2003-04 & 2006-07

TEXT D

A staggering two in three Australians will be diagnosed with skin cancer before the age of 70, which is why it is so important that we learn about skin cancer and sun protection.

The major cause of skin cancer is overexposure to the sun's ultraviolet (UV) radiation. The more exposure you have over your lifetime, the greater your risk of cancer. Working outdoors will increase skin cancer risk, as will a history of severe sunburns and tanning.

Having a history of skin cancer in the family, a large number of moles, fair skin or red hair may also make a person more susceptible to developing skin cancer.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** The ABCDE guide to identifying mole changes of concern _____
- 2** The rise in the number of cases of skin cancer in Australia between 1982 and 2016? _____
- 3** What makes a person susceptible to skin cancer _____
- 4** Trends in sunscreen use among adolescents? _____
- 5** The age of most people who are diagnosed with skin cancer?

- 6** The most commonly diagnosed cancers among men and women in Australia? _____
- 7** The skin protection is needed to prevent skin cancer? _____

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** What are increasingly available to carry out thorough checks for early changes?

- 9** The use of solariums has resulted in the increased exposure to what type of radiation? _____
- 10** Surveys were carried during summers between 2003-04 and 2010-11 to check what sort of compliance over weekends _____
- 11** What other areas of the body apart from skin and eye can develop melanoma?

12 During surveys carried out of adolescent behaviours, what use was constant at 37% _____

13 What hours should a person seek shade in peak summer hours?

14 Working where will increase the risk of developing skin cancer?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 How much the increase in numbers of people with skin cancer is due to _____ methods, is unknown.

16 Over exposure to ultra violet (UV) light cause _____.

17 Having _____ skin or _____ hair may also make a person more susceptible to developing skin cancer.

18 The incidence of melanoma of the skin rose at around _____ per year during the 1980s.

19 Diameter of the mole is _____ (about the size of a pencil eraser) or is growing larger.

20 Sun protection compliance while outdoors on the weekend during peak UV radiation hours was _____.

Reading Practice - PART A

Answer Keys

Correct Answer		Your Answer	
Q 1	B		-
Q 2	A		-
Q 3	D		-
Q 4	C		-
Q 5	D		-
Q 6	A		-
Q 7	B		-

Correct Answer		Your Answer	
Q 8	mole check cent +		
Q 9	solar ultraviol +		
Q 10	sun protection		
Q 11	ano-rectal area		
Q 12	sunscreen		
Q 13	10am-4pm		
Q 14	outdoors		

Correct Answer		Your Answer	
Q 15	improved detect +		
Q 16	95% of melanoma		
Q 17	fair : red		
Q 18	5.0%		
Q 19	larger than 6mm		
Q 20	relatively low		

Review & Explanations

PART B

TEXT 1

Developing diagnostic tests for respiratory disease using sound measurement and machine learning techniques – clinical trial

This study is designed to develop accurate digital diagnostic tests, used on a smart device, for common respiratory illnesses in children and adults including asthma, croup, bronchiolitis, COPD and pneumonia. These tests can then be used in resource-poor communities, emergency departments or via telehealth applications.

The aim is to develop tests that are as accurate as an expert clinical assessment but do not need a clinical examination or other investigations such as x-rays to be performed.

Patients with a history of a chronic respiratory condition or who have symptoms of an acute respiratory disease are included.

Patients will be excluded if they: have an inability to provide a cough either spontaneously or voluntarily; are unable to provide informed consent or assent; have severe respiratory distress including the use of CPAP or BiPAP or have had abdominal or eye surgery within 3 months.

Question

1. The clinical trial is
 - a) open to people with severe respiratory symptoms
 - b) exploring new ways to confirm a person has a respiratory disorder
 - c) aimed at checking that expert clinical assessments are accurate

TEXT 2

Pregnancy care guideline 2018: Involving an interpreter

It is the responsibility of the health professional to make sure that communication is clear. Accredited healthcare interpreters assist by translating the discussion between the health professional and the woman, communicating with the woman in her preferred language either in person or through a telephone service. Involving an accredited interpreter, preferably with training in medical terminology, is recommended for all antenatal appointments if the health professional and the woman have difficulty communicating.

Interpreters accredited by NAATI (National Association of Accreditation for Translators and Interpreters) have been assessed as having a high level of technical competence in both English and one or more other languages and are bound by a code of ethics including strict confidentiality. However, there is a shortage of accredited interpreters, particularly for languages of new and emerging communities. While involvement of female interpreters is preferable in antenatal care, their availability may also be limited.

Question

2. The guideline highlights

- a) the need for accredited interpreters at all appointments if there is a language barrier
- b) that an accredited interpreter is recommended for all antenatal visits
- c) that health professionals are responsible for translating discussions

TEXT 3

Patient safety advice : medication

It's important you keep track of your medicines – taking the right medicine at the right time will help you get well. Using medicines in the wrong way may cause unwanted side effects. To be medicine-wise in hospital you need to tell staff if you have had an allergic or bad reaction to any medicines or if you have trouble swallowing medicines. You should tell staff straight away if you feel unwell after taking any medicine. Let staff know if you think you should have received some medicines, or the medicines appear different. It is important that all medicines are explained to you before you leave hospital to go home or to another care provider. Always ask your doctor, nurse, or pharmacist if you don't understand your medicine instructions. Questions you should ask about your medicines in hospital are "What is this medicine for?", "Are there any possible side effects?" and "Can they be taken safely with other medicines?"

Question

3. The patient safety leaflet encourages patients to
 - a) check the nurse understands the medicine guidelines
 - b) ask staff to clarify the directions to follow about taking their medications
 - c) be sure the doctors asks questions about your medicines

TEXT 4

Memo to staff re: Escalation process for dispute resolution

If, on review, the inpatient team disagrees that a patient requires admission to the respective specialty, the inpatient team shall make the onward referral to the alternate specialty. The senior inpatient MO shall provide the alternate specialty with a clinical handover and advice of the time frame in which the patient must be reviewed.

In the event of a dispute regarding inpatient team acceptance of the patient, a Consultant level case conference shall be convened. The case conference shall

include the ED, original inpatient and alternative inpatient Consultants. The three Consultants shall make a timely clinical decision regarding the patient's admission.

If the matter remains unresolved, then the case shall be escalated to the Executive Director of Medical Services (EDMS) for a clinical and time urgent decision.

Question

4. If the inpatient team cannot agree on accepting a patient who has to attend a case conference?
 - a) The senior MO from the alternative specialty, the ED consultant and the original inpatient team representative
 - b) The patient, the senior patient medical officer and the Executive director of Medical services
 - c) The patient's first consultant, a consultant from the ED and a consultant from the alternative speciality that the inpatient team suggests

TEXT 5

Taking a blood pressure reading manually

Wrap the cuff around the patient's arm and use the INDEX line to determine if the patient's arm circumference falls within the RANGE area. Otherwise, choose the appropriate smaller or larger cuff.

Palpate/locate the brachial artery and position the BP cuff so that the ARTERY marker points to the brachial artery. Wrap the BP cuff snugly around the arm. Ensure the cuff is located about 2.5 cm above the antecubital fossa (crease of the arm).

Perform a preliminary palpatory systolic blood pressure. To achieve this, palpate the arm at the antecubital fossa to locate the brachial artery with your non-dominant hand and ensure the pressure valve is closed. Inflate the bladder until no blood is flowing through the artery and no pulse is palpated. The pressure reading on the sphygmomanometer at this stage is an estimate of the maximum pressure required to measure the systolic pressure. Deflate the cuff and rest the arm for one to two minutes to allow the blood to be released and recirculate.

Questions 1-6

5. How is a preliminary palpatory systolic blood pressure check performed?
- a) Feeling for a pulse in the crook of the elbow, inflating the sphygmomanometer cuff until the pulse stops and noting the reading after a minute or two
 - b) Feeling for a pulse in the crook of the elbow, inflating the sphygmomanometer cuff until the pulse stops in the non-dominant hand
 - c) Feeling for a pulse in the crook of the elbow, inflating the sphygmomanometer cuff until the pulse stops and noting the reading

TEXT 6

GP Notify guidance for GPs

GP Notify is an automated notification system that informs GPs about: patient admission; patient discharge; patient death. When a patient is admitted, discharged or dies, the automated computer system will inform the patients nominated GP by email or fax of the update. GPs will receive an enrolment confirmation the first time such an event occurs. The form needs to be completed and faxed for automated notifications to continue. To receive notification via fax or email: you will receive an enrolment confirmation form the first time an event occurs on one of your patients. Once the form is completed and returned via fax, automated notifications will commence. If your contact details change, notify the GP Notify Coordinator. Please note that GPs are responsible for ensuring their details are correct and ensuring fax machines or computer systems provide adequate privacy after transmission.

Question

6. When a GPs patients are entered into the GP Notify system, the GP
- a) Needs to complete and return a confirmation form by fax to confirm they have received the information

- b) Gets notification via email or fax, unless it is the first time one of their patients has been entered into the GP notify system
- c) Faxes or emails the GP Notify Coordinator to confirm they have received the message

PART C

TEXT 1

Heart failure patients have weakened hearts, but researchers say an experimental drug used for the first time in humans may repair heart cells and improve heart function. According to the results of a small phase 1 trial, a single intravenous infusion of the drug cimaglermin was safe and, at high doses, improved heart function for at least three months.

"Right now we have many therapies that we use for heart failure, and these patients [in the study] were on all of those therapies and still had significant heart dysfunction," said lead researcher Dr. Daniel Lenihan. He's a professor of medicine and director of Vanderbilt University's heart clinical research program in Nashville.

People with heart failure often take a combination of drugs, Lenihan said. These include medications to lower blood pressure and diuretics to help remove excess fluid that builds up as a result of the heart's labored pumping ability. In addition, some people have implanted defibrillators or pacemakers. Even with all these options, the death rate among these patients is "unacceptably high," Lenihan said.

Heart failure, a condition where the heart can't pump enough blood to meet the body's needs, is among the leading causes of death worldwide. A significant number of heart failure patients don't respond well to current treatments, particularly those patients whose left lower heart chamber, which pumps blood into the arteries, is weak, Lenihan said.

Cimaglermin acts as a growth factor for the heart, helping it repair itself following injury, Lenihan said. Specifically, it binds to the HER2 and HER4 receptors on the surface of heart cells that are important for cellular repair and survival, he explained.

Researchers have tried using stem cells to repair heart muscle in much the same way, he said, but these efforts have not been effective. "You don't see any sustained effect," he added.

A phase 1 trial like this one is designed to see if a new drug is safe, not to test its effectiveness. Before cimaglermin could be used to treat patients, it must prove its worth in a series of progressively larger and challenging trials and then be approved by the U.S. Food and Drug Administration. The process can take several years. Based on these preliminary findings, larger trials are being planned, Lenihan said. "This

drug, although still in an experimental phase, might be an important way to improve heart function in patients with heart failure," he said.

For the study, Lenihan and his colleagues randomly assigned 40 patients to get an infusion of cimaglermin or a placebo. Compared with patients who received a placebo, patients given a high dose of cimaglermin had a sustained increase in the heart's ability to pump blood. The improvement lasted 90 days, with the maximum increase in heart function reached in 28 days, the researchers found. The most common side effects were headache and nausea directly after receiving the drug. One patient who received the highest dose of cimaglermin developed abnormal liver function, which cleared up over a two-week period, Lenihan said.

Despite the encouraging results of this first trial, a lot more testing will be needed before cimaglermin can be considered a standard treatment for heart failure, Bishopric said. "These findings need to be replicated in larger trials, and you have to be able to predict whether improved heart function from cimaglermin will help people live longer and feel better," she noted.

Questions 7-14

7. According to the first paragraph, cimaglermin
 - a) is safe when used in high doses
 - b) improves heart function indefinitely
 - c) is undergoing trials
 - d) is used in high doses

8. Which of the following statements does Dr. Lenihan make?
 - a) Patients who received cimaglermin were on many other therapies
 - b) Patients who received cimaglermin suffered from heart failure
 - c) Patients received cimaglermin at the Vanderbilt University
 - d) Patients who received cimaglermin were not on any other therapies

9. Which of the following therapies are not mentioned by Dr. Lenihan?

- a) Therapy to decrease high blood pressure
- b) Diuretics to get rid of excess fluid
- c) Defibrillators and pacemakers
- d) Therapy to increase low blood pressure

10. What is heart failure?

- a) A condition in which the lower heart chamber is weak
- b) A condition in which the left lower chamber pumps blood into the arteries
- c) A condition in which the blood pumped by the heart can't satisfy the body's requirements
- d) A condition for which there is no treatment

11. According to Dr Lenihan, cimaglermin

- a) replaces receptors on the heart's surface.
- b) helps to enhance the size of the heart.
- c) aids the heart's recovery process.
- d) has not been as effective as stem cells.

12. Which of the following considerations is not part of the approval for a new drug?

- a) Safety
- b) Effectiveness
- c) Challenging trials
- d) Planned trials

13. Which of the following results were achieved with cimaglermin?

- a) Cimaglermin didn't produce any results

- b) The effects of cimaglermin were adverse
- c) Cimaglermin improved the heart's ability to pump blood
- d) Cimaglermin produced several side-effects

14. Which of the following is the most suitable title for the article?

- a. A promising new drug treats heart failure
- b. A new heart failure drug shows promise in first human trial
- c. A new heart failure drug shows promise when tested on animals
- d. A promising new drug that treats heart failure has a bright future ahead

TEXT 2

After years of studies that seemed to swing between dire warnings and cheery promises about what our favourite caffeinated beverages do and don't do, much of the recent science regarding coffee and tea is generally positive.

The WHO's International Agency for Research on Cancer recently took coffee off its list of suspected carcinogens, and some research suggests it could help keep colon cancer from coming back after treatment. Other studies suggest drinking coffee might stave off Alzheimer's and Parkinson's diseases.

Various studies have pointed to tea drinkers having lower odds of skin, breast, and prostate cancers.

Researchers are still trying to pinpoint the exact ways that happens. But tea, particularly green tea, is rich in compounds like antioxidants, which can limit cell damage and boost the immune system; and polyphenols, which have been shown to lower blood pressure and cholesterol. It also may help stave off Alzheimer's disease through a polyphenol known as EGCG which prevents the formation of plaques that are linked to that brain-damaging illness.

Is one better for you than the other? Experts say that's hard to say. That's because it's difficult to separate out their different ingredients, their role in your diet, and their effects on different body systems. "I think people are looking at both coffee and tea and how they affect everything, including cancer and GI disease and cardiovascular diseases," says Elliott Miller, MD, a critical care medicine specialist at the National Institutes of Health.

Miller and his colleagues recently looked at signs of heart disease in more than 6,800 people from different backgrounds across the country. About 75% drank coffee, while about 40% reported drinking tea. Drinking more than one cup of tea regularly was linked to less buildup of calcium in arteries that supply blood to the heart, a development that can lead to heart disease. Coffee didn't have an effect either way on heart disease, but that was significant in itself, Miller says. "Very often patients will ask their doctors, 'Hey, doc, I've got coronary artery disease, or I've got risk factors like high blood pressure or cholesterol. Is it safe for me to drink coffee?'

Because everyone thinks drinking coffee makes your heart excited and is potentially bad,” Miller says.

“So finding that it’s neutral, I think, is pretty important.”

Researchers say it’s hard to pinpoint exactly how both drinks affect health. Both coffee and tea are “complex beverages” that contain a variety of ingredients. They include caffeine, polyphenols, and antioxidants -- compounds researchers are studying for their potential cancer-fighting properties, says Lisa Cimperman, a clinical dietitian at University Hospitals Case Medical Center. “It’s more of a dynamic interaction than one single compound,” Cimperman says. Some people have tried to isolate one element in tea or coffee that they think is the secret to one effect or another, “and then they realize that it doesn’t have the same effect.” Cimperman said drinking tea has been linked to lower risks of cancer and heart disease, improved weight loss, and a stronger immune system. Meanwhile, studies point to coffee as a potential way to head off not just Parkinson’s but type 2 diabetes, liver disease, and heart problems, Cimperman says.

Another recent study, led by Charles Fuchs, MD, director of the Gastrointestinal Cancer Center at Boston’s Dana-Farber Cancer Institute, found regular coffee drinking may help prevent colon cancer from coming back after treatment. In his study of nearly 1,000 patients, Fuchs says, there was a “significant and linear” association between drinking coffee and lower risk of colon cancer returning in those who drank four or more cups a day. “The more coffee they drank, the lower risk of recurrence.” But the researchers aren’t clear on which element of the drink contributed to that result, and there didn’t seem to be any effect from drinking tea, he says. “I think you can have two or more cups a day without any concern, and certainly that may benefit you,” Fuchs says. But what about for those who don’t drink coffee? “If it was somebody who hates the stuff and asks, ‘Should I drink it?’ I’d say no. I’d counsel them about diet and exercise and avoiding obesity as measures I think would have a similar benefit.”

Other researchers are asking questions about what role genetics and lifestyle play into the effects of drinking coffee or tea. For instance, coffee and cigarettes once went together like ... well, like coffee and cigarettes, which cause cancer and heart disease. Some people's bodies process coffee differently than others, says Martha Gulati, MD, head of cardiology at the University of Arizona College of Medicine in Phoenix. Meanwhile, a preference for tea over coffee might reflect other healthier behaviors, she says. "Does someone who drinks tea do yoga or meditation more?" Gulati says. "I'm not necessarily saying they're associated, but do they exercise more? Are they drinking things like green tea to maintain their weight better than other types of drinks?"

Questions 15-22

15. What did past studies suggest about coffee and tea?

- a) They produced clear evidence that they were harmful.
- b) They produced clear evidence that they were beneficial
- c) They produced evidence that they could be either harmful or useful.
- d) They didn't produce clear evidence on whether they are harmful or useful.

16. According to some research

- a) Coffee can be used as an effective treatment for Alzheimer's and Parkinson's diseases.
- b) Alzheimer's and Parkinson's diseases can be caused by coffee.
- c) Alzheimer's and Parkinson's diseases can effectively be prevented by coffee.
- d) Drinking coffee may help with the prevention of Alzheimer's and Parkinson's diseases.

17. Which of these effects is not attributed to drinking tea?

- a) Decreasing the risk of breast cancer.
- b) Increasing blood pressure.
- c) Strengthening the immune system.
- d) Decreasing the risk of developing Alzheimer's disease

18. What does fourth paragraph suggest?

- a) That it's easy to determine the positive effects of coffee and tea
- b) That it's not easy to determine the positive effects of coffee and tea.
- c) That it's not easy to determine which one is healthier: coffee or tea.
- d) That the positive effects of coffee and tea are determined by their role in our diet

19. What is Dr. Miller's opinion on the findings?

- a) Coffee increases the risk of heart disease in 75% of the people who took part in the research.
- b) Coffee has no effect on heart disease.
- c) The fact that coffee has no effect on heart disease is noteworthy.
- d) From now on, patients will stop asking doctors if coffee is safe to drink.

20. What makes coffee and tea beneficial, according to Cimperman?

- a) A secret ingredient found in coffee and tea.
- b) The relations among the complex ingredients coffee and tea contain
- c) One single compound isolated by some scientists
- d) The fact that they help to fight many different diseases

21. What does “linear” suggest in seventh paragraph?

- a) That the more coffee people drank the lower the risk of colon cancer was
- b) That coffee intake had no significant impact on colon cancer
- c) That drinking coffee is proportional to the reoccurrence of colon cancer
- d) That drinking coffee is proportional to the prevention of the reoccurrence of colon cancer

22. Which of the following points is not made in eighth paragraph?

- a) Coffee can affect people differently.
- b) People’s lifestyles may play a part in whether they drink tea or coffee.
- c) There is no connection between genetics and whether people drink tea or coffee.
- d) Research has shown that there is a direct connection between drinking coffee and smoking

TEST-2

Total Knee Replacement PART A

TEXT 1

Total knee replacement (TKR) surgery is a major but routine operation performed to reduce the effects of a damaged, worn or diseased knee with an artificial joint. Damaged cartilage and bone are removed from the surface of the knee joint and replaced with metal and plastic. The parts may be held in place with cement or new bone growth. It is generally considered a safe and effective procedure that can relieve knee pain and allow people to be more active. Osteoarthritis is the main reason for a TKR. Other reasons include rheumatoid arthritis, knee injury and knee deformity.

TKR is the last resort because it is a major operation and the artificial joint has a limited life expectancy. Medication, physiotherapy and lifestyle changes are the first choices for managing the knee symptoms. Smaller operations may be considered to delay the need for a TKR. Options include: arthroscopic washout and debridement; osteotomy to realign the knee joint and mosaicplasty to repair the damaged bone surface. A partial knee replacement (PKR) where only one side of the joint is replaced in a smaller operation, is an alternative option to full joint replacement.

The life expectancy of a TKR is about 20 years. Any adult can be considered for a knee replacement, but most people are aged between ages of 60 and 80. Increasingly, younger people are having TKR as it improves their quality of life and their ability to carry on working. However, this also increases the likelihood that parts of the artificial joint will wear out or the joint will loosen. This means further surgery may be required and a revision of a TKR is a more complex procedure and the patient may not be in good health.

TEXT 2

Clinical assessment

- Complete a thorough medical history check to ensure fitness for anaesthetic and rehabilitation
- Knee x-rays
- Physiotherapy assessment of mobility and gait
- Check for symptoms of:

1. severe pain, swelling and stiffness in knee joint
2. mobility is reduced
3. knee 'giving way'
4. knee locking or clicking
5. inability to straighten the knee
6. interference with quality of life and sleep
7. difficulty with, or unable to carry out everyday tasks, such as shopping or getting out of the bath
8. feeling depressed because of the pain and lack of mobility
9. unable work or have a normal social life

TEXT 3

Potential complications

- Infection and blood clots are the most common complications following knee replacement surgery.
- Infection of the wound which is usually treated with antibiotics, but occasionally the wound can become deeply infected and require further surgery; in rare cases it may require replacement of the artificial knee joint.
- The operated leg will be bruised but patients are advised to seek medical help if they find the leg is becoming hot, reddened, hard or painful. Also patients are advised to seek help if they experience chest pains as this may indicate a pulmonary embolism.

Other potential complications include:

- unexpected bleeding into the knee joint
- ligament, artery or nerve damage in the area around the knee joint
- fracture in the bone around the artificial joint during or after surgery
- excess bone forming around the artificial knee joint and restricting movement of the knee
- excess scar tissue forming and restricting movement of the knee
- the kneecap becoming dislocated
- numbness in the area around the wound scar
- allergic reaction to the bone cement

TEXT 4

Post-operative care

Post-operative care focusses on pain relief, deep veined thrombosis prevention and early knee mobility. Bolus analgesia is given during surgery and opioid analgesia is administered using a patient controlled analgesia system (PCAS) for 24-48 hours post

operatively. Oral medication is commenced as soon as possible. Non-steroidal anti-inflammatory drugs are not given as this impacts on the bone healing process.

Deep veined thrombosis prevention options include use of an intermittent foot sole pump, administration of low molecular weight heparin and wearing anti-embolic stockings. Most patients are mobilised with two crutches or a walking frame within 12-18 hours of surgery.

Knee flexion and extension exercises are started as soon as possible, usually on post-operative day one. Physiotherapy exercises are encouraged several times a day to increase muscle strength and gain knee mobility. The knee will be very swollen at this stage so movement will be minimal. Analgesia before exercising is recommended. Some patients need to use a continuous passive motion (CPM) machine to start getting flexion back in their knee.

At discharge, most patients are able to:

- bend their knee well, preferably to a minimum of a 90-degree angle
- walk 10 meters and go up and down stairs with a walking aid
- manage their own personal care with minimal assistance e.g. a shower stool to reduce standing while showering

Timeline to full recovery

Timeline	Activity
Week 1–3	Walking and taking stairs for longer distances without an assistive device. Continuing with exercises to increase strength and range of motion.
Week 4–6	Walking and taking stairs independent of assistive device. Resuming household chores and activities. Returning to work and driving. Continuing with exercise to increase or maintain strength and range of motion.
Week 7–12	Walking for longer periods, using a stationary bike, and continuing prescribed exercises at home. Returning to low-impact activities.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1 The symptoms to check for during clinical assessment? _____

2 The timeline to full recovery? _____

3 Smaller operations as alternatives to knee replacement? _____

4 Post operative care following knee replacement? _____

5 Potential complications? _____

6 The reasons a patient needs to seek medical help after discharge?

7 What a total knee replacement is? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What type of analgesia is administered using a patient controlled analgesia system (PCAS)? _____

9 What do the initials PKR stand for? _____

10 What are the most common complications following knee replacement surgery?

11 During the clinical assessment, what does a Physiotherapist assess?

12 What is the main reason a TKR is carried out for? _____

13 What is the minimum knee bend a patient should have on discharge from hospital? _____

14 During the clinical assessment, a patient's medical history is checked to ensure they are fit for what? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 Deep veined thrombosis prevention options include use of an _____ pump.

16 A person may report knee _____ or clicking during the clinical assessment

17 The parts of an artificial knee joint may be held in place with _____ or new bone growth

18 On discharge, most patients are able to walk _____ and go up and down stairs with a walking aid

19 _____ tissue forming and restricting movement of the knee is a potential complication following knee replacement surgery.

20 _____ and lifestyle changes are the first choices for managing the knee symptoms

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	B	-	Q 8	opioid		Q 15	intermittent fo	
Q 2	D	-	Q 9	partial knee re	+	Q 16	locking	
Q 3	A	-	Q 10	infection, blo	+	Q 17	cement	
Q 4	D	-	Q 11	mobility, gait		Q 18	10 meters	
Q 5	C	-	Q 12	osteoarthritis		Q 19	Excess scar	
Q 6	C	-	Q 13	90 degree angle		Q 20	Medication, phy	+
Q 7	A	-	Q 14	anaesthetic; re	+			

Review & Explanations

PART B

TEXT 1

Risks associated with rubella infection in pregnancy

Maternal rubella infection can result in spontaneous miscarriage, fetal infection, stillbirth, or fetal growth restriction (Reef et al 2000). Congenital infection is most likely if the maternal infection occurs in the first 16 weeks of pregnancy, with congenital rubella syndrome occurring in all fetuses infected before the 11th week and in 35% of those infected at 13–16 weeks (Miller et al 1982). If infection occurs after 16 weeks of pregnancy, the risk of fetal damage is negligible. Features of congenital rubella syndrome include cardiac defects, deafness, ocular defects, thrombocytopenic purpura, haemolytic anaemia, enlarged liver and spleen, and inflammation of the meninges and brain (Sanchez et al 2010). Pneumonitis, diabetes, thyroid dysfunction and progressive panencephalitis are other late expressions of the syndrome (Weil et al 1975; Cooper et al 1995).

Question

1. If a baby is born with visual problems, what stage of pregnancy was it most likely to have been infected with rubella?
 - a) As all babies infected with rubella in their first 11 weeks develop congenital rubella syndrome, the first trimester is when the infection most likely occurred.
 - b) As over a third of all foetuses infected with rubella in weeks 13-16 develop congenital rubella syndrome, the second trimester is when the infection most likely occurred.
 - c) As all babies infected with rubella after their first 16 weeks do not develop congenital rubella syndrome, the third trimester is when the infection most likely occurred.

TEXT 2

Memo to staff re: presence of latex in Fluad trivalent influenza vaccine

During the 2018 influenza season, natural rubber latex is present in the sheath covering the needle of trivalent influenza vaccine Fluad. Fluad is available for people aged 65 years and over through the National immunisation Program (NIP). While reactions to latex are rare, anyone who has a severe allergy to latex should not receive Fluad. Patients aged 65 years and over can still be safely vaccinated with the alternative trivalent influenza vaccine, Fluzone High-Dose through the NIP.

Please note there are no safety concerns regarding the Fluad vaccine itself.

The sponsor of Fluad, Seqirus, commits to update the Product Information and Consumer Medicines Information for this vaccine accordingly.

Action for vaccination providers:

If you are vaccinating patients with Fluad, please be aware of this issue and advise patients accordingly. Reassure patients that reactions to latex are rare.

Before administering Fluad, confirm with patients that they are not allergic to latex.

Anyone who has a severe allergy to latex should not receive Fluad. Patients aged 65 years and over can still be safely vaccinated with the alternative trivalent influenza vaccine, Fluzone High Dose, available through the NIP.

As with all vaccinations, be prepared to treat immediate allergic reactions, including potential anaphylaxis.

Question

2. The statement below highlights that
 - a) latex is being found in the Fluad vaccine itself
 - b) the Fluad vaccine itself contains latex
 - c) latex is not being found in the Fluad vaccine itself

TEXT 3

Spray on skin for diabetic foot ulcers: an open label randomised controlled trial

A novel intervention towards achieving these goals is the use of spray-on skin for patients with DFU. The application of 'spray-on' autologous skin grafting aids epithelial regeneration and wound healing and has been used successfully for the treatment of burns to improve healing. In this research project we will compare spray-on skin with standard care in an open label randomised controlled trial in patients presenting to hospital with DFU. We hypothesise that spray-on skin will shorten the time for the ulcer to heal completely. In doing so, this approach can be expected to prevent amputations and recurrent ulceration whilst improving quality of life.

Outpatient costs for dressings, home nursing visits and outpatient appointments are key cost drivers for DFU. If spray-on skin is effective, large cost savings to WA Health will be realised immediately through a shortened time to healing, and through a higher proportion of patients achieving complete healing. Any economic benefits are likely to be amplified across Australia and other similar demographic settings where aging populations with increased diabetes rates are considered major future challenges.

Question

3. The aim if this trial is to
 - a) check if burns heal at the same rate as diabetic foot ulcers when the same dressing is used
 - b) see if a spray-on skin dressing saves money and improves patient outcomes
 - c) see if patients getting care in outpatient departments heal quicker than those having care provided in their own home

TEXT 4

Injection technique

As the amount of subcutaneous fat varies between patients, individual patient assessment is vital before carrying out the procedure. It is important to avoid inadvertently injecting the drug into muscle, as intramuscular injection can affect drug absorption; for example, inadvertent administration of insulin into the muscle can lead to accelerated insulin absorption and lead to hypoglycaemia.

A lifted skinfold technique (pinching or bunching the skin) can be used to lift the subcutaneous layer away from the underlying muscle. This method reduces the risk

of inadvertent intramuscular injection when undertaken correctly; however, releasing the skin too quickly before the injection is completed or lifting it incorrectly can increase that risk.

Needles need to be long enough to inject the drug into the subcutaneous tissue. They come in lengths of 5cm, 6cm and 8cm. It is suggested the required needle length can be estimated by pinching the skin using the lifted skinfold technique and selecting a needle that is 1.5 times the width of the skinfold.

Question

4. What does this information on injection technique tell us about the lifted skinfold technique
 - a) the lifted skinfold technique allows for a suitable needle choice to be made
 - b) the lifted skinfold technique ensures medication to be administered into the muscle layer
 - c) the lifted skinfold technique involves releasing the skin quickly to reduce the risk of bruising

TEXT 5

Discharge medications

The Discharge Prescription must be completed by the Medical Officer with reference to the current medication chart. The discharge medications section of the Electronic Discharge Summary (EDS) should be used for this process with the prescription being printed and then forwarded to the Hospital Pharmacy. The EDS Discharge Prescription, or the Discharge Medication form, must be forwarded to Pharmacy at least 1 hour prior (or 3 hours for complicated discharges) to the patient being discharged from hospital. If amendments or corrections are made after sending the EDS Discharge Prescription to Pharmacy, it is the responsibility of the Medical Officer who completed the EDS Discharge Prescription to make the amendments to the Discharge Summary within the EDS as soon as possible to ensure the GP receives accurate information regarding their patient's medications on discharge.

Questions 1-6

5. What does this article tell us about the discharge medication dispensing process?
- a) All prescriptions have to be at the Hospital pharmacy an hour before the patient is due to go home
 - b) The prescription goes to the hospital pharmacy after being written up by the medical officer
 - c) The Medical Officer must let the Hospital Pharmacy know if there are any amendments or corrections to be made

TEXT 6

Report on a patients unexpected death

Staff working in the mental health ward of the Townsville Hospital have been sent a directive about correct clinical restraint procedure after a patient died unexpectedly.

A 44-year-old man died in the high dependency unit of the acute mental health ward at about 9pm on July 7.

It is believed the man may have died from a heart attack as a male security guard and female nurse attempted to restrain him.

The staff memo said it was “essential” that when a restraint occurs in an inpatient setting, best practice guidelines and procedures were followed.

The memo also said it was “essential” that the shift co-ordinator was physically present during all restraint and provide clinical over sight to ensure the safety of the patient team.

A source said staff in the hospital had also been advised having five people to restrain a patient was best practice.

Question

6. In this article what were staff on the ward directed to do?
- a) Clinical restraint is essential in acute mental health wards.

- b) When restraint occurs five people need to be present to assist, one of whom must be the shift coordinator.
- c) There must be a shift coordinator available and correct processes must be followed whenever clinical restraint is needed

PART C

TEXT 1

Surgery that preserves the lung, when combined with other therapies, appears to extend the lives of people with a subtype of the rare and deadly cancer mesothelioma, a new study suggests. Tracking 73 patients with advanced malignant pleural mesothelioma -- which affects the lungs' protective lining in the chest cavity - researchers found that those treated with lung-sparing surgery had an average survival of nearly three years. A subset of those patients survived longer than seven years. Mesothelioma patients treated with chemotherapy alone, which is standard care, live an average of 12 to 18 months, the researchers said.

Study participants received lung-sparing surgeries and another treatment called photodynamic therapy that uses light to kill cancer cells. Ninety-two percent of the group also received chemotherapy. The study volunteers achieved far longer survival times, said study author Dr. Joseph Friedberg. "When you take the [entire] lung out, it's a significant compromise in quality of life," said Friedberg. He's director of the University of Maryland Medical Center's Mesothelioma and Thoracic Oncology Treatment and Research Center in Baltimore.

"For all intents and purposes, this [lung-sparing surgical approach] is the largest palliative operation known to man, since chances of curing mesothelioma are vanishingly small," said Friedberg. He completed the research while at his previous post at the University of Pennsylvania. "Plus, most of these patients are elderly, so preserving quality of life was really the goal," he added.

About 3,000 Americans are diagnosed with mesothelioma each year, the American Cancer Society says. Many of these people were exposed to the mineral asbestos in industrial occupations, according to the U.S. National Cancer Institute (NCI). Used in products such as insulation, building shingles and flooring, asbestos dust fibers can be inhaled or swallowed, settling in the lungs, stomach or other body areas. Often, it takes decades after exposure for mesothelioma to develop, the NCI says.

Friedberg and his team performed the lung-sparing surgeries on study participants between 2005 and 2013. Overall average survival was 35 months, the study showed. But survival time more than doubled to 7.3 years for 19 patients whose cancer had not spread to their lymph nodes. Most of the patients in the study had stage 3 or stage 4 cancer. Typically, Friedberg said, only about 15 to 20 percent of mesothelioma patients are treated with surgery, which often removes an entire lung as well as the diaphragm and the sac surrounding the heart.

Friedberg said that between 20 and 40 percent of pleural mesothelioma patients with the epithelial subtype might be eligible for lung-sparing surgery. He explained that this surgery removes all visible traces of cancer. It typically has fewer complications and a lower risk of dying in the 90 days following the 10- to 14-hour procedure.

"It's still relatively new that people do lung-sparing surgery for this disease, and it's not established that this is what we need to do," said Friedberg. "I would say this is one of the most lethal cancers known to man. There's a pressing need for new and innovative treatments," he noted.

Dr. Daniel Petro, a medical oncologist/hematologist at the University of Pittsburgh Medical Center, said lung-sparing surgery for mesothelioma is also done at academic centers such as his, and he was not surprised by the study's results. "This [surgical approach] is a step forward with this particular terrible cancer," Petro said, "and we've got to keep coming up with better options to eradicate it."

Questions 7-14

7. According to the information in first Paragraph

- a) Mesothelioma patients have benefited from chemotherapy alone
- b) Mesothelioma patients have benefited from lung-sparing surgery alone
- c) Mesothelioma patients have benefited most from lung-sparing surgery and additional therapies
- d) Only a subset of mesothelioma patients can survive

8. Why is lung-sparing surgery better than removing the entire lung?

- a) Because patients survive longer
- b) Because patients have a better life
- c) Because patients' lives are compromised
- d) Because cancer cells are completely eradicated

9. What does "palliative" refer to in third Paragraph

- a) Soothing
- b) Life-saving
- c) Routine
- d) Delicate

10. Which of the following are involve the use of asbestos?

- a) Insulation
- b) Dust fibers
- c) Building shingles
- d) Flooring

11. The patients whose cancer had not spread to their lymph nodes

- a) Were eligible for the lung-sparing surgery
- b) Lived 7.3 years longer than the other patients
- c) Lived more than double the average expectancy
- d) Were among the 15 to 20% of the patients treated with surgery

12. Lung-sparing surgery

- a) Can be done on all mesothelioma patients.
- b) Can be done on less than a quarter of the patients.
- c) Can be done on less than half of mesothelioma patients.
- d) Can be done on 40 % of mesothelioma patients

13. In seventh paragraph, Friedberg draws the reader's attention to

- a) All the positive effects of lung-sparing surgery.
- b) The fact that mesothelioma is a deadly cancer.
- c) The fact that there is a new and highly effective treatment.
- d) The necessity of implementing ground-breaking treatments

14. Dr. Daniel Petro believes that the study

- a) Is a step forward towards the discovery of a new treatment.
- b) Confirms his previous research.
- c) Produced results he would have expected
- d) Will contribute to eradicating mesothelioma.

TEXT 2

People with the reading disability dyslexia may have brain differences that are surprisingly wide-ranging, a new study suggests. Using specialized brain imaging, scientists found that adults and children with dyslexia showed less ability to "adapt" to sensory information compared to people without the disorder. And the differences were seen not only in the brain's response to written words, which would be expected. People with dyslexia also showed less adaptability in response to pictures of faces and objects. That suggests they have "deficits" that are more general, across the whole brain, said study lead author Tyler Perrachione. He's an assistant professor of speech, hearing and language sciences at Boston University. The findings, published in the Dec. 21 issue of the journal *Neuron*, offer clues to the root causes of dyslexia.

Other studies have found that people with dyslexia show differences in the brain's structure and function. "But it hasn't been clear whether those differences are a cause or consequence of dyslexia," Perrachione explained. The chicken-and-egg question is tricky, because years of reading, or years of reading disability, affect brain development. Perrachione said his team thinks it has discovered a cause of dyslexia - partly because the reduced adaptation was seen in young kids, and not only adults.

A researcher who was not involved in the study called it "groundbreaking." "Frankly, researchers have struggled with understanding the brain bases of dyslexia," said Guinevere Eden, director of the Center for the Study of Learning at Georgetown

University Medical Center in Washington, D.C. Scientists have known that brain structure and function look different in people with dyslexia, Eden said, but they haven't known why. "This study makes an important step in that direction," she said. "It gets to the true characteristics of the properties of the neurons [cells] in these brain regions, not just their outward appearance."

People with dyslexia have consistent problems with language skills, especially reading. According to the International Dyslexia Association, as much as 15 percent to 20 percent of the population has symptoms of dyslexia -- including "slow" reading, poor spelling and writing skills, and problems deciphering words that are similar to each other.

The new study aimed to see whether "neural adaptation" might play a role. Adaptation is how the brain improves its efficiency. Perrachione offered an example: When you speak to someone for the first time, the brain needs a little time to get used to that person's voice, speaking rhythms and pronunciation of words, for instance.

But then the brain adapts and stops working so hard to process the other person's speech. In people with dyslexia, however, that adaptation seems to be hindered. "Their brains are working harder to process these sensory inputs," Perrachione said.

The new findings are based on functional MRI scans of adults and children with and without dyslexia. The scans were used to capture the study participants' brain activity as they performed a series of tasks. In one experiment, the participants listened to a series of words, read either by a single speaker or several different ones. Overall, the researchers found, people without dyslexia adapted to the single voice, but not to multiple speakers. In contrast, people with dyslexia showed much less adaptation in their brain activity, even when listening to a single speaker. The same pattern was seen when study participants viewed written words.

But the differences went beyond words: People with dyslexia showed less brain adaptation in response to images of faces and objects. That's "surprising," Eden said, since the disorder does not involve apparent problems with recognizing faces or objects. Perrachione speculated on a reason for the findings: The reduced brain adaptation may only "show up" when it comes to reading, because reading is such a complex skill. The brain has no dedicated "reading" area. "Reading is a tool, or technology, that we've invented," Perrachione pointed out. Learning to use that technology requires a complex orchestration of different brain "domains," he explained. And yet, because everyone is expected to read, most people probably do

not realize what an accomplishment it is, Perrachione said. Eden agreed. "Learning to read is an astonishing feat and one that we often take for granted," she said.

Questions 15-22

15. The author describes dyslexic people's brain differences as

- a) More varied than many people realize
- b) Easier to understand than people expect
- c) having wide-reaching effects
- d) Absolute

16. What new information did the study reveal?

- a) That their brains are more plastic.
- b) That their ability to recognize faces and objects is underdeveloped.
- c) It confirmed that they have problems recognizing written words.
- d) That their brains are less flexible in several ways not considered previously

17. What is implied by the chicken-and-egg question in Paragraph 2?

- a) The impossibility of discovering the true cause of dyslexia.
- b) The fact that many studies constantly reveal various findings about the differences in the brain structure
- c) The fact that it is impossible to identify whether alternations are the cause or consequence of dyslexia.
- d) The fact that the brain's structure and function are both the cause and consequence of dyslexia.

18. Why has the study been called "groundbreaking"?

- a) Because previously scientists hadn't been able to understand the brain bases of dyslexia.
- b) Because so far scientists couldn't understand as before why dyslexic brains are different.
- c) Because it reveals the brain basis of dyslexia.
- d) Because it makes progress in understanding the basis of dyslexia in the human brain

19. What is the main problem with people with dyslexia according to Perrachione?

- a) There is no neural adaptation.
- b) Their brains take much longer to adapt.
- c) Adaptation occurs instantly.
- d) There is no sensory adaptation

20. What does the experiment in seventh paragraph reveal?

- a) That people with dyslexia experience difficulties when listening to multiple speakers.
- b) That it was difficult for both people with and without dyslexia to adapt to multiple speakers.
- c) That people with dyslexia have difficulty processing listening as well as written words.
- d) That brains of people with dyslexia have difficulty adapting to listening as well as reading.

21. When is dyslexia most apparent?

- a) During the process of reading.
- b) In face recognition.
- c) In face and object recognition.
- d) When reading is used as a tool or technology

22. Which of the following is not true about reading?

- a. It is a very complex skill.
- b. It can be located in one centre of the brain
- c. It requires communication among different areas in the brain.
- d. It is an amazing achievement

TEST-3

Clinical Depression PART A

TEXT 1

Clinical depression is a prolonged period of feeling sad, hopeless and lacking interest in activities that used to be enjoyable. It is normal for people to experience these symptoms at some stage, but if they are continuous for more than two weeks, then it is likely that the person is suffering from depression.

Levels of depression are considered to be mild, moderate and severe. A person with severe depression may need admitting to hospital for treatment and to reduce the risk of self-harm.

Suicidal thoughts may be experienced by someone with clinical depression, especially if it is severe.

TEXT 2

Clinical assessment

- Carry out a detailed mental health assessment
- Check for psychological symptoms and the duration they have been experienced for
- Symptoms include
 1. feeling sad, down or empty
 2. reduced enjoyment from normal activities
 3. becoming withdrawn from friends
 4. feelings of worthlessness or guilt
 5. suicidal thoughts
 6. tearfulness for no reason
- Check for physical symptoms which include:
 1. low energy, fatigue and reduced activity
 2. difficulty sleeping
 3. increased alcohol intake
 4. loss of appetite or weight
 5. trouble concentrating
 6. slowed thinking
 7. hand-wringing
 8. digestive problems

Management

- Consider blood tests to eliminate anaemia and hypothyroidism

- For mild depression
 1. encourage joining a self-help group
 2. starting or increasing a programme of exercise

For mild to moderate depression

1. advise options recommended for mild depression
 2. refer for counselling for cognitive behaviour therapy (CBT)
- For moderate to severe depression
 1. anti-depressant medication
 2. CBT
 3. talking therapies
 4. referral to mental health service

TEXT 3

Medication options

- Selective serotonin reuptake inhibitors (SSRIs) help increase the level of a natural “good mood” chemical in the brain called serotonin.
- Tricyclic antidepressants (TCAs) work by raising the levels of the chemicals serotonin and noradrenaline in the brain to help lift the mood.
- Serotonin-noradrenaline reuptake inhibitors (SNRIs) change the levels of serotonin and noradrenaline in your brain
- Lithium carbonate or lithium citrate can be used if all other medication options have not worked

Medication withdrawal symptoms

- Medications should not be stopped suddenly unless there are bad side effects
- Medications should be reduced gradually. This is to prevent withdrawal symptoms that include:
 1. an upset stomach
 2. flu-like symptoms
 3. anxiety
 4. dizziness
 5. vivid dreams at night
 6. sensations in the body that feel like electric shocks

TEXT 4

Non-medication options

- A talking therapy is very helpful for most people with clinical depression.

1. Cognitive behavioural therapy (CBT) helps people understand their thoughts and behaviour. It teaches people how to stop negative thoughts. It can be done online or sitting with a therapist.
2. Interpersonal therapy (IPT) focuses relationships with others and on problems the person may be having, such as coping with illness or not getting on with a family member.
3. Psychodynamic or psychoanalytic psychotherapy encourages a person to say whatever is going through your mind. This helps them become aware of hidden meanings or patterns and the effect they have on the person's well being.
4. Counselling is a form of therapy that helps a person think about their problems and come up with new ways of dealing with them.
5. Mindfulness involves paying closer attention to the present moment, and focusing on thoughts, feelings, bodily sensations and the world around to improve a person's mental wellbeing.

Electroconvulsive therapy (ECT) is a very controversial, invasive type of brain stimulation that's sometimes recommended for severe depression if all other treatment options have failed, or when the situation is thought to be life-threatening. It is always carried out in a specialist hospital unit by trained medical staff.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1 mindfulness? _____

2 physical indications of depression? _____

3 therapy looking at interactions with others? _____

4 pharmacological support? _____

5 the most effective therapies? _____

6 blood tests that might be done? _____

7 the effect of stopping medication abruptly? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What does ECT stand for? _____

9 What might cause vivid dreams at night? _____

10 What type of gathering that might help someone with depression?

11 What two conditions that need a blood test to confirm if they are present?

12 What type of ideas might a person with clinical depression have?

13 What type of non-medicine treatment teaches people how to consider the here and now? _____

14 What group of symptoms of depression include the person feeling worthless or guilty? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 Difficulty_____and increased_____intake are possible physical signs that someone is suffering from depression.

16 A person with extreme signs of depression may need admitting_____for treatment.

17 A person may be clinically depressed if they are feeling sad all the time for_____weeks or longer.

18 Psychodynamic psychotherapy helps a person be more sensitive to hidden_____and how they make them ill.

19 Tricyclic antidepressants (TCAs) increase_____and_____to help the person's frame of mind improve.

20 During a clinical assessment the doctor conducts a thorough review of the persons_____

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	D	-	Q 8	Electro Convuls +		Q 15	sleeping, alcoh +	,
Q 2	B	-	Q 9	Stopping medica +		Q 16	to hospital	
Q 3	D	-	Q 10	Self-help group		Q 17	two	
Q 4	C	-	Q 11	Anaemia and hyp +		Q 18	meanings or pat +	
Q 5	D	-	Q 12	Suicidal		Q 19	serotonin; nora +	,
Q 6	B	-	Q 13	Mindfulness		Q 20	mental health	
Q 7	C	-	Q 14	Psychological				

Review & Explanations

PART B

TEXT 1

Device recall: Firmware Update to Address Cybersecurity Vulnerabilities Identified in Abbott's (formerly St. Jude Medical's) Implantable Cardiac Pacemakers: FDA Safety Communication for Health professionals

The FDA has reviewed information concerning potential cybersecurity vulnerabilities associated with St. Jude Medical's RF-enabled implantable cardiac pacemakers and has confirmed that these vulnerabilities, if exploited, could allow an unauthorized user to access a patient's device using commercially available equipment.

The FDA and Abbott do NOT recommend prophylactic removal and replacement of affected devices.

Discuss the risks and benefits of the cybersecurity vulnerabilities and associated firmware update with your patients at the next regularly scheduled visit. As part of this discussion, it is important to consider each patient's circumstances, such as pacemaker dependence, age of the device, and patient preference, and provide them with Abbott's Patient Guide.

For pacing dependent patients, consider performing the cybersecurity firmware update in a facility where temporary pacing and pacemaker generator can be readily provided.

Print or digitally store the programmed device settings and the diagnostic data in case of loss during the update.

After the update, confirm that the device maintains its functionality, is not in backup mode, and that the programmed parameters have not changed.

The firmware update process is described in Abbott's Dear Doctor Letter issued on August 28, 2017.

Question

1. What course of action must a healthcare professional take after reading this device recall notice?

- a) The health professional should contact their patient so they can discuss issues raised in the recall notice
- b) The health professional should discuss the pros and cons resulting from the firmware update at their next routine appointment
- c) The health professional should discuss the possible effects of the firmware update and schedule replacing the device as soon as is convenient.

TEXT 2

Research into incidence of hospital treated Deliberate Self Harm

The incidence of hospital-treated DSH from institutional data was higher in females than males, both in Australia (148 vs 87per 100,000) and in New Zealand (86 vs 46per 100,000).

The incidence of hospital-treated DSH was highest among those aged 15–24 years in both

Australia and New Zealand predominantly due to higher rates in adolescents (15–19years). However, the peak of highest incidence of hospital-treated DSH in adolescents

and young adults in Australia is less marked when stratified by Aboriginal and Torres Strait

Islander status in Australia; rates of DSH among Aboriginal and Torres Strait Islander people aged 25–49years (males and females) are higher than for other Australians in the same age group. In Australia, the incidence of hospital-treated DSH among Aboriginal and Torres Strait Islander people was substantially higher than for the general population (292 vs 117per 100,000), males (259 vs 87 per 100,000) and females (325 vs 148per 100,000)

Question

2. According to this research report which group of the Australian population are most likely to be treated in hospital for deliberate self-harm?
 - a) Women aged 15-19 from the Aboriginal and Torres Strait Island population

- b) Women aged 15-24 from the Aboriginal and Torres Strait Island population
- c) Women aged 25-49 from the Aboriginal and Torres Strait Island population.

TEXT 3

Article on clinical inertia in diabetes

The term “clinical inertia” implies that the provider is to blame when a patient does not receive adequate treatment intensification. In reality, multiple factors related to physicians, patients, and systems contribute to this phenomenon. Intensifying diabetes treatment can require multiple oral and injected medications, fingerstick glucose monitoring, and significant lifestyle modifications. These patients also often have numerous comorbidities, which require additional medications. These circumstances can lead to distress and resistance to new medications, particularly to injectables (e.g., insulin, glucagon-like peptide 1 [GLP-1] receptor analogs), which many initially refuse to use. Rising out-of-pocket costs of diabetes medication can lead patients and providers to feel reluctant to initiate new medications. Patients may thus prefer to first try lifestyle modification as a response to elevated blood sugars.

Question

3. What reasons does the article give for patients preferring lifestyle changes if their blood sugars are raised?
 - a) Ongoing suffering with existing illnesses, opposition to injectables and increased expense.
 - b) Lack of engagement with injectables and lack of understanding of the rationale for using them
 - c) Pain from using injectables and a reluctance by the physician to start new treatment.

TEXT 4

Planning for subsequent antenatal visits

At all visits, opportunities should be provided for the woman to share her expectations and experiences as well as discuss any issues and/or concerns that may have arisen since her last visit, including psychosocial support and mental health issues. Women should also be offered information on aspects of health in pregnancy and early parenthood (e.g. nutrition, alcohol, smoking, symptom relief if conditions common in pregnancy are being experienced, breastfeeding, reducing the risk of sudden and unexpected death in infancy [SUDI]). A woman's confidence in her ability to labour, give birth and look after her new baby should be supported throughout antenatal care and antenatal education should also support her in preparing for changes to her life and her relationship with her partner and understanding the physical and emotional needs of the baby. The woman's needs should dictate the type of information and support provided (e.g. while many women will benefit from written information, other forms of information such as audio or video are sometimes more suitable). The woman should also direct the type of issues and questions discussed.

Question

4. What are the main topics women should be given information on during their ante natal period?
 - a) Managing her expectations, knowing which babies are at risk of SUDI, and how to be confident in her ability as a new Mum
 - b) Managing her health during and after pregnancy, life changes, partnership relationships and baby's needs.
 - c) Managing psychosocial and mental health issues, understanding labour and giving birth.

TEXT 5

Case study on Community acquired pneumonia

Mr Carbury's signs and symptoms suggested that he had community-acquired pneumonia (CAP). The definition of CAP varies in the literature but includes signs and symptoms related to the respiratory tract and the patient's general health (British Thoracic Society, 2009). Mr Carbury's baseline vital signs were recorded as part of a holistic assessment; his RR was at the upper end of normal – 20 breaths per minute (bpm) measured over a full 60 seconds, the depth of his breaths was shallow and the rhythm was regular, and his oxygen saturation (SpO2) was 93%. He was sitting on a

chair but leaning slightly forwards, with his hands on his knees and reported chest discomfort on the right side. However, his chest movement appeared symmetrical and there was no paradoxical movement observed: in normal breathing the chest and abdomen should move in the same direction, while paradoxical breathing occurs when the chest and abdomen move in opposite directions.

Question

5. Mt Carburys chest movements were
- a) Equal and in sync with his stomach
 - b) Unequal and out of sync with his belly
 - c) Uneven and out of timing with his tummy

TEXT 6

The distribution of discharge summaries

Once finalised, the Discharge Summary should be printed and given to the patient (and/or carer as appropriate) so they understand and can remember their follow-up instructions.

For patients referred to Outpatient Clinics for follow up, include the Clinic name and fax number in the CC section using the predetermined list contained within the EDS module of the Clinical Portal, and the referring doctor (if different to GP).

Electronic distribution of the Discharge Summary to the patient's nominated GP, any additional recipients noted in the CC section, and the Clinical Record Information System (CRIS), will occur automatically after "Finalisation" within the EDS module. The method of distribution of electronic discharge summaries is determined by the details listed for each GP/ Practice in ACTPAS, and the consent (to share information with the GP) recorded for the patient in ACTPAS at the time of discharge summary finalisation. If the patient has consented to participate in the Commonwealth Government *My Health Record* system, a copy of their Electronic Discharge Summary will also be distributed to their *My Health Record* at this time.

Handwritten Discharge Summaries received in the Clinical Record Service with the inpatient notes are manually faxed to the GP.

Question

6. Which of the following lists correctly includes the various people or places that always receive a Discharge Summary?
- a) Patient/carer, Outpatient clinic, patients nominated GP
 - b) Patient/carer, patient's My Health Record.
 - c) Patient/carer, patient's nominated GP

PART C

TEXT 1

We see such studies in magazines, in the newspaper, and on websites all the time, that pets can help ease things like depression or anxiety. They also can aid in your quest for love. Guys who are dog lovers have long known that a trip to the dog park can get you a date, and one study backs this up. Another finds that positive interactions with your dog can help you make more friends and improve on the friendships you have.

Pets are even thought to help the sick and the elderly—particularly those with Alzheimer's, and patients whose prognosis is grim. The way we love our pets, like one of the family, makes us feel deep in our gut that such findings must be true. Yet, a new study finds that there is actually little evidence to support these claims.

Animal-based therapy isn't new. Sigmund Freud often had a dog present in sessions with his patients as a calming presence. Social workers today use animals as an ice breaker, particularly when working with adolescence or children. These latest findings are particularly important nowadays however, as more and more hospitals, nursing homes, universities, prisons, and other such facilities, are using animal-assisted therapy as a way to calm residents and improve their condition. According to a review published in the Journal of Clinical Psychology such initiatives may have been enacted prematurely. Though it makes sense that contact with pets would have a positive impact, the truth is, from strictly a scientific standpoint, we don't really know for sure.

Human-animal interaction has been a subject of research for decades. Even so, the evidence is a wash, according to psychologist Molly Crossman, who conducted the

review. Some studies find that interacting with pets has benefits, others do not. Another problem is the quality of such studies. For instance, those that look at animal-assisted therapy have only examined the short-term impact. Also, there isn't usually a control group to compare results to in these studies. So whether the impact is significant is in question and if so, whether or not the benefit is long-term. It is clear that human–animal interaction (HAI) has a small to medium impact in the near-term, but beyond that, we just don't know. This kind of research may be tainted for another reason. In many of these settings, a trainer brings the animal in. But what effect is the trainer having on these outcomes? The positive benefits might derive not from interacting with the animal but its handler.

Owning a pet has also been determined to have health and psychological benefits. And although we can see those who have a dog or a cat at a young age are less likely to develop allergies, with the psychological advantages, since these were observational studies, the results are indeterminable, at least for now.

There are several theoretical reasons why interacting with friendly animals on a regular basis may improve our condition. Animals and humans can share positive emotions, for instance. Say someone is terminally ill and suffering from chronic pain. They take part in animal-assisted therapy. Playing or cuddling with a cute dog or cat would improve their mood. Since mood has been proven to dampen or heighten pain, a better mood would hypothetically improve their condition. Yet, there is such a thing as the placebo effect.

Questions 7-14

7. In first paragraph, what does the word “quest” imply mean?

- a) Search
- b) Mission
- c) Act
- d) Think

8. Why do social workers use animals nowadays?

- a) To relieve tension between people, or start a conversation
- b) To help people clean the ice and snow
- c) To protect people

d) To help people to learn how to calm down

9. A review published in the Journal of Clinical Psychology claims that

- a) Animals help people feel better
- b) Animals don't have any impact on the positive feelings of people
- c) The decision to incorporate animal therapy in various facilities may have been made hastily.
- d) Incorporating animal therapy in various facilities was a good decision.

10. What does the psychologist Molly Crossman mean when she says that the evidence is a wash?

- a) That the evidence is of no benefit to anyone
- b) That the evidence is weak
- c) That the evidence is bad
- d) That the evidence is convincing

11. Which of these things are not mentioned in the review?

- a) The quality of the studies that have been conducted
- b) The Lack of control groups
- c) The long-term significance of human-animal interaction
- d) The fact that only the short-term impact was considered

12. What does the writer claim about the physical and psychological benefits of owning a dog?

- a) There is evidence that people owning a pet since early childhood are less likely to develop allergies.
- b) There is evidence that there are many mental health benefits
- c) The evidence is conclusive.
- d) The evidence is dubious.

13. How can the terminally ill benefit from animal-assisted therapy?

- a) They can play or cuddle with an animal
- b) They may start feeling better
- c) Their pain can be heightened
- d) Their illness can be treated

14. Why is the author feeling skeptical in sixth Paragraph regarding the benefits from animal-assisted therapy?

- a) Because there is a lack of solid evidence.
- b) Because sometimes the belief of the patient in the positive outcome is beneficial
- c) Because there are high chances that the mood can be hypothetically improved
- d) Because there are only theoretical grounds for the claim.

TEXT 2

Ingesting a nightly Advil or Tylenol is protocol for many. The general body aches, joint inflammation, and, especially, headaches that tag along on a stressful workday seem mitigated with a few hundred milligrams of pain relief. Of course, this only masks what could be a serious underlying condition.

We all have triggers for those sharp (or for some, dull) pains that attack the inside of our skull. Within moments a bottle is popped, two pills washed down in one gulp. The mistake we may be making, however, is confusing the trigger for the cause. The headache would most likely still have occurred, says Christopher Gottschalk. The trigger merely exacerbates the pain while giving your mind a supposed 'reason' to latch onto. Most of the things that people have thought of as triggers—stress, weather fronts, diet—when we have studied them, we've come up empty-handed every time. It's not that they cause headaches, but that when you're getting a headache, weather changes and perfumes feel worse.

As director of Yale Medicine's Headache and Facial Pain Center, Gottschalk has developed another approach: treat every headache as a migraine. This, he says, will offer more of an opportunity at long-term success in treating and even potentially eliminating the problem. This will take some work from a consumer standpoint, so accustomed we are to popping pills. Add to this that the cognitive distance between the freely perused aisles and the pharmacist's counter is not steep. When we can walk into our local pharmacy and pay a few dollars for a hundred pills we tend to think of those drugs as safe and practically benign.

One 2001 study shows that over 30 billion over-the-counter doses of NSAIDs (non-steroidal anti-inflammatory drugs) were purchased in the United States alone that year. Yet the safety of NSAIDs is of ongoing concern. One report states that roughly 110,000 people are injured or die due to acetaminophen (an analgesic in Tylenol, for one) each year. Given that there were 27 billion doses of that specific substance sold in 2009, that equals one complication in every quarter-million doses—not an epidemic, though nothing to gloss over either.

Since migraines are a chronic condition, Gottschalk is taking the long view on headaches. He recommends preventive options such as anti-seizure medications, low doses of tricyclic antidepressants, or higher doses of beta blockers. He's also a fan of nose sprays or injections given the tendency of certain pharmaceuticals to disrupt the digestive process. A common complaint of migraines is nausea, so feeding that intestinal fire means replacing one problem with another.

Looking into the future, Gottschalk is heading into research that helps the immune system address a protein triggered by migraines. The hopes are that an antibody infusion targeting this protein would stop common headaches before they start. Migraines have been shown to be progressively crippling with potential cardiovascular, cerebrovascular, and long-term neurologic effects. Experiencing even

intermittent headaches, according to Gottschalk, could be the result of the same process that causes migraines.

Questions 15-22

15. Which of the following conditions can't be treated with Advil or Tylenol?

- a) Body aches
- b) Headaches
- c) Joint inflammation
- d) Stress

16. According to second paragraph, which of the following has been found to be a trigger for headaches?

- a) perfume
- b) changes in the weather
- c) stress
- d) none of the above

17. Why does Gottschalk suggest treating every headache as a migraine?

- a) In order to eradicate the headache.
- b) To avoid buying potentially harmful medicine.
- c) To achieve a durable effect.
- d) To achieve a better effect.

18. What makes it so easy for people to pop pills?

- a) Their availability at the chemist's.
- b) The fact that they are very cheap.
- c) The fact that they offer fast relief.
- d) The fact that they are benign.

19. What is the author's opinion on NSAIDs?

- a) They are deadly
- b) They are not completely safe
- c) They cause major injuries
- d) They cause complications.

20. What does the author imply when he says "nothing to gloss over either" in fourth Paragraph?

- a) That's nothing to show off about
- b) That's something that shouldn't be ignored or avoided
- c) That's nothing to be happy about.
- d) That's something that shouldn't be discussed

21. Which of these medications is considered to be safe for the stomach according to Gottschalk?

- a) beta blockers
- b) anti-seizure medications
- c) injections

d) tricyclic antidepressants

22. What kind of therapy is Gottschalk hoping will help prevent migraines in the future?

- a) Immunotherapy
- b) Protein therapy
- c) Neurological therapy
- d) Meditation

Reading Practice Test 3 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	B	-
Q 2	C	-
Q 3	A	C
Q 4	B	-
Q 5	A	-
Q 6	C	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	A	-	Q 15	D	-
Q 8	A	-	Q 16	D	-
Q 9	C	-	Q 17	C	-
Q 10	A	-	Q 18	A	-
Q 11	C	-	Q 19	B	-
Q 12	D	-	Q 20	B	-
Q 13	B	-	Q 21	C	-
Q 14	B	-	Q 22	A	-

Review & Explanations

TEST-4

Norovirus PART A

TEXT 1

Norovirus is a virus that is commonly known as the 'Vomiting bug' because vomiting is the main symptom. It also causes diarrhoea and it cannot be treated with antibiotics.

Most people do not have norovirus for very long but it can lead to complications in children, sick people or elderly people.

Norovirus spreads very easily by eating food or drinking liquids that are contaminated with norovirus; touching surfaces or objects contaminated with norovirus, and then placing their hand in their mouth; having direct contact with another person who is infected and showing symptoms (for example, when caring for someone with illness, or sharing foods or eating utensils with someone who is ill). **Handwashing** is the best way to stop the norovirus starting. It must be with handwashing with soap and water not alcohol gels. Alcohol gels do not stop the spread of norovirus.

TEXT 2

Signs and symptoms of norovirus are:

- Vomiting (being sick)
- Nausea (feeling sick)
- Diarrhoea (very, loose, frequent bowel movements)
- High temperature
- Headache
- Aching arms and legs

Managing norovirus

- People with signs and symptoms of norovirus must stay at home until 24 hours after they had diarrhoea or vomiting.
- People can eat anything they want but it does not matter if they do not eat anything for one or two days.
- Taking simple painkillers such as paracetamol can ease the fever and pain
- To stop spreading the virus everyone in the home must:

must be very strict about washing their hand

- every time they go to the toilet
 - every time they vomit
 - before they prepare or handle food for others
 - every time they help someone who is having diarrhoea or vomiting
1. thoroughly clean toilet areas and food preparation surfaces, using a bleach-based household cleaner
 2. immediately remove and wash clothing or linens that may be contaminated with virus after an episode of illness
- Rest is very important to help people get better
 - Drinking fluids is the most important thing people need to do. If they do not drink, they may become dehydrated. This can lead to them becoming very ill

If a person does become very dehydrated, they may need to be admitted to hospital and treated with intravenous fluids.

TEXT 3

Looking after a baby or child with norovirus

- Most people suffer with the norovirus for one or two days and recover quickly. This means they probably will not visit the doctor.

- Babies and young children can become very quickly dehydrated, due to diarrhoea and vomiting so need close monitoring
- A baby should see a doctor if they have signs of dehydration, which :-
 1. sunken eyes
 2. a sunken soft spot (fontanelle) on a their head
 3. few or no tears when they cry
 4. a dry mouth
 5. fewer wet nappies
 6. dark yellow urine
 7. are less responsive than usual
 8. have a temperature
 9. have blood or mucus in their bowel motion
 10. A child should see a doctor for dehydration if they:
 - The reasons why a baby should be taken to hospital include:
 1. are less responsive than usual
 2. have a temperature
 3. have blood or mucus in their bowel motion
 4. A child should see a doctor for dehydration if they:
 5. have dry, cracked lips and a dry mouth.
 6. do not pass as much urine as usual or none for eight to 12 hours,
 7. are drowsy or irritable.
 8. are very weak or limp.
 9. have no tears when crying.
 - Other reasons why a child should see a doctor include if they have:

1. 8-10 watery bowel motions or 2-3 larger motions a day
2. diarrhoea and vomiting at the same time
3. bowel motions have blood or mucus in them
4. diarrhoea that lasts for longer than 10 days

TEXT 4

Elderly person, or ones with a chronic illness and norovirus

- This group of people can become very ill if they have norovirus because they can become dehydrated
- Signs and symptoms that an adult is dehydrated include:
 1. having a dry mouth and/or dry skin in the armpit
 2. a high heart rate (usually over 100 beats per minute)
 3. a low systolic blood pressure
 4. weakness
 5. delirium (new or worse-than-usual confusion)
 6. sunken eyes
 7. passing urine less often
 8. dark-coloured urine
- Blood tests can be carried out to diagnose if the person is dehydrated and how badly
 1. the blood tests check to see if there is a:
 2. raised plasma serum osmolality
 3. raised creatinine and blood urea nitrogen
 4. electrolyte imbalances, such as abnormal levels of blood sodium

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1 signs of dehydration in babies? _____

2 how norovirus is transmitted between people? _____

- 3 what to use in the hand hygiene procedure? _____
- 4 what people can eat if they have norovirus? _____
- 5 cleaning surfaces? _____
- 6 the tests that might be done to see if a person is dehydrated? _____
- 7 how long a person with norovirus must stay at home for? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8 What is most vital activity for people to do if they do have norovirus?

- 9 The majority of people have norovirus for how long? _____
- 10 A heart rate would usually be considered high, if it was at least how many beats per minute in an adult? _____
- 11 What is the medical term for feeling sick? _____
- 12 What is the best indicator that a person has norovirus? _____
- 13 What hospital treatment is likely to be started if a person is admitted with norovirus? _____
- 14 What should hands be washed with if norovirus is about? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 15 A person may need a _____ to see if they have an electrolyte imbalance due to dehydration

16 A person with norovirus might have aching_____

17 A doctor should be seen if a child has_____ watery bowel motions a day.

18 People with norovirus can eat_____

19 Norovirus can lead to more serious conditions in children,
_____or_____people.

20 An adult is probably dehydrated if they have a_____ blood pressure.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	C	-	Q 8	Drink fluids		Q 15	blood test	
Q 2	A	-	Q 9	One or two		Q 16	arms and legs	
Q 3	A	-	Q 10	Over 100		Q 17	8-10	
Q 4	B	-	Q 11	Nausea		Q 18	anything they w	
Q 5	B	-	Q 12	(They are) vomit		Q 19	sick people; el	
Q 6	D	-	Q 13	Intravenous flu		Q 20	low systolic	
Q 7	B	-	Q 14	Soap and water				

Review & Explanations

PART B

TEXT 1

Blood cells could hold the key to understanding schizophrenia

Australian scientists have made a remarkable discovery about what is lurking in the brains of people with schizophrenia, giving them new clues about what might cause the illness.

Using new molecular techniques, Professor Cyndi Shannon Weickert showed a particular type of immune cell was causing inflammation in the brain. Her research found the cells could "squeeze through the cracks in the blood-brain barrier" to get into tissues, and start affecting how the brain functions. That was confirmed when she did blood tests on people with schizophrenia. Those who were suffering psychosis had double the level of inflammation compared to patients who were not psychotic at the time of testing. "We can study the white blood cells in living patients and try to find how this relates to their symptoms, their hallucinations, their delusions and their cognitive problems," she said.

Question

1. What discovery have scientists made about possible causes of schizophrenia?
 - a) Certain cells cannot move across the blood-brain barrier affecting how the brain functions in relation to hallucinations and delusions.
 - b) Certain cells cause inflammation in the blood affecting how the brain functions in relation to hallucinations and cognition.
 - c) Certain cells move from the blood into the brain to alter how it functions in relation to cognition and delusions.

TEXT 2

What are the effects of alcohol?

As soon as we have a drink, our brains start producing dopamine which makes us feel euphoric. They also produce a neurochemical called GABA, which makes us feel relaxed. Once we have had a couple of drinks, the inhibitory parts of our brains are

affected too, so it is much harder to make the decision to stop drinking. For some people it is especially hard to stop once their blood alcohol content is above a certain amount, and that is based on their genetics and personality. Often what happens is that we are used to having a drink to bring on those feelings of happiness, or to take away feelings of sadness/loss/boredom/anxiety, but then it is hard to stop there. That is a really common issue and one that takes time to change.

We really like the buzz from the dopamine and those feelings of calm from the increase in GABA, but that doesn't last as we become more affected by the alcohol itself.

Question

2. According to this article what makes it especially hard to stop drinking even if their alcohol level is high?
 - a) A person's reason for starting drinking and dopamine levels.
 - b) A person's characteristics and hereditary traits.
 - c) A person's inhibitions and GABA levels.

TEXT 3

MEMO re: Requirements for Inpatient Admission Facilitation

The Director of Emergency Medicine and ED Nurse Unit Manager (NUM) (or equivalent) shall ensure that:

Patients presenting to the ED are to be discharged, transferred to another facility or admitted to an inpatient unit within four hours of arrival where clinically appropriate;

Appropriate service delivery models, according to site specific demographics and local casemix, are utilised;

A Senior ED Medical Officer (MO) (Registrar or Consultant) is involved early in the care and treatment of all patients to facilitate definitive decision making; The relevant inpatient team shall be notified as soon as it has been identified that a patient requires admission

The hospital Bed Manager shall be notified as soon as it has been identified that a patient requires admission;

All potential admissions to an Inpatient Unit shall be reviewed and approved by the ED Medical Shift Coordinator; and

Any increase in ED activity which impacts on ED capacity is identified early and local escalation procedures activated in a timely manner.

Question

3. In the memo below, what must the ED shift coordinator do?
 - a) Consider the decision to admit a patient to the hospital and agree it is appropriate
 - b) Look at the diagnosis made about all the patients and decide the planned care is what is needed.
 - c) Make sure all patients arriving in ED are admitted to an inpatient facility within 4 hours.

TEXT 4

New Horizons: The review of alcohol and other drug treatment services in Australia

The review of alcohol and other drug treatment services in Australia (the Review) sought to clarify current Australian drug and alcohol treatment funding processes; current and future service needs; the gap between met and unmet demand; and planning and funding processes for the future. The review included two reports: New Horizons: The review of alcohol and other drug treatment services in Australia and the Review of the Aboriginal and Torres Strait Islander Alcohol, Tobacco and Other Drugs Treatment Service Sector: Harnessing Good Intentions. New Horizons identified a lack of clarity around the roles and responsibilities of the Commonwealth and state and territory governments and a need for further analysis and feasibility work to be conducted collaboratively.

The department has commenced consultations with the states and territories to progress this collaborative approach.

Question

4. What was a key finding of the New Horizons review
 - a) States and territories worked in collaboration to decide their responsibilities
 - b) There is no clear understanding among states and territories about their roles.
 - c) States and territories were clear about their roles.

TEXT 5

Monitoring fetal movements

A Cochrane review assessed the effect of formal fetal movement counting and recording (eg using kick charts) on perinatal death, major morbidity, maternal anxiety and satisfaction, pregnancy intervention and other adverse pregnancy outcomes (5 RCTS; n=71,458) (Mangesi et al 2015). The review did not find sufficient evidence to inform practice. In particular, no trials compared fetal movement counting with no fetal movement counting. Only two studies compared routine fetal movements with standard antenatal care. Indirect evidence from a

large cluster-RCT (Grant et al 1989) suggested that more babies at risk of death were identified in the routine fetal monitoring group but this did not translate to reduced perinatal mortality. Therefore it is recommended that Maternal concern about decreased fetal movements overrides any definition of decreased fetal movements based on numbers of fetal movements. It is also recommended not to advise the use of kick charts as part of routine antenatal care.

Question

5. Whose definition of reduced baby movements is endorsed by this review?
 - a) Cochrane's

- b) Mangesi et al's
- c) he mother

TEXT 6

Concern grows over hernia mesh

Royal Australasian College of Surgeons (RACS) said General Surgeons Australia was investigating the feasibility of establishing a mesh audit. According to RACS, nearly 100,000 Australians are hospitalised for hernia each year. Last month Australian surgeons told the ABC that complications were rare for hernia-mesh surgery, usually about five per cent of patients. But in the United States where there are 54,000 hernia-mesh lawsuits pending, a surgeon who has developed mesh-free hernia surgery has rejected claims made in Australia that the mesh is safe. RACS has previously outlined concerns in a submission to a Therapeutic Goods Administration's consultation on surgical mesh. More than a dozen people from across Australia have now contacted the ABC to share post-surgery outcomes they say have left them in agony and depressed. They are mainly men who have said they wanted to share their stories to give comfort to others who have been affected.

Question

6. In Australia
- a) Less than 10% of hernia-mesh surgery operations cause unexpected problems.
 - b) More than a dozen people have had complications following surgery
 - c) There are 54,000 lawsuits underway as a result of hernia-mesh surgery.

PART C

TEXT 1

Now, researchers at the Salk Institute in La Jolla, California have discovered a way to turn back the hands of time. Juan Carlos Izpisua Belmonte led this study, published in the journal *Cell*. Here, elderly mice underwent a new sort of gene therapy for six weeks. Afterward, their injuries healed, their heart health improved, and even their spines were straighter. The mice also lived longer, 30% longer.

Today, we target individual age-related diseases when they spring up. But this study could help us develop a therapy to attack aging itself, and perhaps even target it before it begins taking shape. But such a therapy is at least ten years away, according to Izpisua Belmonte.

Many biologists now believe that the body, specifically the telomeres—the structures at the end of chromosomes, after a certain time simply wear out. Once degradation overtakes us, it's the beginning of the end. This study strengthens another theory. Over the course of a cell's life, epigenetic changes occur. This is the activation or depression of certain genes in order to allow the organism to respond better to its environment. Methylation tags are added to activate genes. These changes build up over time, slowing us down, and making us vulnerable to disease.

Though we may add life to years, don't consider immortality an option, at least not in the near-term. "There are probably still limits that we will face in terms of complete reversal of aging," Izpisua Belmonte said. "Our focus is not only extension of lifespan but most importantly health-span." That means adding more healthy years to life, a noble prospect indeed.

The technique employs induced pluripotent stem cells (iPS). These are similar to those which are present in developing embryos. They are important as they can turn into any type of cell in the body. The technique was first used to turn back time on human skin cells, successfully.

By switching around four essential genes, all active inside the womb, scientists were able to turn skin cells into iPS cells. These four genes are known as Yamanaka factors. Scientists have been aware of their potential in anti-aging medicine for some time. In the next leg, researchers used genetically engineered mice who could have their Yamanaka factors manipulated easily, once they were exposed to a certain agent, present in their drinking water.

Since Yamanaka factors reset genes to where they were before regulators came and changed them, researchers believe this strengthens the notion that aging is an accumulation of epigenetic changes. What's really exciting is that this procedure alters the epigenome itself, rather than having the change the genes of each individual cell.

In another leg of the experiment, mice with progeria underwent this therapy. Progeria is a disease that causes accelerated aging. Those who have seen children who look like seniors know the condition. It leads to organ damage and early death. But after six months of treatment, the mice looked younger. They had better muscle tone and younger looking skin, and even lived around 30% longer than those who did not undergo the treatment.

Luckily for the mice, time was turned back the appropriate amount. If turned back too far, stem cells can proliferate in an uncontrolled fashion, which could lead to tumor formation. This is why researchers have been reticent to activate the Yamanaka factors directly. However, these scientists figured out that by intermittently stimulating the factors, they could reverse the aging process, without causing cancer. The next decade will concentrate on perfecting this technique.

Since the threat of cancer is great, terminally ill patients would be the first to take part in a human trial, most likely those with progeria. Unfortunately, the method used in this study could not directly be applied to a fully functioning human. But researchers believe a drug could do the job, and they are actively developing one.

Questions 7-14

7. How does the author feel about the results of the study in first Paragraph?
 - a) Optimistic
 - b) Pessimistic
 - c) Neutral
 - d) Unbiased

8. Which of the following causes is considered to be a reason for aging?
 - a) Degradation of telomeres

- b) Genetic changes
- c) Depression
- d) Vulnerability to diseases

9. Izpisua Belmonte's ideas, as expressed in fourth Paragraph, are:

- a) Scientifically oriented
- b) Selfish
- c) Virtuous
- d) Success oriented

10. Which of the following is not true about Yamanaka factors?

- a) They are genes
- b) They can turn into any type of cell in the post-natal period
- c) They can reset genes
- d) They can be used to reverse the aging process.

11. What does "leg" mean in sixth and eighth paragraphs?

- a) Stage
- b) Concept
- c) Limb
- d) Period

12. In eighth paragraph, the experiment that involved mice with progeria that underwent the treatment showed that:

- a) Progeria can be treated
- b) Progeria inevitably leads to early death
- c) The lifespan of mice could be extended
- d) The quality of life of mice could be at stake

13. In the ninth paragraph, the writer claims that the best way to activate Yamanaka factors is:

- a) Directly
- b) Occasionally
- c) Appropriately
- d) Reversibly

14. What will the positive outcomes of this method be as delineated in last paragraph?

- a) It will be of great benefit to cancer patients.
- b) It will be of great benefit to progeria patients.
- c) It will be of great benefit to terminally ill patients.
- d) It will lead to the development of a new drug.

TEXT 2

Brain tests at the age of three appear to predict a child's future chance of success in life, say researchers. Low cognitive test scores for skills like language indicate less developed brains, possibly caused by too little stimulation in early life, they say. These youngsters are more likely to become criminals, dependent on welfare or chronically ill unless they are given support later on, they add. Their study in New Zealand appears in the journal, *Nature Human Behaviour*.

The US researchers from Duke University say the findings highlight the importance of early life experiences and interventions to support vulnerable youngsters. Although the study followed people in New Zealand, the investigators believe that the results could apply to other countries. They followed the lives of more than 1,000 children. Those who had low test scores for language, behavioural, movement and cognitive skills at three years old went on to account for more than 80% of crimes, required 78% of prescriptions and received 66% of social welfare payments in adulthood.

It is known that disadvantaged people use a greater share of services. While many of the children in the study who were behind in brain development came from disadvantaged backgrounds, poverty was not the only link with poor futures. When

the researchers took out children below the poverty line in a separate analysis they found that a similar proportion of middle class children who scored low in tests when they were three also went on to experience difficulties when they were older.

The researchers stress that children's outcomes are not set at the age of three. The course of their lives could potentially be changed if they receive support later in life, for example through rehabilitation programmes when they are adults. Prof Terrie Moffitt, from Duke University in North Carolina in the US, who co-led the study, said: "The earlier children receive support the better. That is because if a child is sent off on the wrong foot at three and not ready for school they fall further and further behind in a snowball effect that makes them unprepared for adult life".

Prof Moffitt said nearly all the children who had low scores in cognitive assessments early on in life went on to fall through "society's cracks". "We are able to predict who these high cost service users will be from very early in life. Our research suggests that these were people who, as very young children, never got the chance that the rest of us got. They did not have the help they needed to build the skills they need to keep up in this very complicated and fast-paced economy". She said society should rethink their view of these people who are often condemned as "losers" and "dropouts" and instead offer more support.

Prof Moffitt conducted the study with her husband, Prof Avshalom Caspi, from King's College London. He said he hoped that the study would persuade governments to invest in those in most need early on in life. "I hope what our study does is not feed into prejudice. I hope that our research will create the public compassion and political will to intervene with children and more importantly offer services to families of children so they can get a better start in life".

Successive governments have invested in expanding nursery education in the UK over the past 20 years. According to Josh Hillman, who is the director of education for the Nuffield Foundation, policy makers already realise the value of early years' education. "But this new research suggests that they may have underestimated its importance," he said. "The issue now in the UK is to provide more high quality nursery provision and to consider targeting it to those disadvantaged groups that would benefit the most."

Participants were members of the Dunedin longitudinal study, an investigation of the health and behaviour of a representative group of the population of 1,037 people born between April 1972 and March 1973 in Dunedin, New Zealand. As adults these people account for only 20% of the population - but they use 80% of public services

in an analysis of a group of people in New Zealand whose lives were tracked for 40 years

Questions 15-22

15. Researchers in first Paragraph suggest that cognitive tests at the age of three

- a) Can show the development of language skills
- b) Can influence children's future life prospects.
- c) Can give us an idea of how children will fare in the future
- d) Can predict chronic illness.

16. Researchers in second Paragraph think that the results of the study

- a. Can help problematic children
- b. Point to the importance of early-life development
- c. Highlight the vulnerability of some children
- d. Point to the necessity of conducting studies in other countries

17. Another study in third Paragraph showed that

- a) Disadvantaged children take advantage of social welfare services
- b) Children from disadvantaged families had less developed brains
- c) Scoring low in cognitive tests is not linked solely to poverty
- d) Many of the children from disadvantaged families had less developed brains

18. Which of the following according to fourth Paragraph is not mentioned by Prof Moffitt

- a) underdeveloped children at the age of three are not ready for school
- b) giving support to children increases the possibility of a better adult life
- c) the consequences of a lack of early support can be long-term.

- d) living support to children early in life prepares them better for adulthood

19. The snowball effect

- a) Indicates that underdeveloped children are unprepared for adult life
- b) Means that children who do badly at school do badly later in life
- c) Means that problems which are relatively small in early life gradually worsen as the child grows up
- d) Means that success in adult life is gradually built upon success in earlier stages of life

20. Prof. Caspi

- a) Hopes that the study will eliminate prejudice
- b) Hopes that the research will help build a better life for children who need support most
- c) Is Prof. Moffitt's wife
- d) Believes that the study will stir up public compassion

21. Nursery education

- a) Has expanded in the UK over the course of over the last decade
- b) Has been recently recognized as an invaluable factor for early years' development
- c) Should offer higher standards
- d) Has been a constantly underestimated factor in early years' education

22. Longitudinal means:

- a) The study was conducted between 1972 and 1973
- b) The study investigated health and behavior
- c) The study was conducted over the course of 40 years.

- d) he study showed a great disproportion between the people who use public services and the overall population

TEST-5

Diabetes

PART A

TEXT 1

Diabetes is the name given to a group of different conditions in which the body cannot maintain healthy levels of glucose (a type of sugar) in the blood. It is usually a lifelong condition that causes a person's blood sugar level to become too high. Blood glucose levels that are too low can lead to a person falling into a diabetic coma and death. High levels, over a long period of time can lead to blindness, kidney failure and lower limb amputation.

Type 1 diabetes most commonly affects children and adults aged under 30. Less than 10% of diabetics have Type 1 diabetes. The pancreas cannot produce insulin so the body cannot process glucose. There is a high hereditary link. Type 1 cannot be prevented or managed with diet and exercise. It is thought to be caused by the auto immune system attacking the pancreas. Type 1 diabetes develops quickly and can make a person very ill.

Type 2 mostly affects adults, especially older people. The body does not produce enough insulin to keep blood glucose levels at a healthy level. Type 2 diabetes is strongly linked to being overweight and having a lifestyle with a poor diet and limited exercise. This means it can be prevented and also, the effects of it reduced if lifestyle changes are made. Type 2 diabetes develops slowly, often over many years.

Gestational diabetes develops during pregnancy, usually in the second half. It can cause problems for baby and mother if it is not well managed. A large birthweight baby is the most common side effect. It usually disappears after giving birth but there is an increased risk of the mother developing Type 2 at some point in the future.

Pre-diabetes is when a person has higher than normal blood sugar readings but not high enough to be diagnosed as being diabetic. If a person changes their diet, they can stop or delay the development of Type 2 diabetes.

TEXT 2

Clinical assessment

- Obtain patient history including family history of diabetes and possibility of pregnancy
- Ask if they are experiencing any of the following signs and symptoms:
 1. feeling very thirsty
 2. passing urine more frequently than usual, particularly at night
 3. feeling very tired
 4. weight loss and loss of muscle bulk
 5. itching around the penis or vagina, or frequent episodes of thrush
 6. cuts or wounds that heal slowly
 7. blurred vision
- Check pulse, blood pressure, urinalysis and do a capillary blood glucose test (cBGT)

Further investigations to confirm diagnosis

- Random blood glucose test, with a result of 11.1 mmol/L or higher
- Fasting blood glucose test with a result of 7 mmol/L or higher on two separate tests
- Oral glucose tolerance test with a result 11.1 mmol/L or higher

Glycosylated haemoglobin (Hb1Ac) test might also be performed. This measures the average blood glucose level over the past 2–3 months. A result of 6.5% or higher on two separate tests indicates diabetes.

TEXT 3

Managing diabetes

- Type 2 and gestational diabetics are often managed with oral medication initially.
- Over time they may need to start insulin replacement therapy to keep their blood glucose levels within the optimum range.
- Type 1 diabetics always need insulin replacement therapy.
 1. Insulin can only be taken as subcutaneously. Patients or a relative need to be taught to do this themselves by injection or using a pump.
 2. Type 1 diabetics have fluctuating blood glucose levels so need to monitor their own cBGT levels before meals and the amount of insulin they give themselves, depends on the result.
 3. Type 2 and gestational diabetics are usually more stable. If they are taking oral medication, they may only test their cBGT daily or less frequently.
 4. Once on insulin, Type 2 and gestational patients need to monitor their cBGT frequently until stable and then once or twice daily is usually enough. Their insulin dose usually becomes stable, with the same amount been administered once or twice a day.

- Some diabetic patients, especially Type 1's, struggle to keep their levels stable. These patients may have their insulin administered steadily across 24 hours through an insulin pump. These patients may also have a monitor that checks their blood glucose levels.
 - A healthy lifestyle is an important part of diabetes management
1. Carbohydrate counting is important for Type 1 diabetics to help control their blood glucose levels
 2. Eating healthy foods (low-fat high-fibre foods) and maintaining a healthy bodyweight.
 3. Exercising regularly (at least 30 minutes on most days of the week)

TEXT 4

Monitoring a person

- Regular checks to monitor several risk factors are encouraged to increase the chance of living a full and long life. These include:
 1. annual diabetic eye screening is a 30 minute eye examination that can quickly detect if there is any damage to blood vessels in the eye as a result of poor blood glucose control
 2. annual foot check for ulcers and infections or change in feeling in your feet
 3. annual blood pressure, cholesterol and kidney function checks
 4. 3 monthly Glycosylated haemoglobin (Hb1Ac) until stable and then 6 monthly. It is generally recommended that Hb1Ac levels should be maintained below 7%.
- Encourage appointments with their doctor if they
 1. have cut, blister or graze to a lower limb
 2. increased blurred vision
 3. a urinary tract infection
 4. signs of hyperglycaemia (high blood glucose),
 5. signs of hypoglycaemia (low blood glucose)

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1 Annual eye screening? _____
- 2 The condition that precedes diabetes? _____
- 3 The acceptable readings for a person's HB1Ac? _____
- 4 How insulin replacement is administered? _____
- 5 The habits needed for good diabetes management? _____
- 6 The most common side effect of gestational diabetes? _____
- 7 The indicators that a person might be diabetic? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8 What is the prevalence of diabetics who have Type 1? _____
- 9 Which organ is not working correctly leading to diabetes? _____
- 10 What is the most usual treatment approach for people with Type 2 diabetes?

- 11 How often is it appropriate for a woman with gestational diabetes to test her cBGT if she is taking oral medication? _____
- 12 How are the most common effects of having diabetes monitored?

- 13 If a person cannot keep their blood glucose levels stable, what device might they use to administer their insulin? _____
- 14 Which type of diabetes can be prevented by making changes to diet and taking up an exercise regime? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 An oral glucose tolerance test with a result of _____ or higher confirms a diagnosis of diabetes.

16 A person with diabetes ought to see a doctor if they have a cut to a _____.

17 A diabetic eye screening is a half hour long eye _____ that can pick up any early signs of damage to _____ in the eye.

18 A routine diabetes check involves pulse and blood pressure checks, urine analysis and a _____ test.

19 Type 1 diabetes is most predominant in children and adults aged _____

20 Type 1 diabetic patients check their cBGT before _____ and administer an amount of _____ based on the cBGT _____.

Reading Practice - PART A

Answer Keys

Correct Answer		Your Answer	
Q 1	D		-
Q 2	A		-
Q 3	D		-
Q 4	C		-
Q 5	C		-
Q 6	A		-
Q 7	B		-

Correct Answer		Your Answer	
Q 8	Less than 10%		
Q 9	The pancreas		
Q 10	oral medication		
Q 11	Daily or less		
Q 12	Regular checks		
Q 13	Insulin pump		
Q 14	Type 2		

Correct Answer		Your Answer	
Q 15	11 mmols/L		
Q 16	lower limb		
Q 17	examination; bl		,
Q 18	capillary blood		
Q 19	under 30		
Q 20	meals; insulin;		"

Review & Explanations

PART B

TEXT 1

Testing for Nystagmus

Patients with abnormal hearing on history or physical examination or with tinnitus or vertigo undergo an audiogram. Patients with nystagmus or altered vestibular function may benefit from computerized electronystagmography (ENG), which quantifies spontaneous, gaze, or positional nystagmus that might not be visually detectable. Computerized ENG caloric testing quantifies the strength of response of the vestibular system to cool and warm irrigations in each ear, enabling the physician to discriminate unilateral weakness. Different components of the vestibular system can be tested by varying head and body position or by presenting visual stimuli. Posturography uses computerized test equipment to quantitatively assess the patient's control of posture and balance. The patient stands on a platform containing force and motion transducers that detect the presence and amount of body sway while the patient attempts to stand upright. The testing can be done under various conditions, including with the platform stationary or moving, flat or tilted, and with the patient's eyes open or closed, which can help isolate the contribution of the vestibular system to balance.

Question

1. What tests are identified in this article that determine if a person has nystagmus in one or both eyes?
 - a) Computerised electronystagmography
 - b) Checking the effect on the person when they are on a special platform
 - c) Putting water of variable temperature into the auditory canals

TEXT 2

Screening for childhood obesity and vascular conditions

The proposed study includes children aged between 9-10 years from the SCOPE (Screening for Pregnancy Endpoints) study. The SCOPE study was conducted in Adelaide during 2005-2008 and 1164 nulliparous pregnant women, their partners and babies were recruited. Detailed information was collected during pregnancy. Of

the 1164 women in the study, 861 had uncomplicated pregnancies, 93 developed preeclampsia, 118 had gestational hypertension, 95 normotensive women delivered SGA babies and 69 delivered preterm.

We will invite all children of women recruited to the SCOPE study to participate at age 9-10 years. Data will be collected on diet, physical activity, height and weight. Haemodynamic measures will be done noninvasively as follows. Central blood pressure and augmentation index (an indicator of arterial stiffness) will be measured using the USCOMBP+. Cardiac output and systemic vascular resistance will be measured using the USCOM 1A. Microvascular function will be assessed by Post Occlusive Reactive Hyperaemia method using Laser Doppler. The results of children born after complicated pregnancies will be compared with those of children born after uncomplicated pregnancies.

Question

2. What daily activities of the child subjects in this clinical trial will be observed?
 - a) If they had pre-eclampsia.
 - b) Their eating and exercise habits.
 - c) If they have siblings.

TEXT 3

South Australia worst for children's asthma hospitalisations according to new report

A new report has revealed that more South Australian children are hospitalised because of their asthma than anywhere else in the country. Between 2011 and 2013, 361 children per 100,000 on average presented to South Australian hospitals, above the rates for NSW and Victoria. There were only 157 presentations per 100,000 people in Tasmania, which has the highest prevalence at 12.6 per cent. Asthma Australia chief executive Michele Goldman said the number of hospital cases in South Australia was unusual and finding out why was the "magic question". She said the condition was often overlooked in GP appointments, which could make management of it tricky. "There's not enough time currently in the system to provide the education that's required so that people with asthma understand the condition,

understand the role of preventer medication and have the tools they need to self-manage the condition well," Ms Goldman said.

Question

3. What reason is given in this articles that explains why hospitalisation of children with asthma is worse in South Australia?
 - a) Lack of time to upskill asthma sufferers
 - b) Lack of time to upskill GP practice staff about asthma
 - c) Lack of time to upskill hospital staff about asthma

TEXT 4

Risks associated with hepatitis B in pregnancy

Mother-to-child transmission occurs frequently either in the uterus, through placental leakage, or through exposure to blood or blood-contaminated fluids at or around the time of birth (Lee et al 2006). Perinatal transmission is believed to account for 35–50% of hepatitis B carriers (Yao 1996). The risk of perinatal transmission is associated with the hepatitis B envelope antigen (HbeAg) status of the mother. If a woman is both hepatitis surface antigen (HbsAg) and HbeAg positive, 70–90% of her children will develop hepatitis B (Stevens et al 1975; Akhter et al 1992). If the mother is HbsAg positive but HbeAg negative, the risk is reduced (Okada et al 1976; Beasley et al 1977; Beasley et al 1983; Nayak et al 1987; Aggarwal Ranjan 2004). In a cohort study of HbsAg-positive, hepatitis B DNA-positive women in Sydney (n=313) (Wiseman et al 2009), transmission rates were 3% among hepatitis B DNA-positive women overall, 7% among HbeAg-positive mothers and 9% among women with very high hepatitis B DNA levels. It has been estimated that people who are chronic carriers of HbsAg are 22 times more likely to die from hepatocellular carcinoma or cirrhosis than noncarriers (95%CI 11.5 to 43.2) (Beasley & Hwang 1984).

[http://www.health.gov.au/internet/main/publishing.nsf/Content/4BC0E3DE489BE54DCA258231007CDD05/\\$File/Pregnancy-care-guidelines-updated-30-July-2018.pdf](http://www.health.gov.au/internet/main/publishing.nsf/Content/4BC0E3DE489BE54DCA258231007CDD05/$File/Pregnancy-care-guidelines-updated-30-July-2018.pdf)

Question

4. Most children born will develop hepatitis B if their mother
- a) Has hepatitis B envelop antigen
 - b) Is a chronic carrier of Hepatitis surface antigen
 - c) Has hepatitis B envelop and surface antigen

TEXT 5

Memo: revised guideline for managing urethral catheters in adults

The purpose of this guideline is to provide the best practice principles to be applied when inserting and managing urethral catheters for adult patients in NSW Public Health

Organisations (PHOs), to reduce unnecessary catheterisation and the risk of the catheter associated urinary tract infection.

This document is intended to support all trained and credentialed healthcare clinicians who are competent in urinary catheter practice for acute care settings. It is the responsibility of the PHO to ensure clinicians whose role involves the insertion, maintenance or removal of urethral catheters are trained and credentialed.

This guideline addresses the insertion, care and removal of urethral catheters in adults during acute care. This guideline does not address suprapubic catheterisation, acute paediatric catheterisation or care and use of chronic or long term catheters for adult patients. This document provides limited advice for maternity and birth settings however these units should refer to local procedures for further clarification.

Question

1. Which health care professionals is this guideline NOT targeted at?
- a) Medical officers, acute care nurses and gynaecologists
 - b) Emergency department nurses, theatre staff and surgical ward staff
 - c) Aged care sector nurses, , midwives and paediatricians

TEXT 6

Hearing loss

Sensorineural hearing loss is caused by lesions of either the inner ear (sensory) or the auditory (8th) nerve. This distinction is important because sensory hearing loss is sometimes reversible and is seldom life threatening. A neural hearing loss is rarely recoverable and may be due to a potentially life-threatening brain tumor—commonly a cerebellopontine angle tumor. An additional type of sensorineural loss is termed auditory neuropathy spectrum disorder, when sound can be detected but the signal is not sent correctly to the brain, and is thought to be due to an abnormality of the inner hair cells or the neurons that innervate them within the cochlea.

<https://www.msdmanuals.com/en-nz/professional/ear,-nose,-and-throat-disorders/hearing-loss/hearing-loss>

By Lawrence R. Lustig, MD, Howard W. Smith Professor and Chair, Department of Otolaryngology–Head and Neck Surgery, Columbia University Medical Center and New York Presbyterian Hospital.

Question

- 2) Why is it important to diagnose what type of hearing loss they have?
 - a) Sensory hearing loss may improve.
 - b) Neural hearing loss does not improve.
 - c) Sensory hearing loss is caused by brain tumor.

PART C

TEXT 1

Today, neuroscientists can determine one's intelligence through a brain scan, as sci-fi as that sounds. Not only that, it's only a matter of time before they are able to tell each individual's set of aptitudes and shortcomings, simply from scanning their brain. Researchers at Yale led the study. They interpreted intelligence in this case as abstract reasoning, also known as fluid intelligence. This is the ability to recognize patterns, solve problems, and identify relationships. Fluid intelligence is known to be a consistent predictor of academic performance. Yet, abstract reasoning is difficult to teach, and standardized tests often miss it.

Researchers in this study could accurately predict how a participant would do on a certain test by scanning their brain with an fMRI. 126 participants, all a part of the Human Connectome Project, were recruited. The Human Connectome Project is the mapping of all the connections inside the brain, to get a better understanding of how the wiring works and what it means for things like intellect, the emotions, and more. For this study, researchers at Yale put participants through a series of different tests to assess memory, intelligence, motor skills, and abstract thinking.

They were able to map the connectivity in 268 individual brain regions. Investigators could tell how strong the connections were, how active, and how activity was coordinated between regions. Each person's connectome was as unique as their fingerprint, scientists found. They could identify one participant from another with 99% accuracy, from their brain scan. Yale researchers could also tell whether the person was engaged in the assessment they were taking or if they were aloof about it.

Emily Finn was a grad student and co-author of this study. She said, "The more certain regions are talking to one another, the better you're able to process information quickly and make inferences." Mostly, fluid intelligence had to do with the connections between the frontal and parietal lobes. The stronger and swifter the communication between these two regions, the better one's score in the abstract thinking test. These are some of the latest regions to have evolved in the brain. They house the higher level functions, such as memory and language, which are essentially what make us human.

Yale researchers believe that by learning more about the human connectome, they might find novel treatments for psychiatric disorders. Things like schizophrenia vary

widely from one patient to the next. By finding what's unique to a particular patient, a psychiatrist can tailor treatment to suit their needs. Understanding one's connectome could give insight into how the disease progresses, and if and how the patient might respond to certain therapies or medications.

But there are other uses which we may or may not feel comfortable with. For instance, your child could have their brain scanned to track them at school, according to study author Todd Constable. It might be used to say whether or not a candidate is qualified for a job or should pursue a certain career. Brain scans could tell who might be prone to addiction, or what sort of learning environment a student might flourish in. School curriculum could even be changed on a day-to-day basis to fit student's needs. And the dreaded SAT might even be shelved too, in favor of a simple brain scan.

Peter Bandettini is the chief of functional imaging methods at the National Institute of Mental Health (NIMH). He told PBS that barring ethical issues, brain scans could someday be used by employers to tell which potential candidate possesses desirable aptitudes or personality traits, be they diligent, hardworking, or what-have-you. Richard Haier, an intelligence researcher at UC Irvine, foresees prison officials using such scans on inmates to tell who might be prone to violence.

We may even someday learn how to augment human intelligence from studies such as this. It's important to remember that intelligence research is still in its infancy. Yet, according to Yale scientists, we are moving in this direction

Questions 7-14

7) Through a brain scan, neuroscientists today can determine an individual's

- a) abstract reasoning
- b) abilities
- c) weaknesses
- d) academic performance

8) Which of the following does the Human Connectome Project involve?

- a) Mapping networks inside the brain
- b) Understanding how connections in the brain function
- c) Understanding intellect and emotions

d) Assessing researchers

9) In what way is a connectome similar to a fingerprint?

- a) In its strength
- b) In its activity
- c) In its individuality
- d) In its accuracy

10) What does intelligence depend on, according to the study?

- a) On the quality of connections between different regions in the brain
- b) On the processing of information in different regions in the brain
- c) On the way higher levels function in the brain
- d) On the talk between certain regions in the brain

11) Why can the human connectome help patients with schizophrenia?

- a) Because disorders like schizophrenia require unique treatment
- b) Because the disorder is distinctive for every patient, as is the connectome
- c) Because by understanding the connectome doctors can understand schizophrenia
- d) Because individual courses of treatments can be designed for every patient

12) Which of the following won't be achieved by mapping the human connectome?

- a) Tracking children's success at school
- b) Defining someone's career
- c) Determining the right candidate for a job
- d) Identifying drug addicts

13) What does the author imply by saying "the dreaded SAT might even be shelved too"?

- a) That the SAT might be successfully passed.

- b) That the SAT might be suspended.
- c) That the SAT might be carefully organized.
- d) That the SAT might be failed.

14) Studies such as this

- a) Are not ethical
- b) Reveal much about criminal minds
- c) Are only at the early stage of development
- d) Are at the peak of their success

TEXT 2

The largest studies to date, by arguably the most respected institutions up to this point, show that psilocybin, a compound within "magic mushrooms", can be effective in treating depression and anxiety in cancer patients. The studies were carried out on 29 patients by researchers at New York University (NYU) and on 51 patients at Johns Hopkins University.

Notably, up to 40% of all cancer patients are afflicted by psychological issues related to their illness. Around 80% of the cancer patients in the studies got noticeably better after just one dose. And they sustained the psychological gains they made for up to seven months, with few minor side effects. Patients reported improvements in their quality of life, having more energy, going out more and having better relationships with family members. Interestingly, those who had a stronger trip reported the strongest gains in alleviating their depression and anxiety.

A number of authorities in psychiatry and addiction medicine expressed their support for the work. These include Dr. Jeffrey Lieberman, a former president of the American Psychiatric Association and Dr. Daniel Shalev from the New York State Psychiatric Institute, who wrote that the studies are "a model for revisiting criminalized compounds of interest in a safe, ethical way."

The studies were also reviewed by regulators and were described by the New York Times as "most meticulous" to date. The lead author of the Johns Hopkins study, psychiatrist Dr. Roland Griffiths, was optimistic on the possibilities of the new treatment, likening it to a groundbreaking surgery rather than the painstaking work to make feel people better used by traditional psychiatric approaches. "I really don't think we have any models in psychiatry that look like" the effects demonstrated in the two trials, said Griffiths. "Something occurs and it's repaired and it's better going

forward ... very plausibly for more than six months," he added. "In that sense it's a new model."

One of the participants in the study, Sherry Marcy, described the changes in how she felt this way: "The cloud of doom seemed to just lift... I got back in touch with my family and kids, and my wonder at life," said Marcy, who has been battling cancer from 2010. "Before, I was sitting alone at home, and I couldn't move ... This study made a huge difference, and it's persisted."

The particulars of both studies involved randomly offering patients either psilocybin or a placebo at the first session, and then giving them the opposite treatment seven weeks later. This way all participants eventually took psilocybin. In both studies, the success rate of psilocybin versus the placebo was very clear, with 83% getting better on psilocybin and only 14% on the placebo in the NYU study.

Psilocybin, of course, has been banned in the U.S. for over 40 years. While they were allowed to use it for their studies, the scientists warn against self-medicating depression by taking "magic mushrooms". (7) They caution that professionals are needed to control the dosage and provide a supportive environment in which to experience the drug's effects. The psilocybin therapy may also not be appropriate for people with schizophrenia or young adults.

To move forward in this field, Dr. Shalev and Dr. Lieberman (8) see a need to loosen research restrictions, because "there is much potential for new scientific insights and clinical applications." Ideally, the next step would be a trial with a larger sample, perhaps across several centers with hundreds of subjects.

Questions 15-22

15)How many patients took part in the experimental studies?

- a) Twenty-nine
- b) Fifty-one
- c) Eighty
- d) Eighteen

16)What were the outcomes of psilocybin?

- a) It helped all the patients alleviate the pain.
- b) It helped cancer patients deal with depression.
- c) It had continuous and long-lasting positive effects.
- d) Although it relieved pain, it had some side-effects.

17) What do several renowned psychiatrists think of the studies?

- a) That they offer a helpful but addictive way of addressing illnesses.
- b) That such studies lead to criminalization of illegal compounds.
- c) That such studies lead to decriminalization of illegal compounds.
- d) That such studies pave the way for safe use of illegal compounds.

18) Psychiatrist Dr. Roland Griffiths

- a) Compared the treatment to a revolutionary surgery.
- b) Compared the treatment to a painstaking surgery.
- c) Couldn't compare the treatment to anything else done before.
- d) Was convinced of the treatment's effects due to the plausibility of the results.

19) Which of the following is true about Sherry Marcy's experience with the treatment?

- a) She felt like being on a cloud.
- b) She stopped being depressed.
- c) She managed to battle cancer.
- d) She felt relieved.

20) How were the patients chosen to receive either psilocybin or a placebo?

- a) First, 83% of the patients received psilocybin, then a placebo.
- b) First, 83% of the patients received a placebo, then psilocybin.
- c) The patients weren't selected according to a plan, but by chance.
- d) All the patients were first given psilocybin, and then a placebo.

21) Which of the following is not true about psilocybin?

- a) It is not allowed to be used in the USA.
- b) It is not suitable for those suffering from schizophrenia.
- c) It must be used with caution by the patients.
- d) The dose and provision must be provided by a specialist in the area.

22)What does the author imply with the phrase "a need to loosen research restrictions"?

- a) That it is necessary to make the rules surrounding research less strict.
- b) That it is necessary to create a better atmosphere for further research.
- c) That it is necessary to invite more doctors to take part in the research.
- d) That it is necessary to spread the results of the research across several centres.

Reading Practice Test 5 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	C	-
Q 2	B	-
Q 3	A	-
Q 4	C	-
Q 5	C	-
Q 6	A	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	A	-	Q 15	C	-
Q 8	D	-	Q 16	C	-
Q 9	C	-	Q 17	D	-
Q 10	A	-	Q 18	A	-
Q 11	D	-	Q 19	B	-
Q 12	D	-	Q 20	C	-
Q 13	B	-	Q 21	C	-
Q 14	C	-	Q 22	A	-

Review & Explanations

TEST-6

Glandular Fever PART A

TEXT 1

Glandular fever is a very common conditions caused by the Epstein Barr virus (EBV). It is also called Infectious Mononucleosis. It is sometimes called the 'Kissing Disease' because it mostly affects teenagers and young adults. Many people are exposed to the virus but do not show any signs or symptoms.

EBV is virus in the herpes group and it infects 90 to 95% of people by adulthood. Once exposed to the virus, it remains latent in most people, only becoming active during a period when the immune system is weakened. It is believed it remains in the person's body for life, though it usually does not cause further illness.

The infectious period of glandular fever is extensive. The incubation period is between 30 to 50 days before symptoms appear but the person is unknowingly infectious at this point. It is believed the virus still contagious in the saliva for up to a year after the illness, and intermittently, indefinitely.

TEXT 2

Diagnosing glandular fever

- Up to 50% who have glandular fever show the symptoms
- Young children can often show little or no sign of illness. Adolescents and young adults usually have more severe symptoms
- Symptoms are similar to most throat infections, including:
 - Loss of appetite
 - Chills
 - Mental and physical fatigue/weakness
 - Aching muscles.
- These symptoms are usually followed 2-3 days later by:-
- a high temperature (fever) of 38°C or above

- swollen lymph glands, usually in the neck
 - sore throat
 - headache
 - sore, reddened throat with enlarged tonsils
 - swollen glands in the neck, armpits and/or groin
 - rash, especially on chest
 - Differential diagnosis
 - unwell for two to three weeks
 - Mono spot blood test: note this may not detect antibodies to the EBV in the earlier stages of the disease
 - debilitating tiredness and fatigue
- NOTE: tiredness and swollen glands may last for two to three months, or even longer
- Rare symptoms
 - anaemia
 - in a small percentage of cases a blotchy red rash can occur
 - enlargement of the spleen (a large organ in the upper left side of the abdomen) – this usually doesn't cause symptoms, although in rare cases the spleen may rupture
 - a swollen and painful liver, which can lead to the skin becoming yellow through jaundice.

TEXT 3

Treating glandular fever

- As glandular fever is caused by a virus, antibiotic therapy is not usually required
- Occasionally the sore throat present with glandular fever can be associated with strep throat, caused by streptococcal bacteria. Antibiotics may be prescribed in these cases in order to combat that bacteria.
- **The main treatment includes:** getting plenty of rest drinking lots of fluids simple analgesia pain such as paracetamol. steroid medication (e.g.: prednisone) may be prescribed to reduce pain and swelling of the lymph nodes avoid alcohol as the liver may be compromised from the virus Contact sports and heavy lifting should be avoided for the first month after illness because of risk of damage to the spleen, which often is enlarged during acute infection Most people make a full recovery. Possible complications include inflammation of body systems including: the heart muscle (myocarditis) the sac that surrounds the heart (pericarditis) the nervous system, including the membranes surrounding the brain and spinal cord (meningitis), the nerves (Guillain-Barre syndrome), and the brain itself (encephalitis) Other complications include: Bell's palsy, which is a temporary paralysis of the muscles on

one side of the face Pneumonia, which may require treatment with antibiotics and/or hospitalisation Destruction of red blood cells leading to anaemia Some people believe glandular fever can lead to chronic fatigue syndrome

TEXT 4

Preventing glandular fever

- There is no vaccine to prevent glandular fever
- The EBV is very easily spread in saliva and remains active outside the body, provided it is in a moist environment.
- Actions to reduce the risk of developing glandular fever include:
 - Good hand hygiene is the best way to avoid getting glandular fever
 - Washing children's bedding and toys if they may have glandular fever - Not sharing drinking bottles, glasses and eating utensils
 - Good cough and sneeze hygiene
 - Avoid kissing people, especially if glandular fever is known to be in the community
 - Taking time off work or school is not necessary to prevent the spread of the virus, but is needed to help recovery

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1 The signs that someone might have glandular fever?

2 Unexpected difficulties that might arise from having glandular fever? _____

3 The time it takes for glandular fever to fully develop? _____

4 Ways to avoid getting glandular fever? _____

5 The ways glandular fever can be confirmed? _____

6 The EBV virus _____

7 The prevalence of glandular fever _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What does EBV stand for? _____

9 Which lymph gland are most likely to be swollen if a person has glandular fever?

10 What is the most effective way to stop the spread of glandular fever?

11 What type of sport should not be played for 4 weeks after having glandular fever? _____

12 How many days will person with glandular fever be poorly for?

13 Which bacteria can sometimes cause a sore throat requiring the glandular fever sufferer to need antibiotic therapy? _____

14 What sort of environment does the EBV need to survive outside a human body?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 A person with glandular fever experiences debilitating_____

16 Glandular fever is sometimes called the_____ because it is most prevalent among young people.

17 The EB Virus will have infected_____ of people by the time they reach adulthood.

18 Staying away from work or school is_____ to stop spreading glandular fever.

19 There is_____ to stop people getting glandular fever.

20 Getting plenty of_____ and_____ are the best ways to manage glandular fever.

Reading Practice - PART A

Answer Keys

Correct Answer		Your Answer	
Q 1	B		-
Q 2	C		-
Q 3	A		-
Q 4	D		-
Q 5	B		-
Q 6	A		-
Q 7	B		-

Correct Answer		Your Answer	
Q 8	Epstein Barr Vi +		
Q 9	The neck		
Q 10	Good hand hygi +		
Q 11	Contact		
Q 12	14-21		
Q 13	streptococcal		
Q 14	moist		

Correct Answer		Your Answer	
Q 15	tiredness and f +		
Q 16	kissing disease		
Q 17	90-95%		
Q 18	not necessary		
Q 19	no vaccine		
Q 20	rest, drinking +		

Review & Explanations

PART B

TEXT 1

The shocking abuse in our nursing homes

ACCORDING TO THE most recent 2011-2012 Report On the Operation of the Aged Care Act 1997, there are 187,941 people living in 2725 residential aged care facilities across Australia.

there are many cases -- too many -- where deaths in nursing homes are hastened by the poor standard of care. In recent years, there have been instances of residents starved, burnt, drowned or even strangled while in care. Part of the problem is the lack of supervision of the residents by caring, capable staff. It stands to reason that a patient could not have fallen into the water fountain at her nursing home if she had been properly supervised. Likewise, it would be more difficult for evil people to prey upon residents if there were plenty of staff on hand, keeping an eye on things. Lynda Saltarelli, of the community-based Aged Care Crisis team, says many people are amazed, when embarking on the frustrating process of trying to find a nursing home bed, to discover there is no mandatory staff-to-resident ratio. All that is required under the Aged Care Act 1997 is that each centre provides what is known as an "adequate number of appropriately skilled staff" or "sufficient" staff for the number of residents.

Question

- 1) Abuse in nursing homes has several causes but what issue does this article concentrate on?
 - a) Too many people are living in nursing homes rather than staying in their own homes.
 - b) Insufficient experienced and considerate staff and no compulsory staffing level requirements.
 - c) Too many hazards that allow the residents to be injured, such as water features and heaters.

TEXT 2

The need to increase indigenous nursing population

The Congress of Aboriginal and Torres Strait Islander Nurses (CATSIN) was founded in 1997 with the express intention of increasing the representation of indigenous people in the nursing profession. Chaired by Sally Goold, CATSIN challenges the western model of health care and seeks to ensure nursing education in indigenous health is culturally appropriate, as well as facilitating indigenous students into nursing. Looking at the broad picture, Alice Springs Hospital nursing director Professor Ged Williams says it is pointless to talk about getting indigenous people into nursing until something is done to address the education of indigenous students in schools. 'There aren't too many Aboriginal nurses in Australia because Aboriginal education is appalling,' Professor Williams said. He argues it is important to begin by getting a critical mass of Aboriginal people into the health sector in jobs such as ward clerks, cleaners, or patient carers where they can support each other. 'Once we have a large critical mass in the system, then we can hope that this will form a pool of potential employees to go on and do other health work like nursing,' Professor Williams said. But Professor Williams adds it is crucial to ensure that early education is meeting the needs of indigenous people, and keeping them in the system.

Question

- 1) What two ideas are there about education that could increase the number of nurses from the Aboriginal and Torres Strait Islander populations?
 - a) There need to be apprenticeships for indigenous people to work in different healthcare settings and nursing needs to give extra educational support to Aborigine and Torres Strait islander nursing students.
 - b) There need to be special pathways into nursing education for indigenous people and more Aborigine and Torres Strait Islanders working in early education
 - c) Nurse education around the health of Aborigines and Torres Strait Islanders needs to be culturally appropriate and early education is needed to increase the educational achievement of indigenous people.

TEXT 3

Memo re: outbreak of circulating vaccine-derived poliovirus in Papua New Guinea.

The Government of Papua New Guinea is working with partners, including WHO and UNICEF, to take appropriate outbreak response measures including contact tracing, testing and vaccination. Polio is very unlikely to spread in Australia because of high rates of vaccine coverage, good sanitation, and the quality and ability of the health system to respond to cases.

Australian residents planning to visit PNG for less than 4 weeks should be up to date with their polio vaccination. For adults, this is a 3 dose primary course, with a booster within the last 10 years. For children, a 3 dose primary course with a booster at 4 years old is currently recommended. These recommended vaccines may be given before arrival in PNG.

Australian residents travelling to PNG intending to stay for longer than 4 weeks should have a documented polio booster within 4 weeks to 12 months prior to the date of departure from PNG. The booster may be given before arrival in PNG, as long as it is given within 4 weeks to 12 months prior to leaving PNG.

Question

- 1) This memo outlines the advice for people travelling to PNG about polio boosters and vaccines. What is the risk of a polio outbreak unfurling in Australia?
 - a) The risk is low because Australia has good border control measures in place
 - b) The risk is high because the population is widespread
 - c) The risk is low because Australia has the means to manage cases of polio

TEXT 4

The effect of healthy lifestyle changes on stroke risk in adults with cardiovascular disease

Our aims of the study are:

1. Increase awareness and adoption of healthy lifestyle factors for people with cardiovascular risk factors, to prevent cardiovascular morbidity and mortality

2. To conduct a feasibility study in the primary care setting to promote the Life's Simple 7 approach, developed by the American Heart Association, for people at risk of stroke

3. To implement a 6 month study to confirm the feasibility, and to provide preliminary data on effectiveness of such an approach.

1. Our overarching hypothesis is that targeted and individualised, longterm support will result in significantly reduced rate of, and risk factors for, morbidity and mortality from stroke and other causes, compared to current practice

2. The 6 month intervention, based on the Life's Simple 7 (LSS) approach, will result in a reduction in stroke risk, measured using the LSS score

3. We hypothesise that the intervention will be feasible, acceptable for participants and we will also collect cost data to determine if the approach is cost effective.

Question

1) What approach will the research team take to try and reduce the frequency and effects of stroke on a person?

- a) They will provide personalised and specific assistance to people at risk of stroke
- b) They will give seven simple medications to people at risk of stroke and monitor the effects
- c) They will have people at risk trying our different treatments to see which is the most cost effective in the long run

TEXT 5

Superbug strains resistant to all known antibiotics discovered by Melbourne researchers

A new superbug that is resistant to all known antibiotics has been discovered by Australian scientists, prompting renewed calls for hospitals to do more to prevent the rise of untreatable infections. The completely resistant strains were found in Europe. "Often it just colonises the skin," Doherty Institute researcher Ben Howden, who was involved in the discovery, said. "It doesn't necessarily lead to infection. But in a smaller number of people it can lead to a serious,

invasive infection requiring complex treatment." Antibiotic resistance is a major and growing global health threat. These recent examples show how dangerous it can be.

The superbug was particularly threatening to those in hospital with a weakened immune system, recovering from surgery or who had implanted medical devices. Researchers identified the superbug after studying examples from 78 institutions in 10 countries around the world, including Victorian hospitals. The inappropriate prescription of antibiotics is a contributing factor to the rise of superbugs. One in four antimicrobial drugs were inappropriately prescribed in Australian hospitals and half the prescriptions given in nursing homes were inappropriate, according to recent research by the Australian Government.

Question

- 1) According to this article how often are antimicrobial drugs inappropriately prescribed in Australian nursing homes?
 - a) 10% of the time
 - b) 25% of the time
 - c) 50% of the time

TEXT 6

Testing for Human papilloma virus

Cervical cancer is one of the most preventable and curable cancers. Cells in the cervix show changes or 'abnormalities' before any progression to cancer, which takes around 15 years. Most low-grade abnormalities regress without treatment. High-grade abnormalities may occur after persistent infection with human papilloma virus (HPV), which is a sexually transmitted infection that generally has no symptoms and resolves within 2 years. In a small number of women, persistent infection with a high-risk type of HPV may eventually lead to cervical cancer (AIHW 2012). HPV 16 and 18 are high-risk types that are detected in 70–80% of cases of cervical cancer in Australia (AIHW 2012). A program of vaccination against HPV types 16 and 18 (as well as types 6 and 11, which have a lower risk of causing cancer but are associated with 90% of genital warts) was introduced in Australia in 2007 (DoHA 2013).

Question

- 1) The HPV Vaccination programme reduces the incidence of what?
 - a) Sexually transmitted diseases

- b) Genital warts
- c) Herpes

PART C

TEXT 1

One of humankind's great advantages has been that we band together to accomplish more than we can on our own. You might expect that our bodies would have evolved in such a way as to promote social interaction, and you'd be right. Studies are revealing that those outside the pack — the lonely — are 25% more likely to die prematurely, according to writer . The chronically lonely wind up with more heart disease, more metastatic cancer, and with higher risks of stroke and neurodegenerative diseases like Alzheimer's.

Lately scientists have been learning about the biological ramifications of loneliness. Steve Cole is a professor of medicine, psychiatry and biobehavioral sciences at the David Geffen School of Medicine at UCLA, and director of the UCLA Social Genomics Core Laboratory. He's been working on this at the molecular level since the early 2000s, aided by data from the Human Genome Project. His interest had been triggered by a report on the survival rates of gay men with HIV. It appeared that closeted men died much more quickly than those who had come out. The main discernible difference between the groups was that closeted subjects were more worried about ostracization and rejection.

John Cacioppo of the Center for Cognitive and Social Neuroscience at the University of Chicago, is an expert on the physical effects of social disconnection on cellular mechanisms with a book called *Loneliness: Human Nature and the Need for Social Connection*. He and Cole teamed up for a study of how gene expression varied between people who were lonely and those who weren't. According to Cole, "We found the key antiviral response driven by so-called Type 1 interferon molecules was deeply suppressed in the lonely people relative to the non-lonely people. In addition — and ominously — Cole and Cacioppo also found "... that there was another block of genes that was not suppressed — in fact, it was greatly activated — and this block

of genes was involved in inflammation.” Inflammation can be the engine underlying atherosclerosis, Alzheimer’s, and cancer.

Are people isolated socially being further undermined by their own biology? There are two plausible evolutionary explanations. Cole says, “The best theory is that this pattern of altered immunology is a kind of defensive reaction mounted by your body if it thinks you are going to be wounded in the near-future,” with no one there to protect or help you. Another theory is that if you feel bad, you may be more inclined to seek out others for help. With this notion, the emotional pain of loneliness acts much like a physical pain that tells you something is wrong that needs to be addressed.

In any event, both ideas may have made more sense historically than they do now, and the number of people describing themselves as lonely is growing quickly — Cole refers to it as an “epidemic.”

Part of it is the aging baby boomers who’ve seen their children grow up and move out, and whose friends are dying more frequently due to age. Another factor may be the way “our culture is changing in ways that invite us — in fact, almost require us — to be more lonely and disenfranchised,” according to Cole. Among those is our migration to online relationships that just aren’t individually as powerful due to the lack of . We text, we don’t call, and our families tend to be far-flung, no longer residing in physical communities together.

Questions 7-14

- 1) Which of the following is not attributed to the lack of socialization ?
 - a) Evolved bodies
 - b) Premature death
 - c) Risk of heart disease
 - d) Risk of neurodegenerative diseases

- 2) What aspect does Professor Steve Cole consider when he investigates the consequences of loneliness?
 - a) Psychological
 - b) Biobehavioural
 - c) Molecular
 - d) Medical

- 3) What did John Cacioppo and Steve Cole work together on?
- a) Research on the effects of loneliness on a physical and molecular level.
 - b) Research on social disconnection.
 - c) Research on the variation of gene expression in both lonely and non-lonely people.
 - d) Research on the suppression of certain genes in lonely people.
- 4) What was found to be much lower in lonely people?
- a) The risk of developing Alzheimer's
 - b) Incidences of inflammation
 - c) Their antiviral response
 - d) Gene expression
- 5) What does Cole imply when he says that socially isolated people are being "undermined by their own biology" in paragraph 4?
- a) That they are biologically prone to be undermined.
 - b) That at a biological level they are prone to disease.
 - c) That they are prone to wounds.
 - d) That they are biologically prone to seek out omit this help.
- 6) What does Cole refer to as an epidemic?
- a) To the socially isolated people.
 - b) To the quick growth of socially isolated people.
 - c) To the growing number of people who see themselves as lonely.
 - d) To the loneliness that prevails around us.
- 7) What is the author addressing in sixth Paragraph?
- a) The reasons for the rapid growth of lonely people.
 - b) The factors that influence feelings of loneliness.
 - c) The factors for the pain that lonely people feel.

d) The reasons that influence feelings of loneliness.

8) What does the author say about relationships in sixth paragraph?

- a) That they used to be more powerful.
- b) That they have transformed into online relationships.
- c) That there is no personal contact anymore.
- d) That there is an absence of human contact in online relationships.

TEXT 2

In 1998, Andy Clark and David Chalmers that a computer operates together with our brains as an “extended mind” potentially offering additional processing capabilities as we work out problems, as well as an annex for our memories containing information, images, and so on. Now a professor of biomedical engineering at the University of Southern California, Theodore Berger, is working to bring to market human memory enhancement in the form of a prosthetic implanted in the brain. He’s already testing it attached to humans.

The prosthetic, which Berger has been working on for ten years, can function as an artificial hippocampus, the area in the brain associated with memory and spatial navigation. The plan is for the device to convert short-term memory into long-term memory and potentially store it as the hippocampus does. His research has been encouraging so far.

Berger began by teaching a rabbit to associate an audio tone with a puff of air administered to the rabbit’s face, causing it to blink. Electrodes attached to the rabbit allowed Berger to observe patterns of activity firing off in the rabbit’s hippocampus. Berger refers to these patterns as a “space-time code” representing where the neurons are in the rabbit’s brain at a specific moment. Berger watched them evolving as the rabbit learned to associate the tone and puff of air. He told *Wired*, “As the space-time code propagates into the different layers of the hippocampus, it’s gradually changed into a different space-time code.” Eventually, the tone alone was enough for the hippocampus to produce a recallable space-time code based on the latest incoming version to make the rabbit blink.

The manner in which the hippocampus was processing the rabbit’s memory and producing a recallable space-time code became predictable enough to Berger that he

was able to develop a mathematical model representing the process. Berger then built an artificial rat hippocampus — his experimental prosthesis — to test his observations and model. By training rats to press a lever with electrodes monitoring their hippocampuses, Berger was able to acquire the corresponding space-time codes. Running that code through his mathematical model and sending it back to the rats' brains, his system was validated as the rats successfully pressed their levers. "They recall the correct code as if they've created it themselves. Now we're putting the memory back into the brain," Berger reports.

It's maybe this last statement that's so intriguing. Does the brain have some kind of master memory index? Has it somehow integrated the artificial hippocampus's memories into the rats' directory? Will it also happen in humans?

Dustin Tyler, a professor of engineering at Case Western Reserve University, cautioned *Wired* "All of these prosthetics interfacing with the brain have one fundamental challenge. There are billions of neurons in the brain and trillions of connections between them that make them all work together. Trying to find technology that will go into that mass of neurons and be able to connect with them on a reasonably high-resolution level is tricky.

Still, Berger himself is optimistic, telling *IEEE Spectrum*, "We're testing it in humans now, and getting good initial results. We're going to go forward with the goal of commercializing this prosthesis".

What he envisions bringing to market based on his research is a brain prosthetic for people with memory problems. The tiny device would be implanted in the patient's own hippocampus from where it would stimulate the neurons responsible for turning short-term memories into long-term memories. He hopes it can help patients suffering from Alzheimer's, other forms of dementia, stroke victims and people whose brains have been injured.

Rats and monkeys — the prosthetic improved the memories of rhesus monkeys attached to their prefrontal cortex — are one thing. The greater number of neurons in human brains is a big issue that needs to be grappled before Berger's implant will work well for humans: It's difficult to gain a comprehensive view of what's going on with larger brains due to their greater number of neurons. (Rat brains have about 200 million neurons; humans have 86 billion.) Berger warns, "Our information will be biased based on the neurons we're able to record from," and he looks forward to tools that can capture broader swaths of data going forward. It's anticipated that they'll need to pack a greater number of electrodes into prostheses

Questions 15-22

15) What would be one of the potential benefits of a computer working together with our brain as an extended mind?

- a) Storage of our memories
- b) Solving problems
- c) Recreating images
- d) Annexing information

16) Berger's prosthetic

- a) Has shown unfavourable results
- b) Helps people convert short-term into long-term memory
- c) Has the potential to store memory
- d) Has shown promising results

17) Which of the following is not associated with Berger's experiment?

- a) Electrodes were attached to the rabbit
- b) Activity of the rabbit's hippocampus could be observed
- c) The rabbit learned to associate an audio tone with a puff of air
- d) In the end, the rabbit produced the desired reaction based on the puff of air

18) What was the outcome of Berger's experiment?

- a) He managed to transfer other rats' information from the prosthesis to the rat's brain and they interpreted it as their own memory
- b) He managed to transfer information from the artificial rat prosthesis to the rat's brain and they interpreted it as their own memory
- c) He managed to transfer human information from the prosthesis to the rat's brain and they interpreted it as their own memory
- d) He managed to transfer other rats' information from the prosthesis to the rat's brain and they acquired it as their own memory.

19) How does Dustin Tyler feel about the whole project?

- a) Enthusiastic

- b) Skeptical
- c) Optimistic
- d) Pessimistic

20) What could pose a problem when the prosthetics interface with the brain?

- a) The complexity of the neural connections in the brain.
- b) The existence of numerous masses of neurons
- c) Working out the way the neurons work together
- d) Lack of technology that would function on the neural level

21) Who does Bergen envisage would benefit most from this prosthetic?

- a) Alzheimer's patients
- b) People with memory problems
- c) People with dementia
- d) People who have had strokes

22) How are human's brains different from those of rats and monkeys?

- a) They are not yet adapted to accept the prosthetic.
- b) They are far more comprehensive.
- c) They contain a greater number of neurons.
- d) They are able to process different amounts of data.

Reading Practice Test 6 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	B	-
Q 2	A	-
Q 3	C	-
Q 4	A	C
Q 5	C	-
Q 6	B	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	A	-	Q 15	A	-
Q 8	C	-	Q 16	D	-
Q 9	C	-	Q 17	D	-
Q 10	C	-	Q 18	B	-
Q 11	B	-	Q 19	B	-
Q 12	C	-	Q 20	A	-
Q 13	A	-	Q 21	B	-
Q 14	D	-	Q 22	C	-

Review & Explanations

TEST-7

Angina

PART A

TEXT 1

Angina (angina pectoris) is a type of chest pain caused by reduced blood flow to the cardiac muscles. The arteries supplying blood to the heart muscles are narrower due to atherosclerosis.

Stable angina is the most common type of angina. Medical attention is needed if it is a first event. The attacks usually occur when the heart is required to work harder than usual such as during physical exertion. Emotional stress or a large meal can also provoke an attack. The pain generally passes within 5 minutes. The pain may ease sooner if the patient rests or takes prescribed medication. Many patients who have been diagnosed as having angina, know what activities are likely to cause an attack. This means they can modify the activity or take medication in advance to reduce the severity of the attack.

Unstable angina is more serious and requires urgent medical attention. It is a sign atherosclerosis has developed to a point that a coronary artery has become too narrow to carry sufficient nutrient to the cardiac muscle or that a fatty plaque has ruptured causing a clot for and occlude an artery. The type or severity of pain may be different to a patient's usual angina attack. It will be unexpected and may occur while the patient is resting. It is usually a stronger pain and typically lasts longer than stable angina attacks, often 30 minutes or more. The attack will continue even if the patient rests or takes medication. An unstable angina attack may be a lead to prolonged oxygen deprivation of the heart, causing a myocardial infarction.

Variant angina (Prinzmetal's angina) is a very rare, genetic condition. In variant angina, the coronary arteries spasm, temporarily reducing the blood flow to the heart. It typically occurs when the patient is resting. The pain is usually severe but may be relieved by medication. Cocaine use, stress and smoking may trigger this type of angina.

TEXT 2

Clinical assessment

Take thorough family history of heart disease to establish genetic risk

Ask about lifestyle activities including tobacco and alcohol intakes, exercise and diet

☐ Check for presenting chest pain characteristics of:

- feeling tight, dull or heavy
- pain in the centre of chest
- duration of pain
- onset trigger
- what relieved pain

☐ Check for other symptoms of:

- radiation to left arm, neck, jaw, stomach, back
- shortness of breath with or without chest discomfort.
- cold sweat
- nausea or
- light-headedness

☐ Female patients more likely than males to report:

- shortness of breath
- nausea/vomiting
- back or jaw pain

TEXT 3

Diagnosing angina

Unstable angina diagnosis focusses eliminating if Myocardial Infarction imminent or has occurred

Stable and variant angina diagnosis starts with:

- Clinical assessment
- an electrocardiogram (ECG)
- Blood tests to identify aggravating factors such as anaemia

Test to be arranged:

- a CT coronary angiogram or invasive coronary angiography to identify blocked blood vessels and potential areas for blockage

- Functional imaging tests to see how the heart works under stress
- Exercise ECG to see how the heart works while walking on a treadmill or using an exercise bike – this is not appropriate for variant angina

Treatment options for angina

- Lifestyle changes can ease variant and unstable angina symptoms and eliminate stable angina symptoms
- Refer to cardiac rehabilitation programme
- Attack management medication:
 - Glycerine Tri Nitrate (GTN) to relieve attacks
- Attack prevention medications options are:
 - beta-blockers
 - calcium channel blockers, first choice for variant angina
- Complication prevention options, especially for non-stable angina patients include:
 - low-dose aspirin
 - Clopidogrel
 - stat injection of a blood-thinning medicine on diagnosis
 - statins to reduce
 - ACE inhibitors
- Surgical options for stable angina not managed by medication and lifestyle change
 - coronary artery bypass graft (CABG) percutaneous coronary intervention (PCI)

TEXT 4

Self-help strategies to ease the symptoms of angina and reduce risk of complications include:

- eating a healthy, balanced diet
- regularly gently exercising such as walking on flat ground or treadmill, swimming, gentle cycling or using exercise cycle

- cutting down on alcohol
- stopping smoking
- losing weight if overweight
- avoiding large meals avoiding stress and learning stress-reduction techniques
- rest as soon as symptoms start take medications as prescribed

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1 Self-help strategies?

2 How long pain from unstable angina might last? _____

3 What angina pectoris is? _____

4 The surgical options to treat angina? _____

5 The symptoms of angina that women might experience more than men? _____

6 The characteristics of chest pain that might be experienced in a person with angina? _____

7 Exercise options for a person with angina? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What type of angina is the most common? _____

- 9 What type of coronary angiography might be used to identify blocked blood vessels? _____
- 10 If the heart suffers prolonged oxygen deprivation, what other condition may develop? _____
- 11 Which arm might pain radiate to in a person experiencing an angina attack?

- 12 What should a person with angina do as soon as symptoms start?

- 13 Which medications are the first choice for variant angina sufferers?

- 14 What can be established from taking a thorough family history?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

In Prinzmetal's angina, the _____ spasm.

- 16 Avoiding _____ meals can ease symptoms of angina.
- 17 Taking _____ aspirin can reduce the risk of developing complications of having non-stable angina.
- 18 Ask patients about their lifestyle activities including _____ and _____ exercise and diet.
- 19 Learning _____ can ease symptoms and prevent complications of angina.
- 20 Unstable angina is serious and requires _____

PART B

TEXT 1

Cost of gastroenteritis in Australia: A healthcare perspective

Using 2016 prices, the estimated annual direct per capita cost of acute gastroenteritis illness was AUD\$14.87 (USD\$10.71), equating to AUD\$20.27 (USD\$14.59) per case. The estimated overall economic burden in Australia was AUD\$359 million (USD\$258 million; AUD\$1.5 million per 100,000 people). The major contributors to this cost were hospital admissions (57.1%), emergency department visits (17.7%), and general practitioner consultations (14.0%). Children under five years of age have the highest per capita rates of acute gastroenteritis illness; however, service utilisation rates vary by age group and both young children and older adults accounted for a substantial proportion of the overall economic burden attributable to acute gastroenteritis illness. Conclusions: Although chronic diseases comprise a large cost burden on the healthcare system, acute illnesses, including acute gastroenteritis illness, also impose substantial direct healthcare system costs. Providing data on current cost estimates is useful for prioritizing public health interventions, with our findings suggesting that it would be ideal if targeted interventions to reduce hospitalisation rates among young children and older adults were available.

Questions 1-6

- 1) In which group of the Australian population is acute gastroenteritis most prevalent?
 - a) It affects all groups equally
 - b) Adults aged 57 or over
 - c) Children aged 5 years or less

TEXT 2

Millions of Australians suffering from combined physical, mental ill health, new report finds

A report compiled by the Australian Health Policy Collaboration (AHPC) at Victoria University, found 2.5 million people are living with both mental and physical conditions. It found clear links between the two — people living with chronic physical health issues are at a higher risk of developing mental health conditions,

while those suffering ill mental health were also more likely to develop physical illness. A chart showing that 28 per cent of men with back problems, and 27 per cent of women, also suffer mental health issues. It said the combined health conditions affected welfare and education, health services and costs, productivity, employment, and social participation. The research revealed men with mental health conditions were 52 per cent more likely to report having a circulatory system disease than the general population, while women with ill mental health were 41 per cent more likely.

Men with mental health conditions were 28 per cent more likely to report having back pain, and women were 68 per cent more likely.

"Improving the physical health of people living with mental health conditions, and conversely, the mental health of people living with physical health conditions, must become a priority to improve the health of all Australians," the report said.

Questions 1-6

- 2) According to this report were women with mental illness more prone to back pain or were men with back pain more prone to mental illness?
- a) It is more likely that a man with back pain will have a mental illness than it is for a woman with mental illness to develop back pain
 - b) It was more likely that a woman with mental illness would develop back pain than it was for a man with back pain to have a mental illness.
 - c) It is more likely that a woman with back pain will have a mental illness than it is for a man with mental illness to have back pain

TEXT 3

Incontinence management using an electronic continence pad.

Urinary incontinence is an embarrassing and disabling problem, and a major determining factor for nursing home placement. It affects 2 million Australians at a cost of \$1 billion per annum. In residential aged-care facilities, incontinence detection and documentation and accurate assessment of voiding patterns are necessary to implement effective best practice continence management. However, current processes are labour and time intensive. Current practices also frequently breach infection control standards for staff and threaten the dignity and privacy of the incontinent person. There is a lack of valid instrumentation to assess

incontinence and monitor continence care. The AEGIS Continence Management System comprises a sensor inserted in a pad liner which detects and monitors wetness. It then transmits the information, via radio link, to a nurse's pager and to a tailored software application that creates continence assessment charts. The project will include AEGIS I which is the existing prototype detecting wetness, the AEGIS II that monitors degree and volume of wetness and the AEGIS III that detects chemical markers of infection or other clinical conditions.

Questions 1-6

- 3) What is a key problem facing staff trying to assess and manage incontinence?
- a) There are no effective equipment available to monitor levels of incontinence
 - b) Radio links break so nurses do not know a patient has pressed their call bell
 - c) Patients always have a urine infection

TEXT 4

Quality of Life in the Elderly - Health related quality of Life: influences

Some of the factors that influence health-related quality of life (eg, institutionalization, reduced life expectancy, cognitive impairment, disability, chronic pain, social isolation, functional status) may be obvious to health care practitioners. Practitioners may need to ask about others, especially social determinants of health (ie, the social, economic, and political conditions that people experience from birth to death and the systems put in place to prevent illness and treat it when it occurs). Other important factors include the nature and quality of close relationships, cultural influences, religion, personal values, and previous experiences with health care. However, how factors affect quality of life cannot necessarily be predicted, and some factors that cannot be anticipated may have effects. Also, perspectives on quality of life can change. For example, after a stroke that caused severe disability, patients may choose treatment (eg, life-saving surgery) to sustain a quality of life that they would have considered poor or even unacceptable before the stroke.

Questions 1-6

- 4) There are many factors that affect a person's quality of life. Which particular features of a person's life are most apparent to a physician?
- a) Friendship, spirituality and employment history
 - b) Changing attitudes to quality of life, effect of a stroke and effects of surgery
 - c) Limited life span ongoing pain and loneliness

TEXT 5

Chronic kidney disease

Chronic kidney disease (CKD) refers to all conditions of the kidney, lasting at least 3 months, where a person has had evidence of kidney damage and/or reduced kidney function, regardless of the specific diagnosis of disease or condition causing the disease. Evidence of kidney damage manifests as either urinary protein or albumin (a type of protein that is a more sensitive and specific marker of kidney disease), blood in the urine, or scarring detected by imaging tests. CKD is categorised into 5 stages according to the level of reduced kidney function and evidence of kidney damage. Stages of CKD are measured by the glomerular filtration rate, which is the amount of blood the kidneys clear of waste products in one minute. Because this rate cannot easily be measured directly, current practice is to estimate it by applying a formula based on age, gender and creatinine (a breakdown product of a molecule found in muscle that is important for energy storage) in the blood. An individual can move up and down through the first four stages of severity, but once they reach stage 5, their kidney function does not usually improve.

Questions 1-6

- 5) The severity of kidney disease is diagnosed by checking what?
- a) The amount of blood that passes through the kidneys
 - b) The amount of waste in the blood that is cleared by the kidneys
 - c) The amount of urine that passes through the kidneys

TEXT 6

Memo: Caspofungin 70mg powder for concentrate for solution for infusion

Cadiusun Pharma GmbH has informed us that there is an error on the patient information leaflet for the above product.

The error is on the portion of the leaflet which is intended for Healthcare Professionals, in the sections providing instructions for preparation of both 70mg/m² and 50mg/m² infusion for paediatric patients >3 months of age.

In the incorrect version of the leaflet, the final concentration of the solution is given as 5.2mg/ml for both dilutions. The correct final concentration is 7.2mg/ml, however in addition to the error on the Patient Information Leaflet, there is also an error on the Summary of Product Characteristics for the product.

The heading for preparation of the 70mg/m² infusion for paediatric patients > 3 months and the heading for preparation of the 50mg/m² infusion for paediatric patients > 3 months incorrectly state in brackets 'using a 50-mg vial'. The headings should read as follows:

Preparation of the 70 mg/m² infusion for paediatric patients > 3months of age (using a 70-mg vial)

Preparation of the 50 mg/m² infusion for paediatric patients > 3months of age (using a 70-mg vial)

Steps have already been taken to correct the errors on both the patient information leaflet and the Summary of Product Characteristics.

Please note, the patient information leaflet and Summary of Product Characteristics for the 50mg presentation of the product are correct.

Questions 1-6

6) What is the issue being raised in this memo?

- a) There are errors in one part of the patient information leaflet
- b) There are errors in two parts of the patient information leaflet
- c) There are errors in three parts of the patient information leaflet

PART C

TEXT 1

In an aging world the rampant destruction of families and emotions due to Alzheimer's disease is increasing. Around the planet forty-seven million people are living with the disease at an annual cost of \$236 billion (when other forms of dementia are factored in). The estimated 5.4 million Americans suffering from this disease require 18.1 billion hours of unpaid care each year—an especially daunting reality during a time when Medicare and Social Security are under attack, as many other costs exist. By 2050 Alzheimer's treatment and care is expected to top \$1 trillion, bumping Medicare costs up by 360 percent.

Eli Lilly's recent attempt at a cure has set the company back hundreds of millions of dollars and may result in thousands of layoffs. High hopes for its drug, solanezumab, were quashed after a recent failed late-stage clinical trial; the company's stock has fallen 11.6 percent in the last two weeks.

Solanezumab targeted beta amyloid deposits in the brain. For the last twenty-five years researchers have speculated that deposits of the amyloid- β peptide "initiates a sequence of events that ultimately lead to AD dementia." Thus far companies like Eli Lilly and Biogen have failed to produce potential remedies using this hypothesis. This has not stopped research, however. Earlier this week neuroscience professor Li-Huei Tsai, who is also the director of MIT's Picower Institute for Learning and Memory, announced the results of a potentially game-changing early-stage trial.

In a forthcoming *Nature* article Tsai and her co-authors report LED lights flickering at a frequency that stimulates gamma oscillations at 40 hertz substantially reduces beta amyloid plaques in mice. Targeting the brain's hippocampus, a critical region in memory formation and retention, researchers temporarily suppressed these proteins. As with all such trials the researchers remain cautiously hopeful. Tsai says, "It's a big 'if,' because so many things have been shown to work in mice, only to fail in humans. But if humans behave similarly to mice in response to this treatment, I would say the potential is just enormous, because it's so noninvasive, and it's so accessible."

Forty hertz is the typical gamma oscillation in human brains; these waves range from twenty-five to a hundred hertz. (Trials in Tsai's lab at other levels were not effective.)

Since 2009 researchers have been able to induce gamma waves in mice using optogenetics, a technique first dreamed up by Francis Crick in the late nineties that utilizes light in order to influence cellular activity in living tissues. It is now considered a breakthrough method for changing neuronal behavioral patterns.

Tsai and her team found that an hour of stimulation at forty hertz resulted in a 40-50 percent reduction in plaques while enhancing gamma oscillations. The proteins returned to their original levels within a day, leaving researchers to wonder if longer courses of treatment will more effectively, or even permanently, reduce protein growth. While the rush for a cure continues, preventive measures are becoming increasingly important. Meditation has been shown to influence gamma activity, while listening to binaural beats might yield positive results.

Movement is among the most important facets of memory control and retention. One of exercise's main benefits is neurogenesis, which includes the production of IGF-1 and BDNF, proteins that are protective against dementia. Keeping active significantly increases your brain's hippocampal volume, resulting in improved memory. One of our age's most damaging effects is the outsourcing of memory that instant access to technology offers. Studies have shown using GPS reduces hippocampal volume; checking the internet for every question instead of thinking through ideas and seeking alternate means is another memory-draining habit that has become commonplace.

Perhaps most interestingly is the importance of cognitive variety, flexibility, and cultivating an ability to consider opposing beliefs. As with bones, our brains operate best when met with resistance. Social interactions and daydreaming are essential for memory retention as well. Engaging in creative pursuits and flow states are ways to strengthen your memory system, as are learning new languages and instruments—basically, anything that challenges your current skill set.

Questions 7-14

7) What's the author's concern in the first paragraph?

- a) That Alzheimer's disease is on the rise.
- b) That the costs for the treatment for Alzheimer's disease are on the rise.
- c) That the unpaid care for Alzheimer's disease is on the rise.
- d) That families are being destroyed by Alzheimer's disease.

8) What brought about the financial difficulties of Eli Lilly?

- a) he high hopes that the company had for their drug for Alzheimer's

- b) The inefficiency that the drug demonstrated in the final stage of the clinical trial
- c) The effort to introduce a cure for Alzheimer's
- d) The fact that their stocks plummeted by 11.6%

9) Eli Lilly and Biogen have:

- a) based their efforts to develop a cure for Alzheimer's on the same assumption
- b) spent 25 years carrying out research in order to find a cure for Alzheimer's.
- c) not given up their hopes for discovering a cure for Alzheimer's.
- d) failed in their attempt to produce a common cure for Alzheimer's.

10) What is Tsai referring to in fourth Paragraph when she says "It's a big if"?

- a) To the certainty of developing a new drug that would work on people.
- b) To the uncertainty of developing a new drug that would work on people.
- c) To the uncertainty of achieving the same results in humans as in mice
- d) To the certainty of applying a treatment on humans that has already worked on mice.

11) What is optogenetics?

- a) A dreamed technique.
- b) A technique that uses light to impact activity in cells.
- c) A technique that uses light to impact activity in certain cells.
- d) A technique that changes neuronal behavioral patterns.

12) Which of the following has not given any results in the treatment and prevention of Alzheimer's?

- a) Gamma waves stimulation
- b) Meditation
- c) Listening to binaural beats
- d) Reducing protein growth

13) Instant access to technology

- a) increases human memory
- b) has damaging effects
- c) reduces the risk of Alzheimer's
- d) increases the risk of Alzheimer's

14) Which of the following hasn't shown results in the prevention of Alzheimer's?

- a) Social interaction
- b) Daydreaming
- c) Learning a new language
- d) Developing resistance.

TEXT 2

There is an old saying in education that girls cry tears while boys cry bullets. In other words, females are allowed in our society to express their vulnerability and less pleasant emotions such as sadness. While boys must remain stoic and shoulder the burden quietly or else get angry, and express their pain not through outbursts of emotion, but instead through action. Might there be a biological phenomenon behind these culture-based roles? A study published online in the journal *Depression and Anxiety* suggests so. It found that boys' brains react differently than girls in the aftermath of a highly stressful event.

Researchers at Stanford University discovered this by scanning the insula, or insular cortex, of boys and girls who had PTSD (Post-traumatic Stress Disorder) and comparing them to those who didn't. This is a region deep within our brain responsible for integrating emotions. Feelings and the sensation of pain are processed here. It is also where empathy emanates from. The insula takes in data from other parts of the body and related areas of the brain, and incorporates it all into emotions and actions.

This is the first study to denote a gender difference in how the brain reacts to PTSD. In girls, the insula developed or aged rapidly. The same process was not observed in

boys. Victor Carrion, MD was the study's author. He is a professor of psychiatry and behavioral sciences at the university. Dr. Carrion told the *Stanford Medicine News Center* that this particular region played a key role in PTSD's development. "The difference we saw between the brains of boys and girls who have experienced psychological trauma is important because it may help explain differences in trauma symptoms between sexes," he said.

This is a substantial breakthrough, as it could help neurologists develop personalized treatment options for PTSD sufferers depending on gender. With children and teens, while some who are exposed to a serious trauma do develop PTSD, others don't. Researchers still aren't sure why that is. They do know that girls are more likely to develop it than boys.

In this study, the brains of 59 participants, each between the ages of 9 and 17, underwent MRI brain scans. 30 of them had PTSD. Another 29 took part as a control group. 16 boys and 14 girls had suffered trauma, while another 14 boys and 15 girls had not. Normal participants showed no differences in insula structure, regardless of gender. Of those who were traumatized, 5 participants experienced one episode of severe trauma, while 25 had been exposed to two or more episodes. Researchers tried to match up participants and controls, comparing those of a similar age and IQ.

The area within the insula that changes due to severe trauma is known as the anterior circular sulcus. Researchers discovered that while in traumatized boys this area grows larger than normal, with girls it shrinks. Another way to look at it is its development accelerates. These changes in structure are thought to be integral to PTSD's development. The shrinkage seen in female brains may be the reason why girls are more prone to PTSD.

Dr. Megan Klabunde was the study's lead author. She told the *BBC*, "Our findings suggest it is possible that boys and girls could exhibit different trauma symptoms and that they might benefit from different approaches to treatment." She added that high levels of stress might lead to early puberty in girls, as some previous studies suggest. This breakthrough may also help neuroscientists understand how each gender processes emotions.

Questions 15-22

15) **A new study has recently**

- a) explained an old saying
- b) sustained a belief considered to be a biological phenomenon
- c) sustained a belief considered to be a cultural concept

- d) provided biological grounds for something considered to be a cultural concept

16) Which of the following is not a characteristic of the insula?

- a) It integrates emotions
- b) It processes feelings of pain
- c) Empathy originates there
- d) It relates areas of the brain

17) Why is this study significant?

- a) Because it defines gender differences.
- b) Because it may lead to an explanation of gender differences
- c) Because it may lead to an explanation of traumatic experiences
- d) Because it may lead to an explanation of variations in trauma symptoms between boys and girls

18) What are girls more likely to develop than boys according to the information in paragraph 4?

- a) Serious traumas
- b) PTSD
- c) Treatment options
- d) Personalized treatment options

19) In the control group

- a) 16 boys had suffered trauma
- b) 14 boys had suffered trauma
- c) Neither the boys nor girls had suffered trauma
- d) Five participants experienced one episode of severe trauma

20) Which of these factors was not considered in the study?

- a) Gender
- b) Age

- c) IQ
- d) Background

21)The changes in the structure of the insula suggest

- a) That the traumas are very severe
- b) That traumatized boys develop PTSD.
- c) That girls are more prone to PTSD than boys.
- d) Why girls are more likely than boys to develop PTSD.

22)What could be the outcomes of the study as suggested in paragraph 6?

- a) Development of different methods of approach towards boys and girls.
- b) Development of different treatment methods for boys and girls.
- c) Development of new techniques to treat trauma symptoms.
- d) Development of new treatments to prevent trauma.

Reading Practice Test 7 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	C	-
Q 2	B	-
Q 3	A	-
Q 4	C	-
Q 5	B	-
Q 6	B	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	B	-	Q 15	D	-
Q 8	C	-	Q 16	D	-
Q 9	A	-	Q 17	D	-
Q 10	C	-	Q 18	B	-
Q 11	B	-	Q 19	C	-
Q 12	D	-	Q 20	D	-
Q 13	D	-	Q 21	D	-
Q 14	D	-	Q 22	C	-

Review & Explanations

TEST-8

Gallstones

PART A

TEXT 1

Gallstones (Cholelithiasis) are hardened deposits of digestive fluid that can form in the gallbladder. The stones can be the size of a grain of sand or a golf ball. Gallstones do not cause problems in approximately 70% of cases. Problems occur if a stone, or stones, becomes trapped in the cystic duct or the common bile duct tract that carries the digestive fluids from the gallbladder to the bowel. There are no single causes of gallstones but they are more common in women, overweight people and those with a family history of gallstones.

Biliary colic is a sudden, intense abdominal pain and fever that usually lasts between one and five hours. It occurs when a stone moves into the cystic duct (neck of the gallbladder) leading to obstruction.

Cholangitis (inflammation of the bile ducts) occurs when a bile duct becomes blocked by a gallstone and the bile becomes infected. This causes pain, fever, jaundice and rigors.

Cholecystitis (inflammation of the gallbladder) is a common complication from gallstones. It can cause persistent pain, fever, nausea and vomiting.

Jaundice develops if a gallstone blocks a bile duct leading to the bowel. This means trapped bile enters the person's bloodstream instead of the digestive system. Jaundice is painless but can cause itchiness. Bile pigments cause the person's skin and eyes to turn yellow and their urine may also turn orange or brown.

TEXT 2

Clinical assessment

- Take a thorough family history
- Ask about lifestyle activities including exercise, diet and recent weight loss

Take symptom history including:

1.

- onset of symptoms
- a link between symptoms and eating
- **presence of a fever?**
- severity and duration of symptoms
- any symptom relievers or antagonists
- **Perform abdominal examination including Murphy's test**
- **Diagnostic tests include:**
- plain abdominal x-ray
- ultrasound scan – the most common diagnostic test for gallstones
- endoscopy
- magnetic resonance imaging (MRI)
- cholangiography
- CT Scan
- endoscopic retrograde cholangiopancreatography (ERCP) – conclusive test if diagnosis unclear in other tests and can include removal of gallstones during the procedure
- Blood tests may be performed to check:
- Liver function
-
- For infection

TEXT 3

Treatment for gallstones

Active monitoring for asymptomatic patients

Simple analgesia for biliary colic

- continue indefinitely if episodes mild and/or infrequent
- Surgical options are the preferred choice of treatment

Laparoscopic cholecystectomy (lap.choly) carried out to:

- remove the gallbladder and any stones in the cystic duct
- remove any stones seen in the bile ducts

Open abdominal surgery to remove gallbladder (cholecystectomy) if patient is:

- in last three months of pregnancy
- extremely overweight

- has atypical gallbladder or bile duct physiology

Non-surgical treatment options

Medication to dissolve calcium free gallstones

- rarely effective
- significant side effects

A well balanced diet

- to ease, not cure, symptoms

Lithotripsy

- rarely used for gallstones, but still widely used for kidney stones
- soundwaves that shatter the gallstones.
- suitable for people with small and soft stones

ERCP

- Can be used for diagnosis
- if stones are found, the bile duct is widened with a small incision or an electrically heated wire
- stones are removed or left to pass naturally

TEXT 4

Ways to prevent gallstones

- Avoid eating too many foods with a high saturated fat content such as:
 - meat pies
 - sausages and fatty cuts of meat
 - butter, ghee and lard
 - cream
 - hard cheeses
 - cakes and biscuits
 - food containing coconut or palm oil
- Drinking small amounts of alcohol may also help reduce risk of gallstones
- Gradual weight loss if obese
 - there's evidence that low-calorie, rapid-weight-loss diets can disrupt the bile

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- Gradual weight loss if obese
 - there's evidence that low-calorie, rapid-weight-loss diets can disrupt the bile chemistry and increase the risk of developing gallstones

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** How gallstones are confirmed? _____
- 2** The signs a person has jaundice? _____
- 3** What type of food to avoid to prevent gallstones? _____
- 4** The surgical options available for ladies with gallstones in their third trimester? _____
- 5** Biliary colic? _____
- 6** Pharmaceutical options for managing gallstones? _____
- 7** Key evidence that suggests gallstones that can be captured when taking the patients history? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** What type of cheese should be avoided to reduce the risk of developing gallstones? _____
- 9** What test is usually carried out to check for gall stones? _____

10 If a person has cholecystitis, what part of their body is affected?

11 The gallbladder and what else might be removed during a laparoscopic cholecystectomy? _____

12 Diets that are low calorie, leading to what, should be avoided?

13 A blood test may be taken from a person suspected of having gallstones to check the functioning of which organ? _____

14 Which non-surgical treatment is now mostly used to treat kidney stones?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 In approximately _____ of cases, people with gallstones are asymptomatic

16 Low intake of _____ can keep the risk of developing gallstones low.

17 During surgery, if gallstones are found, the bile duct is expanded with a minor cut or an _____

18 A person may appear yellow if a gallstone occludes a _____ connecting to the bowel.

19 During the clinical assessment of a patient presenting with possible gallstones, an abdominal examination is carried out including _____

20 Cholelithiasis are solid lumps of _____ that can grow in the gallbladder.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	B	-	Q 8	hard		Q 15	70%	
Q 2	A	-	Q 9	Ultrasound scan		Q 16	alcohol	
Q 3	D	-	Q 10	Gallbladder		Q 17	electrically he	+
Q 4	C	-	Q 11	stones		Q 18	bile duct	
Q 5	A	-	Q 12	Rapid weight lo	+	Q 19	murphy's test	
Q 6	C	-	Q 13	Liver		Q 20	digestive fluid	
Q 7	B	-	Q 14	Lithotripsy				

Review & Explanations

PART B

TEXT 1

Focus: wound/infection control. New solutions for wound healing

Treating skin tissue damaged by burns and other trauma, diabetes or vascular disease is a major challenge and a burden on healthcare systems. Several advanced skin graft treatments exist but they are costly, come with risks such as host rejection, excessive scarring and potentially disease transmission, and are limited to treating shallow wounds where formation of blood vessels is less important. The emergence of 3D printing tools and techniques, biofabrication of tissue materials from biologically compatible materials offers the possibility of not only reducing availability and cost of treatment issues, but also the prospect of treating deep wounds comprising several tissue layers. The layered fabrication method could also accommodate the use of wound healing proteins, stem cells and anti-inflammatory drugs during the printing process, as well as creating more complex tissue structures that could eventually include vascular networks that facilitate oxygen and nutrient exchange to hair follicles and sweat glands. Even though the biological complexities of human skin are relatively well understood, appropriate repair mechanisms are scarce and often costly.

Question

- 1) In this article what might eventually be made on a 3D printer?
 - a) Skin with its various layers, components and blood supply
 - b) Skin including nerve cells, stem cells and sweat follicles
 - c) Skin, with muscles, tendon and blood vessels

TEXT 2

Therapeutic Objectives in the Elderly

When treatments are very likely to achieve benefits and very unlikely to have adverse effects, decisions are relatively easy. However, assessing the relative importance of these quality of life factors to each patient is important when treatments may have discordant effects. For example, aggressive cancer therapy may prolong life but have severe adverse effects (e.g.,

chronic nausea and vomiting, mouth ulcers) that greatly reduce quality of life. In this case, the patient's preference for quality vs duration of life and tolerance for risk and uncertainty help guide the decision whether to attempt cure, prolongation of life, or palliation. The patient's perspective on quality of life may also affect treatment decisions when different treatments (e.g., surgical vs drug treatment of severe angina or osteoarthritis) may have different efficacies, toxicities, or both. Practitioners can help patients understand the expected consequences of various treatments, enabling patients to make more informed decisions.

Questions 1-6

- 2) What three options are generally available for discussion when a person is faced with making a decision about what treatment pathway to take for their illness?
 - a) Modifying their lifestyle to deal with the illness, accepting that they cannot be cured, taking all treatment options available.
 - b) Managing the side effects of treatment, consider declining treatment, revising their expectations of whether they will be cured
 - c) Extending life-expectancy, treating the symptoms to lessen their effect, remedying the condition

TEXT 3

Treatment or management of chronic kidney disease

Early CKD is usually asymptomatic and must be actively sought to be recognised. Kidney function is measured by the glomerular filtration rate (GFR), which is the amount of blood the kidneys clear of waste products in one minute. As GFR cannot be measured directly, current practice is to estimate GFR (eGFR) by applying a formula that includes age, gender and creatinine levels in the blood. Kidney function can also be tested by measuring the levels of albuminuria (type of protein) in the urine, but this testing requires follow-up, as CKD is diagnosed where albuminuria is seen to be persistent in the urine for at least three months.

General practitioners (GPs) are the usual source of initial assessment and diagnosis of CKD and have a variety of options available for treating the condition, including the ordering of imaging and pathology tests, prescribing of medications and, where necessary, referral to a specialist. Best practice management of CKD utilises a

collaborative effort, involving at least the individual and their GP, but also including practice nurses and/or allied health professionals as appropriate.

Questions 1-6

- 3) Why does a urine test to check for Chronic Kidney Disease have to be followed up on?
- a) The urine test has to be compared with the blood test results so a formula can be applied to confirm CKD
 - b) The urine is tested for a protein that has to be present for 90 days to confirm CKD.
 - c) The urine test has to be repeated to ensure the results are accurate before CKD can be confirmed.

TEXT 4

Position Statement on Medicinal Cannabis

It is the position of the NSW Nurses and Midwives' Association that:

Access to cannabis for therapeutic purposes should be supported where patients, in consultation with their treating health professionals, receive some benefit or symptoms are alleviated. Clinical trials should be conducted to develop the evidence base. Approved pharmaceutical cannabis products should be accessible and affordable. A legal framework must be established so that patients or their carers who are in possession of cannabis for personal, therapeutic purposes should have a complete legal defence from arrest or prosecution. A legal framework must be established so that approved cannabis products can be developed and sold for the purposes of therapeutic use. Cannabis misuse should be approached primarily as a health issue rather than a criminal issue and we support an appropriate harm-reduction response.

Questions 1-6

- 4) What is the NSW Nurses and Midwives Association stance regarding carers found in possession of cannabis?
- a) Carers should be protected from legal action against them
 - b) Carers should be arrested and prosecuted
 - c) Carers should be referred to health professionals

TEXT 5

Memo to all staff re: Mandatory FONT requirements bulletin

Completion of all components of the FONT program is mandatory for all NSW Health maternity clinicians (including Obstetricians, General Practitioner Obstetricians, Trainees in Obstetric Medicine, Registered Midwives and midwifery students).

This is to occur in three yearly cycles and will consist of:

- 16 hours to complete the online K2MS Perinatal Training Programs (once every three years) which includes:
 - Fetal Monitoring Training System (including 15 Cardiotocograph (CTG) 'Training Simulator' cases)
 - Maternity Crisis Management Training System
- 2½ hours to complete two (2) K2MS - Fetal Monitoring Training System 'Training Simulator' CTG case studies (every year within the cycle where the entire Fetal Monitoring Training System is not completed)
- 16 hours to complete both face-to-face education sessions (once every three years)
 - Fetal Welfare Assessment ('F')
 - Obstetric emergencies and Neonatal resuscitation Training (ONT').

Chief Executives of Local Health Districts – are responsible for:

- Supporting:

- FONT Facilitators to provide the face-to-face sessions
- Maternity clinicians to complete the mandatory FONT requirements
- Monitoring FONT completion compliance.

Information relating to the granting of Recognition of Prior Learning (RPL) will be available in the FONT Curriculum. The changes outlined in this Information Bulletin will be incorporated into relevant NSW Ministry of Health Policy Directives during 2013.

Questions 1-6

- 5) How much cardiocograph training is must be completed to complete the FONT programme?
- a) Two face to face training sessions and fifteen case studies every three years
 - b) Sixteen hours online training every three years and two and half hours annually
 - c) Two case studies and fifteen simulator cases every three years

TEXT 6

Choice of meter for the individual with diabetes

The choice of a blood glucose meter for the person with diabetes will depend on a variety of factors including ease of use, size and portability, type of strip (e.g. canister, individual foil-wrapped strip or strip-free), amount of blood required, suitability for alternate site testing and other additional features such as memory and download capability, alarms and back lights. Individuals with sight or dexterity problems will need a meter that accommodates these issues. Many people with type 1 diabetes use more than one blood glucose meter, and may require a meter to measure blood ketone levels, a smart meter that assists in insulin bolus calculations or a meter that relays blood glucose levels to their insulin pump. Capillary blood samples are best taken from the side of the finger, but avoiding close proximity to the nail bed, particularly when blood glucose levels are changing rapidly. Some blood glucose meters allow the measurement of glucose levels from small samples of blood from the forearm and other sites.

Questions 1-6

- 1) Where is the best place to take blood from to monitor blood glucose levels?
 - a) Near the nail bed
 - b) The side of the finger
 - c) The forearm

PART C

TEXT 1

Cancer, of course, is not new. Throughout the ages societies have defined and dealt with it variously. Ancient Greeks employed the term to describe tumors: mass, burden. It is an apt translation of what cancer does inside of our bodies. In *Emperor of All Maladies* Siddhartha Mukherjee travels to the root of onkos.

Nek is an Indo-European term that represents an active form of “load.” It means to carry, to move the burden from one place to the next, to bear something across a long distance and bring it to a new place. It is an image that captures not just the cancer cell’s capacity to travel—metastasis—but also Atossa’s journey, the long arc of scientific discovery—and embedded in that journey, the animus, so inextricably human, to outwit, to outlive and survive. As Atul Gawande recently expressed in conversation with musician Andrew Bird—Bird asked the doctor how cancer forms, given a severe bout his wife had recently undergone—we all grow cancer cells every day. Fortunately, our bodies are designed to not let them metastasize. Then a mutant gets through, our body under attack.

Certain cancers are genetic—my testicular cancer two years ago is one such case, given a childhood condition that predisposed me to it. Yet many are environmental. More importantly, where genetics and environment meet is either a breeding ground or defense system for cancer. Cigarettes have always been the former, like letting streptococcal bacteria loose in a sauna.

A recent study published in *Science* reminds us just how dangerous cigarettes are. It turns out that hundreds of DNA cells are in danger of mutation in what scientists believe is impactful enough to leave an “archaeological record.” Smoking a pack a day leads to the following number of potential mutations every single year 150 in the

lungs; 97 in the larynx or voice box; 23 in the mouth; 18 in the bladder; six in the liver. Each mutation increases the risk of cells becoming cancerous.

Every cigarette contains at least 60 carcinogens, which is why six million people die because of cigarette-related (and thus avoidable) cancers every year. Tobacco has been implicated in 17 types of cancer. While the researchers remind us that smoking is “mechanistically complex,” and there are multiple factors when considering cancer, they conclude: Although we cannot exclude roles for covariate behaviors of smokers or differences in the biology of cancers arising in smokers compared with nonsmokers, smoking itself is most plausibly the cause of these differences.

Forty-five percent of American adults puffed tobacco 60 years ago. This is partly the result of the hundreds of millions of dollars manufacturers were pouring into advertising. The 1955 introduction of the Marlboro Man increased sales by whopping 5,000 percent over eight months. Peak consumption hit in the early sixties, with sales of nearly \$5 billion in America alone.

It’s been a long, slow withdrawal. In 1956, Richard Doll and Bradford Hill began questioning the role of cigarettes in lung cancer. Today we view the Mad Men-esque nonchalance of lighting up as a romantic throwback to a better time. Yet since January 1, 1971, cigarette ads have been banned on television. That decade marked a profound turn in our understanding of just how dangerous cigarettes are.

Still, addictions persist. While Utah is just over 12 percent smokers, and California in second at 15 percent, Kentucky leads the charge with 30.2 percent of its population. West Virginia and Mississippi follow closely behind. That means over 1.3 million people still smoke in Kentucky. Even though that state doubles the percentage of California smokers, around 5.8 million people still light up on the west coast.

Mukherjee, whose brilliant book on cancer is lucid and frightening, has spent a lot of time in cancer wards around the world. In one passage he describes a fraction of the devastation: “An ebullient, immaculately dressed young advertising executive who first started smoking to calm his nerves had to have his jawbone sliced off to remove an invasive tongue cancer. A grandmother who taught her grandchildren to smoke and then shared cigarettes with them was diagnosed with esophageal cancer. A priest with terminal lung cancer swore that smoking was the only vice that he had never been able to overcome. He then describes that even while going through this, many patients refuse to surrender their vice: “I could smell the acrid whiff of tobacco on their clothes as they signed the consent forms for chemotherapy.

Questions 7-14

- 7) What point does the writer make about the term the Ancient Greeks used for cancer?
- a) That it is a suitable translation that reflects the way in which cancer behaves
 - b) That it is an appropriate description of the outcomes of cancer.
 - c) That it describes precisely the root of the disease.
 - d) That the word cannot describe the disease properly.
- 8) How does the author describe cancer in terms of its Indo-European implication?
- a) As a carrier
 - b) As a load
 - c) As a journey
 - d) As a discovery
- 9) Which of the following can be a cause of cancer?
- a) Environment
 - b) Breeding
 - c) Childhood conditions
 - d) Streptococcal bacteria
- 10) What does “the former” refer to in third Paragraph?
- a) Genetics
 - b) Environment
 - c) A Breeding ground
 - d) A Defense system
- 11) Why do cigarettes cause cancer?
- a) They can cause mutations in DNA cells.

- b) They cause mutations in the organs of the body.
- c) They turn cells into cancerous tissues.
- d) They increase the risk for lung, larynx, mouth, bladder, and liver cancer.

12)When considering cancer, scientists

- a) Are not sure that it is caused by smoking because cancers are complex
- b) Are sure that smoking causes cancer
- c) Believe that smoking is the most likely cause for cancer in smokers as opposed to nonsmokers
- d) Believe that smoking is the most likely cause for differences that are visible in the cancers that occur in smokers compared to nonsmokers

13)What happened in the seventies?

- a) New ways of advertising
- b) A different attitude towards smokers
- c) A Decrease in sales of cigarettes
- d) People began to view cigarettes as a health hazard

14)What does Mukherjee say about smokers who got cancer?

- a) He is worried about them.
- b) He is disgusted by them.
- c) He was disgusted by the smell of tobacco on their clothes.
- d) He knows that many of them haven't quit smoking.

TEXT 2

According to the National Institutes of Health, we spend about 26 years of our life asleep, one-third of the total. The latest research states that between 6.4 and 7.5 hours of sleep per night is ideal for most people. But some need more and others less. A contingent out there, mostly women, who do surprisingly well on just six hours. There is even some data to suggest that a slim minority, around three percent of the population, thrive on just three hours sleep per night, with no ill effects. Of course, most people need much more. Even though in general, Americans are getting far less sleep today than in the past.

Cutting out needful rest could damage your health, long-term. A recent study showed that sleep is essential to clearing the brain of toxins that build up over the course of the day. It also helps in memory formation and allows other organs to repair themselves. Our professional lives and our natural cycles don't always mesh. Often, they are at odds. What if you are insanely busy, like ten times the norm? Say you are going to medical school, earning your PhD, or are trying to get a business off the ground. There may not be enough hours in the day for what you have to do. One thing you can do is rearrange your sleep cycle to give yourself more time.

Paleoanthropologists espouse that our ancestors probably didn't sleep for seven hours at a clip, as it would make them easy prey. Instead, they probably slept at different periods throughout the day and night, and you can too. What we consider a "normal" sleep cycle is called monophasic. This is sleeping for one long period throughout the night. In some Southern European and Latin American countries, the style is biphasic. They sleep five to six hours per night, with a 60-90 minute siesta during midday. There is a historical precedent too. Before the advent of artificial light, most people slept in two chunks each night of four hours each, with an hour of wakefulness in-between. That's also a biphasic system.

Then there is polyphasic sleep. This is sleeping for different periods and amounts of time throughout the day. Certain paragons of history slept this way including Leonardo Da Vinci, Nikola Tesla, Franz Kafka, Winston Churchill, and Thomas Edison, among others. The idea gained popularity in the 1970's and 80's among the scientific community. Buckminster Fuller, a famous American inventor, architect, and philosopher of the 1900's, championed this kind of slumber. So what's the science behind this radical system? Unfortunately, no long-term research has been conducted, yet. One 2007 study, published in the *Journal of Sleep Research*, found that most animals sleep on a polyphasic schedule, rather getting their sleep all at once. This also begs the question, how much sleep does the human brain need to function properly? The answer is unknown.

Sleep is broken into three cycles. There is light sleep, deep sleep, and rapid eye movement (REM) sleep. The last one is considered the most important and restful of phases. We don't stay in any one phase for long. Instead, we cycle through these constantly throughout the night. So with polyphasic sleep, the idea is to experience these three phases in shorter amounts of time, and wake up rested.

We don't know the exact purpose of these phases. Sleep is still something of a mystery. Without a good understanding, it's difficult to quantify the impact a polyphasic schedule has. One question is whether such a schedule allows for enough

REM sleep. Polyphasic practitioners say they are able to enter the REM phase quickly, more so than with a monophasic style. Jost, for example, claimed he could enter REM sleep immediately. This quick entry into the REM state is known as “repartitioning.” The deprivation of sleep may help the body enter REM quickly, as an adaptation.

Questions 15-22

15) In first paragraph, the author

- a) Introduces the phases of sleep
- b) Introduces general facts about sleep
- c) Discusses the amount of time people sleep.
- d) Discusses the period of time when people sleep.

16) The author uses the word “contingent” in the first Paragraph to refer to the women as:

- a) An exception
- b) A group
- c) Resilient.
- d) Brave.

17) What does the author imply when he/she says that our professional lives and natural cycles are at odds?

- a) That they are in congruence.
- b) That they are in conflict.
- c) That they correspond to one another.
- d) That they cannot be compared.

18) What does the author suggest about our ancestors?

- a) That they had monophasic sleep.
- b) That they didn’t sleep for 7 hours.
- c) That they didn’t have monophasic sleep.
- d) That they had biphasic sleep.

19)What is a siesta?

- a) A long afternoon sleep.
- b) A 60-90 minute afternoon sleep.
- c) A 60-90 minute sleep which follows a regular 5-6 hour sleep.
- d) Part of the biphasic sleeping schedule in Latin America.

20)What do scientists suggest about sleep in fourth paragraph?

- a) That in order to be successful you need to have polyphasic sleep.
- b) They can't explain why many animals have polyphasic sleep.
- c) They have no answers about what proper sleep is as no studies have been carried out
- d) They have no specific answers about the amount of sleep we require?

21)What does the author say about the way we sleep in fifth paragraph?

- a) That we all experience polyphasic sleep.
- b) That we all sleep in cycles.
- c) We cycle when we go to sleep
- d) That polyphasic sleep makes us feel rested.

22)Why are some scientists skeptical about the success of polyphasic sleep?

- a) Because they believe that it does not allow for enough REM sleep.
- b) Because they believe that it allows people to enter the REM phase more quickly
- c) They suspect that it may not allow for enough REM sleep.
- d) They don't believe that the repartitioning phase can be achieved.

Reading Practice Test 8 - PART B & C

Answer Keys

Part B		Part C			
		Correct Answer		Your Answer	
Q 1	A			-	
Q 2	C			-	
Q 3	B			-	
Q 4	A			-	
Q 5	C			-	
Q 6	B			-	
Part C		Correct Answer		Your Answer	
Q 7	A			-	
Q 8	C			-	
Q 9	A			-	
Q 10	C			-	
Q 11	A			-	
Q 12	D			-	
Q 13	D			-	
Q 14	D			-	
		Correct Answer		Your Answer	
Q 15	C			-	
Q 16	B			-	
Q 17	B			-	
Q 18	C			-	
Q 19	B			-	
Q 20	D			-	
Q 21	B			-	
Q 22	C			-	

Review & Explanations

TEST-9

Overweight and Obesity

PART A

TEXT 1

Overweight and obesity are defined as abnormal or excessive fat accumulation that may impair health. Originally this was found to be an issue in western societies, but obesity levels are now rising in Africa and Asian states. Worldwide, obesity has nearly tripled since 1975. In 2016, more than 1.9 billion adults were overweight. Of these over 650 million were obese. 39% of adults were overweight in 2016, and 13% were obese. Worryingly, in 2016, over 340 million children and adolescents aged 5-19 were overweight or obese. Becoming overweight or obese is entirely preventable. This means conditions linked to being overweight and obese are also preventable or at least deferred from causing problems.

Body mass index(BMI) is a simple index of weight-for-height that is commonly used to classify overweight and obesity in adults. It is defined as a person's weight in kilograms divided by the square of his height in meters (kg/m^2). The World Health Organisation defines adults as overweight if their BMI greater than or equal to 25; and obese if their BMI is greater than or equal to 30.

Waist circumference measurement is increasingly used as a measure of obesity. BMI can be misleading. A highly muscular athlete or someone of non-European ethnicity can present with a raised BMI but this does not reflect their body fat levels and their lack of health risks. However, people with very large waists – generally, 94cm (37in) or more in men and 80cm (about 31.5in) or more in women – are more likely to develop obesity-related health problems.

Significant health, social and economic impacts are linked to people being overweight or obese. Being overweight or obese increases the risk of suffering from a range of health conditions, including coronary heart disease, Type 2 diabetes, some cancers, knee and hip problems, and sleep apnoea. In 2008, the total annual cost of obesity to Australia, including health system costs, loss of productivity costs and careers' costs, was estimated at around \$58 billion.

Eating too much and moving too little causes is an imbalance between calorie intake and calorie expenditure. For this reason, overweight and obesity are generally considered to be the individual persons 'fault' and they need to solve the issue.

TEXT 2

Medical checks

Take a thorough history is essential to eliminate contributing factors of:

- family history of diabetes, heart conditions and obesity
- an underactive thyroid gland
- Cushing's syndrome
- Polycystic ovary syndrome (PCOS)

medicines for:

- epilepsy
- diabetes
- depression
- schizophrenia
- corticosteroids
- recent smoking cessation
- chronic mobility limiting pain or injury

TEXT 3

Managing overweight and obesity

Treatment focuses on strategies to bring about the lifestyle changes of:

- Eating a healthy balanced diet
- Increasing activity levels

Cognitive behaviour therapy may be offered to help a person change their lifestyle

‘green prescriptions’ can increase opportunities to exercise

Medication – Orlistat is the only prescription medication

Surgical options include:

-lap band surgery: an adjustable ring is put around the top part of the stomach to create a very small pouch that increases the time food remains in the top part of the stomach

-regastric bypass: a small stomach pouch created by stapling is joined directly to the small intestine after some of the intestine has been removed. Food bypasses most of the stomach and fewer calories are absorbed

-gastric sleeve surgery: most of the stomach is removed, including the part that makes a hormone which makes you feel hungry.

TEXT 4

Individual or societal responsibility?

- Social, political and economic factors help to create and maintain an overweight and obese population.

- Daily activity to burn calories is reducing due to:

- an increasingly sedentary nature of many forms of work

- changing modes of transportation

- policy decisions that promote car use and do not ensure adequate open, safe streets

- increasing urbanisation

- Increased intake of food of high calorie, low nutritional value, i.e. foods that are high sugar, high fat, high salt, low fibre is due to:

- convenient, fitting into 'time poor' family lives where both parents work,

- heavily marketed

- widely available in supermarkets, fast food outlets, petrol stations, cafes, etc.

- Political impact low due to:

- food producers arguing for 'personal choice' and resisting political change
- companies being largely unregulated, or at best self-regulated, on the amount of sugar, salt and fat they can include
- reluctance to impose change on large companies who generate significant employment and tax income through their productions
- lack of legislation to reduce number of eating and fast food outlets
- Economic impacts on obesity include:
 - low cost, low nutritional value foods are frequently available in low socio-economic areas
 - the food industry are massive contributors to a country's business, through direct employment and purchasing supplies locally

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** Surgical options to manage obesity? _____
- 2** The political impact on obesity? _____
- 3** The lifestyle changes that obesity management focusses on? _____
- 4** Why the aily activity needed to burn calories is reducing _____
- 5** Waist circumference measurement? _____

6 The medications that might contribute to the risk of obesity? _____

7 What eating too much and moving too little does? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What factors help to create and maintain an overweight and obese population _____

9 What type of prescriptions can increase opportunities to exercise?

10 In 2016, how many children and adolescents aged 5-19 were overweight or obese _____

11 What is found in high levels, in food that is of high calorie but low nutritional value _____

12 What change has happened to many forms of work that makes it difficult for people to burn calories? _____

13 Stopping what activity can lead to a person becoming overweight or obese?

14 An adult with a BMI greater than or equal to 25, is considered to be what _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 A _____ involves a small stomach pouch being created by stapling and then directly joining it to the small intestine after some of the intestine has been removed.

16 Overweight and obesity are defined as abnormal or _____ that may impair health.

17 Food producers argue for _____ and resist political change.

18 Being overweight or obese increases the risk of suffering from a range of health conditions, including coronary heart disease _____ and some cancers

19 A(n) _____ gland can lead to being overweight or obese.

20 Orlistat is the only _____ available to help manage obesity.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	C	-	Q 8	Social, politi +		Q 15	gastric bypass	
Q 2	D	-	Q 9	green		Q 16	excessive fat a +	
Q 3	C	-	Q 10	over 340 millio +		Q 17	personal choice	
Q 4	D	-	Q 11	Sugar fat salt		Q 18	type 2 diabetes	
Q 5	A	-	Q 12	Increasingly se +		Q 19	underactive thy +	
Q 6	B	-	Q 13	smoking		Q 20	prescription me +	
Q 7	A	-	Q 14	overweight				

Review & Explanations

PART B

TEXT 1

Memo to public health workers: A reminder about the safe and correct use of condoms

A batch of Durex Real Feel condoms was recently recalled internationally because of their tendency to burst before their expiry date. This has prompted a reminder about the safe use of condoms, including making sure you're aware of the expiry date.

As a product ages, it has a higher potential to fail, so it is important to always check the use-by date before using a condom and make sure you know how to use a condom correctly. Taking the right precautions will help manage the spread of disease and reduce the likelihood of unwanted consequences, such as pregnancy.

Durex Real Feel condoms from batch 1000432443 (expiry January 2021) should not be used. No other condoms are affected by this issue.

Question

- 1) What guidance do staff need give to their patients to limit the chance of condoms failing?
 - a) Open the packaging to see if the condom has burst
 - b) Check the condom expiry date and put one on correctly
 - c) Discard the condom if it is a Durex Real Feel

TEXT 2

Directive: compliance is mandatory – Introduction

SA Health is responsible for ensuring Local Health Networks provide comprehensive, timely and high quality health services in South Australian (SA) public hospitals. Approximately 46,000 elective surgery procedures are performed each year in South Australian (SA) metropolitan public hospitals and approximately 16,000 elective surgery procedures in country hospitals. Patients requiring elective surgery have been assessed by a medical practitioner or authorised delegate as needing surgery

for a medical condition, but for which admission can be delayed for at least 24 hours. These patients are placed on a booking list and treated according to the clinical urgency category assigned by the treating medical practitioner. *SA Health Elective Surgery Policy Framework and Associated Procedural Guidelines (Policy Framework)* has been developed to provide a consistent, structured approach to support the provision of elective surgery within SA Health public hospitals

Question

- 2) What is the purpose of the SA Health Elective Surgery Policy Framework and Associated Procedural Guidelines?
- a) To ensure all patients have an operation within 24 hours of being referred for publicly funded surgery
 - b) To make sure GPs know the process for referring patients for publicly funded surgery
 - c) To ensure all patients in SA get equal access to the publicly funded operation they need

TEXT 3

Women at lower risk of caesarean section when they're induced

For most women, labour naturally starts between weeks 37 and 42 of pregnancy. When a woman is induced, doctors break her waters and make the uterus contract more strongly to bring on labour deliberately. There's a variety of reasons why a woman might be induced in this way including that the baby is overdue, that the baby doesn't seem to be growing well, or that the mother has a health condition that can make labour trickier, like high blood pressure. But the women in the study didn't have any of those problems, and none had given birth before. The researchers wanted to see whether inducing labour was better than waiting in women who didn't otherwise need to be induced for health reasons. Half of the pregnant women were randomly sorted into a group that would be induced at 39 weeks, while the other half would instead go through "expectant management", where they simply waited for labour to come. The researchers found that women who had a labour induction were 15 per cent less likely to need a caesarean delivery.

Question

- 3) What was the selection criteria for women to be included in this research project?
- a) 42 weeks pregnant and baby has stopped growing
 - b) No problems and a first time mum
 - c) A first time mum whose waters had broken

TEXT 4

Drug categories of concern in the elderly: anti-coagulants

Age may increase sensitivity to the anticoagulant effect of warfarin. Careful dosing and routine monitoring can largely overcome the increased risk of bleeding in elderly patients taking warfarin. Also, because drug interactions with warfarin are common, closer monitoring is necessary when new drugs are added or old ones are stopped; computerized drug interaction programs should be consulted if patients take multiple drugs. Patients should also be monitored for warfarin interactions with food, alcohol, and OTC drugs and supplements. The newer anticoagulants (dabigatran, rivaroxaban, apixaban) may be easier to dose and have fewer drug-drug interactions and food-drug interactions than warfarin, but still increase the risk of bleeding in elderly patients, particularly those with impaired renal function.

Questions 1-6

- 4) From this article, what is the most important role of health professionals when they are caring for an elderly person taking anti-coagulants?
- a) Ongoing and regular reviews
 - b) Frequently checking for bleeding
 - c) Prescribing anticoagulants other than warfarin

TEXT 5

Musculoskeletal conditions

There are more than 150 forms of arthritis and musculoskeletal conditions, but the more common conditions include:

Osteoarthritis: a degenerative joint condition affecting the weight-bearing joints such as the hips, knees and ankles and well as the hands and spine.

Rheumatoid arthritis: an autoimmune disease where the body's immune system attacks its own tissues, and thus differs from osteoarthritis which is characterised by wear and tear of joints. It is more severe than osteoarthritis and while not only confined to the joints, the hand joints are the most commonly affected.

Osteoporosis: a largely preventable condition whereby there is a progressive loss of bone density and decrease in the strength of the skeleton, such that even a minor bump or accident can cause serious fractures. Often people are not aware they have osteoporosis because the condition lacks obvious symptoms. The condition is much more common in females than in males.

Questions 1-6

- 5) Which of these conditions is most associated with wear and tear?
- a) The practically avoidable condition
 - b) An auto immune condition
 - c) The deteriorating joint condition

TEXT 6

Pressure injuries are not inevitable

Some aged care providers suggest anyone can use available risk assessment tools to identify and rate a person's potential for pressure injury and implement treatment. However, while such tools may assist decision making, they cannot be relied on in isolation of knowledge and clinical assessment conclusions. This is precisely where the importance of having the right skills mix in aged care comes into play. Research commissioned by the ANMF in 2016, in conjunction with Flinders University and the University of South Australia, provides evidence that a skills mix of registered nurses – 30%, enrolled nurse – 20% and personal care workers – 50%, is the minimum requirement “to ensure safe residential and restorative care” (Willis et al. 2016). Registered nurses are equipped with the knowledge, backed by an evidence base, to undertake a comprehensive assessment of potential for pressure injury and

commence appropriate preventative measures, and/or, to institute required wound management systems for established pressure injuries.

Questions 1-6

- 6) What do the researchers identify as the lowest safe ratio of staff in an aged care setting?
- a) Just under a third RNs, half personal care workers and the rest ENs
 - b) Mostly personal care workers and an equal split of ENs and RNs
 - c) Fifty percent are personal care workers with thirty percent ENs and 20% RNs.

PART C

TEXT 1

Have you ever experienced déjà vu? If so, you are among the 60-70% of the population who has. The majority of those who report déjà vu are between the ages 15 and 25.

Though some radical notions have in the past been connected to this strange feeling, such as déjà vu being a momentarily aligning with a past life or another you in a parallel universe, scientists now believe it has a neurological basis.

Unfortunately, the feeling is here one minute and gone the next, making it difficult to study. Even so, there are quite a few theories on what causes it. One traditional hypothesis, posited by psychiatrists, is mismatched brain signals. For a second it feels as though we are transported to a moment in the past and we mistake it for the present. This may be why it has been associated with the idea of reincarnation.

Another theory is that déjà vu is our brain trying to piece together a situation on limited information. A third states that it is a misfiring in the parts of the brain that recall memory and decipher sensory input. Sensory information, rather than taking the proper channels, leaks out of the short-term memory and into the long-term one. In this way, current experiences seem to be connected to the past. Some studies even suggest that familiar geometric shapes give us a sense of knowing something about a place that is, in reality, totally unfamiliar to us.

Since we are completely aware of everything that's going on when we experience déjà vu, this suggests that every part of the brain need not participate for the

sensation to take place. Psychologist Anne M. Cleary at Colorado State University, in a study in 2008, found that *déjà vu* followed patterns we associate with memory, specifically recognition memory“. This is the kind that gets us to understand that we are confronting something that we have seen or experienced before. If you have ever recognized a landmark, a friend from across the room, or a song on the stereo, you have experienced recognition memory.”

Familiarity-based recognition is associated with recognition memory. Here, we have that feeling of familiarity, but we can't quite place where we've seen this person, place, or thing. For instance, you recognize someone across the street, but can't remember their name or where you know them from. Prof. Cleary conducted several studies which found that *déjà vu* is a form of familiarity-based recognition. Her work suggests that our memory stores items in fragments. When there is a certain overlap between old and new experiences, we have strong feelings about the connection, which we interpret as *déjà vu*.

Recent studies looking at epileptic patients made impressive breakthroughs in our understanding of the phenomenon. Epileptics with certain intractable conditions require electrodes to be placed inside their brains in order to locate the source of their seizures. During this procedure, some neurologists have had patients experience *déjà vu*. They soon discovered that the phenomenon takes place in the medial temporal lobe, which is responsible for memory. The electrodes are usually placed within the rhinal cortex—the most important piece of which is the hippocampus, the structure responsible for long-term memory formation. French scientists have found that firing current into this cortex can trigger an episode of *déjà vu*.

The French study, published in the journal *Clinical Neurophysiology*, measured EEG wave patterns from patients with epilepsy who experienced *déjà vu* through electrical stimulation. The areas of the brain they examined included the amygdala, which is responsible for emotion and the hippocampus. Researchers found that electrical patterns, emanating from rhinal cortices and the amygdala or the hippocampus, caused *déjà vu* to occur. These neuroscientists believe that some sort of electrical phenomenon in the medial temporal lobe activates the memory in such a way that it causes *déjà vu* to occur.

Stranger still, scientists in the UK have actually found patients who experience “chronic *déjà vu*.” In this case, experts identified four senior citizens who encounter the feeling on a consistent basis. What is the impact of such a phenomenon? It made them feel as if they were clairvoyant. All four refused to go to the doctor, believing

they already knew what the physician would say, and avoided watching the news, thinking they already knew the outcome. That's because each time they took part in either activity that was the result they came to.

Questions 7-14

- 7) What does the first paragraph infer about déjà vu?
- a) Only young people get déjà vu
 - b) As the people get older, their déjà vu episodes will stop
 - c) Some people may never get déjà vu
 - d) None of the above
- 8) In the second paragraph, which of the following is new information about déjà vu?
- a) It is a demonstration of a parallel existence of some experience from the past in the present
 - b) It demonstrates the existence of a being in another parallel world
 - c) Neuroscientists may be able to offer an explanation
 - d) It has given rise to some extreme interpretations
- 9) In the third paragraph, the feeling of déjà vu is described as:
- a) Fleeting
 - b) Quick
 - c) Difficult
 - d) Unreal
- 10) What is the author talking about in the fourth paragraph?
- a) Parapsychological explanations for déjà vu.
 - b) Scientific theories about déjà vu.
 - c) Sensory theories about déjà vu.
 - d) Brain theories about déjà vu.
- 11) Which example of recognition memory does the Psychologist Anne M. Cleary not mention?

- a) Recognizing a landmark.
- b) Recognizing a friend from across the room.
- c) Recognizing a song.
- d) Recognizing patterns that we associate with memory.

12) How does déjà vu come into existence according to Prof. Cleary in the sixth paragraph?

- a) When we experience recognition memory.
- b) When we experience familiarity-based recognition.
- c) When there is overlap between old and new experiences.
- d) When we connect old and new experiences.

13) In the seventh paragraph, scientists have discovered that:

- a) Transmitting current into the rhinal cortex of epileptic patients may produce déjà vu.
- b) Transmitting current into the rhinal cortex of epileptic patients produces déjà vu
- c) Current that is transmitted into the brain of epileptic patients produces déjà vu.
- d) Epileptic patients experience déjà vu when they are subjected to electrode treatment.

14) In paragraph 9, chronic déjà vu

- a) turns people into fortune tellers
- b) cannot be cured by doctors
- c) only affects the elderly
- d) affects its sufferers' decisions

TEXT 2

Imagine healing the body without drugs or surgery, each of which can have nasty side effects. Instead, a physician uses the body's own building blocks to heal you.

Instead of returning again and again, or having to take medication continually, one shot does it all. These are the promises of gene therapy.

The concept is easy to grasp. Genes control proteins that in turn control all of our body's functions. When a faulty gene, usually due to a mutation, malfunctions and causes disease, all that would have to be done is to "knock out" or replace the faulty gene. Once the correct protein enters the system, the disease is finished. It is, however, the replacement process that is complicated.

One problem is exactly how to deliver a gene to a patient's DNA. To do that, scientists create a custom virus that infects a target cell, yet flies under the immune system's radar. By doing so, the virus leaves its own genetic material inside the cell. That cell begins to reproduce, carrying the gene with it, and spreading it throughout the body.

Gene therapies are not currently approved by the FDA. Dozens of clinical trials are ongoing, however. This cutting-edge therapy is approved to treat one particular disorder in Europe—lipoprotein lipase deficiency, where the patient cannot break down fat. Another use will soon be approved, to treat combined immune deficiency, or the "bubble boy" disease.

Other conditions it is expected to someday treat include heart disease, diabetes, some forms of cancer, muscular dystrophy, immune disorders, genetic disorders, AIDS, hemophilia, and certain blindness-causing conditions. With AIDS, gene therapy will be used in a different way. The HIV virus camouflages itself from the immune system. Gene therapy can make its presence known, allowing it to be recognized and destroyed.

The things researchers look at when evaluating a new therapy is its safety profile, how effective it is, and what a proper dosage may look like. Just like any therapy, things can go wrong. For instance, altered viruses could change back into their original form, causing infection. Sometimes the wrong cell is approached by the virus. Or the virus places the gene in the wrong place within a cell's DNA. In this last case, healthy cells may become damaged or cause illness, even develop into a tumor.

There have been stumbling blocks along the way. Keep in mind that all clinical trials are monitored by the FDA and the National Institutes of Health (NIH). Even so, gene therapy almost went bust in 1999 when a volunteer, 19-year-old Jesse Gelsinger, died during testing. The Arizona teen's immune system reacted violently as a result of the treatment. Gene therapy lost its innocence and many young, promising

scientists decided to put their efforts elsewhere, setting the field back. A year after that, during a French trial, some participants developed leukemia.

After these incidents, dozens of clinical trials ceased and funding was pulled. Researchers learned a lot from these disturbing tragedies, and put stringent safety controls in place. They have since discovered how to deliver genes using viruses in a safe and effective manner, that doesn't set off the immune system.

Researchers have also implemented guidelines that help monitor patients and administer to side effects. A few successes then brought gene therapy back from the brink. In 2008, some blind subjects reported improvements in vision. Shortly after, in another experiment, 80% of "bubble boy" children regained immune system function.

The efficacy of gene therapy today is not constant throughout, but varies from one condition to the next. A recent study using the therapy to treat muscular dystrophy saw impressive results. A 2013 study was even more dramatic, where a small clutch of patients with leukemia were cured. Other studies on hemophilia and one cause of blindness, retinitis pigmentosa, have also seen remarkable results.

There have been other trials however that have not been so encouraging. One for congestive heart failure ended in "disappointing" results, and another for Parkinson's ended in what researchers called a "mixed bag." One problem that must be overcome, the immune system sometimes does recognize and take out the viral messenger

Questions 15-22

15) Which of these can be attributed to gene therapy in the process of healing?

- a) Using drugs
- b) Performing surgery
- c) Taking medication continually
- d) Using the body's building blocks

16) How are genes delivered to a patient's DNA?

- a) Through viruses that manage to escape the body's immune system.

- b) By creating a custom virus that carries the necessary genetic material and leaves it in the target cell.
- c) By using a virus that carries the necessary genetic material and leaves it in the target cell in the body.
- d) Through cells that reproduce, carry the gene, and spread it throughout the body.

17) Which of these diseases is eligible for gene treatment?

- a) A disease in which the patient cannot break down fat.
- b) The “bubble boy” disease.
- c) Muscular dystrophy.
- d) Blindness-causing conditions.

18) Which of the following poses a danger to the gene treatment therapy?

- a) Proper dosage of therapy
- b) Its effectiveness.
- c) Controlling and predicting virus’ efficiency.
- d) Cell damage.

19) What is the author referring to when he says that “there have been stumbling blocks along the way”?

- a) The obstacles the researchers faced during clinical trials
- b) The obstacles created by the FDA and the National Institutes of Health (NIH).
- c) The fact that some patients died during the clinical trial.
- d) The fact that some patients developed leukemia during the clinical trial.

20) What did these incidents contribute to?

- a) Discovering how to use viruses to deliver genes.
- b) Finding safer methods and following protocols for monitoring patients
- c) Finding a way to improve sight in blind people.
- d) Finding a way to treat “bubble boy” disease.

21)What does the author imply about gene therapy in tenth paragraph?

- a) That all diseases can be treated with equal success.
- b) That not all diseases can be treated with equal success.
- c) That the leukemia treatment is more successful than the treatment for hemophilia.
- d) Some diseases can be completely cured.

22)What is the author describing in the final paragraph?

- a) Failures of gene therapy.
- b) Disappointing results that gene therapy produced for Parkinson's disease.
- c) The mixed results that gene therapy has produced.
- d) The gene therapy trials that have not produced promising results.

Reading Practice Test 9 - PART B & C

Answer Keys

Part B		Part C			
	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	B	-	Q 7	C	-
Q 2	C	-	Q 8	C	-
Q 3	B	-	Q 9	A	-
Q 4	A	-	Q 10	B	-
Q 5	C	-	Q 11	D	-
Q 6	A	-	Q 12	C	-
			Q 13	A	-
			Q 14	D	-
			Q 15	D	-
			Q 16	B	-
			Q 17	A	-
			Q 18	C	-
			Q 19	A	-
			Q 20	B	-
			Q 21	B	-
			Q 22	C	-

Review & Explanations

TEST-10

Chronic obstructive pulmonary disease

PART A

TEXT 1

Chronic obstructive pulmonary disease (COPD) is serious, progressive and disabling damage to the lungs which makes breathing difficult. There are several conditions that are considered to be COPD. It is most commonly caused by cigarette smoking but several working environments are known to lead COPD. Irritants such as silica dust, coal dust and chemical fumes are key occupational hazards that can lead to COPD.

Emphysema is when the air sacs in the lung are damaged, making it increasingly difficult to get sufficient air exchange.

Chronic bronchitis is long term inflammation of the bronchi and bronchioles. These become narrower and produce more mucous, making it harder to breathe and causing excessive coughing.

Chronic asthma is long-term respiratory condition that causes inflammation and tightening of the airways, making breathing difficult. It can be caused by an overreaction to a stimulus such as exercise or cold air. Asthma affects people of all ages, unlike other COPD conditions which are mostly age-related.

TEXT 2

Clinical assessment

- Take a thorough history including checking for allergies
- Ask about lifestyle activities including smoking and work environment
- Symptoms to check for:

breathlessness

a new or persistent cough, especially in a morning

production of a lot of mucous, which is swallowed or coughed up.

wheezing

chest tightness

recurrent chest infections, increasing in frequency

Diagnostic tests:

- Spirometry

- Chest x-ray

- Blood test to eliminate anaemia

- Other test options include:

electrocardiogram (ECG)

echocardiogram

peak flow test

blood oxygen test

computerised tomography (CT) scan –

sputum sample

Differential diagnosis of COPD to Asthma

- Spirometry – key asthma diagnostic test

- Peak flow – diagnosis of asthma

- Age – COPD more common in over 50's, asthma any age

- Response to medication - COPD slow to respond to medications, if at all. Asthma very responsive

- Disease progression – COPD gradually worsening, asthma often flares up and then settles

Allergy history – Asthma often secondary to allergies such as hay fever

TEXT 3

Management of COPD

COPD cannot be cured but symptoms can be managed

- Stopping smoking is the most important way to ease symptoms
- Pulmonary rehabilitation – a specialised programme of exercise and education
- Having an annual influenza vaccination and a possible one-off pneumococcal vaccine
- Acting quickly if symptoms flare-up
- Home-based oxygen therapy in severe cases
- Lung surgery is rare and suitable for only a small number of people

Inhalers and medications

- Short acting broncho-dilators, or relievers taken to ease symptoms
- Long acting broncho-dilators taken long-term to control symptoms and prevent flare-ups
- Flare-up (exacerbation) medication

Steroid tablets such as prednisone may be prescribed as a short course for 1 or 2 weeks if you have a sudden flare up of your COPD symptoms

- These medications may be administered via nebuliser when symptoms become severe and no longer responsive to inhalers

- Oral medications

Theophylline is a tablet that relaxes and opens up the airways. It's usually taken twice a day. Sometimes a similar medication called aminophylline is also used

Possible side effects include:

- feeling and being sick
- headaches
- difficulty sleeping (insomnia)

- noticeable pounding, fluttering or irregular heartbeats (palpitations)

Regular blood tests are needed if these medications are prescribed

Mucolytic medication called carbocisteine to thin out mucous

TEXT 4

Potential complications of COPD

- Susceptibility to respiratory infections
- Pulmonary hypertension
- An increased risk of heart disease
- Lung cancer (for smokers)
- Depression
- Frequent visits to hospital
- Reduced quality of life

Prevention of COPD

- Not smoking is the best way to avoid COPD
- If working in a high risk environment
- wear Personal Protective Equipment
- follow health and safety procedures

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1 How to prevent COPD? _____

2 The tests that may be carried out to diagnose COPD? _____

3 The different inhalers that might be prescribed to manage COPD?

4 How emphysema affects a person? _____

5 The link between asthma and allergies? _____

6 The oral medication options for COPD patients? _____

7

The possible complications of COPD? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What age are most people who are diagnosed with COPD? _____

9 What part of the body is most affected by COPD? _____

10 What is the specialised programme of exercise and education that COPD sufferers might benefit from? _____

11 What should people working in high risk environments wear, to reduce the risk of developing COPD? _____

12 What is the key diagnostic test for asthma? _____

13 Which condition causes airways to become narrower and produce more mucous?

14 What is the most important way to reduce the symptoms of COPD?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 Pulmonary _____ is a potential complication of COPD.

16 Patients with COPD are often _____ to respond to medication, if at all.

- 17** Emphysema is when the _____ in the lung are damaged.
- 18** Having an annual influenza vaccination and a possible one-off _____ can help with COPD management.
- 19** COPD is _____ and disabling damage to the lungs.
- 20** Production of a lot of _____ which is swallowed or coughed up is a symptom of COPD.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	D	-	Q 8	Over 50		Q 15	hypertension	
Q 2	B	-	Q 9	Lungs		Q 16	slow	
Q 3	C	-	Q 10	Pulmonary rehab	+	Q 17	air sacs	
Q 4	A	-	Q 11	Personal Protec	+	Q 18	pneumococcal va	+
Q 5	B	-	Q 12	Spirometry		Q 19	serious, progre	+
Q 6	C	-	Q 13	Chronic bronchi	+	Q 20	mucous	
Q 7	D	-	Q 14	Stopping smokin	+			

Review & Explanations

PART B

TEXT 1

Policy Statement – Management of Elective Surgery Booking lists

Booking List Hospitals are part of a national information system that collects indicator data on waiting times for elective surgery. Administrative guidelines (Booking List Information System Guidelines, Health Information Service, 2010) must be used for the administration and submission of data from the automated booking list systems at Booking List Hospitals.

BLIS provides an indicator of demand for surgical services across the system by speciality and procedure and can be used to review caseload. It enables SA Health to provide information to Booking List Hospitals about their comparative situation within the state and to plan across the system.

Some procedures are excluded from the calculation of booking list statistics. Hospitals should exclude records with procedure indicators of 181 (Check Cystoscopy) and 999 (Dental or Obstetric surgery) and non-surgical treatment (eg endoscopic treatment).

Questions 1-6

- 1) What is a key benefit of BLIS?
 - a) SA health gets an overview so they can see which hospitals have capacity to perform dental extractions.
 - b) Patients can see which hospitals are able to provide surgical services to them.
 - c) SA health gets an overview so they can see which hospitals have capacity to admit patients for particular operations.

TEXT 2

Heart scare a sign of things to come, expert fears

One of Australia's leading cardiologists fears that heart complaints in athletes will become more common as the demands of professional sport grow. "I fear that with higher levels of sporting excellence, that this may come at a cost — that cost being

more athletes presenting with heart conditions," said Chris Semsarian, the head of the molecular cardiology program at the Centenary Institute. Professor Semsarian said athletes were at higher risk of developing heart conditions because of their occupation. "They're putting their heart under great strain during sports. That puts pressure on the heart," he said. But he said heart problems were also more likely to be diagnosed among athletes than among the general public due to the advanced screening procedures used in professional sport. He said the competitive nature of athletes meant persuading them to prioritise their health over sporting ambition was challenging. "I am forever amazed at the way athletes will never give up," he said.

"You tell them that they have a major heart condition that could kill them and they say, 'What do I need to take so I can get back on the field?'"

Questions 1-6

2) What discovery consistently astounds Professor Semsarian?

- a) Athletes still want to exceed at their game even if it has a negative effect on their health
- b) Athletes have access to incredibly sophisticated diagnostic assessment tools, as professional sportsmen
- c) Athletes are increasingly at risk of developing serious illnesses despite being super fit.

TEXT 3

Programmes and initiatives to address musculoskeletal disorders

Programs that support management and treatment of musculoskeletal conditions include:

The Medicare Benefits Schedule, which provides subsidies for patient care and includes Medicare items for the planning and management of chronic and terminal conditions. Eligible patients can also be referred by a GP for up to five Medicare subsidised allied health services that are directly related to the treatment of their chronic condition, including musculoskeletal conditions.

The Pharmaceutical Benefits Scheme continues to provide subsidies for medicines used in the treatment of musculoskeletal conditions and pain management.

The *National Health and Medical Research Council (NHMRC)* receives significant investments for research into arthritic and rheumatoid conditions. Improving the care of patients with multiple and complex chronic diseases, including musculoskeletal conditions, has also been identified by the NHMRC as a major focus in its 2013-15 Strategic Plan.

Arthritis Australia get funding to improve consumer awareness and to build and implement a local exercise program on a national level.

Osteoporosis Australia receives funding to maintain and update resource materials, improve osteoporosis management in primary care and deliver an exercise program focussing on building bone strength and density.

Questions 1-6

- 3) Which of these initiatives is mostly involves in depth study of one or more musculoskeletal disorder
- a) Osteoporosis Australia
 - b) Medicare benefits schedule
 - c) NHMRC

TEXT 4

Drug categories of concern in the elderly: digoxin

Digoxin, a cardiac glycoside, is used to increase the force of myocardial contractions and to treat supraventricular arrhythmias. However, it must be used with caution in elderly patients. In men with heart failure and a left ventricular ejection fraction of ? 45%, serum digoxin levels > 0.8 ng/mL are associated with increased mortality risk. Adverse effects are typically related to its narrow therapeutic index. One study found digoxin to be beneficial in women when serum levels were 0.5 to 0.9 ng/mL but possibly harmful when levels were ? 1.2 ng/mL. A number of factors increase the likelihood of digoxin toxicity in the elderly. Renal impairment, temporary dehydration, and NSAID use (all common among the elderly) can reduce renal clearance of digoxin. Furthermore, digoxin clearance decreases an average of 50% in elderly patients with normal serum creatinine levels. Also, if lean body mass is reduced, as may occur with aging, volume of distribution for digoxin is reduced. Therefore, starting doses should be low (0.125 mg/day) and adjusted according to

response and serum digoxin levels (normal range 0.8 to 2.0 ng/mL). However, serum digoxin level does not always correlate with likelihood of toxicity.

Questions 1-6

- 4) Which of the following serum digoxin levels can be beneficial in elderly patients?
- a) 0.8 - 2.0ng/mL
 - b) 0.5 - 0.9ng/mL
 - c) 0.5 - 0.9mg/mL

TEXT 5

Memo re: Tuberculosis Control

Hospital and Health Services included in the scope of this directive shall achieve the following outcomes:

- All cases of suspected and confirmed TB are managed in co-operation with an established TB Control Unit (TBCU)
- Implementation of statewide standardised diagnosis, treatment and ongoing management protocols to minimise the risk of drug resistance, treatment failure and/or relapse.
- Follow endorsed state and national guidelines for preventing the transmission of TB in healthcare and community settings and to prevent TB in at-risk children through Bacille Calmette-Guerin (BCG) vaccination.
- Notify the Department of Health of all cases of TB in accordance with the legislative obligations of the Public Health Act 2005.
- Inform the Department of Health (CDB) within one business day of TB cases that pose an increased public health risk, where there is potential for involvement or implication of another jurisdiction, country or other governmental department or non-governmental organisation, and where there is potential for heightened community interest in accordance with the Protocol for the Control of TB

Questions 1-6

5) What key directives are included in this memo?

- a) The need to: know the Protocol for the Control of TB; notify the Department of Health of all cases of TB; administer the BCG to anyone suspected of having TB
- b) The need to: implement the statewide immunisation programme; admit all cases of TB to a specialist TB unit; meet the Public Health Act (2005) requirements
- c) The need to: follow consistent practises relating to identifying, treating and preventing TB; adhere to immunisation programme; follow reporting protocols.

TEXT 6

Advanced Health Directives: Patient guidance

An Advance Health Directive (AHD) is a legal written document, containing your decisions about your future health treatment. Anyone over 18 can prepare an AHD – even healthy people prepare AHDs. If you lose your mental capacity, you are not legally allowed to prepare an AHD and nobody else can do it for you. In WA, the law allows you to write an AHD to say what treatments you want or don't want in specific circumstances. Or you can appoint someone to make medical treatment, personal or lifestyle decisions on your behalf, when you are unable to make or communicate your decision. You can make an AHD in which you either provide consent, or refuse consent, to future treatments. For example, you may say that you want or do not want a certain treatment. An AHD can only be completed while you have the ability – or 'mental capacity' – to make and communicate decisions. As soon as you lose the capacity to make and communicate decisions, you are no longer able to complete or modify an AHD. No one else can complete an AHD for you once you have lost capacity.

Questions 1-6

- 6) What advice should a healthcare professional give adolescents about writing an AHD?
- a) An AHD has to be approved by both parents if the person is under 18.
 - b) An AHD cannot be accepted unless a person is aged 18 or over.

c) An AHD can only be put in place by people aged under 18.

PART C

TEXT 1

Sometimes the Tooth Fairy gives more than a dollar underneath your pillow. In the case of Dr. Alysson Muotri, associate professor of pediatrics and cellular and molecular medicine at UC San Diego School of Medicine and a noted expert on autism, the Tooth Fairy gave new insight into what may make humans social. Through Dr. Muotri's Fairy Tooth Kit Collection campaign, donated baby teeth from both those with autism and those unaffected were collected for research. A tiny brain was then created in a petri dish from the teeth.

These miniature brains may provide a window into the human spectrum of sociability, helping us better understand why certain individuals like those with autism have diminished social skills. It may also help us understand how humans evolved to be as social as we generally are.

Called Brain or Cerebral Organoids, Dr. Muotri and his team were able to create these so-called mini-brains by extracting the pulp cells in the teeth and converting them into brain cells. This is done through the induced pluripotent stem cell (iPS) technique, a reprogramming of cells to be in a stem cell-like state. These neural progenitor cells are able to create networks similar to the developing cortex of a human brain. Dr. Muotri's research showed that the organoids using cells from those with autism had fewer neural connections than those unaffected.

While autism is generally associated with low degrees of sociability, Williams syndrome is a rare genetic disorder where those affected have an extremely high level of sociability to the point of talking with strangers. It is often referred to as the "opposite of autism".

Dr. Muotri and his team of researchers at the University of California San Diego, along with researchers at the Salk Institute of Biological Studies, examined organoids grown from those affected by Williams syndrome. The team noticed that instead of former fewer neural connections like the autism organoids, the organoids contained an abnormally high level of neural connections.

Organoids derived from cells unaffected by a neurobiological disorder were right in the middle. In other words, the level of neural connections in the mini-brains correlated with the sociability of person. The higher the sociability (from autism to

unaffected to Williams Syndrome), the greater the neural connections in the cerebral organoid.

Speaking to New Scientist, Dr. Muotri said: The differences are striking, and go in opposite directions. In Williams syndrome, one of the cortical layers makes large projections linking into many other layers, and these are important for sociality. By comparison, autism-linked brains are more immature, with fewer synapses."

The connection between synapses and sociality was also found by the research team when examining donated brains from those who had autism or Williams syndrome. In addition, another research team working with brain organoids recently found that patients with idiopathic autism overproduced inhibitory neurons.

In the December 2015 issue of the journal Developmental Biology, researchers Madeline Lancaster and Iva Kelava explored both the promise and challenges of cerebral organoids. In the article, Dishing out mini-brains: Current progress and future prospects in brain organoid research , they argue that brain organoids can successfully model neurodevelopmental conditions such as idiopathic autism and the brain organoids "model early human embryonic and fetal brain development to a remarkably high degree." While the work on brain organoids is quite new, there appears to be a great deal of promise in the research with unlocking some of the secrets of the Brain.

Brain organoids (and organoid systems in general), which adequately model tissue development and physiology, are a relatively new development, and the field has exploded in the last several years. Thus, it is easy to envisage that in 10–20 years from now (or even less) we will be able to almost fully mimic development of certain tissues in vitro. In addition, further improvements in the technique might allow us to model adult brain physiology and disorders of the adult and ageing brain."

Questions 7-14

- 7) In what way is the concept of Tooth Fairy related to the research mentioned in first paragraph?
- a) Because teeth for the research were collected from under the children's pillows.
 - b) Because the Tooth Fairy gave new insight into what may make humans social.
 - c) Because baby teeth were donated for the research.
 - d) Because a tiny brain was created from baby teeth.

- 8) The research being performed on the tiny brains may provide us with various pieces of information. Which of the following aspects are not mentioned in second paragraph?
- a) It may provide deeper insight into the human ability to socialize.
 - b) It can contribute to our understanding of autism.
 - c) They can contribute to our understanding of why those affected with autism have a lower ability to socialize.
 - d) They can help us to understand why human beings have developed into social beings.

9) In what way is Williams syndrome different from autism?

- a) Autism is not a genetic disorder.
- b) Autism affects people with fewer neural connections.
- c) Williams syndrome is a genetic disorder.
- d) Patients with Williams syndrome display a very high level of sociability.

10) Williams organoids:

- a) Are produced from those affected by Williams syndrome.
- b) Produce fewer neural connections.
- c) Contain an unusually high quantity of neural connections.
- d) Reach an unusually high level of neural connections.

11) What do we find out about people who don't suffer from either autism or Williams syndrome in eighth paragraph?

- a) That they have a large number of synapses
- b) That they have fewer synapses.
- c) Their neural connections remain unaffected.
- d) They have a moderate number of synapses.

12) What does the information in ninth paragraph suggest?

- a) There has been a lot of research that supports the correlation between neural connection and sociability.
- b) There is other research that supports the correlation between neural connection and sociability, as well.
- c) That the correlation between neural connection and sociability was first discovered many years ago
- d) The correlation between neural connection and sociability was not discovered in brain organoids.

13)What is the advantage of brain organoids?

- a) They are a perfect match to the human brain.
- b) They can be used to model human development.
- c) They are an excellent representation of early human brain development.
- d) They are an excellent guide to the early human brain development.

14)What kind of future is predicted for brain organoids in tenth paragraph?

- a) Bright
- b) Bleak
- c) Foreseeable
- d) Uncertain

TEXT 2

Waking up today, on so-called Blue Monday, the “most depressing day of the year”, you may already be aware that this concept is based on a fraud. An almost deliciously spurious mathematical formula was dreamed up by a PR agency, given the veneer of academic rigour by attaching the name of a lecturer at a further education college, and a media phenomenon was born. More than a decade’s worth of articles and social media memes have at turns reinforced and defied the Blue Monday myth, and it is often now held up as a case study of bad science.

It is, of course, laughable to have a formula where W stands for weather and days-since-Christmas is raised to the power of Q, the days-since-we-quit-our-new-year-resolutions. But the driver behind this mockable maths is a much more sinister lie, one from which many struggle to escape 365 days of the year.

Blue Monday was originally invented for an advertising campaign for Sky Travel. Its sole purpose was to sell holidays with the false promise that spending money would raise flagging spirits. Blue Monday has now been used to sell everything from flowers to Ferraris, takeaways to airport parking. Beat Blue Monday, they tell us (laughing knowingly at the fact that it's made up): buy something new today.

We're told every day by advertisers that buying stuff will make us happy. A new pair of shoes will help us to feel better after a breakup. New beauty products will give us a sense that we're "worth it". A bigger car will give us social status. New toys will make the children happy. Even loans are sold this way: one payday lender is currently running a campaign with a smiling woman, snuggling a mug of tea and feeling happy thanks to a 1,200% APR loan, an implausible scenario if ever there was one. So it's no wonder that on Blue Monday, the day our anxieties and misery are supposed to peak, the advertisers scream that the path out of unhappiness is paved with till receipts.

Sometimes the harm this causes is relatively trivial: overconsumption of stuff we don't need; a bit less money in the bank; a wardrobe that won't close because we're not very good at throwing away our throwaway fashion buys. But the harm can be deadly serious for people struggling with mental health problems. Imagine it: your mood is low. You find yourself crippled by anxiety. You feel like a failure, a burden on your family and friends. This is the moment when a lifetime of being told that stuff will make you feel better takes its toll: in desperation, nine in 10 people with a mental health problems find themselves spending more when they're feeling unwell.

In my work at the research charity Money and Mental Health, I've lost count of the stories I've heard that stop me in my tracks. Mothers spending their way through months of postnatal depression – one who bought nearly 100 buggies over four years. A man who tried to buy a villa in a country he didn't have a visa for, convinced it would turn his life around. A young woman who found buying something online was the only way she could stop a panic attack, so she did it almost every day. A husband who felt such a burden he'd buy endless presents he couldn't afford for his wife. And many of them tell us that the boxes sit in the living room, unopened, unwanted, but impossible to ignore.

Because buying stuff is just the first part of the cycle. Next comes the guilt. People with mental health problems are far more likely to be living on a low income, and the financial damage that compulsive shopping can do is extraordinary. The boxes – or if they're actually opened, the new possessions – are a constant reminder of the mistakes made. Too often, people don't return unwanted goods: nearly half tell us

that's because they just want to pretend it never happened. The guilt brings their mood lower and then buying something for a temporary buzz feels like the only way back up.

We have just launched a new tool, the Shopper Stopper, which is helping people to curb their night-time shopping in particular by allowing them to set the opening hours of online shops. But it's going to take a wider societal shift to really shake us out of these habits for good.

Questions 15-22

15) What does the author imply about the concept of Blue Monday?

- a) It is a non-existent phenomenon.
- b) It was conceived by a PR agency.
- c) It was promoted by a university lecturer.
- d) It is considered to be a result of bad research.

16) What is the underlying purpose of Blue Monday?

- a) To make people sell holidays to feel better.
- b) To make people feel better
- c) To make people buy things under the pretense of feeling happier.
- d) To make people sell things under the pretense of feeling happier.

17) On Blue Monday, advertisers

- a) Take advantage of people's unhappiness
- b) Convince people that their supposed unhappiness will disappear with shopping.
- c) Try to sell as much as possible.
- d) Convince people that as many receipts as possible will make them happy.

18) What are the consequences of Blue Monday for people who suffer from anxiety?

- a) They are unimportant.

- b) They feel like failures
- c) They feel desperate.
- d) They spend more money.

19)What has the writer heard that stopped him / her in his / her tracks?

- a) Large numbers of upsetting stories that left him feeling shocked.
- b) Large numbers of upsetting stories that make him want to stop working for the charity.
- c) Advertisements which encourage people to buy things they don't need.
- d) Occasional stories which have had happy endings.

20)Which of the cases below have not been recorded by the author in his/her work at the research charity?

- a) A depressed mother who bought 100 prams.
- b) A husband who wanted to buy numerous presents for his wife.
- c) A husband who believed he could alleviate his stress by buying unaffordable things for his wife.
- d) A woman whose panic attacks were stopped only by buying things online.

21)Apart from the financial damage compulsive shopping can cause, the author mentions another far more devastating issue:

- a) Low income
- b) Guilt
- c) Mistakes they have made
- d) Not returning unwanted goods.

22)What is the purpose of Shopper Stopper?

- a) To control people's night-time shopping.
- b) To stop their night-time shopping.
- c) To allow them to set limits on shopping to create good habits.
- d) To get rid of the habit of shopping.

Reading Practice Test 10 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	C	-
Q 2	A	-
Q 3	C	-
Q 4	B	-
Q 5	C	-
Q 6	B	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	C	-	Q 15	B	-
Q 8	B	-	Q 16	C	-
Q 9	D	-	Q 17	B	-
Q 10	C	-	Q 18	D	-
Q 11	D	-	Q 19	A	-
Q 12	B	-	Q 20	B	-
Q 13	C	-	Q 21	B	-
Q 14	A	-	Q 22	A	-

Review & Explanations

TEST-11

Mumps

PART B

TEXT 1

What is mumps?

Mumps is a contagious viral infection that is most common in children between 5 and 15 years of age. Mumps is most recognisable by the painful swellings located at the side of the face under the ears. Other symptoms include headache, joint pain and a high temperature.

Mumps is a serious disease because it can lead to inflammation of the spinal cord and brain, the brain itself, and the heart. About one in 200 children with mumps will develop brain inflammation. Mumps can damage nerves leading to deafness. Males might get mumps orchitis which is swollen testes. This can cause a reduced sperm count but not enough to cause infertility. Women may miscarry during their first trimester.

Mumps vaccination

Vaccination is the most effective way to control mumps. The Measles, Mumps and Rubella (MMR) vaccine is part of the routine childhood immunisation schedule in Australia. It is given as a course of two injections. The first injection of vaccine is usually given at around 12 months of age. A second booster dose is usually given at the age of four.

The increase in mumps

In 1998, concern was raised that there was a link between the MMR vaccine and autism. This has been totally disproved but it did lead to a drop in the number of children having the vaccine. 27% of all mumps cases have been in adults, in Australia over the past five years. These adults have already received two doses of the vaccination. The increase in cases might be due to the vaccine not lasting as long as expected. It might also be the rise in the mumps virus in the wider community because of reduced immunisation rates.

TEXT 2

How is mumps spread?

Mumps is caused by the mumps virus, which belongs to a family of viruses known as 'paramyxoviruses'. Mumps is spread in the same way as colds and flu. Infected droplets of saliva can be inhaled or picked up from surfaces and transferred into the mouth or nose.

Mumps spreads rapidly because a person is most contagious a few days before the symptoms appear. This means they do not realise they may be spreading the virus. A person is infectious for a few days afterwards too. Anyone who comes into contact with infectious mumps can get mumps, unless they have had mumps before or have been immunised.

Mumps is so infectious that it is a recognised notifiable disease. This means laboratories, school principals and directors of childcare centres must report all cases of mumps to their local public health unit.

A doctor can usually diagnose mumps based on the person's symptoms and signs alone. A blood test or sample from the throat, urine or spinal cord fluid may be taken to confirm the diagnosis and ensure the incidence can be reported.

There is no specific treatment for mumps. Simple medication like paracetamol may reduce pain and fever. Warm or cold packs to the swollen glands may provide relief. Drinking plenty of fluids and resting is essential.

TEXT 3

Symptoms of mumps

Painful swellings at the side of the face in the parotid salivary glands is the most common symptom of mumps. These glands are within the cheeks, near the jaw line, below the ears. This gives the person a distinctive "hamster face" appearance.

The symptoms of mumps usually develop 14 to 25 days after a person is infected with the mumps virus (the incubation period). The average incubation period is around 17 days.

Common symptoms of mumps are:

- A general feeling of being unwell
- High temperature
- Discomfort when chewing

- Headache
- Joint pain
- Feeling sick
- Mild pain in the abdomen
- Feeling tired
- Loss of appetite

The face will be back to normal size in about a week. About one third of infected people do not show any symptoms at all but they can still spread the virus.

The most common complication of mumps is mumps orchitis, according to statistics from the Australian Government, Department of Health. Mumps is usually a more severe illness in people infected after puberty. 7.1 % of people aged 5-14, experienced additional complications from mumps. Only 3.8% of people aged 25 -59 had additional complications compared to 15.4% of those aged 60+. Nobody in the 0-4 and 15-24 age ranges had additional complications.

TEXT 4

Preventing the spread of mumps

- MMR vaccine is the most effective way to stop a person developing mumps.
- People born after 1965 need to make sure they have had two doses of mumps containing vaccine.

People with mumps should:

- Stay away from childcare, school, work and other places where people gather for nine days after the onset of the swelling of the salivary glands
- Wash their hands regularly, using soap and water
- Always use a tissue to cover their mouth and nose when they cough and sneeze.
- Throw the tissue in a bin immediately after use
- Wait till the doctor says they are no longer infectious.

Vaccines have reduced the number of cases of mumps internationally but the rate of mumps outbreaks is increasing. The number of mumps cases does fluctuate. The

World Health Organisation keeps track of the number of cases by country. In Australia, there have been less than a 100 cases in some years since 1997. Between 2007 and 2014, there was an average of 232 cases per year. In 2015 there were 633 cases but in 2016, 800 cases were reported. In 2017, this rose to 806.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** The number of cases of mumps in Australia? _____
- 2** The number of cases of mumps among adults in Australia in the past 5 years? _____
- 3** What causes the hamster face linked to mumps? _____
- 4** The way the mumps virus is spread? _____
- 5** The condition that was linked to the MMR vaccine in 1998? _____
- 6** The incubation period for mumps symptoms to appear? _____
- 7** What people with mumps should do? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** What is the name of the family of viruses that the mumps virus belongs to?

9 What is the most common age range when someone gets mumps?

10 What does MMR stand for? _____

11 Which glands swell up around the face if a person has mumps?

12 Mumps spreads in the same way as what two other conditions?

13 How many cases of mumps were reported in Australia in 2017?

14 At what stage of their pregnancy, might a woman with mumps miscarry her child?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 Mumps spreads rapidly because a person is _____ a few days before the symptoms appear.

16 MMR vaccine is the most effective way to stop a person _____

17 The most common complication of mumps is _____

18 The Measles, Mumps and Rubella (MMR) vaccine is part of the routine _____ schedule in Australia.

19 Directors of childcare centres must _____ all cases of mumps to their local public health unit.

20 Only _____ of people aged 25 -59 had additional complications from mumps.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	D	-	Q 8	paramyxoviruses		Q 15	most contagious	
Q 2	A	-	Q 9	5 and 15		Q 16	developing mump +	
Q 3	C	-	Q 10	Measles Mumps R +		Q 17	mumps orchitis	
Q 4	B	-	Q 11	Parotid salivar +		Q 18	childhood immun +	
Q 5	A	-	Q 12	Colds, flu		Q 19	report	
Q 6	C	-	Q 13	806		Q 20	3.8%	
Q 7	D	-	Q 14	First trimester +				

Review & Explanations

PART B

TEXT 1

Use of 'StoneChecker' for Kidney Stone Evaluation

'StoneChecker' is medical imaging software used to measure key kidney stone parameters to generate a set of textual metrics. It uses an application that analyses the texture in an existing radiological scan. The aim is to help clinicians assess a stone's heterogeneity and any characteristics not visible to the naked eye. In other words, it creates a virtual 'map' of a kidney stone. The technology is designed to be used with conventional CT scans to help clinical decision making about diagnosing and treating people with kidney stones. It imports imaging data and calculates metrics based on individual pixel data points within a given region of interest. These metrics assess stone heterogeneity, including texture irregularity, not provided by conventional CT scans.

Question

- 1) What do we learn about 'StoneChecker' in the first paragraph?
 - a) Kidney stones can be located using this device.
 - b) Such optical equipment measures microscopic surface features of kidney stones
 - c) It is a computer program that helps clinicians visualize kidney stones.

TEXT 2

Postoperative Treatment of Knee Inflammation

A middle-aged man with a history of gout was hospitalized for a surgical procedure. While in postoperative care, he experienced severe pain and swelling in his left knee. An orthopaedic surgeon was consulted, and he prescribed Colchicine to treat the acute gouty inflammation. The pharmacist overrode a high-dose alert, which indicated that the patient should be treated with a maximum dose of 4.8 mg. Instead, the patient received more than 10 mg of Colchicine before he started to show signs of renal failure. The treatment was stopped, but the patient died. In this instance, a policy concerning high drug doses had been in place. The investigation

found that the pharmacist should have consulted with the prescribing physician for alerts.

Postoperative Treatment of Knee Inflammation

Questions 1-6

- 2) In the example described in the paragraph, what went wrong?
- a) The pharmacist ignored critical information.
 - b) The prescription was incorrectly managed by the orthopaedic surgeon.
 - c) The prescribing physician failed to initiate the 'alert' process.

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TEXT 3

Gluten: It's Not for Everyone

Removing gluten from one's diet is no easy feat. Reading labels and understanding where gluten may be hiding takes time and education. Gluten-free products are typically more expensive, and, while varieties of foods and products are expanding, there are still considerably fewer choices for gluten-sensitive consumers. While there is considerable evidence to support the benefits of a gluten-free diet, the medical community warns against it, if it is not deemed medically necessary. Removing gluten from one's diet can not only reduce the intake of essential vitamins and minerals which are found in fortified foods, but it can also reduce the amount of probiotics in the gut which boost the immune system.

Questions 1-6

- 3) What does the writer imply about gluten?
- a) Eliminating gluten from your food intake is risky.
 - b) More people should take time to be educated about gluten.
 - c) Reducing the amount of gluten you consume is beneficial for your probiotics.

TEXT 4

Blood Culture Policy

Any patient in a hospital setting who develops a fever or evidence of sepsis should have blood cultures sent. As sepsis is one of the leading causes of deaths, rapid identification of bloodstream infection is mandatory to perform adequate antibiotic therapy. Multiple sets of blood cultures may be required, and cultures of other sites should also be considered including urine, skin and throat. Smaller volumes are often taken from babies and children. Standard precautions must be taken when taking blood cultures. Ideally the skin should be washed with soap and rinsed with sterile water. This should be followed by the application of an iodine-based solution, which should then be washed off after 30 seconds of drying.

Questions 1-6

- 4) What does this article tell us about blood cultures?
- a) Blood culture should always precede antibiotic therapy.
 - b) The amount of blood to be taken can vary.
 - c) The use of iodine does not interfere with correct blood culture procedures.

TEXT 5

Helmet Removal Procedures

With each passing year, more individuals are playing contact sports and riding motorcycles. Therefore, healthcare workers involved in emergency care should be proficient in the procedure of helmet removal. This procedure, which requires minimal training, is a safe and quick process that gives care providers access to a patient's airway and allows them to stabilize the patient's head and neck. If the airway is unstable, facemask removal is all that is necessary to intubate the patient. According to one study, face mask removal for the potential spine-injured patient is safer than helmet removal for emergent airway access. Face mask removal results in less motion in all three planes (sagittal, frontal, transverse), requires less completion time, and is easier to perform.

Questions 1-6

- 5) What do we learn from this article about helmet removal?
- a) Intubation to stabilize the airway is not possible while a helmet is in place.
 - b) Healthcare workers should focus on the integrity of the sagittal, frontal and transverse planes.
 - c) It is not always necessary to remove a helmet.

TEXT 6

TYM Otoscope for Imaging the External Ear Canal

The TYM Otoscope (Cupris) uses iOS smartphones to let users capture images and videos of the external ear canal and eardrum. These can then be reviewed and shared securely with other healthcare professionals through the Cupris app. To use the device, the user places their smartphone in the case and slides the otoscope attachment over the lens of the phone camera. A speculum is then screwed onto the attachment until locked into place. Images and videos are captured using the smartphone's camera linked to the Cupris app. The Cupris app can be downloaded from the Cupris website. The images and video captured using the device are stored on a secure and encrypted cloud system and can be viewed on most phones or PC computers.

Questions 1-6

- 6) What does this article tell us about the TYM Otoscope?
- a) Healthcare professionals can transfer data from the device to their own smartphones.
 - b) Without a smartphone, the TYM Otoscope cannot function.
 - c) Data can only be shared on iOS platforms.

PART C

TEXT 6

Over the past 50 years, the frequency of allergies and autoimmune diseases has risen rapidly, but it is not clear why. In a study published today in *Science Translational Medicine*, researchers point to a possible culprit: salt. In lab experiments, the authors found that high concentrations of sodium chloride can influence the differentiation of T helper 2 (Th2) cells, the immune cells responsible for allergies, and that elevated levels of salt are present in the affected skin of people with atopic dermatitis, an allergic skin condition.

“These are the sorts of studies that are really appreciated because they’re so completely different, and that’s what we need because really there hasn’t been much progress in understanding this epidemic of allergic disease,” says Charles Mackay, an immunologist at Monash University in Australia who did not participate in the work. In looking “at why we get an allergy or atopic dermatitis, I try to think of the environmental factors,” he adds. This study “is an interesting new angle to Th2 immunity.”

Hay fever and atopic dermatitis have seen a two-fold upsurge since the 1970s, and researchers do not attribute that to greater awareness or diagnosis. This recent rise in the incidence of allergic diseases is much too fast to be explained by genetic changes, so it’s more likely to be due to an environmental or behavioural cause, according to co-author Christina Zielinski of the Technical University of Munich. “One thing that also changed within the last fifty to sixty years is our diet. We are eating much more fast food, and this also includes much more salt, so that’s how we became interested in the question of whether salt can modulate the immune system,” she says.

Zielinski and her colleagues started by upping the levels of sodium chloride in the tissue culture medium used to grow either human CD4-positive memory T cells, which give off a complex set of chemical signals based on previous exposure to antigens, or naïve T cells, which have not been exposed to antigens before. In both cell types, the salt boosted the abundance of cytokines and transcription factors specific to Th2 cells, indicating that high salinity promotes Th2 cell differentiation. The researchers also found that the effects of the salt seemed to enhance Th2-related programs via two salt-sensitive transcription factors.

The authors next compared salt levels in the skin of adults with atopic dermatitis, an allergic condition that causes red, itchy patches of skin. Lesioned skin had sodium

concentrations 30-fold higher than the patients' unlesioned skin and skin from healthy controls. The team also investigated sodium levels in affected and unaffected skin of people with psoriasis, an autoimmune disorder characterized by red, inflamed patches of skin. While both atopic dermatitis and psoriasis are both chronic inflammatory skin conditions, psoriasis is mediated by a different type of T helper cells. They found no difference in salt concentration in psoriatic lesions and unaffected skin, which led them to rule out a role for inflammation in the differences in sodium that they observed in people with atopic dermatitis. "We were very surprised because atopic dermatitis has been . . . thoroughly investigated, and no one has ever considered sodium chloride to play a role," says Zielinski. She adds that the next steps will be to investigate the connection between salt and allergies.

According to Chuan Wu, an immunologist at the National Cancer Institute who did not participate in the work, this connection could involve nearby epidermal and dermal cells, as well as the skin microbiome. "The skin is a big mucosal surface, colonized with a huge amount of microbes, so investigating whether . . . the Th2 response is somehow connected to the skin bacteria could be interesting," he says. The authors discuss in the study that people with atopic dermatitis often have an overgrowth of a salt-loving microbe (*staphylococcus aureus*) on their skin, which has gone unexplained until now.

Mackay points out that the gut microbiome could also be involved, as high salt diets have previously been shown to have effects on intestinal microbes. "Is it salt from the diet somehow affecting atopic dermatitis? Or is there a gut connection here that they haven't explored yet?" he asks. "This connection to the diet is still very speculative. There are some correlations and associations, but the definite proof is still missing," says Zielinski. "It could be that the sodium accumulation in the skin follows some autonomous skin-intrinsic rules, which are completely independent of diet."

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“These are the sorts of studies that are really appreciated because they’re so completely different, and that’s what we need because really there hasn’t been much progress in understanding this epidemic of allergic disease,” says Charles Mackay, an immunologist at Monash University in Australia who did not participate in the work. In looking “at why we get an allergy or atopic dermatitis, I try to think of the environmental factors,” he adds. This study “is an interesting new angle to Th2 immunity.”

Hay fever and atopic dermatitis have seen a two-fold upsurge since the 1970s, and researchers do not attribute that to greater awareness or diagnosis. This recent rise in the incidence of allergic diseases is much too fast to be explained by genetic changes, so it’s more likely to be due to an environmental or behavioural cause, according to co-author Christina Zielinski of the Technical University of Munich. “One thing that also changed within the last fifty to sixty years is our diet. We are eating much more fast food, and this also includes much more salt, so that’s how we became interested in the question of whether salt can modulate the immune system,” she says.

Zielinski and her colleagues started by upping the levels of sodium chloride in the tissue culture medium used to grow either human CD4-positive memory T cells, which give off a complex set of chemical signals based on previous exposure to antigens, or naïve T cells, which have not been exposed to antigens before. In both cell types, the salt boosted the abundance of cytokines and transcription factors specific to Th2 cells, indicating that high salinity promotes Th2 cell differentiation. The researchers also found that the effects of the salt seemed to enhance Th2-related programs via two salt-sensitive transcription factors.

The authors next compared salt levels in the skin of adults with atopic dermatitis, an allergic condition that causes red, itchy patches of skin. Lesioned skin had sodium concentrations 30-fold higher than the patients’ unlesioned skin and skin from healthy controls. The team also investigated sodium levels in affected and unaffected skin of people with psoriasis, an autoimmune disorder characterized by red, inflamed patches of skin. While both atopic dermatitis and psoriasis are both chronic inflammatory skin conditions, psoriasis is mediated by a different type of T helper cells. They found no difference in salt concentration in psoriatic lesions and unaffected skin, which led them to rule out a role for inflammation in the differences in sodium that they observed in people with atopic dermatitis. “We were very surprised because atopic dermatitis has been . . . thoroughly investigated, and no

one has ever considered sodium chloride to play a role,” says Zielinski. She adds that the next steps will be to investigate the connection between salt and allergies.

According to Chuan Wu, an immunologist at the National Cancer Institute who did not participate in the work, this connection could involve nearby epidermal and dermal cells, as well as the skin microbiome. “The skin is a big mucosal surface, colonized with a huge amount of microbes, so investigating whether . . . the Th2 response is somehow connected to the skin bacteria could be interesting,” he says. The authors discuss in the study that people with atopic dermatitis often have an overgrowth of a salt-loving microbe (*staphylococcus aureus*) on their skin, which has gone unexplained until now.

Mackay points out that the gut microbiome could also be involved, as high salt diets have previously been shown to have effects on intestinal microbes. “Is it salt from the diet somehow affecting atopic dermatitis? Or is there a gut connection here that they haven’t explored yet?” he asks. “This connection to the diet is still very speculative. There are some correlations and associations, but the definite proof is still missing,” says Zielinski. “It could be that the sodium accumulation in the skin follows some autonomous skin-intrinsic rules, which are completely independent of diet.”

Questions 7-14

- 7) What did the study published in Science Translational Medicine find?
- a) Th2 cells can cause allergies and autoimmune diseases
 - b) Some people with atopic dermatitis were allergic to salt
 - c) Salt may play an important role in allergies and autoimmune diseases
 - d) Sodium Chloride causes atopic allergies in skinny people
- 8) When diagnosing allergies or atopic dermatitis, Charles Mackay usually considers
- a) The patient’s hereditary conditions
 - b) The patient’s surroundings

- c) The patient's diet
 - d) The patient's susceptibility to skin conditions
- 9) According to the third paragraph, cases of hay fever and atopic dermatitis have
- a) increased slightly
 - b) increased significantly
 - c) stayed the same
 - d) fluctuated two times
- 10) What was the result of the experiments described in paragraph four?
- a) High salt intake causes allergies
 - b) Sodium Chloride helps CD4-positive memory T cells to grow
 - c) Naïve T cells react more quickly to salt than CD4-positive memory T cells
 - d) Elevated salt levels changed Th2 cell behaviour
- 11) According to the fifth paragraph, psoriasis is a condition that
- a) lasts for a long time
 - b) is triggered by T helper cells
 - c) is identical to atopic dermatitis
 - d) causes immunity
- 12) According to Zielinski in paragraph five, what is the situation regarding atopic dermatitis?
- a) there is a clear link with sodium
 - b) more studies are needed to understand the condition
 - c) the condition has been examined in depth
 - d) the condition is surprising
- 13) Chuan Wu thinks that
- a) The skin microbiome is not part of the epidermal system

- b) Epiderma and dermal cells are examples of skin bacteria
- c) the Th2 response produces salt-loving microbes
- d) the link between skin bacteria and Th2 response is worth further investigation

14) According to the information in paragraph seven, the presence of sodium in the skin

- a) may be unrelated to food consumption
- b) could be explained by intestinal microbes
- c) might be due to autonomous skin regeneration
- d) can be seen by checking accumulation patterns

TEXT 2

Travel vaccinations are an essential part of holiday and travel planning, particularly if your journey takes you to an exotic destination or 'off the beaten track'. The risks are not restricted to tropical travel, although most travel vaccines are targeted at diseases which are more common in the tropics. This article discusses the vaccinations that are available and gives some idea of the time you need to allow to complete a full protective course of vaccination.

The rise in worldwide and adventurous tourism has seen a massive increase in people travelling to exotic destinations. This leads to exposure to diseases that are less likely to occur at home. These are sicknesses against **which** we have no natural immunity and against which most people are not routinely immunised. They include insect-borne conditions such as malaria, dengue, yellow fever and Zika virus. In addition, diseases could be acquired from eating and drinking, such as hepatitis A and traveller's diarrhoea. Poor hygiene in some places could put you at risk from hepatitis B and Ebola virus. These are illnesses which might not only spoil your holiday but might also pose a risk to your life.

Before travelling outside your home country, it is important to check whether there are any vaccinations available which could protect you. You can do this by making a travel planning appointment at your doctor's surgery. There are also several websites which aim to offer up-to-date, country-specific advice on vaccinations and on disease patterns.

Vaccination courses need to be planned well in advance. Some vaccinations involve a course of injections at specified intervals and it can take a up to six months to complete course. Some vaccinations can't be given together. Always check with your surgery or online before travelling, particularly to unusual destinations, for local outbreaks of disease which mean other specific vaccinations are advised. If a vaccination certificate is issued keep it and update it over the years so that you have a full record. Your doctor's surgery will have a record of vaccines they have administered to you and can often issue a copy.

It is worth remembering that the protection offered by vaccination is not always 100%. Vaccination will greatly reduce your chances of acquiring the disease and in many cases the protection level offered is extremely high. The protection will also not be lifelong, so you need to seek medical advice before you travel. However, there isn't a vaccine available for every disease - for example, there is none at present against malaria. Even where a vaccine is available, vaccination should not be the only thing you rely on for protection against illness. It is important to know the risks; taking sensible steps to avoid exposing yourself to disease is by far the most useful thing you can do.

No vaccination is available against malaria. People who live permanently in malarial zones have partial protection but they lose this swiftly when they move away. Protection against malaria is through a combination of avoidance of mosquito bites and the use of anti-malarial tablets. Tablets have to be started before entering the malarial zone and continued for some days or weeks after leaving it. The recommended tablet regime varies by area. Again, your doctor will have access to up-to-date advice on recommendations for your journey.

People often at greatest risk when travelling are those visiting a country which they think of as their place of origin, where members of their family live and their roots may be. People often believe that as one-time residents who may have been born and raised there, they have a natural immunity. They feel that they are not on holiday but visiting home and that vaccinations aren't needed. Unfortunately, that is not the case. We acquire natural immunity by living in a place and being constantly exposed to the diseases that are present. When we leave the area for distant shores that protection is rapidly lost and we need the protection of vaccination, together with the other precautions listed above.

This is particularly true of malaria, where visitors 'going back home' may find their relatives puzzled and even amused that they are taking anti-malarial medication. Even so, it's very important **to do so**. It's only by living there all the time that you

acquire your resident relatives' level of immunity. Your immune system has a short memory for this sort of partial immunity.

Remember, vaccinations have to be paid for, although some vaccinations such as hepatitis A are usually free. Some aid workers and healthcare workers are often offered free vaccinations against occupational risks but others have to pay. Anti-malarial tablets are never free and can add a substantial amount to the cost of your trip. Whilst this may seem expensive, it is usually a small sum relative to the costs of your travel. Safeguarding your health should be considered an essential part of any trip.

Questions 15-22

15)The purpose of this article is to inform readers about

- a) A obtainable vaccines and how long you need to be vaccinated
- b) B common tropical diseases and their cures
- c) C vaccines and how many times you have to have them
- d) D vaccines needed in exotic locations and their risks

16)In the second paragraph, the word 'which' refers to

- a) natural immunities
- b) diseases that you could contract in exotic places i
- c) people who don't get vaccinated
- d) insect-borne diseases

17)Paragraph four infers that

- a) all vaccinations can be given in a single visit to the doctor's surgery
- b) most people visit the doctor's surgery six times for their vaccinations
- c) vaccination certificates cannot be issued at the same time as the vaccination
- d) more than one visit to the doctor's surgery is usually necessary

18)If you have been vaccinated against a disease in the past

- a) it is not necessary to be vaccinated again

- b) only the malaria vaccine provides lifelong protection
- c) you need to have a new vaccination against malaria each time you travel
- d) you should check if you need to have another immunisation

19) Any immunity that a person has against malaria

- a) is never lost
- b) is effective if they can avoid insect bites
- c) doesn't last a long time if they change locations
- d) is better than immunity offered by tablets

20) If people are visiting their home countries

- a) they have natural immunities to local diseases
- b) vaccinations may depend on the distance they travel
- c) it is necessary to take precautions
- d) all family members should be immunised

21) In paragraph eight, what does the expression 'to do so' refer to?

- a) having treatment against malaria
- b) being amused
- c) living in the country
- d) going back home

22) What does the last paragraph tell us about the cost of vaccinations?

- a) Almost vaccinations are free in the UK
- b) most people have to pay for their vaccinations
- c) travel vaccinations are covered by travel insurance
- d) everyone has to pay for vaccinations

Reading Practice Test 11 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	C	-
Q 2	A	-
Q 3	A	-
Q 4	B	-
Q 5	C	-
Q 6	B	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	C	-	Q 15	A	-
Q 8	B	-	Q 16	B	-
Q 9	B	-	Q 17	D	-
Q 10	D	-	Q 18	D	-
Q 11	A	-	Q 19	C	-
Q 12	C	-	Q 20	C	-
Q 13	D	-	Q 21	A	-
Q 14	A	-	Q 22	B	-

Review & Explanations

TEST-12

Necrotising Fasciitis

PART A

TEXT 1

Necrotising fasciitis (NF)

NF is a rare but serious bacterial infection that affects the tissue beneath the skin, and surrounding muscles and organs (fascia). It is often called the "flesh-eating disease", although the bacteria that cause it don't "eat" flesh – they release toxins that damage nearby tissue.

NF is caused by bacteria that gain access to the body, often from only a relatively minor injury, such as a small cut. The conditions gets worse very quickly and can be life threatening if it's not recognised and treated early on. Around a quarter of patients with NF will die of their infection, but this varies with the severity of the infection and the underlying health of the patient.

Quite a few different types of bacteria can cause the disease. However, when they cause infection elsewhere, many are only associated with mild disease. These include group A streptococci, a common cause of tonsillitis, and *Clostridium perfringens*, a cause of food poisoning. The infection can also be spread from person to person, but this is very rare.

About 400 cases of NF are diagnosed in Australian hospitals each year, which is similar to the incidence reported in other countries. Anyone can get necrotising fasciitis, including young and otherwise healthy people. It tends to affect older people and those in poor general health

TEXT 2

Contracting necrotising fasciitis

For a person to develop necrotising fasciitis, several factors relating to themselves, the environment and the presence of certain bacteria all have to be present.

- Patient factors that increase their risk if exposed to bacteria include:

- impaired immunity

- obesity
- acne or asthma sufferers
- chronic diseases such as diabetes, peripheral vascular disease
- a breach of the skin such as:
 - surgical wounds
 - accidental wounds
 - intravenous drug use
- Environmental factors that increase risk include:
 - coral cuts in marine environments
 - contaminated surgical environment or equipment
 - contamination of intra venous injected substances
- Bacteria that can lead to issues include:
 - Group A streptococci are commonly found in the throat and on the skin and is the most common bacteria to cause NF
 - Vibrio bacteria are gram-negative bacteria that grow well in salty environments
 - Aeromonas are Gram-negative, anaerobic bacteria that occur in aquatic environments
 - Cleansing wounds, keeping wound covered and good hand hygiene are the main ways to prevent necrotising fasciitis

TEXT 3

Symptoms of necrotising fasciitis

The symptoms of NF develop quickly over hours or days. They may not be obvious at first and can be similar to less serious conditions such as flu, gastroenteritis or cellulitis. It might take 3 or 4 days for symptoms to fully appear.

Skin becoming red, hot and blistered, together with the patient reporting intense pain in the infected area are the main early symptoms.

Patients with NF report pain that is out of proportion to the changes in skin condition. This is a key warning sign. The pain remains intense until the necrosis kills the nerve endings.

Other symptoms include:

- oedema, or swelling
- crackling under the skin
- confusion
- dehydration
- diarrhoea and vomiting
- skin swells and changes colour, turning violet
- areas of tissue turn black and start to die

After 4 or 5 days, septicaemia is likely to develop causing high temperature, dangerously low blood pressure, and they possible loss of consciousness. Without treatment, necrotizing fasciitis is always fatal.

TEXT 4

Treatment and outlook

NF needs to be treated in hospital, usually in the intensive care unit

The main treatments are:

- surgery to remove infected tissue which may repeated several times to ensure all the infected tissue is removed, and occasionally it may be necessary to amputate affected limbs
- antibiotics, usually several different types, administered intravenously
- supportive treatment of blood pressure, fluid levels and organ functions

- People usually need to stay in hospital for several weeks.

NF can progress very quickly and lead to serious problems such as blood poisoning (sepsis) and organ failure and even with treatment, it is estimated that 1 or 2 in every 5 cases are fatal.

People who survive the infection are sometimes left with long-term disability as a result of amputation or the removal of a lot of infected tissue.

They may need further surgery to improve the appearance of the affected area and may need ongoing rehabilitation support to help them adapt to their disability.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** What conditions can develop in a person with blood poisoning? _____
- 2** Ways to stop develop necrotising fasciitis? _____
- 3** The prognosis for people with necrotising fasciitis? _____
- 4** The preponderance of necrotising fasciitis diagnosed in Australian hospitals? _____
- 5** Underlying issues that can make person more susceptible to developing NF? _____
- 6** Operations that can be done to treat necrotising fasciitis? _____
- 7** The timeframe for symptoms of NF to be full blown? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What types of lesions may let bacteria invade the body? _____

9 What type of pain in the infected area do patients experience?

10 What type of infection is necrotising fasciitis? _____

11 Which bacteria is most likely to lead to NF? _____

12 Which part of a hospital are people with necrotising fasciitis usually treated?

13 What might a person cut themselves on in an ocean that could lead to them getting necrotising fasciitis? _____

14 For every 5 people with necrotising fasciitis, how many are likely to die, even with treatment _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 People with necrotising fasciitis say their pain is _____ to what can be seen in the area.

16 Symptoms of NF might be vague at first and more like common conditions such as _____ or cellulitis.

17 People who have conditions such as: _____ obesity and acne or asthma have a greater chance of developing NF if they are exposed to Group A streptococci bacteria.

18 Often multiple antibiotics are _____ to treat NF.

19 NF is more likely to be a problem for _____ and those in poor general health.

20 NF is commonly known as the _____.

Reading Practice - PART A

Answer Keys

Correct Answer		Your Answer	
Q 1	C		-
Q 2	B		-
Q 3	D		-
Q 4	A		-
Q 5	B		-
Q 6	D		-
Q 7	C		-

Correct Answer		Your Answer	
Q 8	Surgical, acid +		
Q 9	Intense		
Q 10	bacterial		
Q 11	Group A strepto +		
Q 12	Intensive care +		
Q 13	Coral		
Q 14	1 or 2		

Correct Answer		Your Answer	
Q 15	out of proporti +		
Q 16	flu, gastroente +		
Q 17	impaired immuni +		
Q 18	administered in +		
Q 19	older people		
Q 20	flesh eating di +		

Review & Explanations

PART B

TEXT 1

Know the Difference: Infiltration vs. Extravasation

Intravenous infiltration is one of the most common problems that can occur when fluid infuses into the tissues surrounding the venepuncture site. This sometimes happens when the tip of the catheter slips out of the vein or the catheter passes through the wall of the vein. If you are concerned an IV is infiltrated, standard procedures should be followed by, for example, discontinuing the site and relocating the IV.

Know the Difference: Infiltration vs. Extravasation

An extravasation occurs when there is accidental infiltration of a vesicant or chemotherapeutic drug into the surrounding intravenous site. Vesicants can cause tissue destruction and / or blistering. Irritants can result in pain at the site and along the vein and may cause inflammation. The treatment for extravasation will vary depending on hospital policy.

Question

- 1) What should you do if you think an IV is infiltrated?
 - a) You should terminate the procedure before trying again
 - b) You should change the catheter
 - c) You should irrigate the surrounding intravenous site

TEXT 2

Arterial Line Placement

Arterial line placement is a common procedure in various critical care settings. Intra-arterial blood pressure measurement is more accurate than measurement by non-invasive means, especially in the critically ill. Intra-arterial blood pressure management permits the rapid recognition of changes that is vital for patients on continuous infusions of vasoactive drugs. Overall, arterial line placement is considered a safe procedure, with a rate of major complications that is below 1%.

Arterial Line Placement

In both adults and children, the most common site of cannulation is the radial artery, primarily because of the superficial nature of the vessel and the ease with which the site can be maintained. Additional advantages of radial artery cannulation include the consistency of the anatomy and the low rate of complications.

Question

- 2) Why is the radial artery usually chosen for cannulation?
- a) its low profile anatomy is ideal for primary cannulation
 - b) the site can be maintained during other non-invasive manipulations
 - c) it has a shallow position

TEXT 3

Clinically Important Symptoms of PTSD

People with clinically important symptoms of PTSD (Post-Traumatic Stress Disorder) refer to those who are assessed as having PTSD on a validated scale, as indicated by baseline scores above clinical threshold, but who do not necessarily have a diagnosis of PTSD. They are typically referred to in studies that have not used a clinical interview to arrive at a formal diagnosis of PTSD and instead have only used self-report measures of PTSD symptoms. Complex PTSD develops in a subset of people with PTSD. It can arise after exposure to an event or series of events of an extremely threatening or horrific nature, most commonly prolonged or repetitive events from which escape is difficult or impossible. The disorder is characterised by the core symptoms of PTSD; that is, all diagnostic requirements for PTSD are met.

Question

- 3) According to this article, people with PTSD
- a) have a high score on a validated scale, which includes complex PTSD factors
 - b) have experienced prolonged or repetitive symptoms
 - c) have not been examined by qualified clinicians

TEXT 4

How to Assess a Peripheral Intravenous Cannula

Most patients need at least one peripheral intravenous cannula during their hospital stay for intravenous fluids and medications, blood products or nutrition.

Complications are common but they can be prevented or minimised by routine assessment. Explanations to patients should be provided, along with education about the treatment. Ensure the patient knows why the treatment is being given, and encourage them to speak up if there are any problems, such as pain, leaking, swelling, etc. The cannula should not be painful. Pain is an early symptom of phlebitis (inflammation of the vein) and could indicate that the cannula is not working well and should be removed. Involving the patient and their family empowers them to voice their concerns, and prompts nurses to address problems and remove

Question

- 4) According to this article, patients experiencing pain at the cannula site should
- a) tell someone
 - b) ask for medication to stop leaking and/or swelling
 - c) remove the cannula in order to avoid phlebitis (inflammation of the vein)

TEXT 5

Japan Approves New Cell Therapy Trial for Spinal Cord Injury

The Japanese government's health ministry has given the go-ahead for a trial of human induced stem cells to treat spinal cord injury. The treatment will be tested in a handful of patients who suffered nerve damage in sports or traffic accidents. Researchers at Osaka University plan to recruit adults who have sustained recent nerve damage in sports or traffic accidents. The team's intervention involves removing differentiated cells from patients and 'reprogramming' them into neural cells. Clinicians will then inject about two million of these cells into each patient's site of injury. The approach has been successfully tested in a monkey, which recovered the ability to walk after paralysis. These tests follow researcher carried out at Kyoto University which used cells to treat Parkinson's disease.

Question

- 5) How many people will be involved in the trial?
- a) around two million
 - b) as many patients as possible
 - c) less than ten

TEXT 6

Steroid Nasal Sprays and Drops

A steroid nasal spray usually works well to clear all the nasal symptoms such as itching or sneezing. It works by reducing inflammation in the nose. A steroid nasal spray also tends to ease eye symptoms although it is not clear how this occurs. However, they can take up to several days to build up to the full effect. Steroid nasal sprays should be used each day over the hay fever season to keep symptoms away. However, once symptoms have gone, the amount of steroid spray can often be reduced to a low maintenance dose each day to keep symptoms manageable. Side-effects or problems with steroid nasal sprays are rare.

Question

- 6) How long can a nasal spray be used?
- a) While symptoms such as itching or sneezing occur and after
 - b) Before and during the time symptoms occur
 - c) Before, during the time symptoms occur and after

PART C

TEXT 1

Heat and ice have been used for many years to treat pain and to reduce swelling, and many people have found them effective. More recently, studies have been done to investigate whether heat and ice really make a difference to healing and the results have been inconclusive. In general, when used sensibly, they are safe treatments which make people feel better and have some effect on pain levels and there are few harms associated with their use.

Heat is an effective and safe treatment for most aches and pains. Heat can be applied in the form of a wheat bag, heat pads, deep heat cream, hot water bottle or heat lamp. Heat causes the blood vessels to open wide (dilate). This brings more blood into the area to stimulate healing of damaged tissues. It has a direct soothing effect and helps to relieve pain and spasm. It can also ease stiffness by making the tissues more supple. If heat is applied to the skin, it should not be hot; gentle warmth will be enough. If excessive heat is applied there is a risk of burns and scalds. A towel can be placed between the heat source and the skin for protection. The skin must be checked at regular intervals.

Heat should not be used on a new injury. It will increase bleeding under the skin around the injured area and may make the problem worse. The exception to this is new-onset low back strains. A lot of the pain in this case is caused by muscle spasm rather than tissue damage, so heat is often helpful. A large-scale study suggested that heat treatment had a small helpful effect on how long pain and other symptoms go on for in short-term back pain. This effect was greater when heat treatment was combined with exercise.

Ice has traditionally been used to treat soft tissue injuries where there is swelling. However, there is a growing body of evidence which suggests that applying ice packs to most injuries does not contribute to recovery and may even prolong recovery. This is related to the fact that reducing the temperature at the site of an injury will delay the body's immune system response. It is the action of the immune system which will heal the injury. In one study, some people who used ice said that it was helpful for managing pain, although this did not translate into a lower use of painkillers. Many people find that ice is helpful when used to manage pain in the short term. It is unlikely that it will have much of a negative effect in the long term when used in this way.

A review of studies into the effectiveness of ice treatment found that most studies were inconclusive and others showed only a small effect. For example, a review of studies using hot and cold therapy for osteoarthritis of the knee found that ice packs reduced swelling and that ice massage improved muscle strength and range of movement. Heat packs had no effect on pain and swelling. No side-effects were reported to either heat or ice. Another study, which looked at a variety of treatments for neck pain, found that neither heat nor cold was effective.

In the later, or rehabilitation, phase of recovery the aim changes to restoring normal function. At this stage the effects of ice can enhance other treatments, such as exercise, by reducing pain and muscle spasm. This then allows better movement. If you are doing exercises as part of your treatment, it can be useful to apply an ice pack before exercise. This is so that after the ice pack is removed the area will still be a little numb. The exercises can also be done with the ice pack in place. This reduces pain and makes movement around the injury more comfortable, although it can also make the muscles being exercised stiffer.

Ice packs can be made from ice cubes in a plastic bag or wet tea towel. A packet of frozen peas is also ideal and can be used very easily. These mould nicely and can go in and out of the freezer. However, frozen vegetables should not be eaten if they have been thawed and re-frozen. Purpose-made cold packs can also be bought from pharmacies. Take care when using ice and cold packs from a deep freeze, as they can cause ice burns quickly if used without care and proper protection.

Ideally, ice should be applied within 5-10 minutes of injury and for 20-30 minutes. This can be repeated every 2-3 hours or so whilst you are awake for the next 24-48 hours. Do not use ice packs on the left shoulder if you have a heart condition. Do not use ice packs around the front or side of the neck. Both heat and ice can be re-applied after an hour if needed.

Questions 7-14

7) What have studies shown about heat and ice treatments?

- a) Results show heat and ice really make a difference
- b) Results are uncertain
- c) Results have not been investigated
- d) Results show they can cause harm

8) What do we learn about heat in the second paragraph?

- a) it increases muscle tissue
- b) it provokes tissue stiffness
- c) it changes the behaviour of the blood flow
- d) it can cause muscle spasm

9) What did the study mentioned in the third paragraph find?

- a) heat made a problem worse
- b) heat triggered muscle spasms
- c) heat increased new-onset low back pain
- d) heat changed the duration of back pain

10) In the fourth paragraph, what have results shown concerning the use of ice?

- a) Ice could lengthen the time it takes to improve
- b) Ice stimulates the body's immune response
- c) Using ice therapies reduces the need for painkillers
- d) Ice causes swelling in soft tissue injuries

11) In the fifth paragraph, the review found that

- a) heat packs had some small side-effects
- b) ice massage had a positive effect on some muscles
- c) heat therapy worked best on cases of osteoarthritis of the knee
- d) heat treatment was more effective than ice treatment

12) In the sixth paragraph, what positive effect of using ice packs is described?

- a) they eliminate the need for other treatments
- b) they make some areas less sensitive to pain
- c) they move the pain to a different area
- d) they restore normal functions to injured muscles

13) In the seventh paragraph, what does the word 'these' refer to?

- a) frozen peas
- b) ice cubes
- c) wet tea towels
- d) ice packs

14) How long can ice be applied to an injury?

- a) for five to ten minutes
- b) no more than half an hour
- c) for two to three hours
- d) for 24 to 48 hours

TEXT 2

We consider low-dose aspirin so innocuous that we call it baby aspirin. Though we don't give it to kids anymore, many adults take it every day (at the recommendation of their doctor) to stave off heart attacks and strokes. But just as we now know not to give babies aspirin, expert opinion has shifted on low-dose aspirin for adults, too. Research in the last few years has made it clear that daily aspirin doesn't help many of the people taking it. If anything, it might hurt them.

New guidelines from the American College of Cardiology (ACC) and the American Heart Association (AHA) say that aspirin, and even baby aspirin should no longer be prescribed. These principles are largely in line with how other major organizations have begun to view aspirin. The 2016 European guidelines on cardiovascular disease prevention don't recommend it as a primary method of heart attacks or stroke prevention, and the U.S. Preventive Services Task Force recommends it only for people in their 50's with elevated cardiovascular disease risk. For the rest of the population, it wasn't clear whether there was a worthwhile benefit.

Aspirin is an antiplatelet drug, which means it prevents blood from clotting as easily. Forming a blood clot is, of course, a crucial capability—if you couldn't clot at all, you'd bleed out from small wounds. But clots that form inside your blood vessels can block flow entirely, causing a heart attack when that blood fails to get back to your heart, or a stroke if the clot cuts off blood to part of your brain. In theory, preventing platelets from doing their job means aspirin should help decrease the risk of both of these problems. And that's true, but only for a select group of people.

The new guidelines note that aspirin is still very much recommended as a secondary treatment, meaning it definitely helps people who have already had a heart attack or stroke. These people are at a significantly higher risk of having another incident, and aspirin can reduce that risk. What physicians are no longer recommending is its widespread use as a primary treatment, for people who have never had a heart attack or stroke before. In other words, if you've never had a heart attack, you probably shouldn't consider it.

So-called baby aspirin may carry a low dose, but patients shouldn't assume that taking it is harmless. Taking a drug that makes your blood less likely to clot puts you at risk. If you start bleeding in your intestines or your brain, for instance, your platelets are supposed to come to the rescue. If you're on daily aspirin, that happens less effectively. A 2009 study in *The Lancet* found that there was a small, but not insignificant increased risk of major bleeds amongst people taking aspirin regularly. A 2016 study found the same thing, as did a 2018 study in *The New England Journal of Medicine*. Those same risks exist if you've had a heart attack already, but the benefits you get from taking aspirin start to outweigh the potential downsides once you're in this category. That trade-off is what the ACC/AHA cite in their revised recommendations. Once your elevated risk of having a heart attack goes over 10 percent, the guidelines note, it becomes favourable to prescribe aspirin daily. That goes for anyone between 40 and 70. There's not enough evidence in people younger than 40, and adults over 70 have such elevated risk of bleeding that most wouldn't do well on daily aspirin regardless of cardiac risk.

The overarching advice for everyone, though, is to discuss with your doctor whether you should take low-dose aspirin before deciding to do so (or deciding to stop). These guidelines note that there are likely to be exceptions, and your physician should be assessing your personal health risks when deciding whether to prescribe daily aspirin. This isn't actually all that new. Though research from the mid-20th century suggested aspirin would help everyone, these changes to official recommendations are based on many years of modern studies, which the ACC/AHA note are far better designed and more rigorous than anything we've had before. If your doctor scoffs and tells you baby aspirin is a great idea for everyone of a certain age, their knowledge is out of date.

Reversals in expert opinion are, unfortunately, inevitable—it's the scientific process at work. Think of it less as flip-flopping and more as a correction to a formerly mistaken belief. And please talk to your doctor before you prescribe yourself baby aspirin.

Questions 15-22

15)The first paragraph informs us that

- a) even babies can have aspirin
- b) the viewpoint of experts has changed regarding aspirin
- c) aspirin can cause strokes
- d) aspirin dosage depends on doctors' recommendations

16)According to the second paragraph, the European guidelines

- a) continue to recommend the use of aspirin
- b) harmonize with the American guidelines
- c) recommend aspirin for people in their 50's
- d) say that aspirin should not be prescribed to babies

17)The third paragraph informs us that aspirin

- a) inhibits blood clotting
- b) helps to heal small wounds
- c) decreases the production of platelets
- d) slows bleeding by stimulating clotting

18)What does the last word of the fourth paragraph refer to?

- a) primary treatment
- b) secondary treatment
- c) aspirin
- d) the new guidelines

19)The 2009 study published in 'The Lancet' found

- a) that aspirin could be a factor in intestinal bleeding
- b) risks that were different to the study in 'The New England Journal of Medicine'
- c) daily doses of aspirin were less effective

- d) that the risk of major bleeds was relevant

20)The revised recommendations in the fifth paragraph are that

- a) people younger than 40 should take aspirin
- b) people older than 70 can take aspirin to elevate risks
- c) it's a good idea for middle aged people to take aspirin if they have a higher risk of heart attack
- d) anyone with a high risk of heart attack should take aspirin

21)What do we learn in the sixth paragraph about modern studies?

- a) they are superior to older studies
- b) they confirm earlier studies about the use of aspirin
- c) they quickly become out of date
- d) they reveal data that doctors don't accept

22)In the last paragraph, what does the writer infer about expert opinion?

- a) experts shouldn't keep changing their opinions
- b) changes in opinion are unavoidable
- c) opinions need to be corrected
- d) some opinions are unscientific

Reading Practice Test 12 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	A	-
Q 2	C	-
Q 3	C	-
Q 4	A	-
Q 5	C	-
Q 6	C	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	B	-	Q 15	B	-
Q 8	C	-	Q 16	B	-
Q 9	D	-	Q 17	A	-
Q 10	A	-	Q 18	C	-
Q 11	B	-	Q 19	D	-
Q 12	B	-	Q 20	C	-
Q 13	A	-	Q 21	A	-
Q 14	B	-	Q 22	B	-

Review & Explanations

TEST-13

Tobacco Smoking

PART A

TEXT 1

Tobacco Smoking Statistics from the Australian Institute of Health and Welfare

Tobacco smoking is the single most important preventable cause of ill health and death in Australia. Tobacco smoke contains over 7,000 chemicals, of which over 70, cause cancer. Lung cancer, chronic obstructive airways disease and coronary heart disease are the 3 main diseases linked to tobacco smoking.

Smoking-related diseases killed 14,900 Australians in the financial year 2004–05. This equals 40 preventable deaths every day. Smoking resulted in over 750,000 days spent in hospital and cost \$670 million in hospital costs in the financial year 2004–05.

Smoking kills more men than women – 9,700 men compared to 5,200 women. Cancer is the number one cause of tobacco-related death in men (57 per cent) and women (51 per cent), with lung cancer accounting for around 75 per cent and 72 per cent of cancers for men and women respectively. Lung cancer currently causes the most cancer deaths in Australia and this is due mainly to smoking.

The trend for tobacco smoking is dropping with 12% of people aged 14 and older smoking daily in 2016, which is a 24% reduction since 1991. The number of young people who start smoking is also reducing. In 2010, the average age when 14–24 year-olds smoked their first full cigarette was 14.2, but it was 16.3 in 2016. In 1995, 31% of adults smoked in a home where there were dependent children. In 2016, this was down to just 2.8%.

TEXT 2

Why do people smoke?

Cigarettes contain nicotine which does not cause the health issues linked to cigarette smoking but is highly addictive. In small amounts, nicotine causes pleasant feelings which makes the smoker want more. It does not take long before the time between cigarettes gets less, because the smoker is keen to get the pleasant feelings they had before. When a person becomes addicted to nicotine they soon start to have bad

feelings like being irritated and edgy when they are ready for another boost of nicotine.

Most smokers started when they were teens and those who have friends and/or parents who smoke are more likely to start smoking than those who don't.

The tobacco industry spends billions of dollars each year to create and market their products that show smoking as exciting, glamorous, and safe. Tobacco use is also shown in video games, online, and on TV. Movies showing smokers are another big influence and studies show that young people who repeatedly see smoking in movies are more likely to start smoking.

Widespread advertising, price breaks, and other promotions for cigarettes have been big influences in the past but now many governments are bringing in a lot of ways to reduce the number of people who smoke.

In Australia, the government:

- does not allow cigarette advertising
- has had cigarettes moved to covered cupboards so they cannot be seen in places like dairies, petrol stations and supermarkets
- has gradually increased the amount of tax added to a packet of cigarettes

TEXT 3

Stopping smoking is not easy

Common symptoms people have when they stop smoking include:

- Cravings for nicotine which may be strong at first but they usually only last a few minutes
- restlessness and trouble concentrating or sleeping
- irritability, anger, anxiety, depression
- increase in appetite and weight gain

- Less common symptoms include:
- cold symptoms such as coughing, sore throat and sneezing
- constipation
- dizziness or light-headedness
- mouth ulcers.

The benefits of quitting smoking are:

- immediate health benefits
- a dramatic reduction the risk of smoking-related diseases

Statistics include:

- Quitting before 30 years of age reduces the risk of lung cancer by 90 per cent
- After 15 years of being a non-smoker, the risk of stroke is reduced to that of a person who has never smoked
- Within two to five years of quitting, there is a large drop in the risk of heart attack and stroke

TEXT 4

Different support to stop smoking in Australia

- 'Cold turkey' is giving up smoking suddenly, without using medications.
- The prescription medications, bupropion (Zyban) and varenicline (Champix) which reduce withdrawal symptoms from nicotine.
- Nicotine replacement therapy including patches, gum and lozenges.
- QuitCoach is an online tool developed to assist in quitting smoking.
- Quitline is a telephone service available to smokers who want to quit.
- Acupuncture involves treatment by applying needles or surgical staples to different parts of the body.
- Hypnotherapy has not been shown to increase the likelihood of quitting in the long term, although counselling or other treatments that may be offered with it can be helpful to some smokers

E-Cigarettes/ Vaping are increasingly being used instead of traditional cigarettes. However, there is limited evidence available on their quality, safety, efficacy for smoking cessation or harm reduction, and the risks they pose to population health.

In March 2015, the Chief Executive Officer (CEO) of Australia's National Health and Medical Research Council (NHMRC) issued a statement stating that: "there is currently insufficient evidence to conclude whether e-cigarettes can benefit smokers in quitting, or about the extent of their potential harms. It is recommended that health authorities act to minimise harm until evidence of safety, quality and efficacy can be produced".

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** E-cigarettes and their role in stopping smoking? _____
- 2** Statistics about smoking in Australia? _____
- 3** The benefits of quitting smoking? _____
- 4** The different support to stop smoking in Australia? _____
- 5** The addictive features of nicotine? _____
- 6** The 3 main diseases linked to tobacco smoking? _____
- 7** The common symptoms people have when they stop smoking? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** What bad feelings might a person have when they are ready for another boost of nicotine? _____
- 9** What therapy includes patches, gum and lozenges? _____

10 How old is a person if their risk of lung cancer reduces by 90 per cent if they stop smoking? _____

11 What percentage of adults smoked in a home where there were dependent children in 2016? _____

12 Who does not allow cigarette advertising? _____

13 Who should act to minimise harm until evidence of safety, quality and efficacy of e-cigarettes can be produced _____

14 Young people are more likely to start smoking if they see what repeatedly?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 12% of people aged 14 and older smoked daily in 2016, which is a _____ reduction since 1991

16 Common symptoms of nicotine withdrawal include _____ and depression

17 Cigarettes contain nicotine which does not cause the _____ linked to cigarette smoking but is highly addictive.

18 Smoking kills more _____

19 The prescription medications, bupropion (Zyban) and varenicline (Champix) which reduce _____ from nicotine.

20 The Government in Australia has _____ the amount of tax added to a packet of cigarettes.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	D	-	Q 8	Irritated, edgy		Q 15	24%	
Q 2	A	-	Q 9	Nicotine replac	+	Q 16	irritability, a	+
Q 3	C	-	Q 10	Under 30		Q 17	health issues	
Q 4	D	-	Q 11	28		Q 18	men than women	
Q 5	B	-	Q 12	Australian gove	+	Q 19	withdrawal symp	+
Q 6	A	-	Q 13	Health authorit	+	Q 20	gradually incre	+
Q 7	C	-	Q 14	Smoking in movi	+			

Review & Explanations

PART B

TEXT 1

The MIST Therapy system for the promotion of wound healing

The MIST Therapy system shows potential to enhance the healing of chronic, 'hard-to-heal', complex wounds, compared with standard methods of wound management. If this potential is substantiated, then MIST could offer advantages to both patients and the hospitals.

However, comparative research has yet to be carried out. Further investigation is necessary to reduce uncertainty about the outcomes of patients with chronic, 'hard-to-heal', complex wounds treated by the MIST Therapy system compared with those treated by standard methods of wound care. This research should define the types and chronicity of wounds being treated and the details of other treatments being used. It should report healing rates, durations of treatment (including debridement) needed to achieve healing, and quality of life measures (including quality of life if wounds heal only partially).

Question

- 1) Why should further research be carried out on the MIST Therapy system?
 - a) To investigate the range of wounds that this system can help to treat
 - b) To make sure that it offers improvements over the usual treatment option
 - c) To discover if this system can avoid deleterious outcomes for chronic wounds

TEXT 2

Assessing Risk and Prevention

Falls and fall-related injuries are a common and serious problem for older people. People aged 65 and older have the highest risk of falling, with 30% of people older than 65 and 45% of people older than 80 falling at least once a year. The human cost of falling includes distress, pain, injury, loss of confidence, loss of independence and mortality. Falling also affects the family members and carers of people who fall.

All people aged 65 or older are covered by all guideline recommendations as they have the highest risk of falling. According to the guideline recommendations, all

people 65 or older who are admitted to hospital should be considered for a multifactorial assessment for their risk of falling during their hospital stay.

Question

- 2) What does the article tell us about the risk of falling?
- a) People over 65 need extra facilities to help them avoid falls
 - b) More people over 65 fall in hospital environments compared with other places
 - c) Only a minority of people over 65 fall at least once a year

TEXT 3

Dealing With Hazmat

One of the most challenging aspects of providing emergency medical care is attending to patients who have been contaminated with hazardous materials. HAZMAT is a term used to describe incidents involving hazardous materials or specialized teams who deal with these incidents. Hazardous materials are defined as substances that have the potential to harm a person or the environment upon contact. These can be gases, liquids, or solids and include radioactive and chemical materials.

The potential for exposure to hazardous materials in the United States is significant. More than 60,000 chemicals are produced annually in the United States, of which the US Department of Transportation considers approximately 2000 hazardous. More than 4 billion tons of chemicals are transported yearly by surface, air, or water routes.

Question

- 3) According to this article, what is HAZMAT?
- a) Liquid, gaseous or solid materials that are bad for the environment
 - b) Events where harmful substances are released and the groups that deal with the aftermath
 - c) Toxic chemicals that are transported by water, land or air.

TEXT 4

How Does Blood Clot?

Within seconds of a blood vessel cut, the damaged tissue causes platelets to become 'sticky' and gather together around the cut. These 'activated' platelets and the damaged tissue release chemicals which react with other chemicals and proteins in the plasma, called clotting factors. A complex series of reactions involving these clotting factors then occurs rapidly. Each reaction triggers the next reaction and this process is known as a cascade.

The final chemical reaction is to convert a clotting agent called fibrinogen into thin strands of a solid protein called fibrin. The strands of fibrin form a meshwork and trap blood cells which form into a solid clot.

Question

- 4) According to this article, what is a cascade?
- a) A series of events
 - b) The process that occurs when platelets become sticky
 - c) The reaction that precedes the formation of fibrin

TEXT 5

At the onset of a migraine attack, the patient should be given a full dose of painkiller. For an adult this means 900 mg aspirin (usually three 300 mg tablets) or 1000 mg of paracetamol (usually two 500 mg tablets). This dose can be repeated every four hours if necessary. Soluble tablets have the advantage of being absorbed more quickly than solid tablets.

Codeine and medicines containing codeine, such as co-codamol, are not recommended for the treatment of migraine. This is because codeine can make feeling sick (nausea) and being sick (vomiting) worse, which can aggravate the migraine. They are also more likely than paracetamol or aspirin to cause a condition called medication-overuse headache if they are used frequently.

Question

- 5) What do we learn about migraine treatment from this article?
- a) Paracetamol doses should not exceed 1000mg
 - b) Aspirin can cause nausea and/or vomiting
 - c) Codeine can provoke conditions other than migraine

TEXT 6

Not all patients can independently move or position themselves in bed and their immobility may be due to a wide range of factors. Positioning patients in good body alignment and changing position regularly are essential aspects of nursing practice. It is vital to provide meticulous care to patients who must remain in bed. Healthcare givers' measures should ensure to preserve the joints, bones and skeletal muscles and must be carried out for all patients who require bed rest.

Positions in which patients are placed, methods of moving and turning should all be based on the principles of maintaining the musculoskeletal system in proper alignment. In addition, the health care provider must also use good body mechanics when moving and turning patients to preserve his or her own musculoskeletal system from injury.

Question

- 6) What information does the article give us about positioning patients?
- a) Healthcare givers should position immobile patients in accordance with the doctor's instructions
 - b) Positioning patients is a fundamental part of a nurse's job

- c) Improper musculoskeletal manipulation can lead to immobility

PART C

TEXT 1

Once the preserve of hippies and activists, veganism has now hit the mainstream. Forgoing meat, dairy and eggs is more popular than ever. While it's positive that people are taking a more ethically-conscious approach to food shopping, what nutrients could vegetarians and vegans put themselves at risk of losing out on? And how can you approach animal-free consumption in a healthy way?

Generally, people choose to be vegetarian or vegan for ethical reasons or because they want to improve their health. A vegan diet is usually low in saturated fats and rich in fruit and vegetables. 'Meat-free Monday' is a UK campaign, launched by Paul McCartney in 2009, to encourage people to reduce their environmental impact and improve their health by having at least one meat-free day each week. More and more people are realising this is far more manageable than they first thought. When you stop thinking a meal needs meat to be complete, vegetarian options start to look a lot more appealing.

If you've decided to give up meat and have vowed to eat more vegetables, that's a good first step. But vegetarians and vegans do have to be careful they're not missing out on nutrients most easily found in meat and dairy sources. Protein is one of them. Protein builds and repairs tissues and is a building block of bone, muscles, skin and blood. It isn't stored in your body, so you need to make sure you're getting enough from your diet.

Unfortunately for vegetarians, meat is a rich source of this macronutrient. "Whilst many plant-based foods contain protein too, they may not contain protein in the correct balance that the body needs. Therefore, vegetarians need to make sure they eat a combination of foods to achieve the right protein balance," says Dr Jan Sambrook, a doctor who specialises in nutrition. Luckily, you can also find protein in grains, pulses and dairy products. "If you eat any two of these, the protein will balance," reveals Sambrook. "This doesn't necessarily need to be within a single meal, as was previously thought. Examples of protein-balanced meals include cereal with milk, or baked potato with beans and cheese."

A balanced vegetarian or vegan diet generally gives you plenty of vitamins. But if you're not sure, there are some foods to look out for when it comes to specific nutrients. "Vitamin A is found in eggs and dairy products. A different form of the

vitamin, called beta carotene, is found in dark green leafy vegetables and in coloured fruits and vegetables such as mango, carrots and red peppers," explains Sambrook. Vitamin D, 'the sunshine vitamin', is also really important. It helps your body absorb calcium and is also needed for our muscles to work properly. More recently, vitamin D deficiency has been associated with numerous conditions, from heart disease, to dementia and multiple sclerosis. "Vitamin D is mainly made in our skin by the action of sunlight. However, it is also found in dairy products, mushrooms and in fortified cereals and margarine," says Sambrook. "Oily fish and eggs are also among the top dietary sources of vitamin D, so if you're adopting a vegan diet you're less likely to be getting enough."

Most of the minerals we need are found in a wide variety of foods and anyone eating a balanced diet can obtain enough of them. However, vegetarians and vegans must make sure they're getting enough calcium and iron. Recently, the National Osteoporosis Society (NOS) warned that the popularity of 'clean eating' and other diets where major food groups are cut out is setting young people up for a future of weak bones. "Without urgent action being taken to encourage young adults to incorporate all food groups into their diets and avoid particular 'clean eating' regimes, we are facing a future where broken bones will become just the norm," said Susan Lanham-New, a nutrition professor and clinical advisor to the NOS.

Vegans, who normally don't consume dairy products, may find it challenging to obtain calcium in their diet. "Calcium is, however, also present in leafy green vegetables, dried figs, almonds, oranges, sesame seeds, seaweed and some types of bean," reveals Sambrook. She explains that if non-dairy calcium is eaten with a source of vitamin D, this will help the body absorb it.

You need iron in order for your blood to carry oxygen around your body. If you don't get enough, you become anaemic. Whether we like it or not, red meat is the richest dietary source of iron. But there are a few meat-free sources too. "Vegetarian sources of iron include pulses such as chickpeas and lentils, sprouted seeds and beans, breakfast cereals and bread. Spinach is famous for containing iron, but it is also found in other green leafy vegetables such as broccoli and kale," explains Sambrook. She adds that your body can absorb iron from food more easily if it is eaten with vitamin C.

Questions 7-14

7) The first paragraph implies that

- a) Becoming a vegan is an ethical choice

- b) Hippies and activists have always been vegan
- c) Food-shopping for animal-free products is essential for vegans
- d) Being a vegan has potential drawbacks

8) Paul McCartney's 2009 campaign

- a) Helped the number of people who understand that meat-free cooking is possible to increase
- b) Was designed to help the environment for animals
- c) Showed that vegetarian meals can be more attractive than meals prepared with meat
- d) Was aimed at reducing saturated fats in processed food

9) What do we learn about protein in the third paragraph?

- a) The human body only has a small reserve of protein
- b) Without protein, bones and muscle tissues may build more slowly
- c) Levels of protein in your body need to be replenished regularly
- d) It is not possible to find protein in meat-free diets

10) According to the fourth paragraph, how can vegans and vegetarians consume the right kinds of protein?

- a) They should stick to basic food groups, such as grains or pulses
- b) They should have a mixture of food types
- c) They should eat vegetables that contain the same macronutrients as dairy products
- d) They can enhance their diet by taking food supplements

11) What does the fifth paragraph tell us about vitamin deficiency?

- a) Vegans can get enough vitamin D from sunlight on their skin

- b) Coloured fruits and vegetables are good sources of vitamin D
- c) Incidence of heart disease, dementia and multiple sclerosis among vegetarians is the same as among vegans
- d) Vegans have a higher than normal risk of vitamin D deficiency

12) What is the National Osteoporosis Society (NOS) concerned about?

- a) Young people can cut themselves
- b) More vegetarians and vegans will develop Osteoporosis
- c) There may be a rise in a specific type of injury
- d) Some people are not eating enough clean food

13) What does 'it' (the last word of the seventh paragraph) refer to?

- a) Calcium
- b) Vitamin D
- c) Vitamin A
- d) Protein

14) What does the last paragraph say about iron?

- a) Vitamin C and iron consumed together is good for iron absorption
- b) Vegans should consider eating red meat
- c) You can become anaemic if you don't eat enough vegetables
- d) For oxygen-carrying blood cells, vitamin C is more important than iron

TEXT 2

Jennifer Millar keeps rubbish bags and hand sanitizer near her tent, and she regularly pours water mixed with hydrogen peroxide on the pavement nearby. Keeping herself and the patch of concrete she calls home clean is her top priority. But this homeless encampment near a Hollywood freeway slip road is often littered with needles and rubbish. Rats occasionally run through, and Millar fears the consequences. “I worry about all those diseases,” said Millar, 43, who said she has been homeless most of her life.

Infectious diseases, including some that ravaged populations in the Middle Ages, are resurging in California and around the country and are hitting homeless populations especially hard. Los Angeles recently experienced an outbreak of typhus in city centre streets, a disease spread by infected fleas on rats and other animals. Officials briefly closed part of the City Hall after reporting that rodents had invaded the building. Hepatitis A, also spread primarily through faeces, has infected more than 1,000 people in Southern California in the past two years. The disease also has erupted in New Mexico, Ohio and Kentucky, primarily among people who are homeless or use drugs.

Public health officials and politicians are using terms like “disaster” and “public health crisis” to describe the outbreaks, and they warn that these diseases can easily jump beyond the homeless population. “Our homeless crisis is increasingly becoming a public health crisis,” California Governor Gavin Newsom said in his State of the State speech in February, citing outbreaks of hepatitis A, syphilis and typhus in Los Angeles.

Those infectious diseases are not limited to homeless populations, Newsom warned. “Even someone who believes they are protected from these infections may not be.” At least one Los Angeles city employee said she contracted typhus in the City Hall last fall. And San Diego County officials warned in 2017 that diners at a four-star restaurant were at risk of hepatitis A. Last month, the state announced an outbreak of typhus in Los Angeles city centre that infected nine people, six of whom were homeless. After city workers said they saw rodent droppings in City Hall, Los Angeles City Council President Herb Wesson briefly shut down his office and called for an investigation.

The infections around the country are not a surprise, given the lack of attention to housing and health care for the homeless and the dearth of bathrooms and places to wash hands, said Dr. Jeffrey Duchin, the health officer for Seattle, Washington State.

“It’s a public health disaster,” he said. In his area, Duchin said, he has seen shigellosis, trench fever and skin infections among homeless populations.

In New York City, where more of the homeless population lives in shelters rather than on the streets, there have not been the same outbreaks of hepatitis A and typhus, said Dr. Kelly Doran, an emergency medicine physician and assistant professor at NYU School of Medicine. But Doran said different infections occur in shelters, including tuberculosis, a disease that spreads through the air and typically infects the lungs. These diseases sometimes get the “medieval” moniker because people in that era lived in squalid conditions without clean water or sewage treatment. People living on the streets or in homeless shelters are vulnerable to such outbreaks because their weakened immune systems are worsened by stress, malnutrition and sleep deprivation. Many also have mental illness and substance abuse disorders, which can make it harder for them to stay healthy or get health care.

One recent February afternoon, Community Clinic physician assistant Negeen Farmand walked through homeless encampments in Hollywood carrying a backpack with medical supplies. She stopped to talk to a man sweeping the sidewalks. He said he sees “everything and anything” in the gutters and hopes he doesn’t get sick. “To get these people to come into a clinic is a big thing,” she said. “A lot of them are distrustful of the health care system.” On another day, 53-year-old Karen Mitchell waited to get treated for a persistent cough by St. John’s mobile health clinic. She also needed a tuberculosis test, as required by the shelter where she was living. Mitchell, who said she developed alcoholism after a career in pharmaceutical sales, said she has contracted pneumonia from germs from other shelter residents. “Everyone is always sick, no matter what precautions they take.”

During the hepatitis A outbreak, public health officials administered widespread vaccinations, cleaned the streets with bleach and water and installed hand-washing stations and portable toilets near high concentrations of homeless people. But health officials and homeless advocates said more needs to be done, including helping people access medical and behavioural health care and affordable housing. “It really is unconscionable,” said Bobby Watts, CEO of the National Homeless Council, a policy and advocacy organization. “These are all preventable diseases.”

Questions 15-22

15) What is the most important thing for Jennifer Miller?

- a) Avoiding diseases

- b) Sanitizing her immediate environment
- c) Finding a permanent home
- d) Stopping rats and other rodents

16) What does the second paragraph tell us about Hepatitis A?

- a) The recent outbreak was not confined to California
- b) Hepatitis A is transmitted by fleas on rats and other animals
- c) More than 1000 contracted the disease through sharing dirty needles
- d) Some people in the City Hall now have Hepatitis A.

17) What problem did California Governor Gavin Newsom highlight?

- a) Hepatitis A, syphilis and typhus have jumped into the homeless population
- b) Politicians are not doing enough to stop the outbreak
- c) The health situation in his state (California) is now a 'disaster'
- d) There is a link between public health and homelessness

18) What statistic is given in the fourth paragraph?

- a) Four percent of restaurants were at risk of hepatitis A
- b) Two thirds of typhus cases in Los Angeles city centre concerned people living on the streets
- c) Nine out of ten people are concerned about the crisis
- d) Ninety percent of people believe they are protected from these infections

19) What does Dr. Jeffrey Duchin think?

- a) The infections are surprising
- b) There are insufficient washing facilities

- c) Shigellosis and trench fever have disastrous consequences
- d) More houses should be built for the homeless

20) What does the sixth paragraph help us to understand?

- a) The conditions in New York shelters are worse than those in Los Angeles
- b) Tuberculosis infections could be due to poor sewage treatment
- c) Homeless people are more susceptible to these diseases for a number of reasons
- d) The pathology of these diseases has not changed since medieval times

21) What problem does Karen Mitchell have?

- a) She has a chronic cough
- b) She lost her job in pharmaceutical sales
- c) She has to go to a new shelter
- d) She has tuberculosis

22) In the final paragraph, what else needs to be done?

- a) A Install more portable toilets and hand-washing stations
- b) B Prevent more diseases
- c) C Give free health care to homeless people
- d) D Give assistance to people who want medical help or an inexpensive place to live

Reading Practice Test 13 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	B	-
Q 2	C	-
Q 3	B	-
Q 4	A	-
Q 5	C	-
Q 6	B	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	D	-	Q 15	B	-
Q 8	A	-	Q 16	A	-
Q 9	C	-	Q 17	D	-
Q 10	B	-	Q 18	B	-
Q 11	D	-	Q 19	B	-
Q 12	C	-	Q 20	C	-
Q 13	A	-	Q 21	A	-
Q 14	A	-	Q 22	D	-

Review & Explanations

TEST-14

Stroke

PART A

TEXT 1

Stroke

A Cerebro-vascular accident (CVA) is commonly called a stroke. It happens when the blood supply to the brain is interrupted. Blood contains oxygen and important nutrients for your brain cells through a network of arteries. Blood flow may be interrupted or stop moving through an artery because the artery is blocked (ischaemic stroke) or bursts (haemorrhagic stroke). When brain cells do not get enough oxygen or nutrients, they die. The area of brain damage is called a cerebral infarct. Brain cells usually die shortly after the stroke starts. However, some can last a few hours if the blood supply is not cut off completely.

Stroke is a largely preventable event and many risks can be reduced by making lifestyle changes. There are other reasons that people have a stroke. If someone in a person's close family, such as a sibling or parent, has had a stroke, then that person has a higher risk of having one too. If the person themselves has had a stroke or heart attack then they are at risk. Men have strokes more often than women and people aged over 65 are more likely to have a stroke than younger people.

Lifestyle factors likely to increase stroke risk are being overweight or obese, doing very little regular exercise or having long term stress. Cigarette smoking is closely linked to stroke but drinking too much alcohol and/or caffeine and having a lot of food that has cholesterol in it, are also key causes of a person having a stroke

TEXT 2

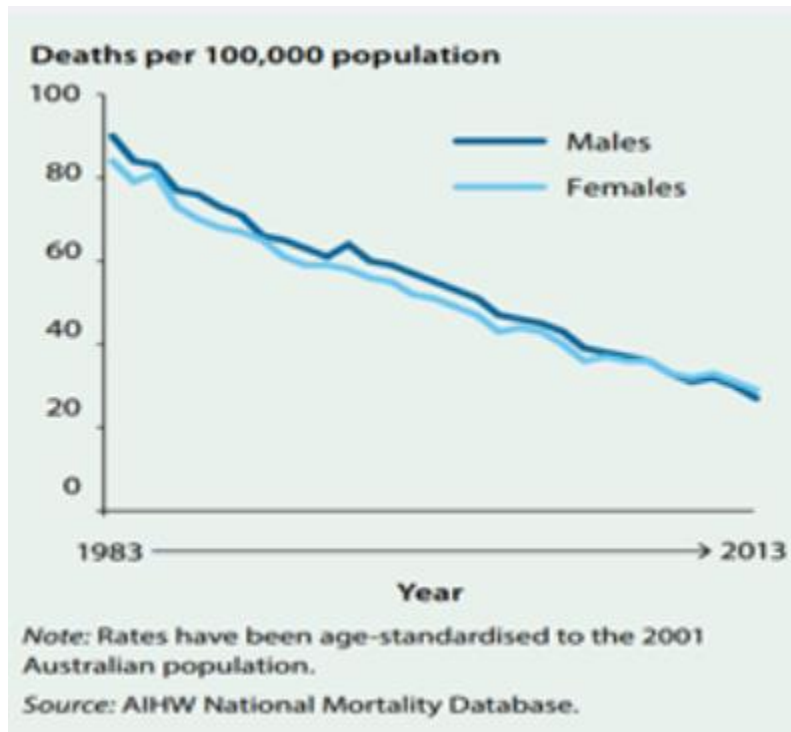
Incidence of stroke in Australia

Information from the Australian Institute of Health and Welfare tells us that:

- in Australia, in 2013–14, there were 37,000 admissions to hospital for acute care of stroke and 28,000 admissions for rehabilitation care for stroke
- the average length of stay in acute hospital care was 8 days, and in rehabilitation care, 14 days
- between 2003–04 and 2013–14, stroke admission rates fell by 15%

- an increasing number of dedicated stroke units in hospitals are showing significant improvements the health outcomes of patients
- two-thirds (67%) of patients received care stroke units in 2015

Death from stroke in Australia is getting less.



TEXT 3

The chances of a full or near complete recovery from a stroke increase significantly if a person gets treatment as soon as possible. This means people need to be able to recognise the signs and symptoms that a person is having a stroke so they can call an ambulance.

The signs and symptoms of a stroke vary from person to person but usually begin suddenly. The symptoms will depend upon the part of your brain affected and the extent of any damage. It is helpful to remember the word (acronym) 'FAST': Face-Arms-Speech-Time to check if someone may be having a stroke.

- Face – the face may have dropped on one side, the person may not be able to smile or their mouth or eye may have drooped.

- Arms – the person with the suspected stroke may not be able to lift one or both arms and keep them there because of arm weakness or numbness.
- Speech – their speech may be slurred or garbled, or the person may not be able to talk at all despite appearing to be awake.
- Time – it is time to call for an ambulance immediately if you see any of these signs or symptoms. When the person gets to hospital they need to have the type of stroke they have experienced diagnosed quickly. Tests that are commonly used include a computer tomography (CT) scan, a magnetic resonance imaging (MRI) and an angiogram.
 - Ischaemic stroke is the most common and this is treated by getting the blood supply returned to the affected area of the brain. This means the clot has to be dissolved by giving the patient a dissolving agent through an intravenous drip.

TEXT 4

Rehabilitation following a stroke

As soon as the person is medically stable, rehabilitation starts. Strokes can cause weakness or paralysis in one side of the body. Many people also have problems with co-ordination and balance, and suffer from extreme tiredness (fatigue) in the first few weeks after a stroke.

Rehabilitation is the therapy and activities that helps the person to re-learn or find new ways of doing things that were affected by the stroke. It aims to stimulate the brain's ability to change and adapt, which is called neuroplasticity. By creating new brain pathways, a person may learn to use other parts of the brain to recover the functions of those parts that were affected by the stroke.

Physiotherapy focuses on setting goals and providing an exercise plan to improve posture and balance. Speech and language therapy helps with problems with communication, including difficulty speaking and understanding others.

The most rapid recovery occurs in the first 3 months after a stroke. Further recovery is possible, but gains are usually slower and may take years.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** The reasons people have a stroke? _____
- 2** The types of rehabilitation a person may have after a stroke? _____
- 3** The signs and symptoms of a stroke? _____
- 4** The death rate from stroke in Australia? _____
- 5** The most common type of stroke? _____
- 6** The role of speech and language therapy in stroke rehabilitation? _____
- 7** The lifestyle factors that are likely to increase the risk of stroke? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** How many people in Australia went into hospital for acute stroke care?

- 9** What part of a stroke rehabilitation programme focuses on setting goals and providing an exercise plan to improve posture and balance?

- 10** What type of stroke occurs if an artery in the brain bursts?

- 11** Which word is useful to check if someone may be having a stroke?

12 What is the brain's ability to change and adapt called? _____

13 What kind of hospital units are showing significant improvements in the outcomes for stroke patients? _____

14 How might a person's speech sound if they are having a stroke?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 If a person is having a stroke you may notice their face has _____ on one side.

16 _____ from stroke in Australia is getting less.

17 The most _____ occurs in the first 3 months after a stroke.

18 A stroke happens when the _____ to the brain is interrupted

19 The average length of stay in rehabilitation care was _____ days.

20 The clot has to be dissolved by giving the patient a _____ through an intravenous drip.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	A	-	Q 8	37,000		Q 15	dropped	
Q 2	D	-	Q 9	physiotherapy		Q 16	Death	
Q 3	C	-	Q 10	Haemorrhagic		Q 17	rapid recovery	
Q 4	B	-	Q 11	FAST		Q 18	blood supply	
Q 5	C	-	Q 12	Neuroplasticity		Q 19	14	
Q 6	D	-	Q 13	Dedicated strok		Q 20	dissolving agen	
Q 7	A	-	Q 14	Slurred or garb				

Review & Explanations

PART B

TEXT 1

E-Cigarettes May Threaten Goal of Achieving Tobacco Endgame

E-cigarettes have become the most popular tobacco products for youth and adolescents in the United States and offer new pathways to nicotine addiction. These products may be beneficial for helping some smokers quit or move to less harmful products, but their long-term effects and net public health impact are unclear. Concern is increasing that use of newer tobacco products may catalyse transition to use of other tobacco products or recreational drugs. Robust U.S. Food and Drug Administration regulation of all tobacco products is needed to avoid the economic and population health consequences of continued tobacco use. The American Heart Association supports minimizing use of all combustible tobacco products first, while ensuring that other products do not addict the next generation of young people.

Question

- 1) What does this article tell us about e-cigarettes?
 - a) Long-term use of e-cigarettes is harmful
 - b) The catalyst used in e-cigarettes can be dangerous
 - c) It is thought that e-cigarettes could be useful in some circumstances.

TEXT 2

Echocardiography

The use of echocardiography as an imaging tool has increased substantially over the past decade. Cardiologists perform most echocardiography studies, with internists being the next most common providers of these studies. Tissue Doppler imaging provides information about movement of cardiac structures. The relation between the dynamics of cardiac structures and the hemodynamics of the blood inside these structures provides information about cardiac diastolic and systolic function.

Dedicated training for competent performance and interpretation of echocardiography is essential. The American College of Cardiology has recommended a set of minimum knowledge and training requirements for the performance and interpretation of echocardiography, including a minimum number

of 150 performed and 300 interpreted examinations for level 2 competency in interpreting echocardiography.

Question

- 2) What do we learn about echocardiology from this article?
- a) Different types of information are combined to give an overall picture
 - b) Only cardiologists with a minimum level 2 qualification should interpret echocardiographs
 - c) Movement of cardiac structures should be kept to a minimum during the imaging process

TEXT 3

Intubation Procedure

This will follow administration of an induction agent (which may be intravenous or inhalational or a combination of both) and a muscle relaxant. Intubation attempts should not last for longer than 30 seconds. Begin by keeping your right hand free - it will be needed to open the mouth, control the head and to use suction, etc. Inspect the mouth for loose teeth or for dentures. If any are found, they should be removed. Once a satisfactory view of the airway is available, the procedure can begin. Hold the laryngoscope in the left hand and introduce the laryngoscope over the right side of the tongue, sweeping the tongue to the midline. Position the tip of the blade in the vallecula and lift upwards and away from yourself until the glottis is visualised.

Question

- 3) What does this article say about intubation attempts?
- a) If the first attempt doesn't succeed, more muscle relaxant can be administered
 - b) More than one procedure can be tried, for a maximum duration of half a minute
 - c) If the first intervention is unsuccessful, try putting the laryngoscope in your other hand for the second attempt

TEXT 4

Most Americans Report Good Health

Most Americans report having excellent or good health and have a usual place to go for medical care, according to a report published recently for the National Health Survey. The researchers note that from January to September 2018, 87.7 percent of people had a usual place to go for medical care, which was not significantly different from the 2017 estimate of 88.3 percent. From January to September 2018, 4.7 percent of persons failed to obtain needed medical care at some time during the previous 12 months, which did not differ significantly from 4.5 percent in 2017. The percentage of people who had excellent or very good health was 66.3 percent for January to September 2018, which showed many similarities to the statistic for 2017, which was 66.4 percent.

Question

- 4) What does the following report say about healthcare in the USA?
- a) There were negligible differences between the reports
 - b) A significant number of people do not have usual medical care
 - c) In terms of healthcare provision (including perceived wellness), the overall trends are positive

TEXT 5

The Management of Self-Harm in Primary Care

Primary care has an important role in the assessment and treatment of people who self-harm. Careful attention to prescribing drugs to people at risk of self-harm, and their relatives, could also help in prevention. When an individual presents in primary care following an episode of self-harm, healthcare professionals should urgently establish the likely physical risk, and the person's emotional and mental state, in an atmosphere of respect and understanding.

All people who have self-harmed should be assessed for risk, which should include identification of the main clinical and demographic features and psychological characteristics known to be associated with risk, in particular depression, hopelessness and continuing suicidal intent. The outcome of the assessment should

be communicated to other staff and organisations who become involved in the care of the service user.

Question

- 5) What does this article tell us about the management of self-harm in primary care?
- a) Primary care providers must not administer drugs if the person appears to be hopeless or suicidal
 - b) Clinical and demographic information is vital in determining risk of further self-harm
 - c) Evaluation results need to be passed on to other concerned agencies

TEXT 6

Current Treatments: Migraines

Paracetamol and aspirin both work well for many migraine attacks. Doses should be taken as early as possible after symptoms begin. The severity of the headache can be significantly reduced if painkillers are taken early enough. A lot of people do not take a painkiller until a headache becomes severe which is often too late for the painkiller to work well.

Strictly speaking, aspirin is an anti-inflammatory painkiller. Recently, as it has become associated with stomach bleeding, aspirin has fallen from favour for the treatment of many painful conditions. However, for migraine, it is often worth a try. Recent studies have confirmed that aspirin either takes away migraine pain, or greatly reduces the pain, within two hours in more than half of the people who take it.

Question

- 6) What does this article say about the use of aspirin for migraine treatment?

- a) Aspirin is no longer prescribed for migraines
- b) The risks of using aspirin have to be weighed up with the potential benefits
- c) Over 50% of sufferers found that aspirin was helpful

PART C

TEXT 1

It's hard to keep up with the message on eggs. Are they good for you or not? In the 1960s, people were told: "Go to work on an egg." But in the 1970s the public was advised to avoid eggs because they were linked to high blood cholesterol. The negative press on eggs continued in the 1980s when raw eggs were linked to salmonella poisoning. The message changed in 1999 when a study, published in a leading medical journal, found no link between egg consumption and the risk of cardiovascular disease. This lack of a relationship between egg consumption and cardiovascular disease was reaffirmed in 2013 in an analysis of 17 reports of studies involving over three million participants. Indeed, eggs seemed to have been rehabilitated as part of a healthy diet, so much so that it became fashionable to keep chickens.

But now the doom merchants are back, warning that eggs can kill. This latest report, published in JAMA, followed nearly 30,000 participants for an average of 17 years. Of these 5,400 had a cardiovascular disease "event" (heart attack or stroke). The researchers found that each egg consumed was associated with a 2.2% greater absolute risk of cardiovascular disease over the follow-up period (roughly 22 extra cases of cardiovascular disease per 1,000 participants). The statistical methods used were robust, and the pooled data from six studies represents the ethnic diversity of the U.S. population and the diets of ordinary Americans.

The study's limitations are the dependence on a single measure of dietary intake at the start of the study and the strong correlation of egg intake with obesity and unhealthy lifestyles, such as smoking, eating lots of red and processed meat, and not eating a lot of fruit and veg. Statistical adjustments were made to correct these confounding factors. However, these corrections are imperfect and invalid when the correlations with egg intake is very strong. For example, in the U.S. eggs are often eaten with bacon, sausages, or burgers, so it's impossible to disentangle the effects on CVD risk of eggs from these fatty meat products. Also, the increased risk was much greater than would be predicted from the known effects of eggs on blood

cholesterol levels. These findings need to be considered in the context of the North American dietary pattern because they may not apply to other dietary patterns, especially Asian.

The average egg consumption in most countries is usually only three or four eggs a week. A medium-sized egg provides 226mg cholesterol and average cholesterol intakes typically range between 200-250mg per day. It is easy to be confused by a high blood cholesterol level, which increases the risk of cardiovascular disease, and its relationship with dietary cholesterol, which is mainly provided by eggs. Very high blood cholesterol levels are usually inherited or caused by a lack of some hormones (such as thyroid hormone). But moderate increases in blood cholesterol are related to diet.

Most adults in North America, Europe, and Australasia have moderately increased blood cholesterol levels as a result of middle-aged spread, saturated fat intake and, to some extent, cholesterol intake. Randomized controlled trials, where participants are fed increasing amounts of eggs, have found that each 200mg of cholesterol from eggs increases the harmful form of blood cholesterol by only a 3% rise.

Between a quarter and a third of the population inherit a version of the APOE gene called e4 that makes them much more sensitive to dietary cholesterol than those who carry the more common e3 version. They can show a 10% increase in LDL cholesterol with dietary cholesterol from eggs. There is also variability on how much cholesterol is absorbed. Most of the cholesterol in the small intestine is derived from bile secreted from the liver rather than from eggs. Plant sterols, which are added to some foods, such as yogurt drinks and margarine, block cholesterol absorption and lower LDL cholesterol by up to 10%. So even people with the e4 gene can eat eggs without increasing their LDL cholesterol if they consume plant sterols in the same meal.

The American diet contains large amounts of meat and eggs, and it seems probable that a high intake of cholesterol (equal to two to three eggs a day) adds to the risk of cardiovascular disease, particularly in people with type 2 diabetes. There is also good reason to caution younger people about the risks of following the fad of high-protein diets that may include eating several eggs a day. Otherwise, eating eggs in moderation (three to four eggs a week) makes a useful contribution to nutrient intake and is harmless.

Questions 7-14

- 7) What does the first paragraph tell us about medical opinions concerning egg consumption?

- a) Although some experts have changed their opinion, eggs are still unsafe
- b) Salmonella poisoning is no longer associated with egg consumption
- c) Over the years, advice has been inconsistent
- d) The risk of cardiovascular disease can be reduced by cutting your egg consumption

8) What does the report in JAMA highlight?

- a) Heart attacks and strokes occurred over a period of 17 years
- b) Even eating just one egg had a quantifiable risk
- c) Patients with robust health had fewer cardiovascular “events”
- d) Egg consumption was not uniform across all ethnic groups

9) What does the third paragraph tell us about the study’s limitations?

- a) Statistical adjustments are able to cancel out the factors of unhealthy diet
- b) It is possible to isolate the clinical effect of egg consumption
- c) Modifying the data does not alter the study’s key findings
- d) People in the study were not permitted to eat a lot of fruit or vegetables

10) The study mentioned in the third paragraph

- a) might have found different results among specific ethnic groups
- b) should have included results among vegetarians
- c) could be misinterpreted due to the known effects of eggs on blood cholesterol levels
- d) should be reanalysed to take into account North American dietary patterns

11) What do we learn about cholesterol from the fourth paragraph?

- a) Cardiovascular disease has a strong correlation to dietary cholesterol consumption
- b) Eggs do not contain the dangerous form of cholesterol
- c) Cardiovascular disease can be inherited
- d) It is not easy to isolate the effect of dietary cholesterol

12) In paragraph five, what were the findings of the randomized controlled trial?

- a) Cholesterol increases from egg consumption were really small
- b) Middle aged people had higher levels of cholesterol
- c) Even just 200mg increase of cholesterol was dangerous
- d) Some participants in the study were allergic to eggs

13) What do we learn from the sixth paragraph?

- a) The e3 gene is more sensitive to dietary cholesterol than other types of the gene
- b) Bile secreted from the liver can cancel out the effects of dietary cholesterol
- c) Less than half of the population get the e4 gene from their parents
- d) Neither the e3 nor the e4 gene reduce sensitivity to plant sterols

14) What does the last paragraph help us to understand?

- a) People with type 2 diabetes should not eat eggs
- b) Eating three to four eggs a week is worse than eating several eggs in one day
- c) Young people are not concerned by high levels of cholesterol consumption
- d) Some fashionable diets can have harmful effects

TEXT 2

By the age of 50, half of all men will have noticeable hair loss. By the age of 60, it will affect around two thirds. The majority of men, therefore, will at some point in their life, have hair loss. It's so common, in fact, going bald could be considered a normal part of being male. It's actually more unusual not to go bald. Yet despite how common male pattern baldness is, it causes untold distress and anguish to men. It's strongly associated with the development of depression, anxiety and poor self-image.

I work in mental health and I see a surprising number of men who confide in me that the distress they experienced from their hair loss has led to their mental health

problems. Yet men rarely discuss openly how much upset their hair loss is causing them. It's a shameful secret. Sometimes the first anyone knows that it's even been on their mind is when they have a hair transplant.

I have one friend, let's call him Steve, who started going bald in his 20s. He was so upset about it, he told me later, that he would sometimes feel unable to leave the house. He had stubble tattooed on to his head (an increasingly popular intervention that looks very convincing) to give him a full hairline. His wife, who he met after the procedure still doesn't know that what she thinks is his hair is actually a clever tattoo. I find it incredible that they've been together five years, yet Steve still doesn't feel able to open up to her about his hair loss and what he's had done but it shows how sensitive this is for men.

Even more take anti-hair loss medication, such as finasteride, on the quiet. It seems astonishing that something that is going to affect the majority of us at some point in our lives is still considered so shameful and embarrassing. Those who try to do something are mocked for being vain, while those who are balding are mocked and ridiculed for being old and unattractive. I know of at least four of my friends taking finasteride, and that's just those who have confided in me. I suspect there are many more, but they feel unable to discuss it because hair loss is such a sensitive issue for so many men.

What's more, it doesn't just affect older men. A fifth of men will experience significant hair loss by the age of 20, meaning that this is something that is an issue affecting many of us, both young and old alike. There's an enduring idea that a man's hair is linked with ideas of strength and power - think of the biblical story of Samson and Delilah, where the source of his strength was his hair until she cut it off while he was sleeping. The image of a fat, balding old man who is mocked because of his looks strikes horror into the heart of many young men who find themselves thinning. Because of its association with ageing, baldness reminds us of our mortality. It speaks on a deep level to how we perceive ourselves and how we think others view us. There is a sense of powerlessness and impotence - our bodies out of control. Anxieties around hair loss often get bound up with other anxieties about our bodies and feed into insecurities about our appearance or low self-esteem.

Of course, many men are able to embrace their thinning hair, such as the actors Bruce Willis and Jason Statham. Some guys adopt a close-shaved look and frame it in terms of evidence of their masculinity and manliness. For those who struggle to accept the change and lament their hair loss, there is a multi-million-dollar industry that, I'm sorry to say, promises a lot but tends not to deliver.

There is no magic pill or miracle surgery to reverse hair loss. Even finasteride only actually causes re-growth of hair in a small number of people who take it, with it simply slowing the rate of loss for most. It only provides about 30% improvement in hair loss over six months. Hair grafts and transplants have limitations and are very costly with results varying considerably. Shampoos and lotions and potions have limited - if any - effect.

That's why I often recommend men who are struggling with going bald see a clinical psychologist to have a course of cognitive behavioural therapy (CBT) to address their distress and help them to change their thinking about their hair loss. It sounds unlikely, but it really does work. The fact is, it is much easier - and cheaper - to learn to accept the hair loss than it is to reverse it.

Questions 15-22

15) In the first paragraph, what does the writer claim about the effects of hair loss?

- a) Two thirds of all men are affected by hair loss
- b) It is linked to an increase in mental health issues
- c) Most men lose as much as two thirds of their hair
- d) Male pattern baldness is noticeable in unusual men

16) What effect of hair loss do we learn about in the second paragraph?

- a) Hair loss in men can be surprising
- b) The best cure for hair loss is a hair transplant
- c) Men seldomly talk about their hair loss
- d) Some men secretly like to be bald

17) What does Steve's wife think?

- a) She thinks his hair looks like a tattoo
- b) She thinks her husband is really clever
- c) She thinks her husband has normal hair
- d) She thinks her husband is particularly sensitive

18) What do we learn about finasteride?

- a) This medication is often taken in secret
- b) Finasteride is the only effective anti-hair loss medication available
- c) Only vain people use this medication
- d) Four out of ten men use finasteride on a regular basis

19) In the fifth paragraph, what does the writer claim about hair?

- a) Only older men are concerned with hair loss
- b) Some hair styles can cause baldness
- c) Hair can make some men feel insecure
- d) People associate hair with strength and power

20) In the sixth paragraph, what does the writer say about the multi-million-dollar industry?

- a) It is used by famous actors, like Bruce Willis and Jason Statham
- b) It is deceptive
- c) It is promising
- d) It promotes values such as masculinity and manliness

21) What do we learn about finasteride in the seventh paragraph?

- a) It causes hair to grow again in some cases
- b) 30% of users are disappointed by its results
- c) Finasteride is a kind of magic pill
- d) Finasteride should not be used for longer than six months

22) In the final paragraph, what is the writer's advice?

- a) It costs less to alter the way you think about the problem
- b) CBT should be taken in combination with finasteride
- c) Reversing hair loss is easier than CBT
- d) Clinical psychologists can give advice of hair loss treatments

Reading Practice Test 14 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	C	-
Q 2	A	-
Q 3	A	-
Q 4	A	-
Q 5	C	-
Q 6	A	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	C	-	Q 15	B	-
Q 8	B	-	Q 16	C	-
Q 9	C	-	Q 17	C	-
Q 10	A	-	Q 18	A	-
Q 11	D	-	Q 19	D	-
Q 12	A	-	Q 20	B	-
Q 13	C	-	Q 21	A	-
Q 14	D	-	Q 22	A	-

Review & Explanations

TEST-15

Ganglion Cyst

PART A

TEXT 1

What is a ganglion cyst?

A ganglion cyst is a collection of synovial fluid in a sac, on or near tendon sheaths and joint capsules. They usually appear on the on the dorsal aspect of hands, fingers and wrists, and can also occur on the feet, ankles and knees. The cyst can range from the size of a pea to the size of a golf ball. The size of a ganglion may increase over time, especially if it near a joint where there are frequent repetitive movements.

About 65% of ganglia of the wrist and hand are dorsal wrist ganglia, followed by the volar wrist ganglion constituting about 20 to 25% of ganglia. Flexor tendon sheath ganglia and mucous cysts arising from the dorsal distal interphalangeal joint make up the remaining 10 to 15%.

Ganglion cysts look and feel like a smooth lump under the skin and the wall of the ganglion is smooth, fibrous, and of variable thickness. The cyst is filled with clear gelatinous, sticky, or mucoid fluid of high viscosity. The fluid in the cyst is sometimes almost pure hyaluronic acid. The cyst is attached to the tendon or joint by a pedicle (stalk).

The cause of them is not known, however it is thought they may be caused by tiny tears in the covering of a tendon or joint. Ganglion cysts are benign and appear in isolation. Around 30 to 50 per cent of ganglion cysts resolve spontaneously without medical intervention, though this can take many years.

Ganglia constitute about 60% of all chronic soft-tissue swellings affecting the hand and wrist. They usually develop spontaneously in adults aged 20 to 50, with a female:male preponderance of 3:1. People who have wear-and-tear arthritis in the finger joints closest to their fingernails are at higher risk of developing ganglion cysts near those joints. Joints or tendons that have been injured in the past are more likely to develop ganglion cysts.

TEXT 2

Diagnosing a ganglion cyst

Ganglia are evident on examination even if they cannot be seen by the naked eye. It is important that cysts are examined by a doctor because there is another type of ganglion on the dorsal wrist that occurs in people with rheumatoid arthritis. A doctor can easily differentiate between them because a rheumatoid cyst is soft and irregular in appearance. Also, a person with rheumatoid arthritis will also have proliferative rheumatoid extensor tenosynovitis.

Most ganglion cysts do not cause symptoms, but the main symptoms people experience are a noticeable swelling or lump. The lump is able to change its size, including going away completely only to return. The lump is usually soft and immobile. In some cases, the lump is painful and aching, particularly those at the base of fingers. The ache and pain is made worse by moving any nearby joints. The affected tendon may cause a sensation of muscular weakness. The back of the hands and wrists are most commonly affected.

A medical examination is generally all that is needed to confirm diagnosis but other tests could include: Aspirating some of the fluid with a syringe An ultrasound to determine if the ganglion is solid or fluid filled X-ray and/or magnetic resonance imaging may be needed if the cyst cannot be seen.

TEXT 3

Passive treatment options for a ganglion cyst

If a cyst is not causing any problems, a passive “watch and wait” approach is recommended. This means the cyst is monitored and action only taken if it increases to a point where it causes symptoms. However, even if there are no symptoms some people prefer treatment for cosmetic reasons.

Temporarily immobilising the joints around a cyst may both slow down the rate at which the cyst grows and reduce the size of the cyst. This may release the pressure on nerves, relieving pain. If a person knows what activity is the likely cause such as starting to play an instrument or using a new piece of equipment, it may be helpful to stop or modify this activity.

Simple over the counter pain relievers and/or anti-inflammatory medications may be required to alleviate pain. In some cases, modifying shoes or how they are laced can relieve the pain associated with ganglion cysts on ankles or feet.

A traditional old home remedy for a ganglion cyst consisted hitting the cyst with the Bible. Thumping a cyst with any heavy object is not recommended because the force of the blow can damage surrounding structures in the hand or foot.

Another TEXT 4

self-help approach is to try and "pop" the cyst by puncturing it with a needle. This is unlikely to be effective and can lead to infection.

Some people advocate herbal remedies that have anti-inflammatory properties such as turmeric and ginger. The true cause of ganglion cysts is not known but they are a bulge in the lining of a structure. This means it is unlikely to be part of the inflammatory process

Active treatment options for ganglion cysts.

If a cyst is causing problems, a needle aspiration performed by a qualified doctor. This simple procedure is carried out in the GP surgery or hospital outpatients department. It involves drawing the liquid contents of the cysts out of the sac via the syringe.

Needle aspiration is usually the first active treatment option offered for ganglion cysts as it is less invasive than surgery. However, nonsurgical treatment fails in about 40 to 70% of patients, necessitating surgical excision.

The cyst may be surgically removed using either open or keyhole approaches.

In open surgery the surgeon makes a medium-sized cut, usually about 5cm (2in) long, over the site of the affected joint or tendon. The sac is removed at the pedicle to reduce recurrence.

Keyhole surgery is often used if the ganglion cyst is near, or in a joint. Smaller incisions are made and a tiny camera called an arthroscope is used by the surgeon to look inside the joint and then pass instruments through the incision to remove the cyst. Excision can be done via arthroscopic or standard open surgery. Recurrence rates after surgical excision are about 5 to 15%.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** The ratio of ganglion cysts between sexes? _____
- 2** The primary dynamic way of removing ganglion cysts? _____
- 3** The investigations that may be done to confirm someone has a ganglion cyst? _____
- 4** How keeping the affected area immobile for a time can reduce the effect of a ganglion cyst? _____
- 5** Another type of ganglion cyst that can develop at the wrist? _____
- 6** The role the bible used to play in managing ganglion cysts? _____
- 7** The contents of a ganglion cyst? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** Where ganglion cysts are usually seen? _____
- 9** What percentage of ganglion cysts come back after a surgical excision?

- 10** Are ganglion cysts more common in men or women? _____
- 11** What can changing shoes achieve for people with ganglion cysts in lower limbs?

- 12** What are the two main complaints people with a ganglion cyst have?

- 13** What type of cysts develop from the fingers? _____

14 What is often the first invasive treatment option offered for ganglion cysts?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 An _____ shows if the ganglion is solid or not.

16 _____ ganglia and mucous cysts in the DIP joints account for a small number of all ganglion cysts.

17 No one really knows why ganglion cysts develop but there is a _____ in the membrane around a structure

18 A surgeon can look into a ganglion cyst around a joint with an arthroscope and then _____ through an additional small cut in the skin to get rid of the cyst.

19 A ganglion cyst on a tendon on can lead to a cause a feeling of _____ weakness.

20 Needle aspiration involves pulling the _____ of the cysts out of the sac with a needle and syringe.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	A	-	Q 8	[Back of] Hands +		Q 15	ultrasound	
Q 2	D	-	Q 9	5% to 15%		Q 16	Flexor tendon s +	
Q 3	B	-	Q 10	women		Q 17	bulge	
Q 4	C	-	Q 11	relieve pain		Q 18	pass instrument +	
Q 5	B	-	Q 12	noticeable swell +		Q 19	muscular	
Q 6	C	-	Q 13	mucous		Q 20	liquid contents	
Q 7	A	-	Q 14	Needle aspirati +				

Review & Explanations

PART B

TEXT 1

What Nurses Need to Know About Celiac Disease and Gluten Sensitivity

Gluten is the group name for two proteins, gliadin and glutenin, which are primarily derived from wheat, barley, rye and triticale. These proteins are responsible for the bonding of particles, giving food its shape. When gluten is consumed, those with an allergy experience an immune response which attacks the small intestine. Once the villi of the small intestine are damaged, nutrients cannot be properly absorbed. While some people may be asymptomatic throughout their lifetime, many experience at least some symptoms.

Recent research shows there is no evidence to support an increased risk of celiac disease when infants are introduced to gluten at an early age (less than 4 months). However, delayed introduction (more than 7 months) to gluten may be associated with an increased risk.

Question

- 1) What does the article say about the causes of celiac disease?
 - a) It can provoke damage to the small intestine
 - b) It could be linked to children's diets
 - c) Normally, children don't suffer from celiac disease

TEXT 2

Aspirin Guidelines

Aspirin should be taken with, or straight after, a meal or snack. This helps to reduce the risk of any stomach irritation. Gastro-resistant tablets (also called enteric-coated or EC tablets) can be taken before food as these have a special coating which will help to protect the stomach from irritation. Gastro-resistant tablets should be swallowed whole, they must not be crushed or chewed. If the patient is using indigestion remedies, aspirin in this form must not be taken for at least two hours before and the two hours after they are used. This is because the antacid in the remedy can affect the way the coating on these tablets works. Melt-in-the-mouth (orodispersible) tablets should be placed on the tongue and allowed to dissolve.

Question

- 2) What do these guidelines say about when to take aspirin?
- a) Aspirin taken close to meal times can irritate the stomach lining
 - b) Some types of aspirin have special indications
 - c) It can be taken in combination with indigestion remedies

TEXT 3

Assessing the Need for a Peripheral Intravenous Cannula

Many cannulas are left in without orders for intravenous fluids or medications. Some patients end up with two, three, or even more concurrent cannulas, despite only needing one in most cases. They are often left in 'just in case' they might be needed. But any catheter leads directly to the bloodstream and can be a source of infection. The need for the cannula must be constantly reassessed.

When a cannula is inserted, a flashback of blood in the chamber confirms it is in the vein. Flushing the cannula with 0.9% saline before and after intravenous medications reduces admixture of medicines and decreases the risk of blockage.

Question

- 3) What does this article say about the use of cannulas?
- a) Cannula usage should be reviewed regularly
 - b) In most cases, concurrent cannula use is justified
 - c) Cannulas can be left in place so long as they are flushed with a 0.9% saline solution

TEXT 4

Description of the 'SecurAcath' Device

SecurAcath' is a single-use device to secure percutaneous catheters in position on the skin. It is intended for use in adults and children who need a central venous

catheter which is a long, thin, flexible tube that is inserted into a vein through the skin.

‘SecurAcath’ has two parts, a base and cover. The base is made up of two foldable metal legs and two securement feet. The feet are placed under the skin at the catheter insertion site and unfolded to make a subcutaneous anchor. The cover then attaches to the catheter shaft and holds it in place when it is clipped onto the base. The device stays in place as long as the catheter is needed and can be lifted off the skin to allow cleaning of the insertion site.

Question

- 4) How should the ‘SecurAcath’ device be used?
- a) The feet can be repositioned in order to clip them to the base
 - b) It should be correctly assembled before attaching the cover
 - c) The flexible tube should be inserted into a vein first

TEXT 5

Assessment of Colorectal Polyps During Colonoscopy

Colorectal polyps are small growths on the inner lining of the colon. Polyps are not usually cancerous, most are hyperplastic polyps with a low risk of cancer. However, some (known as adenomatous polyps) will eventually turn into cancer if left untreated. Detecting and removing adenomas during colonoscopy has been shown to decrease the later development of colorectal cancers. However, removal of any polyps by polypectomy may have adverse effects such as bleeding and perforation of the bowel.

It can take three weeks for a person to get the examination results for polyps that were removed during colonoscopy, and they may feel anxious during this waiting period. Using virtual chromoendoscopy technologies may allow real-time differentiation of adenomas and hyperplastic colorectal polyps during colonoscopy, which could lead to quicker results.

Question

5) What does the article tell us about colonoscopies?

- a) Colonoscopy and polypectomy procedures are thought to be risk-free
- b) Virtual chromoendoscopy technology could speed up the process
- c) Most hyperplastic polyps become cancerous if left untreated

TEXT 6

Osteomyelitis After Traumatic Knee Injury

A 56-year-old woman was admitted to a hospital for the treatment of osteomyelitis following a traumatic knee injury. She received the antibiotic Gentamicin in accordance with the hospital's usual protocol. Kinetics, blood drug levels, and renal function were monitored, and dosage recommendations were made. However, a permanent vestibulopathy (or balance disorder) resulted from the antibiotic.

During the case investigation, the patient testified that she experienced "roaring" in her ears while hospitalized. (The roaring is a form of tinnitus) She further testified that she was not ambulatory; she was restricted to bed rest. No staff member inquired about unusual ear symptoms or told her to report such symptoms. Consequently, a lawsuit was brought against the hospital, specifically against the pharmacists.

Question

6) What went wrong in the treatment of the 56-year-old woman?

- a) The woman was infected by vestibulopathy while in hospital
- b) The correct dosage was not balanced
- c) Staff members failed to take note of the woman's symptoms

PART C

TEXT 1

Many adult hospital inpatients need intravenous (IV) fluid therapy to prevent or correct problems with their fluid and/or electrolyte status. Deciding on the optimal amount and composition of IV fluids to be administered and the best rate at which to give them can be a difficult and complex task, and decisions must be based on careful assessment of the patient's individual needs.

Errors in prescribing IV fluids and electrolytes are particularly likely in emergency departments, acute admission units, and general medical and surgical wards rather than in operating theatres and critical care units. Surveys have shown that many staff who prescribe IV fluids know neither the likely fluid and electrolyte needs of individual patients, nor the specific composition of the many choices of IV fluids available to them. Standards of recording and monitoring IV fluid and electrolyte therapy may also be poor in these settings. IV fluid management in hospital is often delegated to the most junior medical staff who frequently lack the relevant experience and may have received little or no specific training on the subject.

The 'National Confidential Enquiry into Perioperative Deaths' report in 1999 highlighted that a significant number of hospitalised patients were dying as a result of infusion of too much or too little fluid. The report recommended that fluid prescribing should be given the same status as drug prescribing. Although mismanagement of fluid therapy is rarely reported as being responsible for patient harm, it is likely that as many as one in five patients on IV fluids and electrolytes suffer complications or morbidity due to their inappropriate administration.

There is also considerable debate about the best IV fluids to use (particularly for more seriously ill or injured patients), resulting in wide variation in clinical practice. Many reasons underlie the ongoing debate, but most revolve around difficulties in interpretation of both trial evidence and clinical experience. For example, many accepted practices of IV fluid prescribing were developed for historical reasons rather than through clinical trials. Trials cannot easily be included in meta-analyses because they examine varied outcome measures in heterogeneous groups, comparing not only different types of fluid with different electrolyte content, but also different volumes and rates of administration. In addition, most trials have been undertaken in operating theatres and critical care units rather than admission units or general and elderly care settings. Hence, there is a clear need for guidance on IV

fluid therapy for general areas of hospital practice, covering both the prescription and monitoring of IV fluid and electrolyte therapy, and the training and educational needs of all hospital staff involved in IV fluid management.

The aim of these guidelines is to help prescribers understand the physiological principles that underpin fluid prescribing the pathophysiological changes that affect fluid balance in disease states and the indications for IV fluid therapy. In developing the guidelines, it was necessary to limit the scope by excluding patient groups with more specialised fluid prescribing needs. It is important to emphasise that the recommendations do not apply to patients under 16 years, pregnant women, and those with severe liver or renal disease, diabetes or burns. They also do not apply to patients needing inotropes and those on intensive monitoring, and so they have less relevance to intensive care settings and patients during surgical anaesthesia. Patients with traumatic brain injury (including patients needing neurosurgery) are also excluded. The scope of the guidelines does not cover the practical aspects of administration (as opposed to the prescription) of IV fluids. It is hoped that these guidelines will lead to better fluid prescribing in hospitalised patients, reduce morbidity and mortality, and lead to better patient outcomes.

The guidelines will assume that prescribers will use a drug's summary of product characteristics to inform decisions made with individual patients. All patients continuing to receive IV fluids need regular monitoring. This should initially include at least daily reassessments of clinical fluid status, laboratory values (urea, creatinine and electrolytes) and fluid balance charts, along with weight measurement twice weekly. It is important to remember that patients receiving IV fluid therapy to address replacement or redistribution problems may need more frequent monitoring. Additional monitoring of urinary sodium may be helpful in patients with high-volume gastrointestinal losses. Patients on longer-term IV fluid therapy whose condition is stable may be monitored less frequently, although decisions to reduce monitoring frequency should be detailed in their IV fluid management plan. Clear incidents of fluid mismanagement (for example, unnecessarily prolonged dehydration or inadvertent fluid overload due to IV fluid therapy) should be reported through standard critical incident reporting to encourage improved training and practice (see Consequences of fluid mismanagement to be reported as critical incidents).

Questions 7-14

- 7) What does the first paragraph tell us about intravenous (IV) fluid therapy?
- a) Most patients receive a standard composition of fluids
 - b) Electrolyte status should be kept at the optimal level
 - c) It is not easy to decide on the correct volume and speed of delivery of fluids
 - d) It is difficult to correct problems
- 8) What have surveys shown about intravenous (IV) fluid therapy?
- a) There is often a lack of information about correct dosage
 - b) Sometimes, staff mixed up electrolyte fluids with standard IV fluids
 - c) Intravenous (IV) fluid therapy should be delegated to junior medical staff
 - d) Mistakes made in operating theatres were often fatal
- 9) What did the 1999 report highlight?
- a) A small number of patients died because they were prescribed the wrong medication
 - b) Around 20% of patients experience problems due to incorrect IV fluid therapy
 - c) Some hospitals fail to report deaths due to mismanaged procedures
 - d) Not all Perioperative deaths could be linked to IV fluid therapy
- 10) What does the fourth paragraph tell us about IV fluid therapy?
- a) Seriously ill patients generally need more fluids than injured patients
 - b) There are historical reasons to prolong the use of IV fluid therapy
 - c) The best IV fluids are more expensive
 - d) Not everyone agrees on the most suitable fluids to use
- 11) Why is it difficult to perform meta-analyses of trials?
- a) There are not enough qualified analysts
 - b) Trials usually don't take place in different healthcare settings
 - c) The volume of data is too great to analyse

d) More hospital staff need training before the trials take place

12) What do we learn about the scope of the guidelines in the fifth paragraph?

- a) The guidelines are not appropriate for all types of patients
- b) Patients needing inotropes and those on intensive monitoring were included for historical reasons
- c) Pathophysiological patients were excluded because they cannot be given IV fluid therapy
- d) The guidelines only apply to men (that is to say, adult male patients)

13) According to the sixth paragraph, how often should clinical fluid status be reassessed?

- a) Twice a day or more frequently
- b) Once every 24 hours
- c) Twice a week
- d) Never – routine reassessment can be monitored by machine

14) What should be done in the case of fluid mismanagement?

- a) Additional monitoring should be carried out
- b) Rehydration should be prolonged
- c) Information about occurrences should be conveyed to the appropriate authorities
- d) The person or persons involved should be criticised

TEXT 2

A CT scan is a specialised X-ray test. It can give quite clear pictures of the inside of your body. In particular, it can give good pictures of soft tissues of the body which do not show on ordinary X-ray pictures. CT stands for computerised tomography. It is sometimes called a CAT scan. CAT stands for Computerised Axial Tomography. The

CT scanner looks like a giant thick ring. Within the wall of the scanner there is an X-ray source. Opposite the X-ray source, on the other side of the ring, are X-ray detectors. You lie on a couch which slides into the centre of the ring until the part of the body to be scanned is within the ring. The X-ray machine within the ring rotates around your body. As it rotates around, the X-ray machine emits thin beams of X-rays through your body, which are detected by the X-ray detectors.

The detectors detect the strength of the X-ray beam that has passed through your body. The denser the tissue, the less X-rays pass through. The X-ray detectors feed this information into a computer. Different types of tissue with different densities show up as a picture on the computer monitor, in different colours or shades of grey. So, in effect, a picture is created by the computer of a slice (cross-section) of a thin section of your body.

As the couch moves slowly through the ring, the X-ray beam passes through the next section of your body. So, several cross-sectional pictures of the part of your body being investigated are made by the computer. Newer scanners can even produce 3-dimensional pictures from the data received from the various slices of the part of the body being scanned.

A CT scan can be performed on any section of the head or body. It can give clear pictures of bones. It also gives clear pictures of soft tissues, which an ordinary X-ray test cannot show, such as muscles, organs, large blood vessels, the brain and nerves. The most commonly performed CT scan is of the brain to determine the cause of a stroke, or to assess serious head injuries.

Usually, very little preparation is necessary. It depends on which part of your body is to be scanned. You will be given instructions by the CT department according to the scan to be done. As a general rule, you will need to remove any metal objects from your body, such as jewellery, hair clips, etc. It is best not to wear clothes with metal zips or studs. You may be asked not to eat or drink for a few hours before your scan, depending on the part of your body to be scanned.

The CT scan itself is painless. You cannot see or feel X-rays. You will be asked to stay as still as possible, as otherwise the scan pictures may be blurred. Conventional CT scans can take between 5-30 minutes, depending on which part of the body is being scanned. More modern CT scans (helical CT scans) take less than a minute and also use less radiation.

As the scan uses X-rays, other people should not be in the same room. The operator controls the movement of the couch and scanner from behind a screen or in a

separate control room so that they are protected from repeated exposure to X-rays. However, communication is usually possible via an intercom, and you will be observed at all times on a monitor. Some people feel a little anxious or claustrophobic in the scanner room when they are on their own. You can return to your normal activities as soon as the scan is over. The pictures from the scan are studied by an X-ray doctor (radiologist) who sends a report to the doctor who requested the scan.

CT scans use X-rays, which are a type of radiation. Exposure to large doses of radiation is linked to developing cancer or leukaemia - often many years later. The dose of X-ray radiation needed for a CT scan is much more than for a single X-ray picture but is still generally quite a low dose. The risk of harm from the dose of radiation used in CT scanning is thought to be very small but it is not totally without risk. As a rule, the higher the dose of radiation, the greater the risk. So, for example, the larger the part of the body scanned, the greater the radiation dose. And, repeat CT scans over time cause an overall increase of dose. Various studies have aimed to estimate the risk of developing cancer or leukaemia following a CT scan. In general, the risk is small. In many situations, the benefit of a CT scan greatly outweighs the risk.

Questions 15-22

15) What advantage does a CT scan give over a standard X-ray?

- a) It emits less radiation
- b) It can take pictures of bones and soft tissues
- c) It is quieter and uses less electricity
- d) The patient can lie down during the scan

16) What can be seen on a CT scan result?

- a) Tissue thicknesses and densities can be shown using different colours
- b) The computer displays the date, time and patient's name on the result
- c) When this article was written, CT scans could only show shades of grey on the results
- d) Cross-dimensional attributes are shown on the results in colour or shades of grey

17)What does the third paragraph tell us about the CT scans?

- a) Usually, more than one picture is obtained
- b) 3-dimensional pictures provide more information than standard cross-sectional pictures
- c) The CT ring can be programmed to move the couch slowly
- d) Images and scans can be stored on computers for up to a year

18)What type of scan is carried out most frequently?

- a) Scans of the head and neck
- b) Scans of the chest and upper body
- c) Whole body scans
- d) Scans of the head only

19)What should you wear for your CT scan?

- a) Clothing that is free of any metal
- b) A standard hospital gown
- c) There are usually no restrictions on clothing
- d) Some scans require an absence of clothing

20)What can influence the clarity of CT images?

- a) Temperature
- b) Movement
- c) Radiation levels
- d) Levels of pain or discomfort

21)What does the article say about the number of people in the CT room?

- a) A Only the operator will be with you in the CT room
- b) You can ask for one or two people to stay with you during the scan
- c) You can only be accompanied if you feel anxious or claustrophobic
- d) You will be alone in the CT room

22)What does the last paragraph say about the levels of risk?

- a) Generally, the risks are not as significant as the potential advantages
- b) Some people have developed cancer or leukaemia after a CT scan
- c) CT scanners pose a lower risk than standard X-ray machines
- d) There is a high risk of cancer if you have a large body

Reading Practice Test 15 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	B	-
Q 2	B	-
Q 3	A	-
Q 4	B	-
Q 5	B	-
Q 6	C	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	C	-	Q 15	B	-
Q 8	A	-	Q 16	A	-
Q 9	B	-	Q 17	A	-
Q 10	D	-	Q 18	D	-
Q 11	B	-	Q 19	A	-
Q 12	A	-	Q 20	B	-
Q 13	B	-	Q 21	D	-
Q 14	C	-	Q 22	A	-

Review & Explanations

TEST-16

Chicken Pox and Shingles

PART A

TEXT 1

Chickenpox and shingles

Chickenpox is a highly infectious viral illness caused by the Varicella-Zoster virus which can reappear later in life as shingles. It is most commonly known to be a mild childhood illness.

Chickenpox can be associated with severe complications and even death so must be treated seriously in all cases. Infection with chickenpox during pregnancy can cause miscarriage, foetal malformations, skin scarring, and other problems in the baby. Chickenpox in adults and immunosuppressed people can be severe.

Shingles only develops in people who have had chickenpox in the past, usually as a child. While anyone who has recovered from chickenpox may develop shingles, the risk of shingles increases as they get older. About 1 in 3 people will develop shingles at some stage during their lifetime. What causes the virus to reactivate is usually unknown, but reactivation often occurs when a disorder or drug weakens the immune system.

Unlike chickenpox, shingles cannot be spread through person to person contact. However, if a person has never had chickenpox, or received the chickenpox vaccine, they may catch chickenpox from close contact with someone who has shingles, because the shingles blisters contain the chickenpox virus.

A person who develops shingles has not done so because they were exposed to someone with either chickenpox or shingles. This is because after a person recovers from chickenpox, the virus stays in their body, moving to the roots of nerve cells near the spinal cord and there it becomes inactive or dormant. Shingles is caused by the reactivation of the virus. Again, unlike chickenpox, shingles poses no threat to unborn babies.

In Australia, Chickenpox vaccine is now given free as part of the government immunisation program. It is free under the National Immunisation Program Schedule. Before routine vaccination began in November 2005, the incidence of chickenpox appears to have decreased.

Zoster vaccine is also funded under the National Immunisation Program for persons aged 70 years, with catch-up for those aged 71–79 years also funded until October 2021.

TEXT 2

Diagnosing chickenpox and shingles

Chickenpox is so distinctive, a medical diagnosis is not required. Chickenpox starts with red spots that can appear anywhere on the body and the spots are itchy, filled with fluid and may burst. They might spread widely across the body or stay in a small area. The spots scab over and eventually the scabs drop off. More blisters might appear while others scab over. A high temperature above 38C, aches and pains, generally feeling unwell and feeling miserable due to the itchiness of the spots are other symptoms of chickenpox.

People with shingles develop symptoms before a rash appears. The symptoms include: pain; a burning, tingling or itching sensation; a stabbing sensation; a feeling of numbness in the affected area of the body; sensitivity to light; fever and/or headache or fatigue.

Two to three days after the initial symptoms appear, a painful rash will appear on the sensitive area of skin, often on the left or right side of the body. This rash at first consists of painful, red bumps that quickly develop into fluid-filled blisters, which will eventually have a crusty surface.

TEXT 3

Managing chickenpox and shingles

The management of both chickenpox and shingles focusses on managing the rash and reducing the risk of spreading the condition.

To manage the rash the main advice is to keep it as dry and as clean as possible. Applying calamine lotion, cool compresses, baths or ice packs can reduce the urge to scratch the rash as scratching may cause scarring and infection of the blisters. After a bath or shower, skin should be gently patted dry with a clean towel rather than rubbed vigorously to scratch the rash and wearing loose cotton clothes around the parts of the body that are affected can also help. Covering the spots with sticking plasters and using antibiotic creams are not recommended because they may slow down the healing process. If the blisters are open, applying creams or gels is not recommended because they might increase the risk of a secondary bacterial infection.

For chickenpox sufferers, simple paracetamol and plenty of fluids are encouraged to manage the fever.

To reduce the risk of spread of chickenpox either from chickenpox or shingles sufferers need to stay away from school, nursery or work until all the spots have

crusted over, usually 5 days after the spots first appeared. It is important for sufferers to avoid contact with people who have a weak immune system and babies less than 1 month old, along with anyone known to have not been exposed to chickenpox, especially pregnant women. Sharing towels, playing contact sports, or going swimming are also discouraged until the spots have dried up and de-roofed. Minimising scratching of the spots is key to reducing the spread as it is the fluid in them that is infectious and washing hands frequently will also reduce the risk of spreading the virus.

People who are experiencing any symptoms of shingles should see their doctor as soon as possible. If treatment with antiviral medications is commenced within 3 days of shingles starting, it can reduce the condition's severity and the risk of further complications.

TEXT 4

Varicella-Zoster virus is one of eight types of herpes viruses infect humans (Human Herpes Virus [HHV]). After initial infection, all herpes viruses remain latent within specific host cells and may subsequently reactivate. Human herpes viruses do not survive long outside a host; thus, transmission usually requires intimate contact. In people with latent infection, the virus can reactivate without causing symptoms; in such cases, asymptomatic shedding occurs and people can transmit infection.

Common name	Human herpes virus (HHV)	Most common manifestations
Herpes simplex virus Type 1 and Type 2	HHV1 and HHV2	Herpes labialis (cold sores), genital herpes
Varicella-zoster virus	HHV 3	Chickenpox, shingles
Epstein-Barr virus	HHV 4	Glandular fever
Cytomegalovirus (CMV)	HHV5	Congenital CMV
-	HHV6	Roseola infantum Otis media with fever
-	HHV 7	Roseola infantum
Kaposi sarcoma-associated herpesvirus	HHV 8	Kaposi sarcoma and AIDS-related non-Hodgkin lymphomas

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** What can happen if the spots are scratched? _____
- 2** The indications that a person might have chickenpox? _____
- 3** The human herpes virus' ability to be infectious outside the body? _____
- 4** The infectiousness of shingles? _____
- 5** The prevalence of people who will develop shingles? _____
- 6** The most common conditions caused by the human herpes virus? _____
- 7** The signs that someone has shingles before a rash appears? _____

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** How soon should a person who knows they have shingles start taking drugs to ease their symptoms? _____
- 9** Is the loss of feeling where the rash is a symptom of chickenpox or shingles?

- 10** What part of the spots in a person with chickenpox or shingles transmits the virus? _____

11 What vaccines are available to manage chickenpox and shingles in Australia?

12 The virus that causes glandular fever is commonly called what?

13 After about how many days of symptoms, do the fluid filled spots appear for a person with shingles? _____

14 Apart from pregnant women and unborn babies, infection with chickenpox can be significant for which other two groups of people? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 After the first episode of infection, all _____, stay dormant in certain cells.

16 Shingles is _____ problem for foetuses.

17 Chickenpox is so obvious, a _____ is not needed

18 The best guidance for someone with chickenpox is to _____ by keeping it clean and dry.

19 In shingles, the rash starts out sore, with bumpy and red areas but soon develops into _____, which will eventually have a dry top.

20 People with Chickenpox or shingles have to need to stay away from public places until all the spots have dried up, _____ after they started.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	C	-	Q 8	Within 3 days		Q 15	all herpes viru +	
Q 2	B	-	Q 9	shingles		Q 16	not a	
Q 3	D	-	Q 10	fluid		Q 17	medical diagnos +	
Q 4	A	-	Q 11	Chicken pox, zo +		Q 18	manage the rash +	
Q 5	A	-	Q 12	Epstein-Barr (V +		Q 19	fluid-filled bl +	
Q 6	D	-	Q 13	2-3 days		Q 20	usually 5 days	
Q 7	B	-	Q 14	Adults, immunos +				

Review & Explanations

PART B

TEXT 1

Blood Clotting Tests

If a blood clot forms within a healthy blood vessel it can cause serious problems. So, there are chemicals in the blood that prevent clots from forming and chemicals that 'dissolve' clots. There is balance between forming clots and preventing clots. Normally, unless a blood vessel is damaged or cut, the 'balance' tips in favour of preventing clots forming within blood vessels.

A blood sample is taken into a bottle that contains a chemical which prevents the blood from clotting. It is then analysed in the laboratory. There are a number of tests that may be done. These tests measure the time it takes for a blood clot to form after certain activating chemicals are added to the blood sample.

Question

- 1) What do we learn from this article about blood clotting?
 - a) Artificial chemicals exist that can be introduced to 'dissolve' clots
 - b) Most of the time, clots don't form in blood vessels
 - c) It takes time for a damaged or cut blood vessel to form a clot

TEXT 2

Simple snoring is part of a spectrum of breathing disturbance during sleep. The muscles around the upper airway relax during sleep. A narrowed airway can lead to air turbulence, which causes vibration in soft tissues of the oropharynx, generating the snoring sound during inspiration. The specific origin of the noise varies between individuals and may include the soft palate. Snoring may disturb the sleep of the patient and their bed partner and affect relationships.

A surgical procedure may be used for patients with problematic snoring where the soft palate is implicated, and when snoring has not been improved by conservative treatment. The aim of the procedure is to stiffen the soft palate over subsequent weeks as a result of fibrosis.

Question

- 2) What would be a suitable title for this article?
- a) Snoring: Causes and proposed remedy
 - b) Soft palate intervention protocol
 - c) Oropharynx soft tissue syndrome and treatment options

TEXT 3

Apple Watch Detects Irregular Heart Beat In Large U.S. Study

The Apple Watch was able to detect irregular heart pulse rates that could signal the need for further monitoring for a serious heart rhythm problem, according to data from a large study funded by Apple Inc, demonstrating a potential future role for wearable consumer technology in healthcare. Researchers hope the technology can assist in early detection of atrial fibrillation (AF), the most common form of irregular heartbeat. Patients with untreated AF are five times more likely to have a stroke.

Results of the largest AF screening and detection study, involving over 400,000 Apple Watch users who were invited to participate, were presented on Saturday at the American College of Cardiology meeting in New Orleans. Of the 400,000 participants, 0.5 percent, or about 2,000 subjects, received notifications of an irregular pulse.

Question

- 3) What do we learn in the following article?
- a) Atrial fibrillation was found to be uncommon among Apple Watch users
 - b) In the future, everyone will use wearable consumer technology
 - c) Serious heart rhythm problems can be diagnosed by Apple watches

TEXT 4

Review of Computerized Clinical Decision Support in Community Pharmacy

Clinical decision support software (CDSS) has been increasingly implemented to assist improved prescribing practice. Reviews and studies report generally positive

results regarding prescribing changes and, to a lesser extent, patient outcomes. Little information is available, however, concerning the use of CDSS in community pharmacy practice. Given the apparent paucity of publications examining this topic, we conducted a review to determine whether CDSS in community pharmacy practice can improve medication use and patient outcomes.

Most studies showed improved prescribing practice, via direct communication between pharmacists and doctors or indirectly via patient education. Factors limiting the impact of improved prescribing included alert fatigue and clinical inertia. No study investigated patient outcomes and little investigation had been undertaken on how CDSS could be best implemented.

Questions 1-6

- 4) What does the writer imply about CDSS in this article?
- a) CDSS has an important role to play in improving patient outcomes
 - b) Previous studies concerning CDSS recommended changes
 - c) More research is needed to demonstrate its effectiveness

TEXT 5

Workplace Violence: A Serious Problem in Healthcare

There's no industry that's immune to workplace violence, whether it's construction, manufacturing, retail or media. However, healthcare professionals face a greater risk of injury, trauma, emotional distress and even death as they go about their everyday work responsibilities – giving injections, taking X-rays, drawing blood, and prepping patients for surgery.

Violence against healthcare workers occurs in all types of settings, from busy emergency rooms to surgery centres, from walk-in clinics to nursing homes. Too often, it is perceived to be just part of the job and has a long-standing history in the industry. According to American Nurse Today, 67% of all nonfatal injuries caused by workplace violence occur in healthcare, even though the industry represents only 11.5% of the U.S. workforce.

Question

- 5) Which of the following statements is not true?

- a) In the US, less than 20% of the overall labour force works healthcare
- b) Historically, violence is more often directed at nurses rather than doctors
- c) The hazard of violence at work has been found to be more serious for healthcare workers compared to workers in other industries

TEXT 6

Bronchiolitis

Bronchiolitis is an acute viral infection of the lower respiratory tract that occurs primarily in the very young. It is a clinical diagnosis based upon typical symptoms and signs. Bronchiolitis is generally a self-limiting illness, and management is mostly supportive.

There is some discrepancy between the use of 'bronchiolitis' in the UK and in the USA and other parts of Europe, and no universally accepted definition for such a common condition. In the UK, the term describes an illness in infants, beginning as an upper respiratory tract infection (URTI) that evolves with signs of respiratory distress, cough, wheeze, and often bilateral crepitations. In North America, bronchiolitis is used to describe a wheezing illness associated with an URTI in children up to the age of 2 years.

Question

What do we learn about bronchiolitis in this article?

- Not everyone agrees about the exact description of this infection
- Bronchiolitis affects a disproportionately high number of children over the age of two
- Discrepancies in diagnosis are frequent

PART C

TEXT 1

According to Alzheimer's UK, 850,000 people in the UK have dementia, but by 2025 this number is expected to exceed 1 million. The condition has a profound effect on everyday life, both for the person with dementia and for their nearest and dearest. Many people think of dementia as solely being about forgetfulness. While memory loss is one of the main symptoms, dementia is a term describing a number of disorders that cause a loss of brain function, which usually progressively worsens over time. "In the beginning, people might exhibit a number of early signs, such as losing interest in day-to-day activities and becoming apathetic," explains Dr Jane Mullins, dementia nurse specialist and author of the book *Finding the Light in Dementia*. "On top of that, a person's mood might change; they might become easily upset or frustrated and seem to lose confidence."

Often, Mullins, says, it's these emotional and mood-related symptoms that people notice first. Alongside these, a person will commonly have language difficulties. For example, struggling to find the right words and begin forgetting recent events, names and faces. They may become more repetitive, and they may repeat a question or statement after a very short interval. The person may also misplace items or put them in odd places. "We all lose things now and again, but with dementia it's the regularity with which these things happen that is difficult to deal with," Mullins comments.

Commonly, people affected will have noticed themselves that something is happening that just isn't right. This was the case with 91-year-old Salaam Kaffash from North London, who was diagnosed with Alzheimer's disease in 2016. He had been worried about his declining memory for some years before he eventually visited his doctor. "He was aware that he was beginning to forget things, and he was quite bothered by it," explains Salaam's wife, Gill, who is 79. "I was more bothered by the change in his cognitive abilities - it was that slowing down of capacity to think and absorb information."

Mullins advises that, if you're worried someone close to you might have dementia, you should broach the topic very sensitively. "It's not an easy conversation to have, and how you go about it will depend on the relationship you have with the person, everybody has to gauge that for themselves," she says. "The first thing is to slow down, stand back and try to put yourself in their shoes and imagine if someone was approaching you about changes in your memory, your thinking or your mood. How would you feel about it? Empathy helps a lot."

It's important to choose a time when neither of you is tired. Mullins also recommends knowing what you're going to say beforehand, and to avoid the word 'dementia'. Instead, simply tell the person that you're worried because you've noticed changes in their memory or mood, for example, and ask if they have noticed too. "Crucially, make sure you're prepared for any reaction," Mullins comments. "The person might be relieved that you've brought it up, but if they're not ready to face up to what's happening, they might be quite defensive." She adds that you may need to have the conversation several times.

In the first instance the person should visit their doctor, who will try to establish the degree of brain function decline, and importantly, rule out other causes of the symptoms the person is experiencing. These might include depression, vitamin B12 deficiency, thyroid disease, or unstable blood glucose in people with diabetes. If the doctor suspects dementia, they will make a referral to a specialist memory clinic for more extensive tests.

Getting confirmation of dementia can be a very difficult blow for everyone, but for Gill, her husband's diagnosis came as a relief. "It meant we could both get help," she comments. "Salaam's illness is developing all the time, but it helps to have support. We still do things together, like yoga and visiting family, and I also go off and do things on my own." Gill goes to an aqua fitness class every week and takes part in a history club at the local community centre. She says it's also important that she gets plenty of sleep and keeps as healthy as possible.

Mullins points out that remaining as active as possible is key, both for the person with dementia and for their caregiver. Having some respite is equally important. "Try to include friends and family so the person with dementia doesn't only rely on that one person," she suggests. "Getting involved with community initiatives can also be a good way to get out and about, and to enjoy activities both together and separately."

"In my experience working with people with dementia and their families, the people who cope better are those who are more socially active," Mullins says.

Questions 7-14

7) What do we learn about dementia in the first paragraph?

- a) Dementia is a disorder characterized solely by memory loss
- b) Apathetic people are more susceptible to dementia
- c) Dementia can affect how a person feels

d) A symptom of dementia is increased sensitivity to light

8) The word 'struggle' in the second paragraph suggests

- a) A person with dementia may have physical as well as emotional symptoms
- b) Some mental activities become very difficult for people with dementia
- c) Learning a new language may well be impossible for a person with dementia
- d) People with dementia are unable to repeat questions satisfactorily

9) The example of Salaam Kaffash in the third paragraph is given to show

- a) that dementia affects all ethnic groups
- b) that forgetfulness is often accompanied by other symptoms of intellectual decline
- c) that physical slowness can be overlooked as an early sign of dementia
- d) that usually the person with dementia is unaware of any symptoms themselves

10) What advice is given in the fourth paragraph?

- a) you should gauge your relationship with the person first
- b) speaking slowly will enable the other person to understand more readily
- c) only speak to a person with dementia when they are in a good mood
- d) it is important to show compassion when discussing dementia

11) In the fifth paragraph, what may cause a person with dementia to become defensive?

- a) If the person sees a look of pity on your face
- b) The person needs more time before they can accept the new situation
- c) The person doesn't feel as relieved as he should
- d) It's possible that the person is unable to understand because of the effects of dementia

12) What does the sixth paragraph tell us about the medical assessment of dementia?

- a) It may involve more than one assessment
- b) Dementia assessments frequently include blood tests
- c) Vitamin B12 deficiency can be ruled out very quickly
- d) Assessments include an evaluation of body temperature

13) What was one result of Salaam's diagnosis?

- a) Gill could do yoga and aqua fitness
- b) Salaam and his wife Gill were able to obtain assistance
- c) Gill could get medicine for her husband
- d) After many years, Salaam was able to see his family

14) The word 'respite' in the eighth paragraph suggests that

- a) from time to time, people who care for dementia patients need a break
- b) it may be necessary to motivate dementia patients to do more exercise
- c) it is important not to rely solely on medication
- d) there may be specialist facilities available in the community for people who have dementia

TEXT 2

Frostbite, the most common type of freezing injury, is defined as the freezing and crystalizing of fluids in the interstitial and cellular spaces as a consequence of prolonged exposure to freezing temperatures. This article deals with the clinical presentation and treatment of frostbite as a distinct entity. Frostbite may occur when skin is exposed to extreme cold, at times combined with high winds, resulting in vasoconstriction. The associated decrease in blood flow does not deliver sufficient heat to the tissue to prevent the formation of ice crystals.

Because frostbite tends to occur in the same setting as hypothermia, most cases are observed in the winter. Homeless individuals, those who work outdoors, winter sport enthusiasts, and mountaineers are examples of those at risk. The prevalent use of alcohol in colder climates is also a factor. High-altitude mountaineering frostbite, a variant of frostbite that combines tissue freezing with hypoxia and general body dehydration, has a worse prognosis.

Until the late 1950s, frostbite was a disease entity primarily reported by the world's military, which had the most experience in its diagnosis and treatment. Most of the data in the current literature originated from military studies or from Scandinavian countries. However, civilian physicians are becoming more cognizant of the diagnosis and treatment of this condition in urban and rural civilian populations.

The goal of frostbite treatment is to salvage as much tissue as possible, to achieve maximal return of function, and to prevent complications. This may involve both medical and surgical measures as appropriate. The cutaneous circulation plays a major role in maintaining thermal homeostasis. The skin loses heat more easily than it gains heat. Thus, humans acclimatize better to heat than to cold. Cutaneous vasodilation is controlled by direct local effects and decrease of sympathetic vascular tone. Maximum reflex vasodilation occurs when the sympathetic system is blocked. The fingers, toes, ears, and nose—the skin structures most at risk for frostbite—contain multiple arteriovenous anastomoses that allow shunting of blood in order to preserve core temperature at the expense of peripheral tissue circulation. Heat conduction and radiation from deeper tissue circulation prevents freezing and ice crystallization until the skin temperature drops below 0°C. Once tissue temperature drops below 0°C, cutaneous sensation is lost and the frostbite injury cascade is initiated.

Various authors have compared the effects of quick freezing and slow freezing at the microscopic level. Rapid freezing is thought to increase intracellular ice formation superficially, whereas slow freezing causes deeper and more extensive cellular injury by causing freezing of water in the intracellular and extracellular spaces. Because extracellular freezing progresses more rapidly than intracellular freezing, osmotic shifts occur. These shifts cause intracellular dehydration, which decreases the viability and survival of individual cells. As tissue is rewarmed, reperfusion injury becomes prominent. Progressive oedema of the frostbitten area develops over the first 48-72 hours, followed by bleb formation and necrosis of devitalized tissue. Blood flow in the microcirculation resumed at near-normal levels after rewarming, suggesting that the vascular structures were not damaged by freezing.

Generally, recovery is expected and occurs in about 10 days. When external warmth is applied, ischemic insult may occur because perfusion from deep blood vessels tends to return slowly relative to the accelerated tissue oxygen demand. Rapid rewarming is favoured over slow rewarming because it minimizes this discrepancy. Prolonged exposure to cold, refreezing of partially thawed tissue, and slow rewarming predispose the tissue to greater ischemic insult, resulting in greater tissue loss. Frostbite severity and resultant tissue injury are a function of two factors. Both

the absolute temperature and the duration of cold exposure play a role. With regard to these factors, data suggest that the duration of exposure has the greater impact on the level of injury and the amount of tissue damage; however, short-term exposure to extreme cold may produce the same overall injury pattern as excessively prolonged exposure to lesser degrees of cold.

The most commonly affected group includes adult males aged 30-49 years, although all age groups are at risk. In one case series, the mean patient age was 41 years. Younger children have less adaptive behavioural reaction to cold stress; therefore, they have a greater risk of frostbite. Recent US military data indicate a decreasing rate of cold-related injuries in general with increasing age. However, this data set did not specifically address an association of age with frostbite.

Patients should be informed that the frostbitten area may be more sensitive to cold, with associated burning and tingling. Individuals who have sustained a cold-related injury are at a 2- to 4-fold greater risk of developing a subsequent cold-related injury. Therefore, patients with frostbite should be counselled about their increased susceptibility to frostbite injury and about appropriate strategies to avoid it. They should also be given general advice on preparing for cold weather exposure.

Questions 15-22

15) According to the first paragraph, when does frostbite occur?

- a) after skin has had protracted exposure to sub-zero temperatures
- b) only when very low temperatures are combined with high winds
- c) when fluid crystallization precedes vasoconstriction
- d) after an increase in blood flow

16) What do we learn about frostbite in the second paragraph?

- a) contact between alcohol and unprotected skin can cause frostbite
- b) mountaineers have the worst prognosis
- c) patients with frostbite sometimes have other winter-related conditions
- d) some frostbite variants present before hypoxia occurs

17) Before the 1950s

- a) frostbite was unknown outside of Scandinavian countries

- b) expertise in dealing with frostbite was present in the world's armed forces
- c) urban and rural civilian populations became cognizant of the dangers of frostbite
- d) only the Scandinavian army knew how to treat frostbite

18) According to the fourth paragraph, which of the following statements is not true?

- a) it is harder for skin to increase in temperature than it is to decrease
- b) some parts of the human body have a higher risk of frostbite than other parts
- c) up to a certain point, skin is protected by warmth from other parts of the body
- d) ice cascade injuries can prolong loss of cutaneous sensation

19) What do we learn about frostbite in the fifth paragraph?

- a) the rapidity of freezing is thought to be a factor in frostbite recovery
- b) the seriousness of cellular injury depends on the presence of water in the intracellular and extracellular spaces
- c) it is not possible to rewarm damaged tissue
- d) microcirculation can be interrupted by bleb formation

20) What advice is given in the sixth paragraph about treatment?

- a) accelerated oxygen demand should be avoided
- b) duration of exposure to cold is not a relevant factor in subsequent outcomes
- c) warming affected areas quickly is better than doing it slowly
- d) tissue that had short-term exposures should be rewarmed more slowly

21) The figure of 41 is given to demonstrate

- a) the maximum age at which full frostbite recovery has been recorded
- b) the average age of men who had frostbite, according to a series of studies
- c) the world record for the number of individual frostbite injuries that one person has received

- d) the length of time (in years) that the US military has calculated statistics for frostbite related injuries

22) What do we learn about frostbite in the last paragraph?

- a) people who have had frostbite may get more colds in the future
- b) frostbite can recur between 2 and 4 times
- c) there could be some unpleasant after-effects of frostbite
- d) patients should be given advice about moving to a warmer climate

Reading Practice Test 16 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	☒ B	☐
Q 2	☒ A	☐
Q 3	☒ A	☐
Q 4	☒ C	☐
Q 5	☒ B	☐
Q 6	☒ A	☐

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	☒ C	☐	Q 15	☒ A	☐
Q 8	☒ B	☐	Q 16	☒ C	☐
Q 9	☒ B	☐	Q 17	☒ B	☐
Q 10	☒ D	☐	Q 18	☒ D	☐
Q 11	☒ B	☐	Q 19	☒ A	☐
Q 12	☒ A	☐	Q 20	☒ C	☐
Q 13	☒ B	☐	Q 21	☒ B	☐
Q 14	☒ A	☐	Q 22	☒ A	☐

Review & Explanations

TEST-17

Varicose Veins

PART A

TEXT 1

Varicose veins are dilated superficial veins in the lower extremities. They may be blue or dark purple, and are often lumpy, bulging or twisted in appearance. Varicose veins are typically asymptomatic but may cause a sense of fullness, pressure, and pain or hyperesthesia in the legs.

Healthy leg veins have one-way valves to help blood flow back up to the heart but varicose veins may result from primary venous valve insufficiency with reflux or from primary dilation of the vein wall due to structural weakness.

Blood that collects in varicose veins can leak into smaller blood vessels (capillaries), which enlarge and form 'thread veins' or 'spider veins'. These veins are different to varicose veins because they are situated much closer or within the overlying skin. Although they may be unsightly, they are not the same as varicose veins and can be more difficult to treat.

In some people, varicose veins result from chronic venous insufficiency and venous hypertension. Most people have no obvious risk factors. Varicose veins are common within families, suggesting a genetic component. Varicose veins are more common among people who stand a lot, have limited mobility or who are overweight. Menopause and pregnancy affects the oestrogen levels which in turn affects venous structure. Pregnancy increases pelvic and leg venous pressures, or both. People who have had a deep veined thrombosis or suffered trauma to a leg are at increased risk of developing varicose veins.

TEXT 2

Diagnosing varicose veins

Varicose veins may initially be tense and palpable but not necessarily visible. Later, they may progressively enlarge, protrude, and become obvious. They can cause a sense of fullness, fatigue, pressure, and superficial pain or hyperesthesia in the legs. Varicose veins are most visible when the patient stands. Cramping or restless legs, itchiness and swollen ankles are common signs of varicose veins. The symptoms are usually worse during warm weather or after standing up for long periods of time. They may improve with walking or resting with raised legs.

Less common effects of varicose veins include leg ulcers and clotting. For unclear reasons, stasis dermatitis and venous stasis ulcers are uncommon. When skin changes e.g. induration, pigmentation, eczema occur, they typically affect the medial

malleolar region. Ulcers may develop after minimal trauma to an affected area; they are usually small, superficial, and painful.

Varicose veins occasionally thrombose, causing pain. Superficial varicose veins may cause thin venous bullae in the skin, which may rupture and bleed after minimal trauma. Very rarely, such bleeding, if undetected during sleep, is fatal

The Trendelenburg test which compares venous filling before and after release of a thigh tourniquet is no longer commonly used to identify retrograde blood flow past incompetent saphenous valves.

Duplex ultrasonography is an accurate test that uses high-frequency sound waves to produce a picture of the veins in the legs. The picture shows the blood flow and helps locate any damaged valves that might be causing the varicose veins.

TEXT 3

Managing varicose veins

Treatment of varicose veins is only necessary:

- To ease symptoms if they are causing pain or discomfort
 - To treat complications such as leg ulcers, swelling or skin discolouration
 - For cosmetic reasons so usually only available through the private health sector
- 6 months of self-care involving using compression stockings; exercising regularly; avoiding standing up for long periods and elevating the affected area when resting. People with varicose veins that are causing significant symptoms should be referred to a vascular surgeon for further treatment. There are several options available.
- Sclerotherapy which involves injecting special foam into affected veins. The foam scars the veins, which seals them closed.
 - Endovenous laser treatment involves a laser delivering short bursts of energy that heat up the vein and seal it closed. The laser is slowly pulled along the vein using the ultrasound scan to guide it, allowing the entire length of the vein to be closed.
 - Radiofrequency ablation involves heating the wall of the varicose vein using radiofrequency energy. A probe is inserted into the catheter that sends out radiofrequency energy that heats the vein until its walls collapse, closing it and sealing it shut. Once the vein has been sealed shut, blood will naturally be redirected to a healthy vein.

Ligation and stripping is surgery that involves tying off the vein in the affected leg and then removing it. Two small incisions are made. The first is made at the top of the varicose vein and is approximately 5cm in diameter. The second, smaller cut is made further down the leg. The top of the vein is tied up and sealed. A thin, flexible wire is passed through the bottom of the vein and then carefully pulled out and removed through the lower cut in the leg.

TEXT 4

Potential complications from varicose veins

Most people who have varicose veins won't develop complications. If they do, it's usually several years after varicose veins first appear. Some possible complications of varicose veins are:

- Bleeding from the varicose veins near the surface of your skin which may be difficult to stop.
- Blood clots can form in veins located just under the surface of your skin leading to conditions such as:
 - thrombophlebitis, a swelling of the veins in the leg
 - deep vein thrombosis which can cause pain and swelling in the leg, and may lead to serious complications like pulmonary embolism
- Chronic venous insufficiency when the blood in the veins doesn't flow properly, interfering with the way skin exchanges oxygen, nutrients and waste products with blood. It can sometimes cause other conditions to develop, including:
 - varicose eczema that causes skin to become red, scaly and flaky
 - lipodermatosclerosis which causes skin, usually around the calf area, to become hardened and tight, and you may find it turns a red or brown colour
 - venous leg ulcers which develop when there's increased pressure in the veins of the lower leg, which may eventually cause an ulcer

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1 The times when varicose vein are usually more prominent?

2 What happens if blood seeps into the little blood vessels? _____

3 How people can manage varicose veins themselves? _____

4 The changes that can happen to skin colour around the mid lower leg?

5 What varicose veins look like? _____

6 What used to be done to diagnose varicose veins? _____

7 The surgical procedure that might be performed to manage varicose veins?

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8 What is inflammation of the veins in the leg called? _____

9 What is there in the veins that helps blood flow back up to the heart?

10 What treatment involves pushing foam through varicose veins?

11 What condition restricts the way skin moves vital nutrients and waste products to and from blood? _____

12 What conditions change both the hormone levels and the structure of the veins, which can lead to varicose veins? _____

13 How long does it usually take for complications of varicose veins to develop?

14 Approximately, how big is the cut made at the top of a varicose vein, during the surgical procedure? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 If a patient is having endovenous laser treatment, a laser pushes out _____ of energy.

16 Varicose veins just under the skin may cause _____.

17 Varicose veins do not usually cause any problems but may lead to a _____, heaviness, over sensitivity or pain in the legs.

18 To ease symptoms of varicose veins a person should avoid being on their feet for long periods and should _____ the leg when relaxing.

19 Some people have varicose veins because they have chronic venous insufficiency and _____.

20 _____ makes the skin red, scaly and flaky.

Reading Practice - PART A

Answer Keys

Correct Answer		Your Answer	
Q 1	B		-
Q 2	A		-
Q 3	C		-
Q 4	D		-
Q 5	A		-
Q 6	B		-
Q 7	C		-

Correct Answer		Your Answer	
Q 8	thrombophlebiti +		
Q 9	one-way valves		
Q 10	Sclerotherapy		
Q 11	Chronic venous +		
Q 12	Menopause, preg +		
Q 13	Several years		
Q 14	5cm		

Correct Answer		Your Answer	
Q 15	short bursts		
Q 16	thin venous bul +		
Q 17	sense of fullne +		
Q 18	elevate		
Q 19	venous hyperten +		
Q 20	Varicose eczema		

Review & Explanations

PART B

TEXT 1

Urine Dipstick Analysis

There have been many studies evaluating the accuracy of dipsticks tests. These are mostly in relation to their role detecting bacteriuria. A meta-analysis of 26 studies in children, showed wide differences in diagnostic accuracy across studies. This could not be fully explained by differences in age, or by differences in the definition of the criterion standard. The lack of an adequate explanation for the heterogeneity of the dipstick accuracy stimulates an ongoing debate. Overall, the sensitivity of the urine dipstick test for nitrites has been found to be low (45-60% in most situations) with higher levels of specificity (85-98%). The test for nitrites has its highest accuracy in specific populations such as pregnant women, urology patients and elderly people. The test for nitrites may perform better in asymptomatic patients and in patients who are not on antibiotics.

Question

- 1) This article informs the reader that
 - a) there is no clear explanation for the variation in accuracy rates
 - b) use of antibiotics can affect the accuracy of dipstick tests
 - c) dipstick test accuracy was higher for child populations

TEXT 2

Most people want to die at home, surrounded by loved ones and free of pain. However, many of these people do not have advanced directives, nor have they spoken about their wishes to their family. End of life care begins with advanced planning, so it is essential that family members discuss what is important to them before they need end of life care. Putting these wishes in writing can help the family during this most difficult time.

It is important to the healthcare provider to know and recognize when the services that are needed by their patients. Identifying the various symptomology that might be present at end of life will ensure the patient receives the symptom relief he or she needs.

Question

- 2) What would be a suitable title for the following article?
- a) The importance of advanced directives
 - b) End of life care: ensuring needs are met
 - c) How to help your loved one during a most difficult time

TEXT 3

Abnormal Liver Function Tests

Most tests measure hepatocellular damage rather than function, so they are rather misnamed. True liver function tests (LFTs) are those that measure synthesis of proteins made by the liver or the liver's capacity to metabolise drugs. LFTs are not specific to specific systems or disease processes, yet abnormalities may indicate significant or serious disease. Interpreting abnormal LFTs and trying to diagnose any underlying liver disease is a common scenario in Primary Care. Single abnormalities in LFTs are difficult to localise and diagnose. However, the pattern of abnormalities tests helps determine origin of the issue. This usually means dividing the clinical picture into non-hepatic, hepatocellular and cholestatic patterns of abnormality. When this is then combined with a clinical history, medication and drug history and the presence of any current or recent symptoms, it is usually possible to develop a differential diagnosis.

Question

- 3) How can healthcare providers use LFTs?
- a) Along with clinical history and other relevant data, LFT results help to form a diagnosis
 - b) They can determine if hepatocellular damage has occurred
 - c) True LFTs can isolate specific disease processes

TEXT 4

Tissue & Tissue Products

Human cells or tissue intended for implantation, transplantation, infusion, or transfer into a human recipient is regulated as a human cell, tissue, and cellular and tissue-based product. Examples of such tissues are bone, skin, corneas, heart valves, oocytes and semen. CBER does not regulate the transplantation of vascularized human organ transplants such as kidney, liver, heart, lung or pancreas. The Health Resources Services Administration (HRSA) oversees the transplantation of vascularized human organs. Part 1271 requires tissue establishments to screen and test donors, to prepare and follow written procedures for the prevention of the spread of communicable disease, and to maintain records. FDA has published three final rules to broaden the scope of products subject to regulation and to include more comprehensive requirements to prevent the introduction, transmission and spread of communicable disease.

Question

- 4) What is one reason given for the following regulations?
- a) to allow transplantation of vascularized human organs
 - b) to limit the spread of infectious diseases
 - c) to allow the HRSA and the FDA to manage transplants and implants effectively

TEXT 5

WHO Launches New Global Influenza Strategy

The World Health Authority (WHO) today released a Global Influenza Strategy for 2019-2030 aimed at protecting people in all countries from the threat of influenza. The goal of the strategy is to prevent seasonal influenza and prepare for the next influenza pandemic. "The threat of pandemic influenza is ever-present," said WHO Director-General Dr Ghebreyesus. "The on-going risk of a new influenza virus transmitting from animals to humans and potentially causing a pandemic is real. The question is not if we will have another pandemic, but when. We must be vigilant and prepared. The cost of a major influenza outbreak will far outweigh the price of prevention." Every year across the globe, there are an estimated 1 billion cases which result in 290 000 to 650 000 influenza-related respiratory deaths.

Question

- 5) According to this article, which of the following statements is not true?
- a) The WHO strategy is aimed at protecting humans and animals, such as livestock
 - b) The WHO strategy is twofold, prevention and preparation
 - c) It is highly likely that another global pandemic will occur

TEXT 6

What Is Medication-Induced Headache?

Medication-induced headache is caused by taking painkillers too often for headaches of any kind. For example, you may have a series of tension headaches or migraines, perhaps during a time of stress. You take painkillers more often than usual and your body becomes used to the medication. A withdrawal (rebound) headache then develops if you do not take a painkiller within a day or so of the last dose. You think this is another tension headache or migraine, and so you take a further dose of painkiller. When the effect of each dose wears off, a further withdrawal headache develops, and so on. In time, you may have headaches on most days, and you end up taking more and more painkillers.

Question

- 6) According to this article, what is the cause of medication-induced headache?
- a) When you don't take medication
 - b) Taking the wrong type of medication
 - c) Stress and tension are major factors

PART C

TEXT 1

Most snakebites are innocuous and are delivered by non-poisonous species. Worldwide, only about 15% of the more than 3000 species of snakes are considered dangerous to humans. The family Viperidae is the largest family of venomous snakes, and members can be found in Africa, Europe, Asia, and the Americas. The family Elapidae is the next largest family of venomous snakes.

Venom dosage per bite depends on the elapsed time since the last bite, the degree of threat perceived by the snake, and size of the prey. Nostril pits respond to the heat emission of the prey, which may enable the snake to vary the amount of venom delivered. Coral snakes have shorter fangs and a smaller mouth. This allows them less opportunity for envenomation than the crotalids, and their bites more closely resemble chewing rather than the strike for which the pit vipers are famous. Both methods inject venom into the victim to immobilize it quickly and begin digestion.

Evidence suggests that the differences between the venom components of different snake species resulted from a diet directed evolution occurring over time. Venom is mostly water. Enzymatic proteins in venom impart its destructive properties. Proteases, collagenase, and arginine ester hydrolase have been identified in pit viper venom. Neurotoxins comprise around 40% of coral snake venom. Enzyme concentrations vary among species, thereby causing dissimilar envenomations. Copperhead bites generally are limited to local tissue destruction. Rattlesnakes can leave impressive wounds and cause systemic toxicity. Coral snakes may leave a small wound that later results in respiratory failure from systemic neuromuscular blockade.

The local effects of venom serve as a reminder of the potential for systemic disruption of organ system function. One effect is local bleeding; coagulopathy is not uncommon with severe envenomation. Another effect, local oedema, increases capillary leak and interstitial fluid in the lungs. Pulmonary mechanics may be altered significantly. The final effect, local cell death, increases lactic acid concentration secondary to changes in volume status and requires increased minute ventilation. The effects of neuromuscular blockade result in poor diaphragmatic excursion. Cardiac failure can result from hypotension and acidosis. Myonecrosis raises concerns about myoglobinuria and renal damage.

In the United States, more than 40% of victims put themselves in danger by either handling pets or attempting to capture reptiles in the wild. The popularity of keeping exotic species has increased the number of envenomations by non-native species. UTMCK data support this by reporting that 12 of 25 patients were bitten handling snakes; 2 of these were involved in religious ceremonies.

Generally, only localized reporting of international data is available. Most snakebites and deaths due to snakebites are not reported, especially in the developing world. An estimated 1.8-2.5 million venomous snakebites occur worldwide each year, resulting in an estimated 100,000 to 125,000 annual deaths, but this may be underreported. Worldwide, snakebites disproportionately affect low socioeconomic populations in more rural locations. They are often seen as bites to the lower extremities by farmers or workers who step on or disturb a snake in the field or rice paddies, or they can present as a bite to the head or trunk in individuals sleeping outside on the ground.

In the USA, national studies report 50% of patients were aged 18-28 years. 95% of bites were located on an extremity, especially the hand. National studies report a seasonal occurrence of 90% from April to October. In the paediatric population, most snakebites occurred in school-aged children and adolescents around the perimeter of the home during the afternoon in summer months. The most frequent wound sites were the lower limbs.

Full recovery is the rule, though local complications from envenomation may occur. Death occurs in less than 1 bite in 5000. A review of morbidity associated with snakebites from Kentucky, USA was published. Most bites were from copperheads and resulted in 8 days of pain, 11 days of extremity oedema, and 14 days of missed work. A review specifically of copperhead bites in West Virginia described similar outcomes and noted that the peak effects of envenomation were not present until longer than 4 hours after the bite.

Local tissue destruction rarely contributes to long-term morbidity. Occasionally, skin grafting is required to close a defect from fasciotomy, but wounds requiring fasciotomy to reduce compartment pressures from muscle oedema are infrequent. A new web site (Australian Venom Research Unit) based at the University of Melbourne in Australia comprehensively outlines the species, first aid, and treatment of all venomous creatures indigenous to the region. The web site is easily navigated and sectionally divided for the practitioner, interested epidemiologists, snake fanciers, and children of Australia and the Asia/Pacific region.

Questions 7-14

7) Around the world, how many snakes are thought to be dangerous to humans?

- a) About 3000 species of snakes altogether
- b) Less than one in five species
- c) 15% of all snakes
- d) 15% of the family Viperidae

8) In the second paragraph, which of the following is not mentioned?

- a) Various factors determine the venom dosage per bite
- b) Different species of snake have different bite characteristics
- c) Pit vipers deliver a higher amount of venom per bite than coral snakes
- d) One effect of venom is to stop the victim moving

9) What do we learn about venom from the third paragraph?

- a) There is significantly more water in venom than other constituents.
- b) Venom is another term for enzymatic destruction
- c) Pit viper venom is the most destructive, compared to other species
- d) Rattlesnakes are more impressive than copperhead or coral snakes

10) The fourth paragraph informs the reader that

- a) snakebites can have fatal consequences
- b) not all snakebites result in bleeding
- c) lactic acid increases can lead to coagulopathic leakage
- d) neuromuscular blockage can lead to loss of balance

11) The UTMCK data is quoted in order to highlight

- a) Most snake bite victims were among people who hunt snakes in the wild
- b) Twice as many victims in the USA were bitten during religious ceremonies
- c) Non-native species are more dangerous than species native to the USA
- d) More than half of all victims did not deliberately seek contact with snakes

12) According to international data, which group is at the highest risk of snake bites?

- a) Poor people who live outside of cities
- b) Children who are bitten during the summer months
- c) People who keep venomous snakes as pets
- d) All populations are equally vulnerable

13) What do the statistics mentioned in the seventh paragraph point to?

- a) 95% of patients are aged between 18-25 years
- b) In the US, there are seasonal variations in the number of cases
- c) 95% of snakebites are extremely serious
- d) Children are safer at home than outside the home perimeter

14) The study carried out in West Virginia is quoted to highlight

- a) There are many similarities of morbidity among snake species
- b) Most symptoms of copperhead envenomation do not occur straightaway
- c) Death rates from snakebites are very low
- d) Some complications of snakebites are difficult to manage

TEXT 2

The first months of the year can be a miserable time. Holiday festivities are over, money is tight and pressure to keep up your resolutions is mounting. This can lead to feelings of anxiety, depression and even failure. For many, these symptoms begin even earlier as the seasons change from autumn to winter. According to the Royal College of Psychiatrists, around every 3 in 100 people suffer from significant seasonal depression, which can interfere with daily life. We explore why winter leaves so many feeling down and how you can tackle it at home.

Symptoms of seasonal affective disorder (SAD) typically develop between September and November and continue until early spring, often reaching their peak during December, January and February. To determine whether you have SAD or non-

seasonal depression, it can be useful to record your symptoms over time. If you show none of the common signs of depression during the summer months for two years or more, it is likely that you have SAD.

Dr Mark Winwood, director of Psychological Services for AXA PPP Healthcare, explains: "In non-seasonal depression, people commonly sleep less and eat less. Whereas with SAD, they usually sleep more and eat more as if they are in hibernation." Dr Preethi Daniel, clinical director at the London Doctors Clinic, lists other common symptoms of SAD as persistent low mood, lack of energy, difficulty concentrating or achieving deadlines, lack of interest or pleasure in usual activities and feelings of worthlessness compared to others who are coping fine.

It is thought that the lack of sunlight, shorter days and cold temperatures bring on the so-called 'winter blues' in several ways. Levels of melatonin (the hormone that controls your body's sleep-wake cycle) and the brain chemical serotonin (which plays a major role in mood) are affected; we are unable to produce enough vitamin D (the 'sunshine vitamin') naturally and our sleep cycle is triggered early by the dark evenings. There is also a theory that SAD can be triggered by traumatic or upsetting past events that occurred around the same time, although there are not many studies investigating this further.

If you are experiencing a persistent low mood and that does not subside when it reaches the summer months, you could be suffering from depression. Make an appointment with your doctor to talk things through and figure out the best course of treatment.

Your circadian rhythm, also known as the circadian clock or sleep/wake cycle, is a "24-hour internal clock running in the background of your brain", telling you when it's time to wake up and go to sleep, according to the National Sleep Foundation. "Natural daylight supports our circadian clock and helps us sleep at night," reveals Winwood. So, if you're going to work and coming home when it's dark, you're more likely to feel depressed from lack of sunlight and a disrupted sleep pattern.

A way to combat sluggish mornings and low mood is light therapy. This involves using a lamp or light box that is designed to replicate sunlight. "By mimicking natural sunlight, your body is tricked into feeling good and energetic," says Dr Daniel. One study found light therapy to cause a significant reduction in depression symptoms when treating SAD and had similar effectiveness to antidepressants. For light therapy to be an effective treatment for SAD, you are advised to use your light for around 30 minutes a day, preferably in the morning, and will usually feel the effects after 3-4

days. It's important to use the light continually throughout the winter for it to be beneficial. My own experiences with SAD light therapy have been very positive and it has become a key part of my morning routine while I get ready. I use a lamp manufactured by a company called 'Lumie'. Although light therapy can be an effective treatment, the cost of a lamp can be a barrier for many.

However, there are other things you can try without the need to splash out. It's well-known that sunshine provides our bodies with vitamin D, as much as 90% of it! "On cloudy days, we make less of it and deplete our stores quickly. This not only affects our mood, but it also causes general aches and pains," notes Dr Daniel. Taking a daily vitamin D supplement during the winter can also aid good bone health and combat tiredness. Low levels have also been linked to an increased risk of heart disease and even multiple sclerosis. As well as taking supplements, ask to sit by a window in the office if possible, and open the curtains at home even on cloudy days. Daniel also suggests making time for a daily walk in your lunch hour.

It may sound obvious, but having a routine keeps you focused. It can be hard to get up in the mornings when it's cold, so perhaps time the heating to come on for when you need to get up or lay your clothes out the night before. Winter often leaves us wanting to sleep most of the time, except when it comes to bedtime. A good night's sleep is essential for optimal mental and physical well-being, insists Winwood.

Questions 15-22

15) What does the first paragraph say about seasonal depression?

- a) Some people are affected by holiday festivities
- b) The majority of people are unaffected by this problem
- c) Around 13% of people have this type of depression
- d) 3 in 100 people report significantly worse symptoms during winter months

16) According to the second paragraph, how can a person make a self-assessment?

- a) a person does not notice the usual indications of depression in the summer
- b) it is important to keep a record of sleep patterns
- c) a doctor can evaluate symptoms such as mood and/or feelings
- d) a person may notice changes in appetite

17)What does the third paragraph say about the causes of SAD?

- a) it seems there are a number of different contributing factors
- b) an imbalance of melatonin and serotonin occurs
- c) too much sleep can affect the levels of vitamin D
- d) psychological factors are thought to outweigh physiological factors

18)In the fifth paragraph, what is thought to be conducive to healthy sleep?

- a) reducing the amount of light when it is time to sleep
- b) a well-adjusted sleep/wake cycle
- c) making sure that the circadian rhythm is running in the background of your brain
- d) getting enough daylight

19)According to Dr Daniel, why are special light boxes effective?

- a) they imitate ordinary light
- b) they can boost circadian rhythms
- c) they are an inexpensive alternative to antidepressants
- d) they can amplify natural sunlight

20)What does the reader learn about vitamin D from Dr Daniel?

- a) it is not necessary to splash vitamin D out
- b) overuse of vitamin D can lead to general aches and pains
- c) the body does not retain vitamin D for extended periods
- d) Vitamin D does not need to be supplemented if you sit near a window

21)What does the reader learn about sleep in the eighth paragraph?

- a) it is not beneficial to lay on your clothes
- b) optimal sleep is more difficult to achieve during the winter
- c) the temperature has little or no effect on sleep patterns

- d) in the winter, a person's desire for sleep can diminish at certain times of the day

22) What would be a suitable title for this text?

- a) New Treatments to Boost Circadian Rhythms
- b) Seasonal Affective Disorder – Why Your Sleep is to Blame
- c) How to Look After Your Mental Health This Winter
- d) 'Lumie' – A New Cure for SAD

Reading Practice Test 17 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	A	-
Q 2	B	-
Q 3	A	-
Q 4	B	-
Q 5	A	-
Q 6	A	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	B	-	Q 15	B	-
Q 8	C	-	Q 16	A	-
Q 9	A	-	Q 17	A	-
Q 10	A	-	Q 18	D	-
Q 11	D	-	Q 19	A	-
Q 12	A	-	Q 20	C	-
Q 13	B	-	Q 21	D	-
Q 14	B	-	Q 22	C	-

Review & Explanations

TEST-18

Appendicitis

PART A

TEXT 1

Appendicitis is acute inflammation of the appendix, the thin pouch attached to the large intestine on the right side of the abdomen. It is usually about the size of a finger.

The exact cause of appendicitis is not known. Some people think the appendix becomes obstructed during a bout of appendicitis. Others believe it is an obstruction that causes appendicitis. Regardless, the result is an obstruction of the appendiceal lumen, possibly by faeces, foreign body, or even worms.

There are no medically proven ways to prevent appendicitis and there is no known diet to prevent appendicitis. Many people treated for acute appendicitis may have had previous episodes of appendicitis that they did not seek treatment for.

Appendicitis can occur at any age, but is most common in children and young adults. In 2013, Australia's rate of appendectomy was among the highest in the Organisation for Economic Co-operation and Development (OECD). Rates per 100,000 population were 194 in South Korea, 177 in Australia, 168 in Germany, 139 in New Zealand, 105 in Canada and 94 in the United Kingdom. Appendectomy was the most common emergency surgery performed in public hospitals in 2014–15. In 2014–15, approximately 30,000 appendectomies were performed in public or private hospitals as a result of an emergency admission.

TEXT 2

Symptoms and diagnosis of appendicitis

Appendicitis typically starts with a pain in the middle of the abdomen that may come and go. Within hours, the pain travels to the lower right-hand side, where the appendix is usually located, and becomes constant and severe.

Some people's appendix may be located in a slightly different part of their body, such as: the pelvis; behind the large intestine or around the small bowel. The pain may be

worsened by pressing around the area, coughing, or walking. Other symptoms include: nausea and/or vomiting; anorexia; diarrhoea; pyrexia or a flushed face.

Diagnosing appendicitis can be tricky because the typical symptoms are only present in about half of all cases. Some people develop pain similar to appendicitis, but it's caused by something else, such as:

- Gastroenteritis;
- Severe irritable bowel syndrome;
- Constipation,
- Ectopic pregnancy
- A urine infection

History taking and abdominal examination to see if the pain gets worse when pressure is applied to the appendix area are usually sufficient to diagnose appendicitis.

Further tests may involve: a blood test to look for signs of infection; a pregnancy test for women; a urine test to rule out other conditions, such as a bladder infection; an ultrasound scan to see if the appendix is swollen or a computerised tomography (CT) scan.

TEXT 3

Managing appendicitis

Medical advice should be sought for ongoing abdominal pain, and if the pain suddenly gets worse, emergency transfer to hospital is required.

If appendicitis is strongly suspected, the appendix is surgically removed as an emergency, without full investigation rather than run the risk of it bursting. This means some people will have their appendix removed even though it's eventually found to be normal. This is called a negative appendicectomy. Surgery may be laparoscopic or open.

An alternative to immediate surgery is the use of antibiotics to treat appendicitis. However, studies have looked into whether antibiotics could be an alternative to surgery. As yet there isn't enough clear evidence to suggest this is the case.

In some cases where a diagnosis is not certain and symptoms are not too severe, a doctor may recommend waiting up to 24 hours to see if symptoms improve, stay the same, or get worse.

Sometimes appendicitis can lead to the development of a lump on the appendix called an appendix mass. This lump, consisting of appendix and fatty tissue, is an attempt by the body to deal with the problem and heal itself. If an appendix mass is found during an examination, your doctors may decide it's not necessary to operate immediately. Instead, a course of antibiotics is given and an appendicectomy is performed a few weeks later, when the mass has settled.

Without surgery or antibiotics the mortality rate for appendicitis is 50%. With early surgery, the mortality rate is < 1%, and convalescence is normally rapid and complete. With complications such as rupture and development of an abscess or peritonitis and/or advanced age, the prognosis is worse: Repeat operations and a long convalescence may follow.

TEXT 4

Potential complications from appendicitis

The obstruction of the appendix can lead to distention, bacterial overgrowth, ischemia, and inflammation. If untreated, necrosis, gangrene, and perforation occur.

If the appendix perforates or bursts, it releases bacteria into other parts of the body. This can cause peritonitis if the infection spreads to the peritoneum, the thin layer of tissue that lines the inside of the abdomen. If peritonitis isn't treated immediately, it can cause long-term problems and may even be fatal.

Sometimes an abscess forms around a burst appendix. This is a painful collection of pus that occurs as a result of the body's attempt to fight the infection. It can also occur as a complication of surgery to remove the appendix in about 1 in 500 cases.

Abscesses can sometimes be treated using antibiotics, but in the vast majority of cases the pus needs to be drained from the abscess.

Wound infection can occur after surgery. The risk of this is less for people who have a laparoscopic appendicectomy.

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** Where the appendix is usually found? _____
- 2** The adverse situations a person may experience if they have appendicitis? _____
- 3** Appendicitis can be avoided? _____
- 4** An unnecessary appendicectomy? _____
- 5** The surgical approach that keeps infection risk low? _____
- 6** The way a person's body can try and manage appendicitis itself? _____
- 7** The prevalence of appendix removals in Australia? _____

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** The number of people who would die from appendicitis without modern treatments? _____
- 9** How big is a healthy appendix in most people? _____

10 Where does pain usually start if a person has appendicitis?

11 If an appendix ruptures, what condition could develop in the membrane lining of the tummy? _____

12 What drugs can be used instead of surgery if a person has appendicitis?

13 If an inflamed appendix is left alone it might burst and what other conditions might develop? _____

14 What is the appendix usually attached to? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 After a few hours of appendicitis developing, the _____ to the lower right-hand side of the abdomen.

16 An _____ can envelop a burst appendix

17 Some believe an appendix _____ in an episode of appendicitis.

18 A doctor examines a patient's tummy to find out if the _____ when they press around the appendix

19 An accumulation of _____ can develop as the body tries to get rid of any infection.

20 It is not always easy to confirm a person has appendicitis because the _____ only show up about 50% of the time.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	B	-	Q 8	>50%		Q 15	pain travels	
Q 2	D	-	Q 9	Size [of a] fin +		Q 16	abscess	
Q 3	A	-	Q 10	Middle [of the] +		Q 17	obstructed	
Q 4	C	-	Q 11	peritonitis		Q 18	pain gets worse +	
Q 5	D	-	Q 12	antibiotics		Q 19	pus	
Q 6	C	-	Q 13	necrosis, gangr +		Q 20	typical symptom +	
Q 7	A	-	Q 14	large intestine				

Review & Explanations

PART B

TEXT 1

Loneliness Is Harmful to Our Nation's Health

It has long been recognized that social support—through the availability of nutritious food, safe housing and job opportunities—positively influences mental and physical health. Studies have repeatedly shown that those with fewer social connections have the highest mortality rates, highlighting that social isolation can threaten health through lack of access to clinical care, social services or needed support. However, how the subjective sense of loneliness (experienced by many even while surrounded by others) is a threat to health, may be less intuitive. It is important to recognize that feelings of social cohesion, mutual trust and respect, within one's community and among different sections of society, are all crucial to well-being. Perhaps this is especially so at a time of great social polarization exacerbated by contentious politics and vitriolic TV news.

Question

- 1) What does the reader learn about loneliness in the following article?
 - a) a person's sentiments may be more important than objective factors
 - b) feelings of solitude are increasing in modern society
 - c) the government should provide more services in order to reduce social isolation

TEXT 2

Introduction to Recurrent Abdominal Pain

Recurrent abdominal pain (RAP) in children describes recurring abdominal pain without organic cause. It presents commonly in general practice and it causes a great deal of school absence and considerable anxiety. Most cases can be managed in primary care. Medication is not normally needed. The initial approach adopted by primary care doctors is crucial to successful management. It involves thorough history and examination skills, understanding and awareness of red flags which suggest organic pathology, and the knowledge and consulting style that offer a clear and empowering approach to patients, whilst avoiding unnecessary investigation.

RAP is believed to be a functional gut-brain interaction disorder caused by altered feedback mechanisms between the gut and central pain pathways.

Question

- 2) The writer uses the words 'red flags' to indicate
- a) an example of colour codes used in pathology diagnosis
 - b) a patient has a mental illness (an informal term used by healthcare workers)
 - c) symptoms which may point to a more serious medical condition

TEXT 3

Emotional Intelligence and Nursing

The concept of emotional intelligence (EI or EQ) emerged over 20 years ago and still applies today. Emotional intelligence is described as the ability to monitor or handle one's own emotions as well as the emotions of others. Emotional intelligence involves recognizing feelings, self-monitoring or awareness, how emotions impact relationships and how they can be managed. Studies have shown that there is a correlation between emotional intelligence and positive patient outcomes. This includes clinical outcomes, patient satisfaction and the ability to develop therapeutic relationships. Team performance and morale have also been found to be related to emotional intelligence, including positive conflict resolution rather than hostile environments or horizontal violence. Nursing retention, job satisfaction, and engagement have also been associated with emotional intelligence.

Question

- 3) Which of the following statements is not true?
- a) there is a link between emotional intelligence and lower rates of recruitment
 - b) although beneficial for nursing staff, emotional intelligence has little effect on patients
 - c) emotional intelligence is a relatively new idea

TEXT 4

Arsenic Trioxide Recommendations

Arsenic trioxide is recommended, within its marketing authorisation, as an option for inducing remission and consolidation in acute promyelocytic leukaemia in adults with untreated, low-to-intermediate risk disease and for patients with relapsed or refractory disease, after a retinoid and chemotherapy. People with untreated, low-to-intermediate risk acute promyelocytic leukaemia are given ATRA plus chemotherapy. Clinical trial evidence shows that arsenic trioxide plus ATRA is effective for untreated disease. Some assumptions in the model, such as the long-term effect of treatment, lead to the cost-effectiveness analyses being uncertain. Arsenic trioxide is already used to treat relapsed or refractory acute promyelocytic leukaemia. The clinical- and cost-effectiveness evidence for arsenic trioxide in relapsed or refractory disease is uncertain, because the clinical trial was small and did not compare arsenic trioxide with other treatments.

Question

- 4) What is inconclusive about the use of arsenic trioxide?
- a) if it represents good value for money
 - b) if it can be used effectively with intermediate-risk leukaemia patients
 - c) if arsenic trioxide can be used with treatments other than ATRA therapy

TEXT 5

Who Should Not Be Immunised?

Immunisations are generally very safe and effective. The main reasons for a person not to have a vaccine is if they have had a severe allergic reaction to a previous dose of that vaccine or to an ingredient in the vaccine that was also present in a different vaccine. People who have had very severe allergic reactions to egg should not have the yellow fever or flu vaccines other than under specialist care as there may be small amounts of egg protein in these vaccines. Certain vaccines are not usually given to women who are pregnant. They may not be suitable for people who are immunosuppressed. If you are unwell with a high temperature (fever), vaccination is usually put off until you are well again.

Question

- 5) Which of the following statements is not mentioned?
- a) some vaccine components can cause an allergic reaction
 - b) expectant mothers may need to postpone a vaccination
 - c) some vaccines are developed from the yellow part of eggs

TEXT 6

New Drug Class Available for Eczema

The new drug, Eucrisa, is a topical ointment that contains a phosphodiesterase 4 enzyme inhibitor that helps reduce symptoms of itchiness and inflammation caused by atopic dermatitis. Atopic dermatitis (AD), also known as eczema, is a skin condition experienced in 10-12% of children and 0.9% of adults in the United States. Diagnosis almost always occurs in infancy and childhood. Pruritus is considered the hallmark symptom of AD, as there is no objective test or biomarker that is used for diagnosis. Other symptoms include dry skin and erythema. The most common spots for lesions to occur are inside the elbows and knees, and on the hands and feet. It also can present on the skin around the eyes, eyelids, eyebrows and lashes.

Question

- 6) What do we learn about pruritus from the following article?
- a) pruritus is experienced by 0.9% of adults in the United States
 - b) pruritus has been superseded by the new treatment, Eucrisa
 - c) pruritus is the defining characteristic of atopic dermatitis

PART C

TEXT 1

When it comes to summer skincare, most of us feel pretty clued-up. But according to Cancer Research UK, rates of skin cancer are on the rise. Yet, 9 out of 10 cases could be prevented by staying safe in the sun. We look at sun protection mistakes you might be making.

With a variety of products available all promising to keep us safer in the sunshine, it's no surprise that many of us believe sunscreen offers the best protection during the hot weather. However, we need to combine the use of this product with other forms of sun protection. "One of the biggest mistakes people make is to rely on sunscreen alone as their sole protection," says Emma Shields, senior health information officer at Cancer Research UK. "However, it's best to use sunscreen in combination with time in the shade when the sun is strong, wearing a hat, covering up and wearing sunglasses."

Many of us associate a golden glow with good health, but when it comes to sun-tanning, appearances can be deceptive. "There's no such thing as a safe tan. In fact, any change in skin colour is a sign of damage." Shields claims. Consultant dermatologist Dr Daniel Glass of The Dermatology Clinic in London adds "Often, people associate sun-kissed skin with good health, but in fact, UV exposure will account for over 75% of skin ageing. In addition, the extra sun exposure may increase the risk of skin cancer later in life."

So, we get a little burnt, but if we slap on some after-sun lotion, that will repair the skin, right? Well, no. According to Shields, whilst after-sun lotion products "might help to soothe the skin, they don't undo the damage." However, Shields is quick to reassure that skin damage caused by mild sunburn can usually be dealt with by the body's own healing processes. "Your body does have its own repair mechanisms that can fix sun damage," she explains.

When we expose vulnerable areas such as the tops of our ears or our nose, it may be tempting to opt for a total block product. However, whilst such a product may look highly protective and usually offers an impressive level of protection, the name is a little misleading. "There is no such thing as a total block, as no cream can prevent all UV rays," explains Dr Stephanie Munn, dermatology clinical lead at Bupa UK. However, sunblock does provide a good level of protection, when used effectively. "Sunblock is a physical sunscreen such as titanium oxide or zinc oxide which blocks

out the UVB rays by acting as a physical barrier, as opposed to sunscreen which absorbs UVA. Sunblocks are less cosmetically acceptable as they create a chalky layer on the skin but are better tolerated on sensitive skin so are preferable to children," adds Munn.

With the price of sunscreen often on the high side, it can be tempting to dig out last year's bottle and use it up before restocking. But using an out-of-date or badly stored product could mean that your skin isn't fully protected. "You should discard any sunscreen after it has been open for a year," agrees Munn. "Some sunscreens include an expiration date too - so make sure you discard any that go past this." In addition, that bottle of sunscreen you've left in the garden, might not offer the protection it once did. "Leaving your sunscreen in the heat can cause it to break down faster, making it less reliable," explains Munn. "You're putting your skin at risk, as you won't know what the SPF is. Once it's overheated, you won't be as protected so it's important to keep your sunscreen in the shade."

It can be tempting to think that darker skin, or skin that is already tanned, doesn't need protection. However, this is not the case. "Anyone can get sunburn, including dark-skinned people," explains Shields. "Although generally the fairer your skin is, the more you are at risk. The same sun prevention risk applies to everyone, but some people need to be more careful."

It's lovely to feel the sun's rays on your skin, so it's good to know that a little sun exposure can be beneficial to health. Exposure to sunlight can help our bodies to produce vitamin D and avoid deficiency. "We all need the same amount of vitamin D on a daily basis to maintain healthy bones, but the rate our bodies produce the vitamin differs for everyone," explains Munn. "If you've got paler skin, you should aim for a short period in the sun everyday for about 10-15 minutes. Those with darker skin will need a little longer. You will still absorb the necessary rays while wearing sunscreen, but you'll need to stay out for longer."

Questions 7-14

7) In the introduction, what does the writer infer about summer skincare?

- a) Only 10% of people need to use more skincare
- b) Some people need clues to know how much protection to use
- c) People in general don't know enough about it
- d) 9 out of 10 people should use more sun cream

8) What advice does Emma Shields offer?

- a) effective protection involves using the right products and adopting the right behaviour
- b) it is important to choose the right kind of sunglasses
- c) sunscreen is more effective in hot rather than cold weather
- d) you shouldn't forget about protecting the soles of your feet

9) What do we learn in the third paragraph about sun tans?

- a) some kinds of sun tan are perfectly safe
- b) in 75% of cases, sun tans are safe
- c) sun tanning is a safe activity except for a small risk of skin cancer later in life
- d) acquiring a sun tan is a risky activity

10) What does Emma Shields claim regarding after-sun lotion?

- a) it cannot provide any remedial remedies
- b) the body doesn't tolerate after-sun lotion as well as it does sun cream
- c) after-sun lotion can help the body's own mechanisms to heal faster
- d) it can fix some minor damage to the skin

11) What do we learn about sun block in the fifth paragraph?

- a) oxides of titanium or zinc can reflect the sun's rays
- b) it blocks a higher percentage of UV light on young skin
- c) sun block that contains chalky substances can be used on children
- d) it isn't as effective as most people assume

12) What advice does Dr Munn give in the sixth paragraph?

- a) low factor sunscreen can be stored for longer periods than high factor sunscreen
- b) each summer, it is worth buying new sunscreen
- c) sunscreen should be stored in a refrigerator or similar low-temperature environment

- d) restocking sunscreen should take into account expiration dates

13) What danger does the last paragraph highlight?

- a) although they feel nice, the sun's rays always present a risk to health
- b) people with pale skin often do not produce enough vitamin D
- c) exposure to the sun in winter is just as dangerous as during the summer
- d) sunscreen can interfere with normal vitamin D production

14) What would be a suitable title for this article?

- a) The New Dermatological Crisis
- b) Sunscreen, Sun Cream and Sun Block – A User's Guide
- c) How to Use Sun Cream and Sunbathe Safely
- d) Sun Tanning – Changes in Recent Medical Opinion

TEXT 2

With the decreasing global boundaries and increasing activities, travel medicine has become a rapidly evolving field of medicine. Classically, travel medicine focused on individuals traveling to developing countries with prevention and treatment of malaria, traveller's diarrhoea, and general vaccinations as its primary goal. Travel medicine has subsequently become a dynamic multidisciplinary specialty that encompasses aspects of infectious disease, public health, tropical medicine, wilderness medicine, and appropriate immunization. Although these aspects are broad in reach, they are tightly integrated within the realm of travel medicine and require appropriate understanding prior to venturing out. Therefore, whether you are a humanitarian aid worker in Tanzania, a volunteer working in the Ebola-stricken areas of West Africa, a tourist, or a businessperson for a multinational corporation, understanding the dynamics of travel and the interplay of healthcare will minimize the adverse effect of travel-related illnesses and concerns while maximizing enjoyment and success for the trip.

The specialty of travel medicine is dynamic and vast in its medical knowledge requirements, as it focuses on the prevention and management of health issues related to global travel. Areas of expertise include vaccinations, epidemiology, region-specific travel medicine, pre-travel management and travel-related illnesses.

This increasing globalization of travel facilitates increased health exposures in different environments and the potential spread of disease.

Collaborative sentinel surveillance networks specifically to monitor disease trends among travellers offer new supplemental options for evaluating travel health issues. These networks can inform pre-travel and post-travel patient management by providing complementary surveillance information, facilitating communication and collaboration between participating network sites, and enabling new analytical options for travel-related research. TropNetEurop and GeoSentinel represent two major networks currently available. Data obtained from studying health problems among travellers may provide significant benefits for local populations in resource-limited countries. However, given their limitations, they should be considered as complementary tools and not relied on as an exclusive basis for evaluating health risks among travellers.

With a heightened interest in adventure travel, international destinations, and ecotourism, more patients return from vacations with presentations of possible exotic disease that are beyond the scope of a primary care or emergency physician's daily practice. However, many of the illnesses encountered could be eliminated with adequate pre-travel education and preparation. In the circumstance when prophylactic treatment and lifestyle modification fail, physicians need to know what to look for and where to find information on exotic diseases beyond the scope of daily practice. Further information can be quickly and easily accessed through the CDC Yellow Book, an online resource providing country-specific information related to endemic diseases.

Whether the participant is on an excursion to Nepal, is serving at a medical mission in Belize, or is the adventure-seeking traveller, preparation is paramount to a successful venture. All people planning travel should become informed about the potential hazards of the countries they are traveling to and learn how to minimize any risk to their health. Forward planning, appropriate preventive measures, and careful precautions can substantially reduce the risks of adverse health consequences. Although the medical profession and the travel industry can provide a great deal of help and advice, the traveller is responsible to ask for information, to understand the risks involved, and to take the necessary precautions for the journey.

Travellers should ascertain the associated travel health information for their specific itinerary several months in advance of departure. This should include general health information such as vaccine requirements, prophylactic medications, disease

outbreaks, political environment, and medical resources. As can be seen, this includes but is not limited to a pre-travel medical consultation and evaluation.

Improvisation (ie, creative use of unusual supplies for diagnosing, treating, splinting, transporting) is an invaluable skill taught in Wilderness Medical Society (WMS) and other similar courses. Efficient selection and knowledge of medications lightens the medical kit. For example, rather than carrying multiple antibiotics of choice for several possible infections, consider carrying a medication, such as ciprofloxacin, which despite some growing resistancy issues, treats travellers' diarrhoea (TD) as well as respiratory, wound, bladder, and other infections. Another example is diphenhydramine, which is excellent as an injectable local anaesthetic as well as treatment for nausea, allergic reactions, and insomnia.

In anticipation of upcoming travel, it is essential that one is well educated regarding the regions that will be visited and how one's current level of health may be impacted. Vaccinations are a vital part of any preparatory process. Once the regions of anticipated travel are identified, scheduling a visit to one's doctor or a travel medicine provider is essential—ideally 4-6 weeks before the trip because most vaccinations require a period of days or weeks to become effective. Reviewing current recommendations for the region of travel is recommended prior to the scheduled medical appointment. In addition, if uncertain regarding previous immunizations, variable tests are available to identify appropriate titer levels and whether updated boosters are indicated.

Questions 15-22

15) In the first paragraph, the example of Ebola is given to show

- a) an example of a disease that falls under the category of wilderness medicine
- b) not all diseases have a vaccine
- c) an example of a disease that may occur in an area where a travel medicine beneficiary could be present
- d) travel medicine can prepare you for any and all eventualities

16) What is one effect of the globalization of travel?

- a) it has increased the possibilities for health problems
- b) the field of epidemiology has had to develop quickly
- c) it has resulted in better healthcare facilities
- d) it has exposed existing diseases to new environments

17) Increased disease monitoring has led to

- a) specific surveillance of certain disease groups
- b) advantages to both travellers and individual countries
- c) better quality analytical tools for healthcare workers
- d) a vast increase in exploitable medical knowledge

18) When do doctors need to find information on exotic diseases?

- a) when preventative measures are unsuccessful
- b) when travellers remain uneducated
- c) when they have limited access to the CDC Yellow Book
- d) when they are vacation in exotic destinations

19) What does the fifth paragraph inform the reader concerning responsibility?

- a) the medical profession have the responsibility to give specific advice
- b) the onus is on the traveller to investigate possible dangers
- c) excursion organisers are normally responsible for medical hazard analysis
- d) individual countries are responsible to publicize specific health-related hazards

20) Which of the following statements is not mentioned in the sixth paragraph?

- a) it is important to prepare well in advance
- b) before starting their journey, travellers should see a medical professional
- c) travellers should obtain items to ensure safe sexual contact (such as condoms)
- d) travellers should be well-informed about conditions in their destination countries or regions

21)What advice is given about medical kits?

- a) it may be necessary to carry unusual supplies
- b) ciprofloxacin is preferable to diphenhydramine despite resistancy issues
- c) it is a good idea to pack injectable local anaesthetic
- d) preference should be given to versatile medicines

22)What does the eighth paragraph inform the reader about preparations?

- a) trips longer than 4-6 weeks need vaccination boosters
- b) preventative actions need to be taken one to two months before travel
- c) effective vaccines should be used rather than those that require boosters

Reading Practice Test 18 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	A	-
Q 2	C	-
Q 3	B	-
Q 4	A	-
Q 5	C	-
Q 6	C	-

Part C

	Correct Answer	Your Answer		Correct Answer	Your Answer
Q 7	C	-	Q 15	C	-
Q 8	A	-	Q 16	A	-
Q 9	D	-	Q 17	B	-
Q 10	A	-	Q 18	A	-
Q 11	D	-	Q 19	B	-
Q 12	B	-	Q 20	C	-
Q 13	D	-	Q 21	D	-
Q 14	C	-	Q 22	B	-

Review & Explanations

TEST-19

Back Pain

PART A

TEXT 1

Back pain is very common, affecting people of all ages. It usually feels like an ache, tension or stiffness in the back. It usually improves over a few days, but sometimes back pain can last for weeks or months. If no obvious reason is found for the back pain, this is known as non-specific back pain.

Acute back pain can happen if there has been a sudden movement, fall or injury. The pain can be due to a pulled muscle or ligament in the back. It could also be because a disc that sits between each of the spinal vertebrae suddenly bulges out. This is commonly known as a slipped disc and needs immediate medical assessment.

Persistent or chronic back pain is when some part of the back has become damaged and continues for 3 months or longer. People most at risk of chronic back pain include those who frequently lift heavy items, who are overweight or have arthritis.

TEXT 2

Clinical assessment

- Gather previous history of back pain
- Carry out these checks to see if the person has non-specific back pain
 - where along the spine the pain is
 - range of movement
 - possible causes or triggers
 - what type of pain, such as burning or stabbing pain
 - what makes pain worse
 - what make the pain better

Investigations

- If the diagnosis is non-specific back pain, further investigations are unnecessary
- For possible sciatica or slipped disc, an x-ray, magnetic resonance imaging (MRI) or a computed tomography (CT or CAT) scan may be required

Management of non- specific back pain

For acute pain:

- advise patient to keep moving such as walking, light exercise, swimming.
- the temptation is to rest but this lets muscles and tendons tighten up, causing more pain
- avoid lifting heavy items or excessive exercise until pain eases
- simple analgesia and/or anti- inflammatory medications

For persistent back pain:

- supervised exercise, with a physiotherapist or similar professional
- referral to pain management team
- referral to psychologist for cognitive behaviour therapy for guidance on changing poor behaviours and habits that contribute to back pain

TEXT 3

• Causes of non-specific back pain

- Aging causes wear and tear on the spine. This means that people over age 30 or 40 are more at risk for back pain than younger individuals
- An occupation that involves repetitive bending and lifting has a high incidence of back injury. Work that involves long hours of standing or sitting in a chair also puts the person at greater risk
- Lack of regular exercise increases risks for occurrence of lower back pain, and increases the likely severity of the pain
- Being overweight increases stress on the lower back, as well as other joints
- Any type of prolonged poor posture will, over time, substantially increase the risk of developing back pain. Examples include slouching over a computer keyboard, driving hunched over the steering wheel, lifting improperly
- Pregnant women are more likely to develop back pain due carrying excess body weight in the front, and the loosening of ligaments in the pelvic area as the body prepares for delivery
- People who smoke are more likely to develop back pain than those who don't smoke

Self-care of non-specific back pain

Almost everyone suffers back pain at some stage but recover quickly with little or no treatment.

- The most effective treatments people can do themselves include:
 - learning more about back pain – what makes it better, what makes it worse?
 - exercising and staying active as much as possible
 - managing stress
 - managing weight – a healthy weight lessens the strain on your back
 - avoid sitting or standing in one position for a period of time
 - quitting smoking – smoking increases your chances of developing back pain
 - lifting and carrying safely
 - relaxing – relaxation techniques to reduce stress levels and related muscle tension

massage, heat or cold packs and gentle exercise

TEXT 4

Specific conditions that cause back pain

- Sciatica is inflammation of the sciatic nerve, which runs from hips to feet, is irritated. It may get better if treated in the same way as acute back pain.
- A slipped disc is when the soft cushion of tissue between the bones in spine pushes out and presses on nerves. It usually gets better slowly with rest, gentle exercise and painkillers but manual therapy or surgery is sometimes needed. A slipped disc can cause issues that need immediate medical attention. These include:
 - paraesthesia including tingling, pins and needles and numbness around the bottom or genitals
 - can't urinate, or loss of control of bladder and bowel
 - lose feeling in one or both legs
 - pyrexia or feel hot and shivery
 - have a swelling in the back
 - notice the pain is worse at night
- Spinal stenosis is a narrowing of the spaces between vertebrae. This can cause pressure on the spinal nerves

Osteoarthritis is wear and tear of the vertebrae, causing pain and reduced range of movement

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1** Reasons why of back pain with no obvious cause develops? _____
- 2** Issues that may need the person to see a doctor urgently? _____
- 3** Self help strategies for people with back pain? _____
- 4** The tests people may have to find the cause of their back pain? _____
- 5** What back pain feels like? _____
- 6** Why pregnancy increases the chances of a woman suffering from back pain? _____
- 7** The link between spinal stenosis and back pain? _____

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8** What type of back pain can be eased with gentle exercise? _____
- 9** How long has a person been experiencing back pain if it is considered to be persistent? _____
- 10** What sort of pain might a person be suffering from if they are referred to a pain management team? _____
- 11** People over what age are at higher risk of back pain? _____
- 12** What condition might a person have if they are experiencing pins and needles around their bottom? _____

13 Which nerve carries information from the pelvic area to the toes?

14 Being overweight can cause back pain and place stress on what parts of the body?

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 Over time _____ posture will raise the risk of a person developing back pain.

16 Massage and _____ packs can ease acute back pain.

17 If a person has back pain for no detectable reason it is called _____ back pain.

18 A person with acute back pain should keep moving and include light exercise such as _____

19 Surgery or _____ is often needed if someone has a slipped disc.

20 An x-ray, _____ or a computed tomography (CT or CAT) scan are some of the options available to diagnose a back problem.

Reading Practice - PART A

Answer Keys

Correct Answer		Your Answer	
Q 1	C		-
Q 2	D		-
Q 3	C		-
Q 4	B		-
Q 5	A		-
Q 6	C		-
Q 7	D		-

Correct Answer		Your Answer	
Q 8	acute		
Q 9	3 months		
Q 10	persistent		
Q 11	30		
Q 12	slipped disc		
Q 13	sciatic		
Q 14	other joints		

Correct Answer		Your Answer	
Q 15	poor		
Q 16	heat or cold		
Q 17	non specific		
Q 18	walking, swimmi		
Q 19	manual therapy		
Q 20	magnetic resona		

Review & Explanations

PART B

TEXT 1

General Guidelines for Eyedrop Application

These guidelines give general advice on how to use most eye drops. However, you may be given specific instructions which may vary from the advice below. After thoroughly washing your hands, sit or stand in front of a mirror. Bend your head backwards and gently pull your lower eyelid down. Hold the dropper above one eye. Squeeze one drop into the pocket formed by gently pulling down the lower eyelid. Try not to touch your eye, eyelashes, or anything else with the dropper tip in order to keep it clean. Let go of the eyelid and keep the eye closed for as long as possible (2-3 minutes at least) after application of the eye drop, with your head tilted down towards the floor. Wipe away any liquid that falls on to your cheek with a tissue.

Question

- 1) What does the following article say about the application of eye-drops?
 - a) there is more than one way to apply eyedrops
 - b) the dropper tip should be wiped clean after each use
 - c) your eye should be closed for 2-3 minutes before applying eye drops

TEXT 2

Cochlear Implantation Recommendations

This technology appraisal examined the currently available devices for cochlear implantation. Unilateral cochlear implantation is recommended as an option for people with severe to profound deafness who do not receive adequate benefit from acoustic hearing aids. If different cochlear implant systems are considered to be equally appropriate, the least costly should be used. Assessment of cost should take into account acquisition costs, long-term reliability and the support package offered. Simultaneous bilateral cochlear implantation is recommended as an option for children and for adults who are blind or who have other disabilities that increase their reliance on auditory stimuli as a primary sensory mechanism for spatial awareness. Patients should have severe to profound deafness with limited or no adequate benefit from acoustic hearing aids.

Question

- 2) Which of the following statements is not mentioned?
- a) value for money is an important factor in cochlear implantation acquisition
 - b) both adults and children could be possible candidates for cochlear implantation
 - c) simultaneous bilateral cochlear implantation necessitates the use of identical devices

TEXT 3

Apple Watch Detects Irregular Heart Beat In Large U.S. Study

Dr. Deepak Bhatt, one of the study's lead investigators, said "the physician can use the information from the study, combine it with their assessment ... and then guide clinical decisions around what to do with an alert." The study found that 57 percent of participants who received an alert on their watch sought medical attention. For Apple, the data provides firepower as it pushes into healthcare. Its new 'Series 4 Watch' has the ability to take an electrocardiogram to detect heart problems and required clearance from the U.S. Food and Drug Administration. Dr. Bhatt called it an important study as use of this type of wearable technology is only going to become more prevalent.

Question

- 3) What can you infer about Dr. Bhatt's opinion?
- a) he is doubtful that a proper diagnosis can be made
 - b) he thinks this type of technology is likely to become widespread
 - c) he is a proponent of the medical uses of wearable technology

TEXT 4

Violence in the Workplace: A Growing Problem

Workplace violence in any form comes at a high cost to the individual and institutions. It has lasting effects on our healthcare system and deeply impacts

clinicians on a personal and professional level. Workplace violence can have a wide range of impacts, including effects on self-esteem, relationships with co-workers, recruitment and retention. The psychological trauma often makes individuals fearful of returning to work. As healthcare professionals, we must be advocates for not only our patients, but also for ourselves. A safe work environment is built upon a foundation of trust and respect for all. No case should go unreported and zero tolerance should be implemented to ensure that we can protect ourselves as well as deliver safe patient care.

Question

- 4) In this article, what does the word 'advocates' indicate?
- a) sometimes, workplace violence results in legal procedures
 - b) healthcare workers may need to support themselves as well as their patients
 - c) violent patients should be admonished

TEXT 5

Is It Safe to Reinfuse Blood Drawn from a CVAD Via A Syringe When Checking Line Patency or Drawing Blood?

Before withdrawing a blood aspirate from a central venous access device (CVAD), ask yourself, why am I taking this blood aspirate? Are you taking the blood aspirate to assess for device patency prior to hooking up to an intravenous infusion or do you require a blood sample for laboratory analysis? If you are simply checking CVAD patency you only need to pull blood back into the catheter until you see the liquid gold; it never needs to come as far as the catheter hub and needleless connector. The ease of aspiration can still be assessed with this small aspiration technique as you get a different feel when you are aspirating air (more resistance is felt) as opposed to the free-flowing feeling of aspirating liquid.

Question

- 5) What does this article tell the reader about drawing blood?
- a) the procedure of pulling blood back gives rise to more than one sensation
 - b) the catheter should be removed before taking blood samples
 - c) aspiration should be avoided in CVAD procedures

TEXT 6

What Should I Consider Before Participating in a Clinical Trial?

Before participating, you must be provided with an "informed consent" document explaining the risks and potential benefits of the trial. Be sure to read over this information carefully. It is important to fully understand the purpose of the trial and what to expect. You will want to find out if the treatment will interact with any of your current medications or affect any other medical conditions you may have. You should be informed about what tests or procedures, such as biopsies or blood draws, will be performed, and you should consider your comfort level with what will be done. Also, think about whether you are prepared for any anticipated side effects, pain, or discomfort that may be involved.

Question

- 6) What advice is given in this article about participating in a clinical trial?
- a) you should inform a family member of your intentions
 - b) you should give consent to possible side-effects
 - c) you should know about the reason for the trial

PART C

TEXT 1

The ongoing measles outbreaks across the United States and Europe prove definitively that our personal choices affect everybody around us. Although you have a right to your own body, your choice to wilfully be sick ends where another's right to be healthy begins. For that reason, people who "opt out" of vaccines should be opted out of American society. This is America, the Land of the Free. That freedom, however, doesn't mean "I can do whatever I want, whenever I want." When we choose to live in a society, there are certain obligations—both moral and legal—to which we are bound. You cannot inflict harm or infringe on the rights and liberties of those around you.

Those obligations extend even to your constitutional rights. Although we have a First Amendment, you are not allowed to play music as loudly as you want in your apartment. Your neighbours have a legal right to peace and quiet. Even though we have a Second Amendment, you are not allowed to shoot a gun for sport in the middle of a city or town. Stray bullets are not only scary, they're hazardous, and often inadvertently kill people. Finally, your moral and legal obligations to the safety of others can even curtail combinations of your rights. Even though consuming alcohol and driving are both legal activities, they are not legal when performed together. Nearly 11,000 people die every year because people choose to exercise their "rights" inappropriately.

The exact same reasoning applies to vaccination. There is no moral difference between a drunk driver and a wilfully unvaccinated person. Both are selfishly, recklessly and knowingly putting the lives of everyone they encounter at risk. Their behaviour endangers the health, safety and livelihood of the innocent bystanders who happen to have the misfortune of being in their path.

The reasons why are simple and straightforward. Vaccines aren't perfect (e.g., they can wear off over time) and not everyone can be vaccinated. There is one and only one legitimate reason to skip a vaccine: being immune-compromised. Some individuals, because of genetic deficiencies or diseases like cancer, cannot receive vaccines. Other people are too young. Vaccines such as MMR (measles, mumps, rubella) cannot be administered before 12 months of age. These vulnerable people rely on the responsible actions of everyone else in society to protect them, a concept known as "herd immunity."

For their sake, we have a moral—and there should also be a legal—obligation to protect them. Everyone who can be vaccinated must be vaccinated in order to prevent the spread of disease. This is a protection we demand even for our animals: kennels will turn your pet away if they aren't properly vaccinated and on an accepted flea treatment. There are rules we all have to play by and responsibilities we have to live up to if we want to live in a society together.

If this isn't enough to convince a person to become fully vaccinated, then perhaps there is a solution that maintains everybody's freedom: Anti-vaxxers (as people who object to vaccinations call themselves) can opt out of American society. No public or private school, workplace or other institution should allow a non-exempt, unvaccinated person through their doors. A basic concern for the health and safety of others is the price it costs to participate. Is that too harsh? We don't think so. If a person wants to blast their music loudly, shoot guns aimlessly, and drink and drive, they should be allowed to do exactly as they please: so long as it's on their own property, sufficiently isolated from everyone else. Similarly, if you don't want to be vaccinated, perhaps that should be allowed too, so long as you agree to permanently live out in the middle of nowhere.

It is inexcusable that society has reached this point. Many of the deadliest diseases known to mankind are due to bacteria and viruses, and dozens of them are now entirely preventable thanks to the sciences of microbiology and immunology. People falsely believe that diseases like measles have "gone away," but they have not. They're always there, waiting to strike as soon as our collective guard goes down. Not so long ago, smallpox ran the risk of obliterating entire cities, while polio paralyzed large fractions of a generation. We have forgotten this morbid history because public health has been a victim of its own success.

But misinformation abounds. The internet, both a blessing and a curse, has allowed devilish lies, propaganda and a discredited fraud masquerading as science to infect the minds of millions of people. Unfortunately, there's no vaccine that can inoculate someone against a counterfactual, unscientific mindset. There are, however, vaccines that can prevent dozens of harmful diseases. Those who refuse, and recklessly endanger others, should be put in quarantine.

Questions 7-14

7) In the first paragraph, what does the writer infer about freedom?

- a) freedom has its limits
- b) moral and legal freedom is undesirable

- c) an individual's freedom is more important than other's rights and liberties
- d) you cannot 'opt out' of the question of freedom in modern society

8) In paragraph two, which of the following are not true?

- a) the people who live near you have a right to a certain degree of calm
- b) the First Amendment limits your activity in some ways
- c) the Second Amendment does not permit unlimited use of guns
- d) moral and legal obligations extend to other amendments

9) The example of a drunk driver is given to

- a) show that there can be terrible consequences of selfish behaviour
- b) demonstrate that acting freely can lead to accidents
- c) emphasise the link between decisions and risk
- d) some people are innocent of wilful misdemeanours

10) In paragraph four, which of the following would not be exceptions to the idea that everyone should be vaccinated?

- a) people with pre-existing conditions which mean that vaccines would be ineffective
- b) people for whom the effects of vaccines do not wear off over time
- c) very young children
- d) anyone who has a severe problem with their immune system

11) In paragraph five, what does the writer infer about vaccinations?

- a) the need for vaccinations is growing as some animals can spread disease
- b) the society that administers vaccines should be more powerful
- c) there should be a law (or laws) to enforce vaccination for vaccinatable people
- d) fleas can carry diseases that can harm humans

12) What alternative does the writer propose in the sixth paragraph?

- a) more education should be given to encourage vaccination

- b) people who refuse to be vaccinated should live in secluded locations
- c) some institutions could be established for unvaccinated people
- d) harsh laws should be passed to enforce vaccination

13) In the seventh paragraph, the writer's opinion could be summarized as

- a) the sciences of microbiology and immunology are extremely important
- b) collective vaccination is a sign of a well-adjusted society
- c) the present situation cannot be justified
- d) thanks to vaccinations, there are more successes than victims

14) What can the reader infer about the writer's opinion in the last paragraph?

- a) there are advantages and disadvantages in using the internet
- b) internet usage can masquerade as propaganda
- c) counterfactual mindsets ought to be inoculated against
- d) quarantine is the only remedy for certain types of diseases

TEXT 2

Although sports injuries to the knee, ankle, and shoulder have been well documented, injuries to the pelvis, hip, and thigh get little attention because of their low prevalence. Unfortunately, severe consequences may result if these injuries are improperly managed.

Femoral neck stress fractures were mainly seen in military recruits due to a triad of activity that is new, strenuous, and highly repetitive. However, as a result of self-imposed fitness regimens of recreational athletes, over the last 20 years the number of these injuries has been increasing in non-military populations. In contrast, contact sports such as football, rugby, and soccer are usually the cause of most fractures of the hip. Stress fractures occur in normal bone undergoing repeated submaximal stress. As the bone attempts to remodel, osteoclastic activity occurs at a greater rate than osteoblastic activity. When these cumulative forces exceed the structural strength of bone, stress fractures occur. Stress fractures occur mainly at the femoral neck and are classified as either tension (at the superior aspect of the femoral neck) or compression (at the inferior aspect of the femoral neck).

Hip fractures are classified as intracapsular, which includes femoral head and neck fractures, or extracapsular, which includes trochanteric, intertrochanteric, and subtrochanteric fractures. The location of the fracture and the amount of angulation and comminution play integral roles in the overall morbidity of the patient, as does the pre-existing physical condition of the individual. Fractures of the proximal femur are extremely rare in young athletes and are usually caused by high-energy motor vehicle accidents or significant trauma during athletic activity. Other causes may be an underlying disease process such as Gaucher disease, fibrous dysplasia, or bone cysts.

Identification and initiation of treatment is imperative in attempts to avoid complications, such as avascular necrosis (AVN). AVN is more common in patients in the paediatric and adolescent age groups. This outcome is due to the precarious nature of the blood supply to the subchondral region of the femoral head, which does not stabilize until years after skeletal maturity, after which collateral flow develops.

The blood supply to the femoral head has been studied extensively and has been found to change substantially during development. Until the cartilaginous growth plate forms a barrier at age 4 years, the major blood supply comes from the medial and lateral circumflex arteries (metaphyseal arteries), which arise from the deep femoral artery. After age 4 years, the posterosuperior and posteroinferior arterial branches of the medial femoral circumflex bypass the growth plate and form the main blood supply to the femoral head. During adolescence, the growth plate fuses and the metaphyseal vessels again become significant, traveling along the femoral neck. Fractures in this area can disrupt this delicate blood supply, leading to AVN, the most severe complication of this fracture.

An estimated 340,000 hip fractures occur each year. Estimates indicate that in 2040, approximately 500,000 hip fractures will occur. Nine of 10 hip fractures occur in patients aged 65 years and older, and 3 of 4 occur in women. White females have been reported to be twice as likely to fracture their hips than black and Hispanic females. This frequency has been associated with a metropolitan setting, increased caffeine use, alcohol use, sedentary lifestyle, psychotropic drug use, and senile dementia.

Patients with hip fractures may present in a variety of ways, ranging from an 80-year-old woman reporting hip pain after a trivial fall to a 30-year-old man in haemorrhagic shock after a high-speed motor vehicle accident. Stress fractures usually manifest more insidiously, with an otherwise healthy person reporting pain related to activity

and not healing with the conservative treatments suggested by their primary care doctor. Although the classic presentation of a hip fracture is an elderly patient who is in extreme pain, a young, healthy athlete usually has the same presentation. The affected leg is externally rotated and may be shortened. The extremity shortening occurs because the muscles acting on the hip joint depend on the continuity of the femur to act, and when this continuity is disrupted, the result is a shorter-appearing leg. Assessing peripheral pulses and checking Doppler pressures to assure vascular patency is very important.

The patient with a stress fracture may present more subtly, reporting pain in the anterior groin or thigh. This pain increases with activity and can persist for hours afterward. The pain can progress to a point of consistency, even without activity. This pain generally expresses itself in the groin; however, it can also be referred to the knee. An antalgic gait pattern is often present. Signs and symptoms usually involve a diffuse or localized aching pain in the anterior groin or thigh region during weight-bearing activities that is relieved with rest. Night pain is also common. A study by Brännström et al that included 408,000 older adults reported an association between antidepressant medications and hip fracture before and after the initiation of therapy. Further investigations are needed to study this association.

Questions 15-22

15) What do we learn about sports injuries in the first paragraph?

- a) some sports injuries have severe consequences for subsequent sporting careers
- b) improper injury management has been well-documented
- c) low frequency pelvis, hip, and thigh injuries are more difficult to treat
- d) there is more information available for some types of injuries than for others

16) When do stress fractures happen?

- a) they happen in specific situations
- b) they often occur when athletes leave military service
- c) they often happen following abnormal osteoclastic activity
- d) when bones cannot remodel to deal with contact sports injuries

17) According to the third paragraph, which of the following statements is not true?

- a) the exact position of the fracture is relevant for subsequent treatment
- b) a patient's state of health can play a role in the severity and complexity of fractures
- c) high-energy athletic accidents can exacerbate hip fractures
- d) the angle of any distortion is an important factor

18) Why is AVN more common in younger patients?

- a) because treatment initiation usually takes longer in young patients
- b) because blood supply does not become stable until later in life
- c) because skeletal maturity can hinder blood flow
- d) because collateral flow is infrequent during adolescence

19) In the fifth paragraph, which of the following statements about blood supply to the femoral head is not mentioned?

- a) there have been a large number of studies into this topic
- b) a significant development in this area occurs during adolescence
- c) there is a significant change in infants aged four years old
- d) metaphyseal vessels are significant only for infants aged less than four

20) What do the statistics in the sixth paragraph reveal?

- a) a disproportionately high number of elderly women are affected by hip fractures
- b) the number of hip fractures rises by over 340,000 each year
- c) the rise in hip fractures affects all ethnic groups equally
- d) hip fractures can be set more quickly in metropolitan hospitals

21) What do we learn about hip fractures in the seventh paragraph?

- a) stress fractures are more difficult to detect than ordinary fractures
- b) 30-year-old patients have a better likelihood of recovery than patients over 80
- c) primary care doctors may suggest painkillers rather than treatment for a fracture

- d) in some patients, contorted leg muscles may make stress fracture diagnosis more difficult

22) Which of the following is not mentioned in the last paragraph?

- a) the location of a pain is an important factor in correct diagnosis
- b) nocturnal pain can be symptomatic of a stress fracture
- c) pain duration should be taken into account
- d) antidepressant medication can relieve some symptoms of stress fractures

Reading Practice Test 19 - PART B & C

Answer Keys

Part B

	Correct Answer	Your Answer
Q 1	A	-
Q 2	C	-
Q 3	B	-
Q 4	B	-
Q 5	A	-
Q 6	C	-

Part C

	Correct Answer	Your Answer
Q 7	A	-
Q 8	D	-
Q 9	A	-
Q 10	B	-
Q 11	C	-
Q 12	B	-
Q 13	C	-
Q 14	A	-
Q 15	D	-
Q 16	A	-
Q 17	C	-
Q 18	B	-
Q 19	D	-
Q 20	A	-
Q 21	A	-
Q 22	D	-

Review & Explanations

TEST-20

Menopause

PART A

TEXT 1

Menopause (amenorrhea) is the end of menses (menstruation cycles or periods) due to decreased ovarian function.

It is usually part of the natural aging process but can also occur as a result of drug or surgical treatments. The average age for a woman to reach menopause is 51. But any time between 45 and 55 years old is considered normal. Women often start experiencing changes in their menstrual cycles in the 40's.

Perimenopause refers to the years before and the 1 year after the last menses. It is typically the most symptomatic phase because hormones are fluctuating. The length of the perimenopause, the severity of the symptoms and an individual woman's tolerance of the symptoms, vary widely between women.

Post-menopause refers to the time after the first 12 months since the last menses. The first 3-6 years of post- menopause are known as early stage post menopause. Late stage post menopause, is from that point until death.

Premature menopause or premature ovarian insufficiency is when menopause occurs before 40 years of age. This occurs in approximately 1 in 100 women.

Premature ovarian failure (primary ovarian insufficiency) may occur as a result of certain medical conditions, such as autoimmune diseases and genetics conditions. The woman may be in her early 20s or in extreme cases, even still be a child

TEXT 2

Physiology of the menopause

- Ovaries age and become less responsive to pituitary gonadotropins follicle-stimulating hormone (FSH) and luteinizing hormone (LH) initially causing:
 - a shorter follicular phase, with shorter and less regular menstrual cycles
 - fewer ovulations
 - decreased progesterone production

Effect on menstrual cycle

- menstrual periods usually start to become less frequent over a few months or years before they stop altogether.

- sometimes they can stop suddenly.
- many women find that their periods become lighter or less painful during menopause
 - some may experience menorrhagia (heavy bleeding).
- they may also experience an erratic pattern, which may last two to five years before periods stop completely
- pregnancy is still possible during menopause

Wider effects of menopause

- vasomotor overreactions causing hot flushes and night sweats
 - vulvovaginal atrophy leading to vaginal dryness and discomfort during sex, reduced libido
 - neuropsychiatric changes causing low mood or anxiety, problems with memory
 - Increased levels of low-density lipoprotein (LDL) cholesterol leading to atherosclerosis and subsequent coronary artery disease.
- Reduced bone density with up to 20% of bone density loss occurs during the first 5 years after menopause.

TEXT 3

Diagnosing the menopause

- Clinical assessment and comprehensive history is usually sufficient for diagnosis
- Blood test to check levels of follicle-stimulating hormone (FSH) and oestrogen (oestradiol)

Treatment options

- Hormone Replacement Therapy:
 - replaces oestrogen, progestogen, or both and is the most effective treatment for menopausal symptoms. It is used to relieve moderate to severe hot flushes and, when an oestrogen is included, to relieve symptoms due to vulvovaginal atrophy
 - may help prevent certain conditions such as:
 - colon cancer
 - macular degeneration
 - osteoporosis
 - may be linked with a slight increase in the risks of some medical conditions, such as:
 - ovarian and endometrial cancer

- breast cancer
- coronary heart disease
- cerebro-vascular accident
- venous thromboembolism
- can be administered as tablets, skin patches, gels and implants
- can be used as a topical treatment as vaginal oestrogen cream, lubricant or moisturiser for vaginal dryness
- Cognitive behavioural therapy (CBT) can help with low mood and anxiety

TEXT 4

Self-help strategies to ease the symptoms of menopause include:

- eating a healthy, balanced diet
- exercising regularly
 - maintaining a healthy weight and staying fit and strong can improve some menopausal symptoms
- avoiding triggers such as caffeine, alcohol and spicy foods, help reduce hot flushes
- relaxation and exercise can reduce mood swings and anxiety
- using water-based lubricants during intercourse, as this can help with symptoms of vaginal dryness
- reducing stress where possible such as:
 - mindfulness, yoga or meditation,
 - an end-of-the-day hot bath, reading, listening to music,
- cooling the environment (e.g., lowering the thermostat, using fans) and wearing clothing in layers that can be removed as needed may help hot flushes
- Restful sleep can be achieved by:
 - avoiding exercise within two hours of bedtime
 - going to bed at the same time every night
- wearing lighter clothing and keeping your room cool

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1 HRT? _____
- 2 Influences on the menstrual cycle? _____
- 3 The perimenopause? _____
- 4 The tests that may be carried out to confirm menopause? _____
- 5 Strategies to get a restful sleep? _____
- 6 The commonest age of women reaching menopause? _____
- 7 The hormones that the ovaries are less responsive to? _____

Questions 8-14

Answer each of the questions, 8-14, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8 What does staying away certain stimuli help reduce? _____
- 9 Neuropsychiatric changes cause sadness or apprehension. What else?

- 10 Hormone Replacement therapy can reduce the incidence of disorders such as osteoporosis, colon cancer and what other condition? _____
- 11 What is the phase after the first year since the last period called?

- 12 What does reduced functionality of the ovaries cause? _____
- 13 What can being a healthy weight help with? _____
- 14 What menopause treatment option can raise the chance of developing endometrial cancer? _____

Questions 15-20

Complete each of the sentences, 15-20, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15 Menstrual periods generally become _____ for a while before they completely stop.

16 _____ cholesterol levels can increase and there is a greater risk of atherosclerosis developing.

17 _____ can happen if illnesses such as autoimmune diseases and genetics conditions are present.

18 _____ or both.

19 Women often have an irregular menstrual cycle pattern, which may last _____ years before finally ceasing.

20 _____ or _____ can help reduce stress.

Reading Practice - PART A

Answer Keys

	Correct Answer	Your Answer		Correct Answer	Your Answer		Correct Answer	Your Answer
Q 1	C	-	Q 8	Hot flushes		Q 15	less frequent	
Q 2	B	-	Q 9	Problems with m		Q 16	Low-density lip	
Q 3	A	-	Q 10	Macular degener		Q 17	Premature ovar	
Q 4	C	-	Q 11	Post menopause		Q 18	Oestrogen, pro	
Q 5	D	-	Q 12	Menopause		Q 19	two to five	
Q 6	A	-	Q 13	Improve some me		Q 20	Mindfulness, yo	
Q 7	B	-	Q 14	Hormone replace				

Review & Explanations

PART B

TEXT 1

Telehealth: Crossing Barriers

Telehealth, often used interchangeably with the term telemedicine, is an important use of technology in today's healthcare. It is the use of a real time interactive communication between parties. Telemedicine includes remote clinical services, such as between a patient and a provider at another site. Telehealth uses electronic and telecommunications for long-distance healthcare, including education and non-clinical services. Technologies can include the internet, telephonic communication, videoconferencing, imaging, streaming media, and wireless communications. Benefits of telehealth include providing healthcare services to those patients in underserved or remote locations. This technology may also allow more patient independence, improve feelings of social isolation or depression, and promote patient dignity. Challenges can include limited assessment ability of the healthcare professional, inaccurate data provided by the patient, and difficulties with technology, including the patient's comfort with its use.

Question

- 1) What may limit the use of telehealth?
 - a) some people don't know how to use it properly
 - b) the remoteness of the patient
 - c) poor communication links could cause data degradation

TEXT 2

Snoring: Introduction to Treatment Options

Snoring is a noisy inspiratory sound produced by vibration and partial airway obstruction in the pharynx. It is a form of sleep-disordered breathing, and can lead to disrupted sleep, daytime tiredness and poor concentration, both for the person who snores and for anyone sleeping close by. Snoring can be associated with obstructive or central sleep apnoea. However, for the purpose of this overview, the reviewed studies included patients who had an oxygen saturation level no lower than 85%. Conservative treatments involve lifestyle changes, including weight loss, avoiding

alcohol and sedatives, stopping smoking and sleep position training. Physical appliances (such as dental or oral devices) have also been used to maintain normal airflow dynamics during sleep. Procedures available for pharyngeal airway obstruction include laser-assisted uvulopalatoplasty (LAUP) and uvulopalatopharyngoplasty (UPPP).

Question

- 2) Which of the following problems associated with snoring are not mentioned in this article?
- a) it can impact daytime activities
 - b) it can cause inflammation of the pharynx
 - c) it can negatively affect both the snorer and others nearby

TEXT 3

Sickle Cell Trait and Athletic Participation

In recent years there has been heightened attention on cases of athletes with sickle cell trait who have experienced exertion-related illness and, in some cases, sudden death, during or after strenuous athletic training sessions. These cases caused some organizations to mandate that athletes be tested for sickle cell trait as a prerequisite to participation and has led to some confusion about whether or not it is safe for individuals with sickle cell trait to play sports. In January 2012, ASH released a Report on Sickle Cell Trait and Athletic Participation, opposing mandatory sickle cell trait screening as a prerequisite to athletic participation and urging athletics programs to adopt universal preventive interventions in their training. These preventative interventions include drinking adequate amounts of fluid and taking rest breaks as needed.

Question

- 3) Which position does ASH advocate concerning athletes with sickle cell trait?
- a) athletes should continue to be screened for the trait and excluded from competitions if necessary

- b) affected athletes should be able to compete along with healthy competitors
- c) only athletes who have the exertion-related diagnosis should not compete

TEXT 4

Can Blood Draws Cause Anaemia?

A relationship between the volume of blood drawn and iatrogenic anaemia was first described in 2005, when a study found that in adult patients on general medicine floors, the volume of blood drawn strongly predicted decreased haemoglobin and haematocrit levels. For every 100 mL of blood drawn, haemoglobin levels fell by an average of 0.7 g/dL. In 2011, 17,676 patients with acute myocardial infarction were studied across 57 centres and a correlation was found between the volume of blood taken and the development of anaemia. They also found significant variation in blood loss from testing in patients who developed moderate or severe anaemia. The authors believed this indicated that moderate to severe anaemia was more frequent at centres with higher than average diagnostic blood loss.

Question

- 6) What were the findings of the 2011 study?
- a) there appears to be a link between blood draws and anaemia
 - b) decreased haemoglobin and haematocrit levels could be predicted from the volume of blood drawn
 - c) drawing blood can lead to iatrogenic complications

TEXT 5

MGUS: It's About the Protein, Not Just the Marrow

Monoclonal gammopathy of undetermined significance (MGUS) has always been a favourite topic on internal medicine teaching rounds and is sometimes used to challenge residents. It is a relatively uncommon cause of some common laboratory and clinical anomalies. Thus, residents must field questions such as, "What is a cause of a high erythrocyte sedimentation rate with a concurrently normal C-reactive protein level and a low anion gap?" And for internists who love probabilistic assessments, there are now data and flowcharts to help predict the likelihood that a

patient with MGUS will develop myeloma, or other malignant clonal proliferative disorder that will warrant therapy. In the past decade, it has been increasingly recognized that these clonally produced proteins may be directly pathogenic, independent of any pathologic effect of cellular clonal expansion and infiltration.

Question

- 5) Why should healthcare workers know about MGUS, according to this article?
- a) they need to be ready to deal with relatively uncommon anomalies
 - b) they need to understand it so that they can give adequate probabilistic assessments
 - c) they may be asked about it as part of their training

TEXT 6

Shoulder Pain

Shoulder pain is a common symptom in primary care. It can be due to an intrinsic shoulder problem, but pain can also be referred from other structures, such as the neck, diaphragm or the heart. Common shoulder problems share overlapping clinical features. When assessing shoulder pain, it is important to look for any 'red flags' that mean investigation and diagnosis need a more focused or urgent approach. Ligaments and surrounding musculature, including the rotator cuff muscles, contribute to shoulder joint stability. The rotator cuff is composed of the four muscles: supraspinatus, infraspinatus, teres minor and subscapularis that interlock to function as one unit. These muscles help with internal and external rotation of the shoulder and importantly depress the humeral head against the glenoid as the arm is elevated.

Question

- 1) Which of the following are not mentioned in the following article?
- a) complications can be caused by insufficient depression of the humeral head against the glenoid

- b) shoulder pain could be a symptom of something more serious
- c) a lot of patients seek treatment for shoulder pain

PART B

TEXT 1

Researchers have come up with a new method to control brain cells in live animals using specially designed receptor proteins that respond to the drug varenicline. While drug-responsive receptors have been around for sometime, the new incarnations, described today in *Science*, have been structurally optimized, as has the drug itself, to create a novel repertoire of precise and powerful chemo-genetic resources.

“It really is an exciting new development that has great potential not only for basic research but potentially also in translation and applications for human use,” says neuroscientist Dr Christian Lüscher of the University of Geneva who was not involved with the research. “There is a tremendous need for novel medications that have higher selectivity and higher potency at very low doses, and hence fewer side effects. And this technology potentially fits these needs,” adds Dr Lüscher.

The aim of chemo-genetic techniques is to enable researchers to activate or silence specific cell types at will. Applied typically to brain cell manipulations, the techniques employ specially designed receptors that only respond to particular ligands (drugs or molecules). Introducing the receptors into chosen cells thus allows drug-dependant control of those cells’ activities.

One of the principal chemo-genetic systems, called Designer Receptors Exclusively Activated by Designer Drugs, or DREADDs for short, has limitations. For one thing, a commonly used DREADD activator, CNO, was recently found to transform into the drug clozapine, which has widespread effects in the brain. Moreover, DREADDs are based on G-protein coupled receptors (GPCRs), meaning they must associate with ion channels in the cell to have an effect. “So if the cell doesn’t express the [necessary ion] channel it will simply not work,” Lüscher says. Scott Sternson of the HHMI Janelia Research Campus and colleagues’ new system by contrast is based on fusion proteins consisting of an ion channel domain and a receptor domain—specifically that of the acetylcholine receptor. “They are in themselves already the effectors,” says Lüscher, meaning they can work in essentially any cell type regardless of the other ion channels present. “That’s a big advantage.”

Rather than making both a designer receptor and ligand straight away, Sternson's team first focused on creating receptors that would respond to a given FDA-approved drug. "To use chemo-genetics therapeutically you're going to have a gene therapy component, which is the receptor, and then you have a chemical component, which obviously is a small molecule drug, and that, from a practical standpoint, creates certain regulatory challenges" in terms of clinical testing, Sternson explains. Starting with an approved drug would thus more likely result in a system ripe for translation into, for example, therapies for pain or epilepsy.

To that end, the team screened an array of safe, well-tolerated, brain-entering drugs for their ability to interact with a variety of fusion receptors. Varenicline, an anti-smoking drug, stood out as a strong candidate, explains Sternson. To maximize the effect of varenicline the investigators then studied the crystal structure of the drug-receptor interaction and, with a great deal of educated tinkering and patience, tweaked the receptor until they produced optimized versions many times more responsive than the originals. Indeed, in cultured mouse neurons as well as in live mice and monkeys, doses of varenicline substantially lower than that normally required for the drug's nicotine-substitution effect were able to induce or suppress the activity (depending on the ion channel domain spliced to the receptor) of cells presenting the optimized receptors.

Sternson's team has also tinkered with varenicline itself to make versions of the drug that interact with the optimized receptors more specifically, or that offer yet more potency. In mice, one of these varenicline variants, when used at a three-fold lower dose than the original drug, could just as effectively suppress the activity of neurons expressing the engineered receptor and alter the animals' behaviour. This is important for future translation of the system to human use, says Lüscher. "If you can use such low doses of varenicline then I anticipate there should be minimal side effects."

While these varenicline variants, like the fusion receptors, are so far only for research purposes, "I am rather cautiously optimistic that this will change the landscape of our ability to use chemo-genetics in a way that has the potential to fairly quickly be applied in clinical contexts," says neuroscientist Gordon Fishell of Harvard Medical School who was not part of the research team. "We're getting to a point where we can hack the brain."

Questions 7-14

- 7) What developments of drug-responsive receptors are reported in the first paragraph?
- a) drug-responsive receptors have been tested on live animals for the first time
 - b) some proteins have been found to be resistant to drug-responsive receptors
 - c) the chemo-genetic resources have been extensively catalogued
 - d) both the drugs and the receptors have been enhanced
- 8) What is Dr Christian Lüscher's reaction to this new development?
- a) he says this development is unlikely to yield positive results
 - b) he is concerned about possible side effects
 - c) he thinks it could be used to benefit people
 - d) he claims that, if the correct receptors are not identified, this development cannot be used
- 9) In the third paragraph, what is the aim of chemo-genetic techniques?
- a) to increase brain cell production
 - b) to allow scientists to switch cells on or off by using specific drugs
 - c) to make brain cells resistant to particular ligands
 - d) to respond to the need for more brain molecules
- 10) What advantage does the new technique developed at the HHMI Research Campus offer?
- a) it is more versatile than the DREADD system
 - b) it does not require the presence of clozapine
 - c) it can eliminate ion channels
 - d) it can replicate on G-protein coupled receptors (GPCRs) more readily
- 11) Why did Sternson's team use an FDA-approved drug?
- a) the team were given samples of an FDA-approved drug to work with
 - b) only FDA-approved drugs were available
 - c) it is against the law to use a non-FDA approved drug

- d) an existing medication would increase their chances of a successful outcome

12) What was the team able to achieve with varenicline?

- a) they incorporated a new ion channel domain
- b) they made it possible to use it on animals
- c) they improved it
- d) they added a new crystalline structure

13) What additional result is described in the seventh paragraph?

- a) varenicline was produced which had fewer side effects
- b) stronger versions of varenicline were made
- c) a version of varenicline was made that is three times less expensive
- d) a version was created that could engineer new behaviour receptors

14) What is Gordon Fishell's opinion of this research?

- a) it won't be long until doctors can use this development to help their patients
- b) it is likely to remain as research for the foreseeable future
- c) it poses a danger as some may use this research to try to hack a brain
- d) chemo-genetics is the future of clinical contexts

TEXT 2

General anaesthesia is the state produced when a patient receives medications to produce amnesia and analgesia with or without reversible muscle paralysis. An anesthetized patient can be thought of as being in a controlled, reversible state of unconsciousness. Anaesthesia enables a patient to tolerate surgical procedures that would otherwise inflict unbearable pain, potentiate extreme physiologic exacerbations, and result in unpleasant memories.

General anaesthesia is induced and maintained using a combination of intravenous and inhaled agents. A point worth noting is that general anaesthesia may not always be the best choice; depending on a patient's clinical presentation, local or regional

anaesthesia may be more appropriate. Anaesthesia providers are responsible for assessing all factors that influence a patient's medical condition and selecting the optimal aesthetic technique accordingly.

A general aesthetic can be broken down into three distinct phases, the pre-, intra-, and post-operative periods. The pre-operative phase involves patient preparation from the time surgery is scheduled until the patient enters the operating room. A thorough preoperative plan should ensue prior to induction of general anaesthesia and, ideally, in advance of the operative day. Complete history should be attained with attention to any new, ongoing, or worsening medical conditions, previous personal or familial adverse reactions to general anaesthetics, assessment of functional cardiac and pulmonary states, and allergy and medication history. Preoperative evaluation also helps to relieve anxiety of the unknown surgical environment for patients and their families as well as reduce the likelihood of same-day case cancellation.

For induction of general anaesthesia, it is important that the patient is properly fasted to prevent untoward events such as pulmonary aspiration. Of course, emergency cases are not delayed until fasting times are met. Patients should be instructed as to the current guidelines on perioperative fasting. Unnecessarily long fasting times should be avoided to reduce dehydration, postinduction hypotension, and patient dissatisfaction. Recent catastrophes under anaesthesia have focused attention on the interaction between non-prescribed medications and aesthetic drugs, including interactions with vitamins, herbal preparations, traditional remedies, and food supplements. Good information on the exact content of these supplement preparations is often hard to obtain.

The pre-operative period, which is usually conducted in the surgical ward or in a preoperative holding area, originated in the early days of anaesthesia, when morphine and scopolamine were routinely administered to make the inhalation of highly pungent ether and chloroform vapours more tolerable. The goal of premedication is to have the patient arrive in the operating room in a calm, relaxed frame of mind. The most commonly used premedication is midazolam, a short-acting benzodiazepine that has the benefit of providing antegrade amnesia.

For the most part, contemporary practice dictates that adult patients and most children aged at least 10 years be induced with intravenous drugs, this being a rapid and minimally unpleasant experience for the patient. However, sevoflurane, a well-tolerated aesthetic vapor, allows for elective inhalation induction of anaesthesia in adults. The next step of the induction process is securing the airway. This may be a

simple matter of manually holding the patient's jaw such that his or her natural breathing is unimpeded by the tongue, or it may demand the insertion of a prosthetic airway device such as a laryngeal mask airway or endotracheal tube. Various factors are considered when making this decision. The major decision is whether the patient requires placement of an endotracheal tube.

Appropriate levels of anaesthesia must be chosen both for the planned procedure and for its various stages. In complex plastic surgery, for example, a considerable period of time may elapse between the completion of the induction of anaesthesia and the incision of the skin. During the period of skin preparation, the patient is not receiving any noxious stimulus. This requires a very light level of anaesthesia, which must be converted rapidly to a deeper level just before the incision is made. As the procedure progresses, the level of anaesthesia is adjusted to provide the minimum amount of anaesthesia that is necessary to ensure adequate anaesthetic depth. This requires experience and judgment. The specialty of anaesthesiology is working to develop reliable methods to avoid cases of awareness under anaesthesia. As the surgical procedure draws to a close, the patient's emergence from anaesthesia is planned. Experience and close communication with the surgeon enable the anaesthesia provider to predict the time when the operation will be complete.

When inducing general anaesthesia, the patient is no longer able to protect their airway or provide an effective respiratory effort. The goal of care is to provide adequate ventilation and oxygenation during general anaesthesia. Patients are evaluated in the preoperative period for the signs of difficult mask ventilation and/or intubation. Positioning is especially important in morbidly obese patients. The body habitus of these patients can make them difficult to ventilate and intubate.

Questions 15-22

15) Which of the following statements about general anaesthetic is not mentioned in the first paragraph?

- a) patients don't have to suffer unbearable pain
- b) doctors and surgeons can manipulate a patient's limbs more easily
- c) muscles can be paralysed
- d) patients can wake up with pleasant memories

16) According to the second paragraph, general anaesthetic can be administered

- a) by inhalation only
- b) by breathing it in and/or by intravenous routes
- c) by oral medicines
- d) by intravenous route only

17)What can the pre-operative phase achieve?

- a) it can lead to reduced psychological stress
- b) it can produce adverse reactions
- c) it can result in higher than anticipated cancellation rates
- d) it can bring on cardiac and pulmonary complications

18)What has caused critical complications with general anaesthesia?

- a) unnecessarily long fasting times
- b) avoidable cases of dehydration
- c) operations that were delayed until correct fasting times were met
- d) drugs that the patient had taken that the medical team were unaware of

19)What is the main objective of premedication?

- a) to enable a patient to tolerate anaesthetics such as morphine or scopolamine
- b) use a short-acting benzodiazepine to cause drowsiness
- c) to prepare a patient for surgery
- d) to guarantee antegrade preparations

20)What normally follows the introduction of anaesthesia?

- a) prosthetic airway devices are used to manipulate the patient's tongue
- b) priority should be given to the patient's ability to breath
- c) laryngeal movements are monitored and adjusted
- d) sevoflurane vapor is usually used to maintain control of the unconscious patient

21)According to the seventh paragraph, what could complicate the amount of anaesthesia to be given?

- a) the presence of unwanted noxious stimulus
- b) experience and judgment of the anaesthetist
- c) a breakdown of communication between the surgeon and the anaesthetist
- d) the interval between initial induction and the commencement of surgery

22)In the last paragraph, what factor should be evaluated when inducing general anaesthesia?

- a) how much the patient weighs
- b) if the patient has an unprotected airway
- c) the appropriate level of oxygenation
- d) whether the patient has opted for mask ventilation